Linear Algebra for Biomedical Engineering

Tutorial notes, explanations and solutions



Editing and teaching: Chen Gadi

Course website, materials and questions

`https://chengadi.science`

These notes may contain errors, inaccuracies, omissions or imperfect wording.
Use the website linked above for comments, corrections and questions.
The lectures and official course announcements determine the required material.

My sincere thanks to everyone who contributed in any way through comments, corrections, explanations, ideas and support along the way, and to everyone who taught me, offered advice, sent templates, shared exercises or helped in any other way.

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The syllabus and course announcements determine the required material, submission deadlines and assessment.

Use the website system for corrections, questions and messages:

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How to Use These Notes

MOTIVATION AND GUIDING IDEA

Linear algebra provides a language for describing systems: constraints, signals, images, computational models, and maps between vector spaces. These notes are intended for engineering students across disciplines. The engineering examples explain the meaning of the calculations; the definitions and proofs state the conditions under which those calculations are valid.

A Study Routine

1. **Before the tutorial:** Please review the definitions and basic examples from the lectures. Please state the given data, the space in which you are working, and the underlying field.
2. **While solving a problem:** Please attempt the problem before reading its solution. Please write a clear solution using the definitions and results covered in class.
3. **After the calculation:** Please check the result by substitution. Please check dimensions, boundary cases, and the geometric interpretation. A diagram can support understanding, but it does not replace a proof.
4. **At the end of a topic:** Please summarize a definition, a central theorem, an example, and a counterexample. Please distinguish necessary conditions from sufficient conditions.

The Boxes in These Notes

DEFINITION

Formal definition: Concepts, notation, and conditions for their use. Please read the assumptions as carefully as the conclusion.

RECALL AND TOOLS

Lecture reminder: Facts and tools used in the solutions.

NOTE

Mathematical note: A clarification of an assumption, a warning about a common error, or an explanation of a boundary case.

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EXAMPLE

Example: A short illustration of a concept or a calculation. Please check how the definition appears in the example.

NOTATION AND CONVENTIONS

An exercise number gives its tutorial and its position within that tutorial. For example, 4.9 is the ninth exercise in Tutorial 4. Exercise numbering is independent of definition and example numbering. The **FOUNDATION** tag marks practice with definitions and calculations. The **CHALLENGE** tag marks a problem that combines ideas or calls for a proof. The **EXAM QUESTION** tag identifies a question from an examination. Each tutorial begins with concepts, notation and tools, and ends with exercises for foundation practice and further study.

The Website and Course Announcements

Study materials and updates are available through chengadi.science. Please send messages, logistics requests, and questions about the material through the website's system. The detailed communication guidelines and group logistics appear at the beginning of Tutorial 1, in the section entitled ``Course Logistics.'' The website is also available at:

`https://chengadi.science`

The official syllabus and course announcements determine the required material, submission deadlines, assessment rules, and office hours. The accompanying homework booklet retains the original distinction between questions marked for submission and questions for independent practice. Please check the website to find out which questions are actually assigned for submission in the current course.

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Notation and Conventions

- $\mathbb{F}$ denotes a field; $\mathbb{R}$ and $\mathbb{C}$ denote the real and complex fields. Please state any field assumption required by a claim.
- A basis is an ordered list. Column j of a representing matrix contains the coordinates of the image of the jth basis vector.
- In a complex inner product space, the inner product is **linear in the first argument** and conjugate-linear in the second. In particular, the standard inner product is

$$\langle x, y \rangle = \sum_j x_j \overline{y_j}.$$

- For $A \in M_n(\mathbb{F})$, our characteristic polynomial convention is

$$p_A(t) = \det(tI - A).$$

  The alternative $\det(A - tI)$ may also be used, provided that the factor $(-1)^n$ is tracked consistently.

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TUTORIAL 1

Complex Numbers and Induction

1.1 Course Logistics

NOTE

The current International Engineering syllabus and official announcements determine the actual class arrangements, submission deadlines, office hours, and assessment rules. In particular, WeBWorK is optional practice and contributes neither to the grade nor to a protective component in the international course.

1.1.1 Access to Tutorials

Tutorials may be held over Zoom, and recordings may be available. The way links and recordings are made available may differ between staff members; they will usually be found on Moodle. Technical problems may prevent a tutorial from being made available. The course team will try to avoid such problems, but students remain responsible for catching up on any material missed because of a technical failure. Follow the current course announcements for the arrangements that apply to your group.

1.1.2 Homework and Assessment

- For a homework sheet with two parts, the first contains two questions marked for submission, to be solved independently and submitted for grading. The second contains additional questions for independent practice, which are not submitted. Full solutions to some of these questions are provided so that you can assess your own work. Check the current assignment instructions for the questions you must actually submit.
- The homework is based on material from the lectures and/or tutorials.
- Submitting work gives you professional feedback. The additional questions help you consolidate the material and practise solving relevant problems. You are strongly encouraged to attempt as many questions as possible.
- One possible publication schedule is Tuesday morning, with submissions accepted until Saturday evening of the following week (a submission window labelled as 12 days). The current course announcements determine the deadlines for the international program.
- Where WeBWorK sheets are provided, they may be released approximately every week or at a longer interval and may be open for two weeks. International Engineering students may use available WeBWorK questions as optional practice. The material may be in Hebrew; WeBWorK scores do not contribute to the international course grade or to any protective component.

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- The final assessment rules are stated in the current syllabus. Homework instructions appear in the relevant course section. Direct questions about grades and examinations to the course lecturer.

1.1.3 Office Hours

- Office hours with the teaching assistant are intended for questions arising from the tutorial material and homework.
- The teaching assistant sets the time, location, and format of office hours. Office hours are not an additional tutorial and do not replace independent study.

Procedures may change; updates will be announced when necessary.

1.1.4 Study Recommendations

Mathematics is a language, and learning it requires practice. This course is an integral (and differential) part of the mathematical foundations that support your later academic studies. The following suggestions are intended to help you learn the material and manage the course.
Scope of this advice: The recommendations in this section do not make the author responsible for readers' study decisions, understanding, or application of the material. Each reader remains responsible for their own learning and use of the information.

- Attempt all questions marked for submission and all practice questions; they have been selected carefully.
- If you fall slightly behind, use the recordings and tutorials to catch up, and check that you can solve the homework questions on the same topic.
- If you fall substantially behind, prioritize the theoretical foundations and basic practice questions, and try to rejoin the class's pace as soon as possible.
- Consider making a ``three magic sheets'' summary: one sheet for definitions, one for propositions and theorems, and one for useful techniques.
- A competitive atmosphere does not help the group or the individual. Please help one another. Group learning and support for classmates can strengthen both learning and resilience.

1.1.5 Clarifications, Personal Recommendations, and Expectations

- Please send messages and questions through the website system at chengadi.science.

1. Logistics requests should include your full name, student ID, and course name. Requests lacking these details will not be processed. If you have received no response within three days, I may have missed your request; please send a reminder through the system and include or quote the original request.
2. Questions about the material: in previous years and semesters, I received a very large number of questions (at least 20 requests per day). Questions may be sent through the website system, subject to the following guidelines:
   (a) First make an independent effort to understand the issue, using notes, books, the internet, and tools such as ChatGPT. Check mathematical claims independently.

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(b) Keep your message short and specific. Include the course name and a question that can be answered in one or two lines. Attach an image of the relevant material where possible, rather than referring only to a particular theorem.

(c) Do not send requests such as ``Is my method correct?'' or ``Check this part.'' I will not grade homework solutions through messages. You can discuss an approach during office hours or obtain feedback through the homework grader.

(d) You may ask for clarification of the wording of a homework question.

(e) I will try to answer within a few days. Messages that are too long, or that require more than a few minutes to read and answer, may not receive a response. Thank you for your understanding.

- I may upload videos of myself solving problems on YouTube. These are an additional resource for students, rather than part of my formal responsibilities; please do not send requests about their production or availability.
- During tutorials, I will not summarize all the theorems and conclusions covered in lectures. I recommend preparing your own summary. Tutorials are intended to practise the lecture material and work on somewhat more demanding questions that encourage thought and independent learning.
- I will also make sure to include basic exercises.
- Time will be set aside for questions during class. I will not repeatedly answer questions that have already been addressed.
- **The lectures determine the required material.**
  It is not possible to cover every topic in the tutorials or practise every type of question on each topic.
- I strongly recommend exploring the material through the resources available to you: books, online sources, earlier summaries (for example, materials from the students' association), office hours, and similar resources. **Each student is responsible for their own learning.**
- Tutorial pace and presentation: I will try to work at a reasonable pace, write clearly, and use headings. Please let me know if the presentation needs adjustment.
- WeBWorK is optional and recommended as additional practice with feedback. **For the International Engineering Program, WeBWorK does not count toward the final grade and does not provide protective credit.**
- I will try to provide examples, proofs, and counterexamples for each topic.
- I will conduct an anonymous mid-semester teaching survey, and I welcome feedback.
- I will take questions at the designated times rather than in the middle of solving an exercise.
- I will try to upload typed tutorial sheets before the tutorials, but they may contain errors or imprecise statements.
  The material presented in the lectures, both in writing and orally, remains authoritative.
- Come prepared: review the definitions and material covered in the lectures before each tutorial.

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MOTIVATION AND GUIDING IDEA

Cartesian form is well suited to addition and comparison of components; polar form is well suited to multiplication, powers, and roots. Before calculating, choose a representation that fits the requested operation. For induction, first choose the precise statement and the first value for which it is supposed to hold.

NOTATION AND CONVENTIONS

For $z = x + iy \in \mathbb{C}$, the symbols $\operatorname{Re} z = x$ and $\operatorname{Im} z = y$ denote real numbers; in particular, the imaginary part is y, not iy. Both z^* and $\overline{z}$ denote the conjugate $x - iy$. The modulus $|z|$ is a nonnegative real number. When $z \neq 0$, $\arg z$ refers to an argument modulo integer multiples of 2π, whereas $\operatorname{Arg} z \in [0, 2\pi)$ is the fixed choice in these notes. The notation $a \mid b$ means that $b = am$ for some $m \in \mathbb{Z}$. The expression $P(n) \Rightarrow P(n+1)$ is a logical implication; it does not by itself establish the base case.

DEFINITION 1.1

Existence, uniqueness, and conditions of use. Every $z \in \mathbb{C}$ has a unique representation $z = x + iy$ with $x, y \in \mathbb{R}$. Thus two complex numbers are equal if and only if their real parts are equal and their imaginary parts are equal. For $z \neq 0$, its inverse exists and is unique, since $zz^* = |z|^2 > 0$; check this condition before dividing by z. If $r > 0$ and $\theta \in \mathbb{R}$, de Moivre's formula gives

$$(re^{i\theta})^n = r^n e^{in\theta} \qquad (n \in \mathbb{Z}).$$

Negative powers require $r > 0$. For nonnegative integer powers, the formula also holds when $r = 0$, except for 0^0, which is not defined here. When finding nth roots of a nonzero number, for $n \geq 1$, list all n roots rather than selecting a single angle.

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How to use these notes

These notes are intended for students in all engineering disciplines. They combine definitions, guiding ideas, examples, and detailed solutions.

MOTIVATION AND GUIDING IDEA

Objectives. We will convert between Cartesian and polar forms, calculate powers and roots, and interpret operations in the complex plane. We will then formulate an induction proof, stating the domain on which its claim holds.

MOTIVATION AND GUIDING IDEA

A working habit. In every solution, state what is given, what must be established, which definition or result you use, and why every operation is permitted. Correct calculations and clear explanations complement one another.

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1.2 Complex numbers: Cartesian form

DEFINITION 1.2

The imaginary unit i satisfies $i^2 = -1$.
A complex number has the form $z = x + iy$, where $x, y \in \mathbb{R}$.

1. If $\mathrm{Re}(z) = x = 0$, then z is purely imaginary.
2. If $\mathrm{Im}(z) = y = 0$, then z is real.

EXERCISE 1.1

Please solve $z^2 + 2z + 17 = 0$ over $\mathbb{C}$. Please factor it into linear factors over $\mathbb{C}$ and please determine whether such a factorization exists over $\mathbb{R}$.

WORKED SOLUTION

The discriminant is $\Delta = 2^2 - 4 \cdot 17 = -64$. Hence

$$z_{1,2} = \frac{-2 \pm \sqrt{-64}}{2} = -1 \pm 4i, \qquad z^2 + 2z + 17 = (z + 1 - 4i)(z + 1 + 4i).$$

The negative discriminant gives no real root, so there is no factorization into two real linear factors.

DEFINITION 1.3

- **Complex conjugate:** for $z = x + iy$, write $z^* := x - iy$.
- **Modulus:** $|z| = \sqrt{x^2 + y^2} \in \mathbb{R}$.
- **Multiplicative inverse:** for $z \neq 0$, $z^{-1} = \dfrac{z^*}{|z|^2}$.

EXERCISE 1.2

Please prove the following statements. **(a)** $\mathrm{Re}\, z = \dfrac{z + z^*}{2}$ and $\mathrm{Im}\, z = \dfrac{z - z^*}{2i}$. **(b)** $|z| \geq \mathrm{Im}\, z$ and $|z| \geq \mathrm{Re}\, z$ for every $z \in \mathbb{C}$. **(c)** (At home) $(z^*)^* = z$. **(d)** (At home) If z is real, then $z^* = z$.

WORKED SOLUTION

Write $z = x + iy$. Then

$$\frac{z + z^*}{2} = x = \mathrm{Re}\, z, \qquad \frac{z - z^*}{2i} = y = \mathrm{Im}\, z,$$

and

$$|z| = \sqrt{x^2 + y^2} \geq |x| \geq x = \mathrm{Re}\, z, \qquad |z| \geq |y| \geq y = \mathrm{Im}\, z.$$

(c) $(z^*)^* = (x - iy)^* = x + iy = z$.
(d) If z is real, then $y = 0$, and hence $z^* = x - 0i = x = z$.

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1.2.1 Useful properties

1. $|z| = |z^*|$.
2. $z \cdot z^* = |z|^2$ (multiplication by the conjugate gives a real number).
3. $|zw| = |z|\,|w|$.
4. $(zw)^* = z^* w^*$.

EXERCISE 1.3

For $z \neq 0$, please prove that $(z^{-1})^* = (z^*)^{-1}$.

WORKED SOLUTION

Since $z \neq 0$, we also have $z^* \neq 0$ and $|z|^2 > 0$. The inverse formula gives

$$(z^{-1})^* = \left(\frac{z^*}{|z|^2}\right)^* = \frac{z}{|z|^2}, \qquad (z^*)^{-1} = \frac{(z^*)^*}{|z^*|^2} = \frac{z}{|z|^2}.$$

Both sides equal the same expression, so $(z^{-1})^* = (z^*)^{-1}$.

1.3 Polar form

1.3.1 Converting between Cartesian and polar forms

DEFINITION 1.4

For every $z = x + iy \neq 0$, there are $r = |z| > 0$ and an angle θ such that

$$z = r(\cos\theta + i\sin\theta) = re^{i\theta}, \qquad x = r\cos\theta, \quad y = r\sin\theta.$$

Conversely, $r = \sqrt{x^2 + y^2}$. If $x \neq 0$, then $\tan\theta = y/x$, but this ratio alone does not determine the quadrant. **Remarks:**

- The angle is determined up to adding $2\pi k$, where $k \in \mathbb{Z}$. For $z = 0$, we have $r = 0$, and no argument is defined.
- All angles in exponents $e^{i\theta}$ are measured in radians. Degrees are used only for numerical illustration.

EXERCISE 1.4

Please convert $z = 5+5i$ and $w = -2+2i$ to polar form. Please write $a = 10e^{2i\pi/3}$ in Cartesian form. Please choose arguments in $[0, 2\pi)$.

WORKED SOLUTION

We have $|z| = \sqrt{25+25} = 5\sqrt{2}$, so $z = 5\sqrt{2}e^{i\pi/4}$. Similarly $|w| = \sqrt{4+4} = 2\sqrt{2}$. Its second-quadrant position gives $w = 2\sqrt{2}e^{3i\pi/4}$. Euler's identity gives

$$a = 10\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right) = 10\left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = -5 + 5\sqrt{3}i.$$

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1.3.2 Arithmetic operations

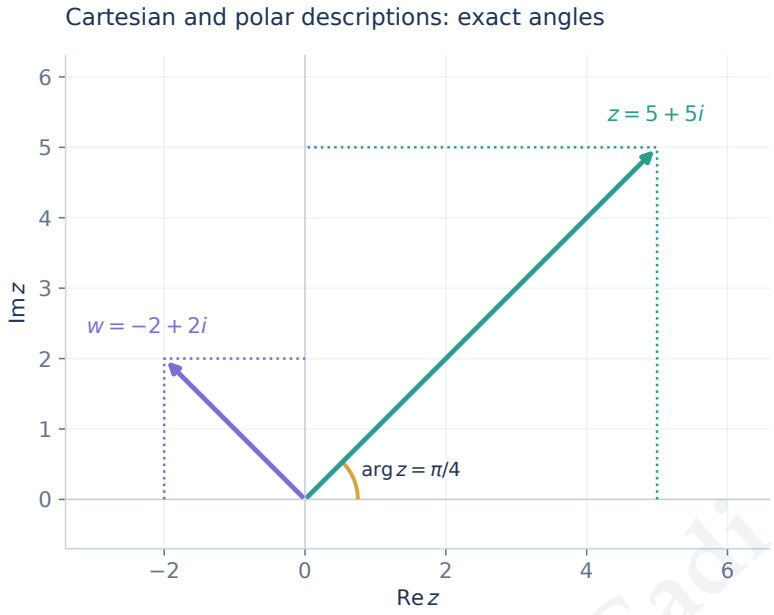


Figure 1.1. Cartesian and polar forms in the same plane: the modulus is the distance from the origin, and the argument is the direction angle.

DEFINITION 1.5

Multiplication: $(r_1 e^{i\theta_1})(r_2 e^{i\theta_2}) = r_1 r_2\, e^{i(\theta_1+\theta_2)}$.

Division (when the denominator is nonzero): $\dfrac{r_1 e^{i\theta_1}}{r_2 e^{i\theta_2}} = \dfrac{r_1}{r_2}\, e^{i(\theta_1-\theta_2)}$.

Conjugation: $(re^{i\theta})^* = r\, e^{-i\theta}$.

Periodicity: $re^{i(\theta+2\pi k)} = re^{i\theta}$, $re^{i(\theta+\pi)} = -re^{i\theta}$.

EXERCISE 1.5

Let $z = 2+2i$ and $w = -3+3i$. Please calculate zw and w/z using polar and Cartesian forms. Please check their moduli as well.

WORKED SOLUTION

Write $z = 2\sqrt{2}e^{i\pi/4}$ and $w = 3\sqrt{2}e^{3i\pi/4}$. Then

$$zw = 12e^{i\pi} = -12, \qquad \frac{w}{z} = \frac{3}{2}e^{i\pi/2} = \frac{3}{2}i.$$

In Cartesian form,

$$(2+2i)(-3+3i) = -6+6i-6i+6i^2 = -12,$$

$$\frac{-3+3i}{2+2i} = \frac{(-3+3i)(2-2i)}{8} = \frac{-6+12i+6}{8} = \frac{3}{2}i.$$

The moduli are $|z||w| = 2\sqrt{2}\cdot 3\sqrt{2} = 12$ and $|w|/|z| = 3/2$, as expected.

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1.3.3 Powers

EXERCISE 1.6

Please calculate $(1-i)^{16}$ using polar form. Please verify the result using $(1-i)^4$.

WORKED SOLUTION

Since $1-i=\sqrt{2}e^{-i\pi/4}$,

$$(1-i)^{16}=(\sqrt{2})^{16}e^{-4\pi i}=2^8=256.$$

For a check, $(1-i)^2=-2i$ and $(1-i)^4=(-2i)^2=-4$, so $(1-i)^{16}=(-4)^4=256$.

1.3.4 Complex roots

DEFINITION 1.6

Let $0\neq z\in\mathbb{C}$, and let $n\geq 1$ be an integer. The equation $w^n=z=re^{i\theta}$ has n distinct roots:

$$w_j=\sqrt[n]{r}\,e^{i\frac{\theta+2\pi j}{n}},\quad j=0,1,\ldots,n-1.$$

Geometric meaning: the roots lie on the circle of radius $\sqrt[n]{r}$ at equal angular intervals of $2\pi/n$. For $n\geq 3$, they are the vertices of a regular polygon; for $n=2$, they are two opposite points, and for $n=1$, there is one point.

EXERCISE 1.7

Please find all solutions of $z^3=8i$.

WORKED SOLUTION

Write $8i=8e^{i\pi/2}$. The three solutions are

$$z_j=2e^{i(\pi/6+2\pi j/3)},\qquad j=0,1,2.$$

Their angles are $\pi/6, 5\pi/6, 3\pi/2$, with radius 2. In Cartesian form they are $\sqrt{3}+i, -\sqrt{3}+i, -2i$. Cubing any of them gives $8e^{i\pi/2}=8i$.

EXERCISE 1.8

Let $n\geq 2$. Please find all nth roots of unity and please prove that their sum is zero. What happens when $n=1$?

WORKED SOLUTION

The roots are $w_j=e^{i\frac{2\pi j}{n}}$, $j=0,\ldots,n-1$. Put $w=e^{i\frac{2\pi}{n}}$. Then $w_j=w^j$, and therefore

$$\sum_{j=0}^{n-1}w^j=\frac{w^n-1}{w-1}=0,$$

because $w^n=1$ and $w\neq 1$ when $n\geq 2$. For $n=1$, the only root is 1, so the sum is 1, rather than zero.

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EXERCISE 1.9

(At home) Consider $a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 = 0$, with real coefficients. Please prove that if z is a solution, then z^* is also a solution.

WORKED SOLUTION

Use the identities $(z_1 + \cdots + z_m)^* = z_1^* + \cdots + z_m^*$ and $(z_1 \cdots z_m)^* = z_1^* \cdots z_m^*$, together with $a^* = a$ for real a. If $a_0 + a_1 z + \cdots + a_n z^n = 0$, conjugating gives $a_0 + a_1 z^* + \cdots + a_n (z^*)^n = 0$. Thus z^* is a solution.

1.4 Further properties of complex numbers

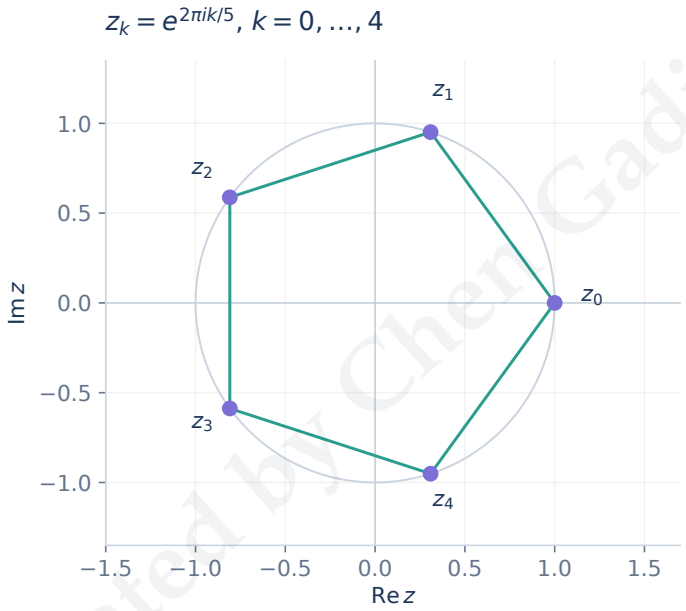


Figure 1.2. The roots of unity are equally spaced around the unit circle; their sum is zero whenever their order is greater than one.

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1.4.1 Multiplication by i

Multiplication by i rotates a complex number through $\frac{\pi}{2}$ counterclockwise.

EXERCISE 1.10

Please prove that multiplication by i gives a rotation through $\frac{\pi}{2}$.

WORKED SOLUTION

Cartesian representation:
Let $z = a + bi$. Multiplication by i gives

$$z \cdot i = (a + bi)i = -b + ai.$$

The vector (a, b) becomes $(-b, a)$. This is exactly a counterclockwise rotation through $\pi/2$ about the origin: the new x coordinate is $-b$, and the new y coordinate is a.

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EXERCISE 1.10 CONTINUED

Polar representation:
For $z \neq 0$, write $z = re^{i\theta}$, where $r = |z|$ and θ is an angle from the positive x axis to z. Then

$$z \cdot i = re^{i\theta} \cdot i = re^{i\theta} \cdot e^{i\frac{\pi}{2}} = re^{i(\theta+\frac{\pi}{2})}.$$

Multiplication by i adds $\frac{\pi}{2}$ to θ, so z rotates through $\frac{\pi}{2}$ about the origin, as required. The origin is fixed, since $0 \cdot i = 0$.

1.4.2 Division in polar form

DEFINITION 1.7

If $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$, with $r_2 \neq 0$, then

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}.$$

EXERCISE 1.11

Please calculate z_1/z_2 for $z_1 = 9e^{5i\pi/6}$ and $z_2 = 3e^{i\pi/3}$.

WORKED SOLUTION

Divide the moduli and subtract the angles:

$$\frac{z_1}{z_2} = \frac{9}{3} e^{i(5\pi/6 - \pi/3)} = 3e^{i\pi/2} = 3i.$$

EXERCISE 1.12

Let $z_1, z_2 \neq 0$. Please prove that an argument of z_1/z_2 is the difference of arguments of z_1 and z_2, up to an integer multiple of 2π.

WORKED SOLUTION

Write $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$, where $r_1, r_2 > 0$. Division gives

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}.$$

Thus $\theta_1 - \theta_2$ is an argument of the quotient. If an argument in a prescribed interval is required, add a suitable integer multiple of 2π.

1.4.3 Choosing an argument in a fixed interval

In these notes, $\mathrm{Arg}(z)$ denotes the unique argument in $[0, 2\pi)$, for $z \neq 0$.

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RECALL AND TOOLS

The equalities $\text{Arg}(z_1 \cdot z_2) = \text{Arg}(z_1) + \text{Arg}(z_2)$ and $\text{Arg}\left(\frac{z_1}{z_2}\right) = \text{Arg}(z_1) - \text{Arg}(z_2)$ do not always hold, because the expression on the right may lie outside $[0, 2\pi)$.

EXERCISE 1.13

Please calculate $\text{Arg}(z_1 z_2)$ for $z_1 = e^{7i\pi/6}$ and $z_2 = e^{11i\pi/6}$.

WORKED SOLUTION

The product is

$$z_1 z_2 = e^{i(7\pi/6+11\pi/6)} = e^{3\pi i} = -1.$$

Since 3π lies outside $[0, 2\pi)$,

$$\text{Arg}(z_1 z_2) = \pi, \qquad \text{Arg}(z_1) + \text{Arg}(z_2) = 3\pi.$$

The two values differ by 2π.

1.4.4 Calculating arguments and identifying quadrants

DEFINITION 1.8

For a complex number $z = a + bi \neq 0$ with $a \neq 0$, every argument θ satisfies

$$\tan(\theta) = \frac{b}{a}.$$

To determine θ precisely, take the quadrant into account:

- If $a > 0, b > 0$, then $\theta = \arctan\left(\frac{b}{a}\right)$.
- If $a < 0, b > 0$, add π to obtain an angle in the second quadrant: $\theta = \pi + \arctan\left(\frac{b}{a}\right)$.
- If $a < 0, b < 0$, add π for the third quadrant: $\theta = \pi + \arctan\left(\frac{b}{a}\right)$.
- If $a > 0, b < 0$, add 2π to obtain a positive angle in the fourth quadrant: $\theta = 2\pi + \arctan\left(\frac{b}{a}\right)$.

If $b = 0, a > 0$, then $\text{Arg}(z) = 0$; if $b = 0, a < 0$, then $\text{Arg}(z) = \pi$. If $a = 0$, the angle depends only on b:

- If $b > 0$, then $\theta = \frac{\pi}{2}$.
- If $b < 0$, then $\theta = \frac{3\pi}{2}$.

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EXERCISE 1.14

Please find the argument in $[0, 2\pi)$ of $z = -\sqrt{3} + i$.

WORKED SOLUTION

The number lies in the second quadrant, and

$$\tan\theta = \frac{1}{-\sqrt{3}} = -\frac{1}{\sqrt{3}}.$$

Its reference angle is $\pi/6$, so $\theta = \pi - \pi/6 = 5\pi/6$.

EXERCISE 1.15

Please find the argument in $[0, 2\pi)$ of $z = -6 - 2\sqrt{3}i$.

WORKED SOLUTION

The number lies in the third quadrant, and

$$\tan\theta = \frac{-2\sqrt{3}}{-6} = \frac{1}{\sqrt{3}}.$$

Its reference angle is $\pi/6$, so $\theta = \pi + \pi/6 = 7\pi/6$.

1.5 The induction principle

DEFINITION 1.9

The principle of mathematical (weak) induction:
Let n_0 be a nonnegative integer, and let $P(n)$ be a statement defined for every integer $n \geq n_0$. Suppose that:

1. **Base case:** $P(n_0)$ is true.
2. **Inductive step:** for every integer $n \geq n_0$, the assumption $P(n)$ implies $P(n+1)$.

Then $P(n)$ is true for every integer $n \geq n_0$.

EXERCISE 1.16

Please prove by induction that $1 + 3 + \cdots + (2n-1) = n^2$ for every $n \geq 1$.

WORKED SOLUTION

For $n = 1$, the identity is $1 = 1^2$. Suppose that for $k \geq 1$,

$$\sum_{j=1}^{k}(2j-1) = k^2.$$

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EXERCISE 1.16 CONTINUED

Adding the next term gives

$$\sum_{j=1}^{k+1}(2j-1) = k^2 + (2k+1) = (k+1)^2.$$

Induction proves the claim for every $n \geq 1$.

EXERCISE 1.17

Please prove by induction that $11 \mid (12^n - 1)$ for every $n \geq 1$.

WORKED SOLUTION

For $n = 1$, $12 - 1 = 11$. Suppose $12^k - 1 = 11m$ for some $m \in \mathbb{Z}$. Then

$$12^{k+1} - 1 = 12(12^k - 1) + 11 = 12 \cdot 11m + 11 = 11(12m + 1).$$

This completes the induction step. As a check, $12 \equiv 1 \pmod{11}$.

EXERCISE 1.18

Please prove by induction that $3^n \geq n^2$ for every $n \geq 2$.

WORKED SOLUTION

For $n = 2$, $3^2 = 9 \geq 4 = 2^2$. Suppose $3^k \geq k^2$ for $k \geq 2$. Then

$$3^{k+1} \geq 3k^2.$$

Moreover,

$$3k^2 - (k+1)^2 = 2k(k-1) - 1 \geq 3 > 0,$$

so $3^{k+1} \geq (k+1)^2$. Induction proves the claim for all $n \geq 2$.

1.6 Supplementary exercises: geometry and signal calibration

EXERCISE 1.19

Please find and please describe geometrically all $z \in \mathbb{C}$ satisfying $|z-2| = |z+2|$. Please find which of them also satisfy $|z| = 5$.

WORKED SOLUTION

Writing $z = x + iy$ and squaring gives

$$(x-2)^2 + y^2 = (x+2)^2 + y^2 \iff x^2 - 4x + 4 = x^2 + 4x + 4 \iff x = 0.$$

This is the imaginary axis, the perpendicular bisector of the segment joining -2 and 2. The extra condition is $y^2 = 25$, giving $z = 5i$ or $z = -5i$.

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EXERCISE 1.20

Let $T(z) = (2 - 2i)z$. Please determine its effect on modulus and angle. Please calculate $T(1 + i)$ and please find an input whose output is $8i$.

WORKED SOLUTION

Since $2 - 2i = 2\sqrt{2}e^{-i\pi/4}$, the map multiplies moduli by $2\sqrt{2}$ and rotates clockwise through $\pi/4$. The origin stays fixed. Directly,

$$T(1 + i) = (2 - 2i)(1 + i) = 2 + 2i - 2i - 2i^2 = 4.$$

To obtain $8i$, divide by the nonzero multiplier:

$$z = \frac{8i}{2 - 2i} = \frac{8i(2 + 2i)}{8} = -2 + 2i.$$

Check: $(2 - 2i)(-2 + 2i) = -4 + 8i + 4 = 8i$.

Additional practice

EXERCISE 1.21 FOUNDATION

Let $z = 5 - 12i$. Please calculate $|z|$, z^* and z^{-1}. Please solve $zw = 22 - 19i$ and please verify by direct multiplication.

WORKED SOLUTION

We have $|z| = \sqrt{25 + 144} = 13$, so $z^* = 5 + 12i$ and $z^{-1} = (5 + 12i)/169$. Hence

$$w = \frac{22 - 19i}{5 - 12i} = \frac{(22 - 19i)(5 + 12i)}{169} = \frac{338 + 169i}{169} = 2 + i.$$

Check: $(5 - 12i)(2 + i) = 10 + 5i - 24i - 12i^2 = 22 - 19i$. The solution is unique because $z \neq 0$.

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EXERCISE 1.22 CHALLENGE

Let $z \in \mathbb{C} \setminus \{-1\}$. Please prove that $(z-1)/(z+1)$ is purely imaginary if and only if $|z| = 1$. Please solve $(z-1)/(z+1) = 3i$ and please check the geometric condition.

WORKED SOLUTION

Multiply by the conjugate of the denominator. Since $z \neq -1$, $|z+1|^2 > 0$ and

$$\frac{z-1}{z+1} = \frac{|z|^2 - 1 + z - z^*}{|z+1|^2}.$$

The term $z - z^*$ is purely imaginary, so

$$\operatorname{Re}\left(\frac{z-1}{z+1}\right) = \frac{|z|^2 - 1}{|z+1|^2}.$$

This vanishes if and only if $|z| = 1$, including $z = 1$ where the quotient is zero. The given equation yields $z - 1 = 3iz + 3i$, hence

$$z = \frac{1+3i}{1-3i} = \frac{(1+3i)^2}{10} = \frac{-4+3i}{5}.$$

This differs from -1 and has modulus $\sqrt{16+9}/5 = 1$. Finally,

$$\frac{z-1}{z+1} = \frac{-9+3i}{1+3i} = 3i.$$

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EXERCISE 1.23 CHALLENGE

Let $a, b, c \in \mathbb{C}$ be pairwise distinct. Please prove that their points form an equilateral triangle if and only if

$$a^2 + b^2 + c^2 = ab + bc + ca.$$

WORKED SOLUTION

Since $b \neq a$, put $w = (c-a)/(b-a)$. The map $z \mapsto (z-a)/(b-a)$ sends the points to $0, 1, w$ and multiplies every distance by $1/|b-a|$, so it preserves equilateral triangles. Substituting $c = a + (b-a)w$ gives

$$a^2 + b^2 + c^2 - ab - bc - ca = (b-a)^2(w^2 - w + 1).$$

Thus the identity is equivalent to $w^2 - w + 1 = 0$, whose roots are $w = 1/2 \pm i\sqrt{3}/2$. Both satisfy $|w| = |w-1| = 1$. Conversely, if $0, 1, w$ form an equilateral triangle and $w = u + iv$, then

$$u^2 + v^2 = 1, \qquad (u-1)^2 + v^2 = 1.$$

Subtracting yields $u = 1/2$ and then $v^2 = 3/4$. These are exactly the two roots above, so the identity follows.

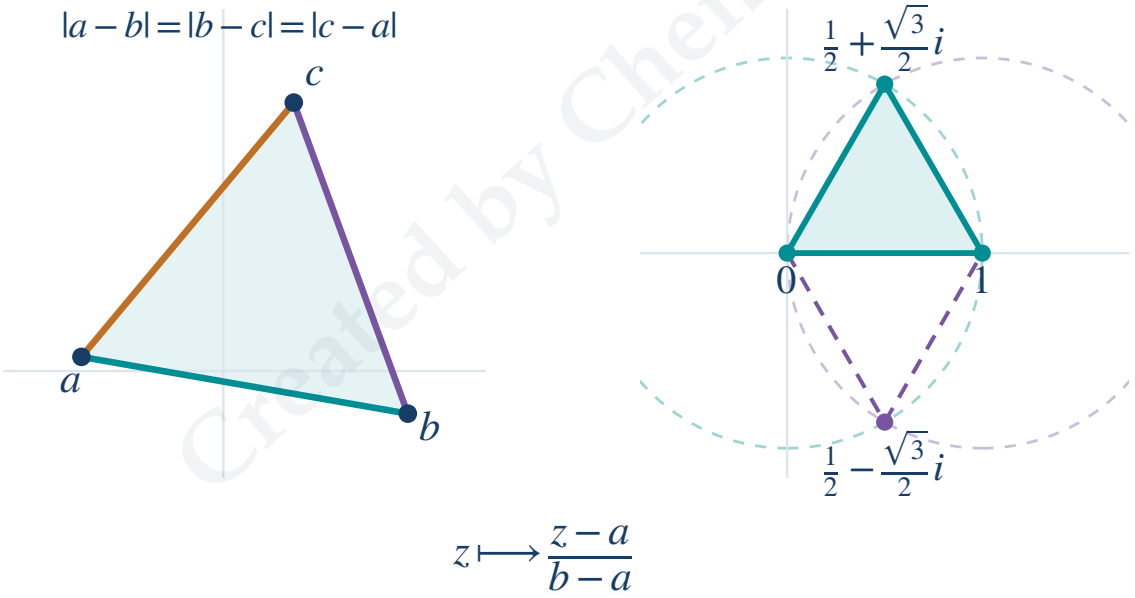


Figure 1.3. The map $z \mapsto (z-a)/(b-a)$ sends the vertices to $0, 1, w$. The two possible third vertices are $w = \frac{1}{2} \pm \frac{\sqrt{3}}{2}i$.

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EXERCISE 1.24 CHALLENGE

Let $a, b, c \in \mathbb{C}$ satisfy $|a| = |b| = |c| = 1$. Please prove that $|a + b + c| = 1$ if and only if at least one of $a + b, a + c, b + c$ is zero.

WORKED SOLUTION

Put $s = a + b + c$. Since each of a, b, c has modulus 1, its conjugate equals its reciprocal. Thus

$$ab + ac + bc = abc\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) = abc\,\overline{s}.$$

Expanding gives

$$\begin{aligned}(a + b)(a + c)(b + c) &= (a + b + c)(ab + ac + bc) - abc \\ &= abc(s\overline{s} - 1) = abc(|s|^2 - 1).\end{aligned}$$

If $|s| = 1$, the product on the left is zero, so at least one factor is zero. Conversely, if for example $a + b = 0$, then $s = c$ and $|s| = 1$. The other two pairs give the same conclusion.

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TUTORIAL 2

Mathematical Language, Sets, and Proofs

2.1 The induction principle

DEFINITION 2.1

The principle of mathematical (weak) induction:
Let n_0 be a nonnegative integer, and let $P(n)$ be a statement defined for every integer $n \geq n_0$. Suppose that:

1. **Base case:** $P(n_0)$ is true.
2. **Inductive step:** for every integer $n \geq n_0$, the assumption $P(n)$ implies $P(n+1)$.

Then $P(n)$ is true for every integer $n \geq n_0$.

EXERCISE 2.1

Please prove by induction that $2 + 4 + \cdots + 2n = n(n+1)$ for every $n \geq 1$.

WORKED SOLUTION

For $n = 1$, $2 = 1 \cdot 2$. Suppose $\sum_{j=1}^{k} 2j = k(k+1)$ for $k \geq 1$. Then

$$\sum_{j=1}^{k+1} 2j = k(k+1) + 2(k+1) = (k+1)(k+2).$$

Induction proves the identity for every $n \geq 1$.

EXERCISE 2.2

Please prove by induction that $17 \mid (18^n - 1)$ for every $n \geq 1$.

WORKED SOLUTION

For $n = 1$, $18 - 1 = 17$. Suppose $18^k - 1 = 17m$ for some $m \in \mathbb{Z}$. Then

$$18^{k+1} - 1 = 18(18^k - 1) + 17 = 17(18m + 1).$$

This is divisible by 17, completing the induction step and the proof.

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EXERCISE 2.3

Please prove by induction that $4^n \geq 3n^2$ for every $n \geq 3$.

WORKED SOLUTION

For $n = 3$, $4^3 = 64 \geq 27 = 3 \cdot 3^2$. Suppose $4^k \geq 3k^2$ for $k \geq 3$. Then

$$4^{k+1} \geq 12k^2.$$

Since $4k^2 - (k+1)^2 = 3k^2 - 2k - 1 = (3k+1)(k-1) > 0$,

$$4^{k+1} \geq 12k^2 \geq 3(k+1)^2.$$

The induction proves the claim for every $n \geq 3$.

2.2 Sets

DEFINITION 2.2

- **Inclusion:** $A \subseteq B$ means that every element of A also belongs to B.
- **Union:** $A \cup B = \{\, x \mid x \in A \text{ or } x \in B \,\}$.
- **Intersection:** $A \cap B = \{\, x \mid x \in A \text{ and } x \in B \,\}$.

$$\Omega = \{2,4,6,8,10\}, \quad A = \{2,4,6\}, \quad B = \{4,6,8\}$$

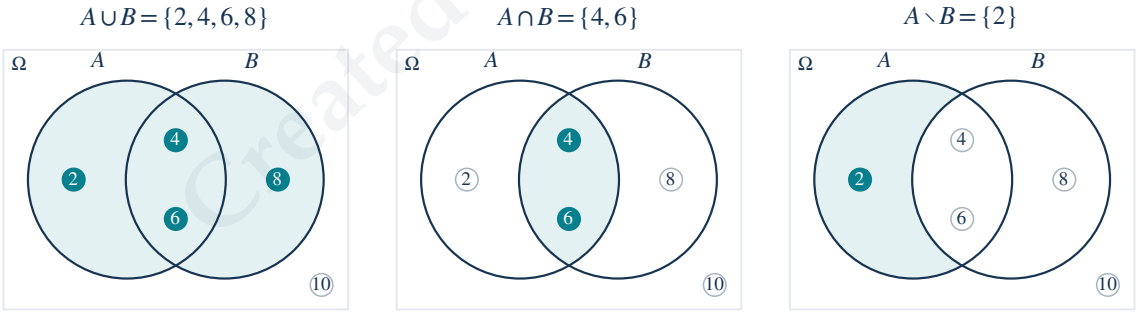


Figure 2.1. For $A = \{2,4,6\}$ and $B = \{4,6,8\}$ in $\Omega = \{2,4,6,8,10\}$, the shaded regions show $A \cup B$, $A \cap B$, and $A \setminus B$, respectively. Each numeral's position records its membership; 10 belongs to the universal set but to neither A nor B.

EXERCISE 2.4

Please prove that $A \subseteq B$ if and only if $A \cup B = B$.

WORKED SOLUTION

First direction: suppose $A \subseteq B$. The definition of union gives $A \cup B \subseteq B$, while $B \subseteq A \cup B$ always holds. Therefore $A \cup B = B$.
Second direction: suppose $A \cup B = B$. Let $x \in A$. Then $x \in A \cup B$, so $x \in B$. Therefore $A \subseteq B$.

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EXERCISE 2.5

Please prove that $A \subseteq B$ if and only if $A \cap B = A$.

WORKED SOLUTION

First direction: suppose $A \subseteq B$. If $x \in A \cap B$, then $x \in A$ and $x \in B$, so $A \cap B \subseteq A$. Conversely, if $x \in A$, then the inclusion gives $x \in B$. Thus $x \in A \cap B$, proving $A \subseteq A \cap B$. Hence $A \cap B = A$.

Second direction: suppose $A \cap B = A$. Let $x \in A$. Then $x \in A \cap B$, so $x \in B$. Therefore $A \subseteq B$.

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TUTORIAL 3

Fields and Finite Fields

MOTIVATION AND GUIDING IDEA

Objectives. We will strengthen proofs by induction and proofs of equivalence, identify the field axioms, and calculate in finite fields. We will distinguish familiar real arithmetic from arithmetic adapted to the specified field.

The first part is a focused review of proof methods and the language of sets. The second part introduces fields and finite field arithmetic; the order used in class will follow the progress of the lectures.

MOTIVATION AND GUIDING IDEA

A proof must work within the given setting: equality of sets is proved by two inclusions, and a field identity is proved from the field axioms. In a finite field, addition, subtraction, multiplication, and division by nonzero elements are permitted; numerical notation does not imply an ordering like that of $\mathbb{R}$.

NOTATION AND CONVENTIONS

The symbol $x \in A$ expresses membership of an element, whereas $A \subseteq B$ expresses inclusion of sets. The quantifiers $\forall$ and $\exists$ mean ``for every'' and ``there exists''; their order matters. The set $A \setminus B$ consists of elements of A absent from B, and $|A|$ is its cardinality when A is finite. For a field F, write $F^\times = F \setminus \{0\}$. The notation $a \equiv b \pmod{p}$ means $p \mid (a-b)$. For prime p, writing $\mathbb{Z}_p = \{0, \dots, p-1\}$ uses representatives of residue classes. The symbol $\mathbb{F}_q$ denotes a field with q elements; when $q = p^n$ with $n > 1$, it must not be identified with arithmetic modulo q.

DEFINITION 3.1

Equality of sets and cancellation. For sets A, B, we have $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$. To prove equality, take an arbitrary element from each side and prove that it belongs to the other side. In a field F, for $a, b, c \in F$ with $c \neq 0$,

$$ac = bc \quad \Longleftrightarrow \quad a = b,$$

because multiplication by c^{-1} is permitted. The condition $c \neq 0$ is essential. The characteristic of a field is the least positive integer p for which $p \cdot 1 = 0$, if such an integer exists; otherwise its characteristic is 0. The characteristic of a finite field is prime.

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3.1 Algebraic structure: fields

3.1.1 The field axioms

RECALL AND TOOLS

A field is a set F equipped with two operations, addition and multiplication. For all $a, b, c \in F$, the operations satisfy the following axioms. In particular, $0_{\mathbb{F}} \neq 1_{\mathbb{F}}$ is required:

- **Closure:** if $a, b \in F$, then $a +_{\mathbb{F}} b \in F$ and $a \times_{\mathbb{F}} b \in F$.
- **Commutativity:** $a +_{\mathbb{F}} b = b +_{\mathbb{F}} a$ and $a \times_{\mathbb{F}} b = b \times_{\mathbb{F}} a$.
- **Associativity:** $(a +_{\mathbb{F}} b) +_{\mathbb{F}} c = a +_{\mathbb{F}} (b +_{\mathbb{F}} c)$ and $(a \times_{\mathbb{F}} b) \times_{\mathbb{F}} c = a \times_{\mathbb{F}} (b \times_{\mathbb{F}} c)$.
- **Distributivity:** $a \times_{\mathbb{F}} (b +_{\mathbb{F}} c) = a \times_{\mathbb{F}} b +_{\mathbb{F}} a \times_{\mathbb{F}} c$ and $(a +_{\mathbb{F}} b) \times_{\mathbb{F}} c = a \times_{\mathbb{F}} c +_{\mathbb{F}} b \times_{\mathbb{F}} c$.
- **Identity elements:** there is $0_{\mathbb{F}} \in F$ such that $a +_{\mathbb{F}} 0_{\mathbb{F}} = a$ for every $a \in F$, and there is $1_{\mathbb{F}} \in F$ such that $a \times_{\mathbb{F}} 1_{\mathbb{F}} = a$ for every $a \in F$.
- **Additive and multiplicative inverses:** every $a \in F$ has an additive inverse $-a$ with $a +_{\mathbb{F}} (-a) = 0_{\mathbb{F}}$; every nonzero $a \in F$ has a multiplicative inverse a^{-1} with $a \times_{\mathbb{F}} a^{-1} = 1_{\mathbb{F}}$.

EXAMPLE 3.1

- **Examples:** $\mathbb{Q}$ (rational numbers), $\mathbb{R}$ (real numbers), and $\mathbb{C}$ (complex numbers) are fields with their usual addition and multiplication.
- **Nonexamples:** $\mathbb{Z}$ and $\mathbb{N}$ are not fields, because most of their elements have no multiplicative inverse; for example, 3 has no inverse in $\mathbb{Z}$.

3.1.2 Notation

A general field is usually denoted by F (sometimes $\mathbb{F}$). When no particular field is intended, we write ``Let F be a field.''
Notation convention: when the field is clear from context, we omit its subscript from the operations and the elements $0, 1$.

EXERCISE 3.1

Let $a \in F$ with $a \neq 0_{\mathbb{F}}$. Please prove that $(a^{-1})^{-1} = a$.

WORKED SOLUTION

Recall that every invertible field element has a unique multiplicative inverse, and in particular $a \times_{\mathbb{F}} a^{-1} = 1_{\mathbb{F}}$. Put $x = (a^{-1})^{-1}$ and calculate:

$$(a^{-1}) \times_{\mathbb{F}} x = 1_{\mathbb{F}} \quad / \times_{\mathbb{F}} a$$

$$a \times_{\mathbb{F}} (a^{-1}) \times_{\mathbb{F}} x = a \times_{\mathbb{F}} 1_{\mathbb{F}}$$

$$1_{\mathbb{F}} \times_{\mathbb{F}} x = a \qquad (*)$$

Thus $x = a$, that is, $(a^{-1})^{-1} = a$.
Remark: step $(*)$ uses $a \times_{\mathbb{F}} a^{-1} = 1_{\mathbb{F}}$.

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EXERCISE 3.2

Let $a \in F$ with $a \neq 0_{\mathbb{F}}$. Please prove that the multiplicative inverse of a is unique.

WORKED SOLUTION

Suppose $b \in F$ satisfies $a \times_{\mathbb{F}} b = 1_{\mathbb{F}}$. Multiply on the left by a^{-1}:

$$a \times_{\mathbb{F}} b = 1_{\mathbb{F}} \quad / \times_{\mathbb{F}} a^{-1}$$

$$(a^{-1}) \times_{\mathbb{F}} a \times_{\mathbb{F}} b = a^{-1} \times_{\mathbb{F}} 1_{\mathbb{F}}$$

$$1_{\mathbb{F}} \times_{\mathbb{F}} b = a^{-1} \qquad (*)$$

Hence $b = a^{-1}$, so the inverse of a is unique. *Remark:* again, step $(*)$ uses $(a^{-1}) \times_{\mathbb{F}} a = 1_{\mathbb{F}}$.

3.2 Further exercises on fields

EXERCISE 3.3

Let a, b be elements of a field $\mathbb{F}$. Please prove that if $a \times_{\mathbb{F}} b = 0_{\mathbb{F}}$, then $a = 0_{\mathbb{F}}$ or $b = 0_{\mathbb{F}}$.

WORKED SOLUTION

We distinguish two cases according to b. **Case 1:** $b = 0_{\mathbb{F}}$, and the conclusion already holds. **Case 2:** $b \neq 0_{\mathbb{F}}$. Since $\mathbb{F}$ is a field, b has an inverse b^{-1}. Multiplying both sides by b^{-1} gives

$$a \times_{\mathbb{F}} b \times_{\mathbb{F}} b^{-1} = 0_{\mathbb{F}} \times_{\mathbb{F}} b^{-1} \Longrightarrow a \times_{\mathbb{F}} 1_{\mathbb{F}} = a = 0_{\mathbb{F}}$$

PROPOSITION 3.1

A field has no nontrivial zero divisors.

EXERCISE 3.4

Let F be a field. On F^3, define

$$(a, b, c) \oplus (d, e, f) = (a + d, b + e, c + f),$$

$$(a, b, c) \otimes (d, e, f) = (ad, be, cf).$$

Please determine whether this is a field.

WORKED SOLUTION

The zero element is $(0, 0, 0)$, but

$$(1, 0, 0) \otimes (0, 1, 0) = (0, 0, 0).$$

Neither factor is zero, since $1 \neq 0$ in F. A field has no nontrivial zero divisors, so this structure is not a field.

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EXAMPLE 3.2

In $\mathbb{Z}_4 = \{0, 1, 2, 3\}$ with addition and multiplication modulo 4, we have $2 \cdot 2 = 0$ although $2 \neq 0$. Thus 2 has no inverse: if $2b = 1$, then $2 = 2(2b) = (2 \cdot 2)b = 0$, a contradiction. Hence $\mathbb{Z}_4$ is not a field. Complex arithmetic, polar form, and Euler notation are covered in Tutorial 1.

3.3 Congruence and Modular Arithmetic

DEFINITION 3.2

Write $\mathbb{Z}$ for the integers. For an integer $m \geq 2$, the notation

$$a \equiv b \pmod{m}$$

means that $a - b$ is divisible by m: there is $k \in \mathbb{Z}$ such that $a - b = km$. The residue class of a is the set

$$\overline{a} = \{a + km : k \in \mathbb{Z}\}.$$

Any two elements in the same class are congruent modulo m. There are exactly m classes, represented by $0, 1, \ldots, m - 1$.

EXAMPLE 3.3

Modulo 13, we have $-5 \equiv 8$ and $34 \equiv 8$. For example,

$$(-5) + 34 \equiv 8 + 8 \equiv 3, \qquad (-5) \cdot 34 \equiv 8 \cdot 8 \equiv 12 \pmod{13}.$$

PROPOSITION 3.2

Addition and multiplication of classes can be computed using representatives:

$$\overline{a} + \overline{b} = \overline{a + b}, \qquad \overline{a}\,\overline{b} = \overline{ab}.$$

The results do not depend on the representatives. Indeed, if $a' = a + km$ and $b' = b + \ell m$, then

$$(a' + b') - (a + b) = (k + \ell)m,$$

$$a'b' - ab = m(kb + \ell a + k\ell m).$$

Both differences are divisible by m.

NOTATION AND CONVENTIONS

For a prime p, the classes with these operations form the field $\mathbb{Z}_p$, also denoted $\mathbb{F}_p$. We normally omit the bars and use the representatives $0, 1, \ldots, p - 1$. Addition and multiplication are modulo p. Division by $a \neq 0$ means multiplication by its inverse a^{-1}, which satisfies $aa^{-1} = 1$ in the field.

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PROPOSITION 3.3

Every nonzero $a \in \mathbb{Z}_p$ has an inverse. A prime dividing a product divides at least one factor. Thus

$$ax = ay \quad \Longrightarrow \quad a(x-y) = 0 \quad \Longrightarrow \quad x = y.$$

Multiplication by a therefore sends the $p-1$ nonzero elements to $p-1$ distinct nonzero elements. One of them is 1, so there is x satisfying $ax = 1$. The same cancellation argument gives uniqueness. The other field axioms follow from the laws of integer arithmetic.

NOTE

For example, $5^{-1} = 8$ in $\mathbb{Z}_{13}$ because $5 \cdot 8 = 40 \equiv 1$. Hence $5x = 7$ gives $x = 8 \cdot 7 = 56 \equiv 4$. Primality matters: modulo 4, we have $2 \cdot 2 = 0$ although neither factor is zero. These operations do not form a field. There are also finite fields whose number of elements is not prime; they are not obtained simply by integer arithmetic modulo that number.

EXERCISE 3.5

Let p be an odd prime. Please prove that every $t \in \mathbb{Z}_p$ can be written as $t = x^2 + y^2$. Please find such an expression for 7 in $\mathbb{Z}_{13}$.

WORKED SOLUTION

Let $Q = \{x^2 : x \in \mathbb{Z}_p\}$ be the set of squares. The notation $|Q|$ is the number of its distinct elements. If $x^2 = y^2$, then

$$(x-y)(x+y) = 0.$$

A field has no zero divisors, so $x = y$ or $x = -y$. For $x \neq 0$, the elements $x, -x$ are distinct: equality would give $2x = 0$, and odd p would imply $x = 0$. Thus the $p-1$ nonzero elements give exactly $(p-1)/2$ distinct squares. Including 0 gives

$$|Q| = \frac{p+1}{2}.$$

The set $t - Q = \{t - q : q \in Q\}$ has the same number of elements. If Q and $t - Q$ were disjoint, their union would contain $p+1$ elements inside a set of size p, a contradiction. Their intersection therefore contains an element: there are x, y such that $x^2 = t - y^2$, giving $t = x^2 + y^2$. In the example,

$$4^2 + 2^2 = 16 + 4 = 20 \equiv 7 \pmod{13}.$$

EXERCISE 3.6

Let p be an odd prime, and define $h(x) = x + x^{-1}$ for $x \in \mathbb{Z}_p \setminus \{0\}$. Please prove that h takes exactly $(p+1)/2$ distinct values. Please find the set of values when $p = 11$.

WORKED SOLUTION

For nonzero x, y,

$$h(x) - h(y) = x - y + \frac{y-x}{xy} = \frac{(x-y)(xy-1)}{xy}.$$

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EXERCISE 3.6 CONTINUED

Division by xy is allowed because both factors are nonzero. Consequently,

$$h(x) = h(y) \quad \Longleftrightarrow \quad y = x \text{ or } y = x^{-1}.$$

The elements equal to their own inverse satisfy $x^2 = 1$, so they are exactly $1, -1$. They are distinct because p is odd. Their values are also distinct: $h(1) = 2$ and $h(-1) = -2$. The remaining $p - 3$ elements split into pairs $\{x, x^{-1}\}$, each giving a value not obtained from another pair. The number of values is therefore

$$2 + \frac{p-3}{2} = \frac{p+1}{2}.$$

In $\mathbb{Z}_{11}$, the pairs and singletons are

$$\{1\},\ \{10\},\ \{2,6\},\ \{3,4\},\ \{5,9\},\ \{7,8\}.$$

Their respective values of $x + x^{-1}$ are $2, 9, 8, 7, 3, 4$, so the set of values is

$$\{2, 3, 4, 7, 8, 9\}.$$

3.4 Finite fields

NOTE

For a prime p, we write $\mathbb{Z}_p = \{0, \ldots, p-1\}$ and perform addition and multiplication modulo p. These integers represent residue classes. The same field is also denoted $\mathbb{F}_p$. Most finite-field examples here use these residue fields.

Definition and notation

DEFINITION 3.3

- A **finite field** is a field with finitely many elements. Write $|\mathbb{F}| = q$.
- Every finite field has $q = p^n$ elements, where p is prime and $n \geq 1$ is an integer. This is a structural theorem about finite fields.
- For a prime p, the set $\mathbb{Z}_p = \mathbb{Z}/p\mathbb{Z}$, with addition and multiplication modulo p, is a field. It is called the **prime field of characteristic** p.
- Addition and multiplication in $\mathbb{Z}_p$ are defined by

$$a +_{\mathbb{Z}_p} b := (a + b) \bmod p, \quad a \times_{\mathbb{Z}_p} b := (a \cdot b) \bmod p.$$

Examples

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EXAMPLE 3.4

- $\mathbb{Z}_2 = \{0, 1\}$, with $1 + 1 = 0$ and $1 \cdot 1 = 1$.
- $\mathbb{Z}_7 = \{0, 1, \ldots, 6\}$, with arithmetic modulo 7; for example, $5 + 4 = 2$ and $3 \cdot 5 = 1$ in the field.
- $\mathbb{Z}_{11} = \{0, 1, \ldots, 10\}$, with arithmetic modulo 11; for example, $8 + 7 = 4$ and $4 \cdot 3 = 1$ in the field.

Basic properties

PROPOSITION 3.4

In every finite field $\mathbb{F}_q$, the set $\mathbb{F}_q^\times$ of nonzero elements is closed under multiplication, and each of its elements has a unique inverse.

PROPOSITION 3.5

In every finite field $\mathbb{F}_q$, each $a \in \mathbb{F}_q$ satisfies

$$a^q = a.$$

PROPOSITION 3.6

Every nonzero a in a finite field $\mathbb{F}_q$ satisfies $a^{q-1} = 1$.

Explanation: Closure follows because a field has no zero divisors. The existence and uniqueness of inverses follow from the field axioms and the proof above. To prove both power identities, let $a \neq 0$ and put

$$P = \prod_{x \in \mathbb{F}_q^\times} x \neq 0.$$

Multiplication by a permutes the nonzero elements: the map $x \mapsto ax$ is invertible, with inverse $x \mapsto a^{-1}x$. Thus

$$P = \prod_{x \in \mathbb{F}_q^\times} (ax) = a^{q-1}P.$$

Since $P \neq 0$, cancellation is permitted and gives $a^{q-1} = 1$, hence $a^q = a$. For $a = 0$, the identity $0^q = 0$ is immediate.

EXAMPLE 3.5

The nonzero inverses in $\mathbb{Z}_{11}$ are

a	1	2	3	4	5	6	7	8	9	10
a^{-1}	1	6	4	3	9	2	8	7	5	10

For example, $2 \cdot 6 = 12 \equiv 1$, $5 \cdot 9 = 45 \equiv 1$ and $7 \cdot 8 = 56 \equiv 1 \pmod{11}$.

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EXERCISE 3.7

Please show that, in every finite field $\mathbb{F}_q$,

$$(a-b)^q = a^q - b^q.$$

WORKED SOLUTION

Use the identity $x^q = x$, proved for every $x \in \mathbb{F}_q$. In particular, it applies to $a-b$, so

$$(a-b)^q = a - b = a^q - b^q.$$

This proves the claim directly, without calculating binomial coefficients.

EXERCISE 3.8

Please work in $\mathbb{Z}_{17}$.

1. Please find the inverses of $2, 3, 5, 7, 8$.
2. Please calculate $E = (7^{-1}+3)(5-2^{-1})^{-1} - 2$.

WORKED SOLUTION

All arithmetic is modulo 17. We have

$$2 \cdot 9 = 18 \equiv 1, \quad 3 \cdot 6 = 18 \equiv 1, \quad 5 \cdot 7 = 35 \equiv 1,$$

$$7 \cdot 5 = 35 \equiv 1, \quad 8 \cdot 15 = 120 \equiv 1.$$

Thus the inverses, in order, are $9, 6, 7, 5, 15$. Now $7^{-1} + 3 = 5 + 3 = 8$, while

$$5 - 2^{-1} = 5 - 9 = -4 \equiv 13 \neq 0.$$

The inverse of 13 is 4, because $13 \cdot 4 = 52 \equiv 1$. Therefore

$$E = 8 \cdot 4 - 2 = 30 \equiv 13 \pmod{17}.$$

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EXERCISE 3.9

Please show that for every nonzero a in a finite field $\mathbb{F}_q$, there is $n \in \mathbb{N}$ with $a^n = 1$.

WORKED SOLUTION

The set $\mathbb{F}_q^\times$ contains exactly $q-1$ elements and is closed under multiplication. Put $m = q-1$ and consider the list

$$a, a^2, a^3, \ldots, a^{m+1}.$$

It has $m+1$ entries but only m possible distinct values, so two entries are equal, say $a^k = a^l$ with $k > l$. Multiplying by a^{-l} gives $a^{k-l} = 1$. Thus every nonzero element has a positive integer power equal to the multiplicative identity.

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3.5 Calculations and applications in finite fields

EXERCISE 3.10

Let $a, b, c, d \in \mathbb{Z}_2$ be four bits. Add a bit e such that $a+b+c+d+e=0$. Please find e and please calculate it for $(a,b,c,d)=(1,0,1,1)$. Please determine whether the check detects one flipped bit and what happens with two flipped bits.

WORKED SOLUTION

In $\mathbb{Z}_2$, $-x=x$, so $e=a+b+c+d$. For the given example, $e=1+0+1+1=1$. Write the five error bits as $\varepsilon_1,\dots,\varepsilon_5$, with 1 indicating a flip and 0 no flip. The received parity is

$$(a+b+c+d+e)+\sum_{j=1}^{5}\varepsilon_j=\sum_{j=1}^{5}\varepsilon_j.$$

A single flip gives parity 1 and is detected. Two flips give $1+1=0$ and are not detected. The check does not identify which bit flipped.

Additional practice

EXERCISE 3.11 FOUNDATION

In $\mathbb{Z}_{11}$, define

$$A=\{x\in\mathbb{Z}_{11}:3x+5=2\},\qquad B=\{x\in\mathbb{Z}_{11}:x^2=1\}.$$

Please find A, B, $A\cup B$, $A\cap B$ and please determine whether $A\subseteq B$.

WORKED SOLUTION

All arithmetic is modulo 11. The first equation gives $3x=8$. Since $3^{-1}=4$, we get $x=4\cdot 8=32=10$, so $A=\{10\}$. The second equation is equivalent to $(x-1)(x+1)=0$. A field has no zero divisors, so $x=1$ or $x=-1=10$. Hence

$$B=\{1,10\},\quad A\subseteq B,\quad A\cup B=\{1,10\},\quad A\cap B=\{10\}.$$

EXERCISE 3.12 CHALLENGE

Let F be a finite field with $q>2$ elements. Please prove that

$$\sum_{x\in F}x=0,\qquad \sum_{x\in F^\times}x^{-1}=0.$$

Please check both statements in the field with two elements. Do not assume an ordering on F.

WORKED SOLUTION

Put $S=\sum_{x\in F}x$. Since $q>2$, choose $c\in F\setminus\{0,1\}$. The map $x\mapsto cx$ is bijective, with inverse

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EXERCISE 3.12 CHALLENGE CONTINUED

$x \mapsto c^{-1}x$. It therefore only rearranges the summands, and

$$cS = \sum_{x \in F} cx = \sum_{x \in F} x = S.$$

Thus $(c-1)S = 0$. Since $c - 1 \neq 0$, field cancellation gives $S = 0$. Removing the summand 0 does not change the sum, so $\sum_{x \in F^\times} x = 0$. The map $x \mapsto x^{-1}$ also permutes $F^\times$, because applying it twice returns x. Therefore

$$\sum_{x \in F^\times} x^{-1} = \sum_{x \in F^\times} x = 0.$$

In $\mathbb{Z}_2$, we have $0 + 1 = 1 \neq 0$, and the sum of the nonzero inverses is $1^{-1} = 1 \neq 0$. Indeed, no element c distinct from $0, 1$ exists there, so the choice used in the proof is unavailable.

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EXERCISE 3.13 CHALLENGE

Let F be a field satisfying $x^3 = x$ for every $x \in F$. Please prove that F has either two or three elements. Please write its operations in each case and please identify them with those of $\mathbb{Z}_2$ or $\mathbb{Z}_3$.

WORKED SOLUTION

For $x \neq 0$, cancellation gives $x^2 = 1$, hence $(x-1)(x+1) = 0$. A field has no zero divisors, so $x = 1$ or $x = -1$. Thus $F \subseteq \{0, 1, -1\}$. If $1 = -1$, then $F = \{0, 1\}$ and $1 + 1 = 0$. Together with multiplication by zero and $1 \cdot 1 = 1$, these are the operations of $\mathbb{Z}_2$. If $1 \neq -1$, all three elements are distinct and present. The sum $1 + 1$ cannot be 0, which would give $1 = -1$, or 1, which would give $1 = 0$ by cancellation. Therefore $1 + 1 = -1$. Writing $2 = -1$, we obtain

$$1 + 2 = 0, \qquad 2 + 2 = 1, \qquad 2 \cdot 2 = 1.$$

These, together with the zero and identity rules, give exactly $\mathbb{Z}_3$. Conversely, direct substitution of $0, 1$ in $\mathbb{Z}_2$, and $0, 1, -1$ in $\mathbb{Z}_3$, verifies $x^3 = x$.

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EXERCISE 3.14 CHALLENGE

Let F be a finite field. Please prove

$$\prod_{x \in F \setminus \{0\}} x = -1.$$

Here $\prod$ multiplies all nonzero elements. Please also prove that, for every $b \in F$,

$$\prod_{a \in F \setminus \{b\}} (b - a) = -1.$$

WORKED SOLUTION

Pair each nonzero x with x^{-1}, except when the two are equal. Each pair contributes 1 to the product. The unpaired elements satisfy $x = x^{-1}$, equivalently

$$x^2 = 1, \qquad (x-1)(x+1) = 0.$$

Since a field has no zero divisors, these are exactly 1 and -1. If they are distinct, their product is -1. If $1 = -1$, only the element 1 remains unpaired, and its value is again -1. This proves the first identity in every characteristic. For fixed b, the values $b - a$ as a ranges over $F \setminus \{b\}$ run through every nonzero field element exactly once. The second product is therefore the same product in a different order.

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TUTORIAL 4

Linear Systems, Gaussian Elimination, and Linear Combinations

4.1 Notation

NOTATION AND CONVENTIONS

Let F be a field. The set F^n consists of all columns $(x_1, \ldots, x_n)^T$ with $x_i \in F$. Here T indicates column notation. Addition and scalar multiplication are componentwise. The set $M_{m\times n}(F)$ consists of matrices with m rows and n columns over F. The entry in row i and column j is a_{ij}. For a system with n unknowns and m equations, append the column of right-hand sides to the coefficient matrix to obtain the augmented matrix.

MOTIVATION AND GUIDING IDEA

Row reduction replaces a problem with an equivalent one that is easier to read. After calculating, return to the question: is the system consistent, which variables are free, and what is the solution set? For a nonhomogeneous system, the difference of two solutions solves the associated homogeneous system. This explains the form ``a particular solution plus a homogeneous solution.''

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4.2 Linear systems and row reduction

4.2.1 A system of linear equations

RECALL AND TOOLS

In general, for a positive integer n, a linear equation in n unknowns (variables) has the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

Here $a_1, a_2, ..., a_n$ and b are **given scalars** in a specified field $\mathbb{F}$, such as $\mathbb{R}$, $\mathbb{C}$, or a finite field, and $x_1, x_2, ..., x_n$ are the **unknowns**, or variables.
The scalar a_i is the **coefficient** of x_i, and b is the **constant term** of the equation.

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A system of m equations in n unknowns has the general form

$$\begin{cases} a_{1,1}x_1 + a_{1,2}x_2 + \cdots + a_{1,n}x_n = b_1 \\ a_{2,1}x_1 + a_{2,2}x_2 + \cdots + a_{2,n}x_n = b_2 \\ \vdots \\ a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,n}x_n = b_m \end{cases}$$

To solve the system, it is convenient to collect the **coefficients** in a matrix, preserving the variable order. To shorten the notation, we write either the matrix of variable coefficients, called the **coefficient matrix**, or the **augmented matrix**, which also includes the constant terms. Following the basic rules consistently makes it clear which coefficient belongs to each variable.

EXAMPLE 4.1

$$\begin{cases} 3x_1 - 2x_2 = 0, \\ -6x_1 + 5x_2 = 2 \end{cases} \longleftrightarrow \left(\begin{array}{cc|c} 3 & -2 & 0 \\ -6 & 5 & 2 \end{array}\right).$$

4.2.2 Elementary operations

DEFINITION 4.1

Elementary row operations:
An elementary row operation modifies a matrix in one of the following ways:

1. Interchange rows i and j, denoted by $R_i \longleftrightarrow R_j$.
2. Multiply row i by a **nonzero** scalar λ, denoted by $R_i \longleftarrow \lambda R_i$.
   (Dividing by a nonzero scalar a is multiplication by $1/a$.)
3. Add λ times row j to a different row i, where $i \neq j$, denoted by $R_i \longleftarrow R_i + \lambda R_j$.

- Elementary row operations preserve the solutions of a linear system.
- Every elementary row operation has an inverse that is also an elementary row operation.

DEFINITION 4.2

Matrices A and B are **row equivalent** (or simply equivalent), denoted by $A \sim B$, if B is obtained by applying a finite sequence of elementary row operations to A.

DEFINITION 4.3

Leading entry (pivot): in every nonzero row of a matrix, the first nonzero entry is called its leading entry.

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DEFINITION 4.4

Row echelon form: each leading entry is strictly to the left of the leading entry in the next nonzero row. Any zero rows appear below all nonzero rows.

DEFINITION 4.5

Reduced row echelon form: a matrix in row echelon form such that, in each nonzero row, the leading entry equals 1, and every other entry above and below it in the same column equals 0. These requirements apply to every column, including the constant column of an augmented matrix.

PROPOSITION 4.1

Every matrix is row equivalent to a unique matrix in reduced row echelon form.

Examples:

1. $\begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix}$ is not in echelon form: its leading entries are in the same column.
2. $\begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 3 \end{pmatrix}$ is in echelon form but not reduced echelon form.
3. $\begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 4 \end{pmatrix}$ is in reduced row echelon form.

4.2.3 Gaussian elimination

RECALL AND TOOLS

1. Start with the first active row and the first column. Look in that column for a nonzero entry in the active row or below it. If none exists, move to the next column.
2. Interchange rows to bring this entry into the active row. Division by it is permitted because it has been checked to be nonzero.
3. Multiply the active row by the inverse of its leading entry to obtain a leading 1, and use it to eliminate the entries below.
4. Move to the next active row and seek a pivot column to the right of the previous one. A pivot column need not have the same index as its row.
5. After reaching row echelon form, eliminate the entries above each pivot, working from the last pivot to the first, to obtain reduced row echelon form. This completes Gauss-Jordan elimination.

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EXERCISE 4.1

Please reduce the augmented matrix over $\mathbb{R}$

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 2 & 1 & 3 & 12 \\ 1 & 3 & 2 & 3 \end{array}\right)$$

to reduced row echelon form and please find the system's solution.

WORKED SOLUTION

Apply $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - R_1$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & -3 & 1 & 10 \\ 0 & 1 & 1 & 2 \end{array}\right).$$

Swap the last two rows and apply $R_3 \leftarrow R_3 + 3R_2$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 4 & 16 \end{array}\right).$$

Divide R_3 by 4, subtract it from R_2, and clear the entries above the last two pivots:

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 4 \end{array}\right).$$

The unique solution is $(1, -2, 4)^T$. Substitution gives $1-4+4 = 1$, $2-2+12 = 12$, $1-6+8 = 3$.

NOTE

To solve a linear system, reduce its **augmented matrix**, including the constant terms. You may stop at row echelon form and use back substitution, or continue to reduced row echelon form and read off the solutions.

EXAMPLE 4.2

Consider the system

$$\begin{cases} x + 2y = 0, \\ y + z = 2, \\ 2x + z = 7. \end{cases}$$

Its augmented matrix is

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 1 & 1 & 2 \\ 2 & 0 & 1 & 7 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 5 & 15 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right).$$

The first step uses $R_3 \leftarrow R_3 - 2R_1 + 4R_2$. Hence $z = 3$, $y = 2 - z = -1$ and $x = -2y = 2$.

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The solution set is $\{(2, -1, 3)\}$.

This example shows how row reduction helps solve a linear system. We will next see that a system need not have a unique solution.

EXERCISE 4.2

Please reduce the augmented matrix over $\mathbb{Z}_{11}$

$$\left(\begin{array}{cccc|c} 1 & 0 & 2 & 1 & 3 \\ 0 & 1 & 3 & 2 & 4 \\ 2 & 3 & 4 & 1 & 5 \end{array}\right)$$

to reduced row echelon form. Please perform the arithmetic modulo 11.

WORKED SOLUTION

Apply $R_3 \leftarrow R_3 - 2R_1 - 3R_2$. The last row has entries $4-4-9 = -9 \equiv 2$, $1-2-6 = -7 \equiv 4$ and $5-6-12 = -13 \equiv 9$. Thus

$$\left(\begin{array}{cccc|c} 1 & 0 & 2 & 1 & 3 \\ 0 & 1 & 3 & 2 & 4 \\ 0 & 0 & 2 & 4 & 9 \end{array}\right).$$

The inverse of 2 is 6. Multiplying R_3 by 6 gives

$$\left(\begin{array}{cccc|c} 1 & 0 & 2 & 1 & 3 \\ 0 & 1 & 3 & 2 & 4 \\ 0 & 0 & 1 & 2 & 10 \end{array}\right).$$

Finally apply $R_1 \leftarrow R_1 - 2R_3$ and $R_2 \leftarrow R_2 - 3R_3$:

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 8 & 5 \\ 0 & 1 & 0 & 7 & 7 \\ 0 & 0 & 1 & 2 & 10 \end{array}\right).$$

All entries use representatives $0, 1, \ldots, 10$.

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EXERCISE 4.3

Please solve the following system over $\mathbb{C}$:

$$\begin{cases} iz + (2-i)w = -3 - i, \\ 2iz + (5-2i)w = -5 - 5i. \end{cases}$$

Please verify the solution in both equations.

WORKED SOLUTION

Subtract twice the first equation from the second to get

$$w = (-5-5i) - 2(-3-i) = 1 - 3i.$$

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EXERCISE 4.3 CONTINUED

Now $(2-i)(1-3i) = -1-7i$, so

$$iz = -3-i-(-1-7i) = -2+6i, \qquad z = 6+2i.$$

The first equation gives $(-2+6i)+(-1-7i) = -3-i$. The second gives

$$2iz + (5-2i)w = (-4+12i)+(-1-17i) = -5-5i.$$

4.3 Solving linear systems

DEFINITION 4.6

After row reduction, a variable whose column in the coefficient matrix contains a leading entry is called a **pivot variable** (or dependent variable). A variable whose column contains no leading entry is called a **free variable**. In a consistent system, every free variable can be assigned an arbitrary value in the field, after which the pivot variables are determined by the equations. Over an infinite field, a free parameter gives infinitely many choices; over a finite field, the number of choices is finite. The number of free variables is the number of degrees of freedom of a consistent system.

DEFINITION 4.7

The rank of a matrix A, denoted by $\operatorname{rank}(A)$, is the number of nonzero rows in a row echelon form. Equivalently, it is the number of leading entries (pivots) in that form.

Remark: if a system with coefficient matrix A and n variables is consistent, its number of degrees of freedom is $n - \operatorname{rank}(A)$. Over $\mathbb{F}_q$, it has exactly $q^{n-\operatorname{rank}(A)}$ solutions.

4.4 Linear Combinations and Spanning in F^n

DEFINITION 4.8

For $v_1, \dots, v_k \in F^n$, a linear combination is a column $\alpha_1 v_1 + \cdots + \alpha_k v_k$, where $\alpha_j \in F$. The set of all such combinations is their span, denoted $\operatorname{span}\{v_1, \dots, v_k\}$. The list spans F^n if every column in F^n is such a combination. The span of the empty list is $\{0\}$, where 0 is the zero column.

EXAMPLE 4.3

For $u = (1,1,0)^T$ and $v = (0,1,1)^T$ over a field F,

$$\alpha u + \beta v = (\alpha, \alpha+\beta, \beta)^T.$$

Their span therefore consists of the columns $(x,y,z)^T$ satisfying $y = x + z$. To test whether a given column belongs to the span, solve for α, β.

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EXERCISE 4.4

Please solve over $\mathbb{R}$ the system

$$\begin{cases} x + 3y - 2z = 7, \\ 2x - y + 3z = 0, \\ 4x + 5y - z = 14. \end{cases}$$

WORKED SOLUTION

Write the augmented matrix and subtract multiples of its first row:

$$\left(\begin{array}{ccc|c} 1 & 3 & -2 & 7 \\ 2 & -1 & 3 & 0 \\ 4 & 5 & -1 & 14 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 3 & -2 & 7 \\ 0 & -7 & 7 & -14 \\ 0 & -7 & 7 & -14 \end{array}\right).$$

Subtract row two from row three, divide row two by -7 and clear its pivot column:

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

Put $z = t$. Then $x = 1 - t$, $y = 2 + t$, so

$$S = (1, 2, 0) + \text{span}\{(-1, 1, 1)\}.$$

There is one free variable and infinitely many solutions. Substitution gives $7, 0, 14$ in the three equations for every $t \in \mathbb{R}$.

EXERCISE 4.5

Please solve the system

$$\begin{cases} 2x_1 - 6x_2 + 4x_3 = 7, \\ x_1 - 3x_2 + 2x_3 = 2. \end{cases}$$

WORKED SOLUTION

Swap the rows, then subtract twice row one from row two:

$$\left(\begin{array}{ccc|c} 2 & -6 & 4 & 7 \\ 1 & -3 & 2 & 2 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & -3 & 2 & 2 \\ 0 & 0 & 0 & 3 \end{array}\right).$$

The last row says $0 = 3$, so there is no solution.

Geometric illustration

We have encountered three possible cases:

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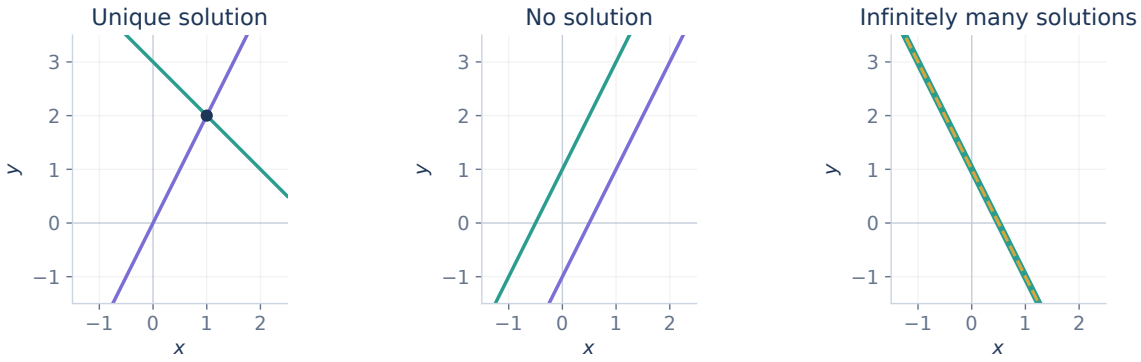


Figure 4.1. Over the real numbers, two equations in two unknowns can describe intersecting lines, distinct parallel lines, or coincident lines. These give, respectively, a unique solution, no solution, or infinitely many solutions.

EXERCISE 4.6

Let $a \in \mathbb{R}$. Please classify the number of solutions and please state the coefficient matrix rank in each case:

$$\begin{cases} x - 2y = 0, \\ 3x + 3z = 2, \\ x + az = 1. \end{cases}$$

WORKED SOLUTION

Subtract three times row one from row two, and row one from row three. Swap the last two rows and subtract three times row two from row three:

$$\left(\begin{array}{ccc|c} 1 & -2 & 0 & 0 \\ 3 & 0 & 3 & 2 \\ 1 & 0 & a & 1 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & -2 & 0 & 0 \\ 0 & 2 & a & 1 \\ 0 & 0 & 3-3a & -1 \end{array}\right).$$

If $a = 1$, there is a contradiction row and rank 2. If $a \neq 1$, the rank is 3 and the solution is unique:

$$z = -\frac{1}{3-3a}, \qquad x = 1 + \frac{a}{3-3a}, \qquad y = \frac{1}{2}\left(1 + \frac{a}{3-3a}\right).$$

No value of a gives infinitely many solutions.

4.4.1 Homogeneous systems

RECALL AND TOOLS

All constant terms are 0. In this case:

- We may write only the coefficient matrix rather than the augmented matrix: the zero column is unchanged by all row operations, so there is no need to keep writing it.
- A solution always exists: no contradiction row can occur, so inconsistency is impossible. We must still determine whether the zero solution is the only solution. Over an infinite field, the existence of a nonzero solution implies infinitely many solutions.
- If there is a unique solution, it is the zero solution, also called the **trivial solution**.

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- The solution set of a homogeneous system in n variables is a vector subspace of $\mathbb{F}^n$ and can always be expressed as a span.

EXERCISE 4.7

Let $a \in \mathbb{R}$. Please classify the number of solutions of this homogeneous system and please find the solution set in each case:

$$\begin{cases} x - 2y = 0, \\ 3x + 3z = 0, \\ x + az = 0. \end{cases}$$

WORKED SOLUTION

The coefficient matrix is the same as in the preceding exercise. Row reduction gives

$$\begin{pmatrix} 1 & -2 & 0 \\ 3 & 0 & 3 \\ 1 & 0 & a \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & 0 \\ 0 & 2 & a \\ 0 & 0 & 3-3a \end{pmatrix}.$$

If $a \neq 1$, the rank is 3 and only the zero solution exists. If $a = 1$, the rank is 2, and $x = -z$, $y = -z/2$. Choosing $z = 2t$ gives

$$S = \{(-2t, -t, 2t) : t \in \mathbb{R}\} = \text{span}\{(-2, -1, 2)\}.$$

There are infinitely many solutions in this case. A homogeneous system is always consistent, so no case has zero solutions.

4.5 Supplementary exercises: measurements and counting solutions

EXERCISE 4.8

Consider the system

$$x_1 + x_2 = 7, \qquad x_2 + x_3 = 11, \qquad x_1 + x_3 = 8.$$

Please find x_1, x_2, x_3. Please find the only value of c for which adding $x_1 + x_2 + x_3 = c$ leaves the solution set unchanged. Please determine what happens for $c = 14$.

WORKED SOLUTION

Add the first two equations and subtract the third:

$$2x_2 = 7 + 11 - 8 = 10, \qquad x_2 = 5.$$

Then $x_1 = 7 - 5 = 2$ and $x_3 = 11 - 5 = 6$. The third equation checks as $2 + 6 = 8$. Adding all three equations gives

$$2(x_1 + x_2 + x_3) = 7 + 11 + 8 = 26.$$

Thus the required value is $c = 13$. Setting $c = 14$ creates a contradiction, so the augmented

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EXERCISE 4.8 CONTINUED

system has no solution.

EXERCISE 4.9

Please solve over $\mathbb{Z}_3$ the system

$$\begin{cases} x + y + z = 1, \\ x + z = 2. \end{cases}$$

Please count its solutions and please compare with the same equations over $\mathbb{R}$.

WORKED SOLUTION

Subtracting the second equation from the first gives $y = -1 = 2$ in $\mathbb{Z}_3$. Choosing $z = t$ gives

$$(x, y, z) = (2 - t, 2, t), \qquad t \in \mathbb{Z}_3.$$

The three solutions are $(2, 2, 0), (1, 2, 1), (0, 2, 2)$. Over $\mathbb{R}$ we get

$$(x, y, z) = (2 - t, -1, t), \qquad t \in \mathbb{R},$$

so there are infinitely many solutions. Both systems have one free variable, but the number of solutions depends on the field size.

Additional practice

EXERCISE 4.10

Please find a spanning set for each solution space over $\mathbb{R}$.

1. $\begin{cases} x + 2y = 0, \\ x - z = 0. \end{cases}$
2. $2y - z = 0$.

WORKED SOLUTION

1. Setting $y = t$ gives $x = -2t$ and $z = x = -2t$. Hence

   $$S_1 = \{t(-2, 1, -2) : t \in \mathbb{R}\} = \text{span}\{(-2, 1, -2)\}.$$

   Substitution verifies that each multiple satisfies both equations.

2. Set $x = s$, $y = t$; then $z = 2t$. Thus

   $$(x, y, z) = s(1, 0, 0) + t(0, 1, 2), \qquad S_2 = \text{span}\{(1, 0, 0), (0, 1, 2)\}.$$

   Both generators satisfy the equation, and the representation includes every solution.

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EXERCISE 4.11 CHALLENGE

Consider a finite linear system over a field F with $n \geq 1$ unknowns. Assume every collection of at most $n+1$ of its equations is consistent. Please prove that the full system is consistent. Please construct an inconsistent system of $n+1$ equations for which every collection of n equations is consistent.

WORKED SOLUTION

Suppose the system is inconsistent and choose an inconsistent collection of minimum size k. Write its equations as

$$\sum_{i=1}^{n} a_{ji} x_i = b_j, \qquad 1 \leq j \leq k.$$

The hypothesis gives $k > n+1$. Consider the homogeneous system in $\lambda_1, \ldots, \lambda_k$ given by

$$\sum_{j=1}^{k} \lambda_j a_{ji} = 0 \quad (1 \leq i \leq n), \qquad \sum_{j=1}^{k} \lambda_j b_j = 0.$$

It has $n+1$ equations and more than $n+1$ unknowns. Row reduction therefore gives a nonzero solution. Choose j with $\lambda_j \neq 0$. These identities express equation j, including its right-hand side, as a linear combination of the other equations. Hence every solution of the other equations also satisfies equation j. Minimality of k says that the other equations have a solution. That solution satisfies the whole collection, a contradiction.
The bound is sharp. Take

$$x_1 = 0, \ldots, x_n = 0, \qquad x_1 + \cdots + x_n = 1.$$

The full system is inconsistent. If the final equation is omitted, zero is a solution. If $x_j = 0$ is omitted, setting $x_j = 1$ and all other unknowns to zero gives a solution. Thus every collection of n equations is consistent.

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TUTORIAL 5

Linear Dependence, Bases in F^n, and Matrix Operations

5.1 Dependence and Bases in F^m

DEFINITION 5.1

The list $v_1, \ldots, v_k \in F^m$ is linearly independent if

$$\alpha_1 v_1 + \cdots + \alpha_k v_k = 0 \quad \Longrightarrow \quad \alpha_1 = \cdots = \alpha_k = 0.$$

It is dependent if there is a relation with coefficients not all zero. A basis of F^m is an independent spanning list. Let e_j be the column with entry 1 in position j and zero elsewhere. The list $(e_1, \ldots, e_m)$ is the standard basis: $(x_1, \ldots, x_m)^T = \sum_{j=1}^{m} x_j e_j$, and each coefficient is determined by its corresponding component.

PROPOSITION 5.1

A list is a basis if and only if every vector has a unique representation as a linear combination of its entries. Spanning gives existence. Independence gives uniqueness by subtracting two representations. Conversely, uniqueness of the representation of zero implies independence.

NOTATION AND CONVENTIONS

For a collection of columns or matrices closed under addition and scalar multiplication, a basis is an independent spanning list. Its dimension, denoted dim, is the number of elements in a basis. In particular, $\dim F^m = m$ and $\dim M_{m\times n}(F) = mn$. The abstract vector-space definition appears in Tutorial 8.

NOTATION AND CONVENTIONS

Let F be a field. The set $M_{m\times n}(F)$ consists of matrices with m rows, n columns, and entries in F; $M_n(F)$ consists of square matrices of order n. The entry A_{ij} is in row i and column j. The column $e_j \in F^n$ has entry 1 in position j and zero elsewhere. For a square matrix, the trace is $\operatorname{tr} A = \sum_{i=1}^{n} A_{ii}$. The identity matrix I_n has diagonal entries 1 and all other entries 0. The zero matrix has only zero entries.

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5.2 Linear Dependence and Independence

EXAMPLE 5.1

The set

$$\left\{ \begin{pmatrix} 0 \\ -4 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$

is linearly dependent, since

$$1 \cdot \begin{pmatrix} 0 \\ -4 \\ 1 \end{pmatrix} - 1 \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + 4 \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 0.$$

Rearranging this equation expresses $\begin{pmatrix} 0 \\ -4 \\ 1 \end{pmatrix}$ in terms of the other vectors.

How Do We Test Dependence?

MOTIVATION AND GUIDING IDEA

- Write a **homogeneous** system whose unknowns are the coefficients in the linear combination.
- Form its matrix and row-reduce.
- **Linear independence:** the only solution is the trivial, or zero, solution.
- **Linear dependence:** there is a nonzero solution. Over an infinite field there are then infinitely many solutions.
- A homogeneous system always has a solution: row reduction cannot produce an inconsistent row.

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EXERCISE 5.1

Let

$$v_1 = (1, 1, 0)^T, \qquad v_2 = (0, 2, 1)^T, \qquad v_3 = (3, -1, -2)^T.$$

Please determine linear dependence, please find all dependence relations, and please express one vector using the others.

WORKED SOLUTION

Write $av_1 + bv_2 + cv_3 = 0$. Its coordinates give

$$a + 3c = 0, \qquad a + 2b - c = 0, \qquad b - 2c = 0.$$

Thus $a = -3c$, $b = 2c$, and the middle equation is $-3c + 4c - c = 0$. All relations are

$$(a, b, c) = t(-3, 2, 1), \qquad t \in \mathbb{R}.$$

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EXERCISE 5.1 CONTINUED

In particular, $-3v_1 + 2v_2 + v_3 = 0$, so $v_3 = 3v_1 - 2v_2$. This is a nontrivial dependence. In $av_1 + bv_2 = 0$, the first and third coordinates force $a = b = 0$, so the first two vectors are independent.

5.2.1 Useful Criteria for Linear Dependence

1. Every set containing the zero vector is linearly dependent.
2. A singleton consisting of a nonzero vector is linearly independent.
3. **Testing a set for linear dependence.**
   - As in the previous tutorial, set a linear combination equal to zero, obtain a homogeneous system, row-reduce, and determine its solutions.
   - Recall: only the zero solution means independence; a nonzero solution means dependence.
   - Another method from the lecture is to place the vectors as the **rows** of a matrix and row-reduce. If a zero row occurs, the original rows are dependent; if there are no zero rows, they are independent.

5.2.2 Spanning Sets

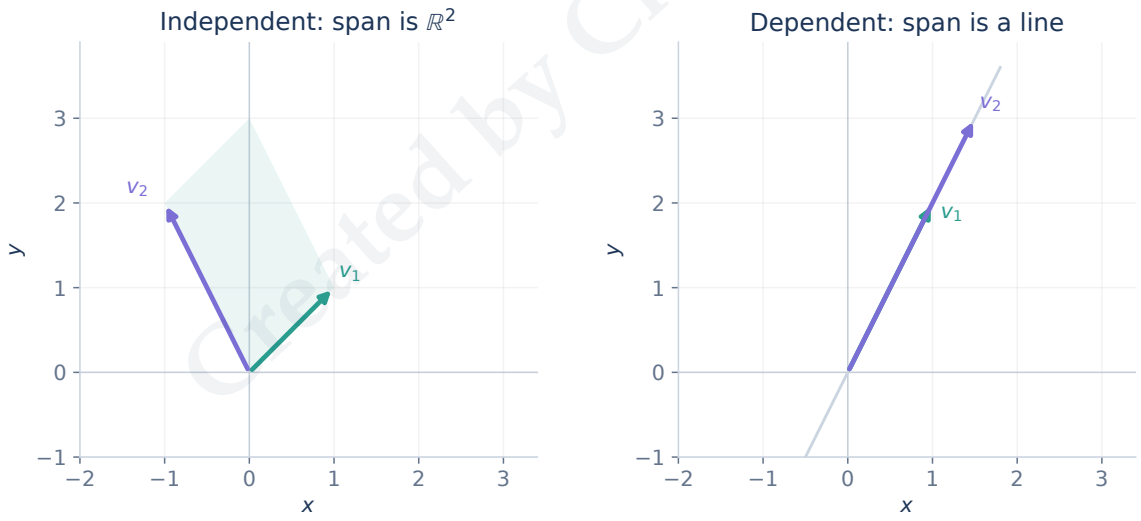


Figure 5.1. Geometric span: a nonzero vector spans a line through the origin; two independent vectors span a plane through the origin.

EXERCISE 5.2 FOUNDATION

Work over $\mathbb{R}$. Let $u = (1, 1, 0)^T$ and $v = (0, 1, 1)^T$.

1. Please describe $U = \text{span}\{u, v\}$ by an equation.
2. Please determine whether $(3, 7, 4)^T$ and $(3, 6, 4)^T$ belong to U, finding coefficients when possible.
3. Please determine whether u, v are independent.

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EXERCISE 5.2 FOUNDATION CONTINUED

WORKED SOLUTION

For $a, b \in \mathbb{R}$,

$$au + bv = (a, a + b, b)^T.$$

Every vector in the span therefore satisfies $y = x+z$. Conversely, if $(x, y, z)^T$ satisfies $y = x+z$, then

$$(x, y, z)^T = xu + zv.$$

Thus $U = \{(x, y, z)^T : y = x + z\}$. For the first vector, $7 = 3 + 4$, and the vector is $3u + 4v$. For the second, $6 \neq 3 + 4$, so it is not in the span. Finally, if $au + bv = 0$, the first and third coordinates give $a = b = 0$. Thus u, v are independent; in particular, the coefficients found above are unique.

EXERCISE 5.3

Please determine whether these vectors form a basis of $\mathbb{C}^3$. If not, please give a dependence relation.

$$v_1 = (1, 0, 2i)^T, \quad v_2 = (1 - i, 1, 0)^T, \quad v_3 = (5 + 2i, 1 + 2i, -2 + 4i)^T.$$

WORKED SOLUTION

The first two vectors are independent: in $av_1 + bv_2 = 0$, the second coordinate gives $b = 0$, and then the first gives $a = 0$. Computing coordinates gives

$$(2 + i)v_1 + (1 + 2i)v_2 = v_3.$$

$$(2 + i) + (1 + 2i)(1 - i) = 5 + 2i, \qquad (2 + i)2i = -2 + 4i.$$

Thus $(2 + i)v_1 + (1 + 2i)v_2 - v_3 = 0$ is nontrivial. The rank is 2, so the triple is not a basis of $\mathbb{C}^3$.

EXERCISE 5.4

Please determine whether these vectors form a basis of $\mathbb{C}^3$. If not, please give a dependence relation.

$$v_1 = (1, 0, i)^T, \quad v_2 = (2 - i, 1, 0)^T, \quad v_3 = (4 - 3i, 2 - i, -1 + i)^T.$$

WORKED SOLUTION

The first two vectors are independent: in $av_1 + bv_2 = 0$, the second coordinate gives $b = 0$, and then the first gives $a = 0$. Computing coordinates gives

$$(1 + i)v_1 + (2 - i)v_2 = v_3.$$

$$(1 + i) + (2 - i)^2 = 4 - 3i, \qquad (1 + i)i = -1 + i.$$

Thus $(1+i)v_1 + (2-i)v_2 - v_3 = 0$ is nontrivial. The rank is 2, so the triple is not a basis of $\mathbb{C}^3$.

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EXERCISE 5.5

Let $\{u, v, w\}$ be a basis of $\mathbb{R}^3$. Please determine which sets are also bases.

1. $\{u, u + 2v, u - 2v\}$.
2. $\{u/3, v/5, w/7\}$.
3. $\{u + v + w, u + 3v, 2v - w\}$.

WORKED SOLUTION

Each set has three vectors in a space of dimension 3, so independence suffices.

1. No: $-2u + (u + 2v) + (u - 2v) = 0$ is a nontrivial dependence.
2. Yes. If $a(u/3)+b(v/5)+c(w/7) = 0$, independence of u, v, w gives $a/3 = b/5 = c/7 = 0$, hence $a = b = c = 0$.
3. No: $(u + v + w) - (u + 3v) + (2v - w) = 0$ is a nontrivial dependence.

5.3 Matrix Arithmetic

Throughout this section, let

$$A = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 2 \\ -1 & 5 \end{pmatrix}.$$

5.3.1 Scalar Multiplication

Scalar multiplication is defined for every matrix. Multiply **every** entry by the scalar: $(\lambda A)_{ij} = \lambda(A)_{ij}$.

EXAMPLE 5.2

Compute

$$-3A = \begin{pmatrix} -3 \cdot 2 & (-3)(-1) \\ -3 \cdot 3 & -3 \cdot 4 \end{pmatrix} = \begin{pmatrix} -6 & 3 \\ -9 & -12 \end{pmatrix}.$$

5.3.2 Matrix Addition and Subtraction

These are defined only for two matrices of **exactly the same shape**. Add or subtract corresponding entries: $(A \pm B)_{ij} = (A)_{ij} \pm (B)_{ij}$.

EXAMPLE 5.3

Compute

$$A + B = \begin{pmatrix} 2 + 0 & -1 + 2 \\ 3 - 1 & 4 + 5 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 2 & 9 \end{pmatrix}.$$

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5.3.3 Matrix Multiplication

A product is defined only when the number of columns of the first (left) matrix equals the number of rows of the second (right) matrix. If A is $k \times m$ and B is $m \times n$, then AB is $k \times n$ and is defined by

$$(AB)_{ij} = \sum_{r=1}^{m} a_{ir} b_{rj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{im}b_{mj}.$$

This is the entry in row i, column j of the product. Follow row i of A and column j of B, each containing m entries: multiply matching entries and add. **Note:** the existence of AB does **not** ensure the existence of BA. With the dimensions above, the latter also requires $n = k$.

EXERCISE 5.6

Please compute AB and BA for the matrices defined above.

WORKED SOLUTION

$$AB = \begin{pmatrix} 2\cdot 0 + (-1)(-1) & 2\cdot 2 + (-1)5 \\ 3\cdot 0 + 4(-1) & 3\cdot 2 + 4\cdot 5 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -4 & 26 \end{pmatrix},$$

$$BA = \begin{pmatrix} 0\cdot 2 + 2\cdot 3 & 0(-1) + 2\cdot 4 \\ (-1)2 + 5\cdot 3 & (-1)(-1) + 5\cdot 4 \end{pmatrix} = \begin{pmatrix} 6 & 8 \\ 13 & 21 \end{pmatrix}.$$

Each entry is the sum of the matching row-column products. Reversing the order changes the result.

NOTE

Important conclusion: matrix multiplication is not commutative. The equality $AB = BA$ need not hold. In the example, $AB \neq BA$ even though both products are defined.

EXERCISE 5.7

Let A be a $k \times m$ matrix with a zero row, and let B be $m \times n$. Please prove that the $k \times n$ matrix AB has a zero row.

WORKED SOLUTION

Let i be the index of a zero row of A, where $1 \leq i \leq k$. Thus $(A)_{ir} = 0$ for every $1 \leq r \leq m$. In row i of AB, for every $1 \leq j \leq n$,

$$(AB)_{ij} = \sum_{r=1}^{m} (A)_{ir}(B)_{rj} = \sum_{r=1}^{m} 0 = 0.$$

All entries in row i of AB are zero, as required. **The zero matrix** is a matrix of any specified shape whose entries are all zero.

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5.4 Diagonal Matrices

A **diagonal matrix** is a **square** matrix whose off-diagonal entries are all zero. Write

$$D = \operatorname{diag}(d_1, \ldots, d_n) = \begin{pmatrix} d_1 & & 0 \\ & \ddots & \\ 0 & & d_n \end{pmatrix}.$$

Equivalently,

$$[D]_{ij} = \begin{cases} 0, & i \neq j, \\ d_i, & i = j. \end{cases}$$

A **scalar matrix** is a diagonal matrix with equal diagonal entries. The **identity matrix** is a diagonal matrix with all diagonal entries equal to 1, hence is a scalar matrix. A scalar matrix can therefore be written aI, where a is its common diagonal entry. Write I_n for the $n \times n$ identity, or simply I when its size is clear. For every $n \times m$ matrix A, $AI_m = I_n A = A$.

EXAMPLE 5.4

$$I_2 A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}, \qquad AI_2 = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}.$$

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EXERCISE 5.8

Let A, B be diagonal matrices of order n. Please prove that:

1. $A + B$ is diagonal.
2. AB is diagonal and $[AB]_{ii} = a_{ii}b_{ii}$ for every $1 \leq i \leq n$.
3. A and B **commute**: $AB = BA$.

WORKED SOLUTION

1. For $i \neq j$, $(A + B)_{ij} = a_{ij} + b_{ij} = 0 + 0 = 0$. Thus $A + B$ is diagonal, with diagonal entries $(A + B)_{ii} = a_{ii} + b_{ii}$.
2. By the definition of multiplication,

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj}.$$

   In row i of A, only a_{ii} can be nonzero, so the sum reduces to $a_{ii}b_{ij}$. If $i \neq j$, then $b_{ij} = 0$, giving $(AB)_{ij} = 0$. If $i = j$, then $(AB)_{ii} = a_{ii}b_{ii}$. Hence AB is diagonal.
3. Similarly BA is diagonal with diagonal entries $b_{ii}a_{ii}$. Scalars in a field commute, so

$$a_{ii}b_{ii} = b_{ii}a_{ii} \qquad (1 \leq i \leq n).$$

   All entries of AB and BA agree; therefore $AB = BA$.

NOTE

Check dimensions before computing. The product $A_{k\times m}B_{m\times n}$ has k rows and n columns. Equality of matrices requires both equal dimensions and equal corresponding entries.

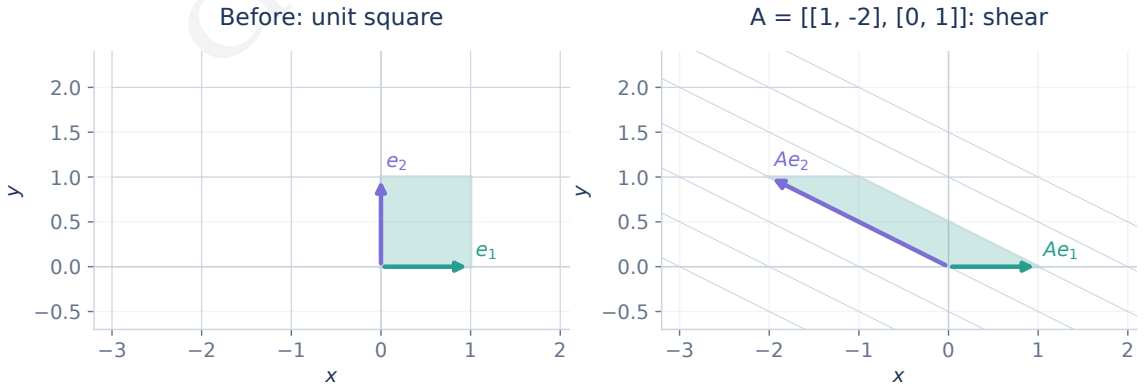


Figure 5.2. A linear map takes a straight grid to a straight grid. The columns of the matrix describe the images of the basis vectors.

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5.5 Transpose

- The transpose is defined for every matrix. The **transpose** A^T satisfies $(A^T)_{ij} = (A)_{ji}$.
- Rows become columns and columns become rows.
- If $A \in \mathbb{R}^{m\times n}$, then $A^T \in \mathbb{R}^{n\times m}$: the numbers of rows and columns are interchanged.

EXAMPLE 5.5

$$C = \begin{pmatrix} 2 & -3 \\ 0 & 7 \\ -4 & 1 \end{pmatrix} \in \mathbb{R}^{3\times 2}, \qquad C^T = \begin{pmatrix} 2 & 0 & -4 \\ -3 & 7 & 1 \end{pmatrix} \in \mathbb{R}^{2\times 3}.$$

The rows of C are the columns of C^T.

5.6 Symmetry and Skew-Symmetry

- A **square** matrix A is **symmetric** if $A^T = A$.
- A **square** matrix A is **skew-symmetric** if $A^T = -A$.
- Every diagonal matrix is symmetric; in particular, every scalar matrix and the identity matrix are symmetric.

The square condition matters: $A \in \mathbb{F}^{m\times n}$ and $A^T \in \mathbb{F}^{n\times m}$ have the same shape only when $n = m$. Intuitively, symmetry means reflection across the main diagonal; skew-symmetry means the same reflection followed by a change of sign.

EXERCISE 5.9

Please classify each matrix as symmetric, skew-symmetric, or neither.

$$A = \begin{pmatrix} 2 & -4 \\ -4 & 7 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 2 & 1 \\ -5 & 0 \end{pmatrix}.$$

WORKED SOLUTION

Computing transposes gives

$$A^T = \begin{pmatrix} 2 & -4 \\ -4 & 7 \end{pmatrix} = A, \qquad B^T = \begin{pmatrix} 0 & -3 \\ 3 & 0 \end{pmatrix} = -B,$$

$$C^T = \begin{pmatrix} 2 & -5 \\ 1 & 0 \end{pmatrix} \neq C, \qquad C^T \neq -C.$$

Thus A is symmetric, B is skew-symmetric, and C is neither.

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EXERCISE 5.10

Please prove or disprove.

1. If A, B are symmetric, then AB is symmetric.
2. If A, B are both nonsymmetric, then AB is nonsymmetric.

WORKED SOLUTION

1. $(AB)^T = B^T A^T = BA$, so symmetry of both factors does not ensure symmetry of the product. For example,

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & -1 \\ -1 & 3 \end{pmatrix}, \quad AB = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}.$$

   Both factors are symmetric, but the product is not.

2. Take $A = B = \begin{pmatrix} 0 & 2 \\ -3 & 0 \end{pmatrix}$. Neither is symmetric, but

$$AB = \begin{pmatrix} 0 \cdot 0 + 2(-3) & 0 \cdot 2 + 2 \cdot 0 \\ (-3)0 + 0(-3) & (-3)2 + 0 \cdot 0 \end{pmatrix} = -6I_2.$$

   The product is symmetric. Thus both assertions are false.

EXERCISE 5.11

Let $A \in M_n(\mathbb{F})$, where $\text{char}(\mathbb{F}) \neq 2$. Please prove that:

1. $A + A^T$ is symmetric.
2. $A - A^T$ is skew-symmetric.
3. A can be written as the sum of a symmetric matrix and a skew-symmetric matrix.

WORKED SOLUTION

1. $(A + A^T)^T = A^T + (A^T)^T = A^T + A = A + A^T$.
2. $(A - A^T)^T = A^T - A = -(A - A^T)$, proving skew-symmetry.
3. Add the matrices from the preceding parts:

$$2A = (A + A^T) + (A - A^T),$$
$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T).$$

   As shown above, the first summand is symmetric and the second is skew-symmetric.

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EXERCISE 5.12

$$D = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \qquad H = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}.$$

Please compute DH and HD entry by entry, please compare their action on e_2, and please find all v satisfying $DHv = HDv$.

WORKED SOLUTION

$$DH = \begin{pmatrix} 2\cdot 1 + 0\cdot 0 & 2(-1) + 0\cdot 1 \\ 0\cdot 1 + 1\cdot 0 & 0(-1) + 1\cdot 1 \end{pmatrix} = \begin{pmatrix} 2 & -2 \\ 0 & 1 \end{pmatrix},$$

$$HD = \begin{pmatrix} 1\cdot 2 + (-1)0 & 1\cdot 0 + (-1)1 \\ 0\cdot 2 + 1\cdot 0 & 0\cdot 0 + 1\cdot 1 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix}.$$

Thus $DHe_2 = (-2, 1)^T$, whereas $HDe_2 = (-1, 1)^T$. For $v = (x, y)^T$,

$$DHv - HDv = (2x - 2y - (2x - y),\ y - y)^T = (-y, 0)^T.$$

The results agree exactly when $y = 0$, on $\mathrm{span}\{e_1\}$.

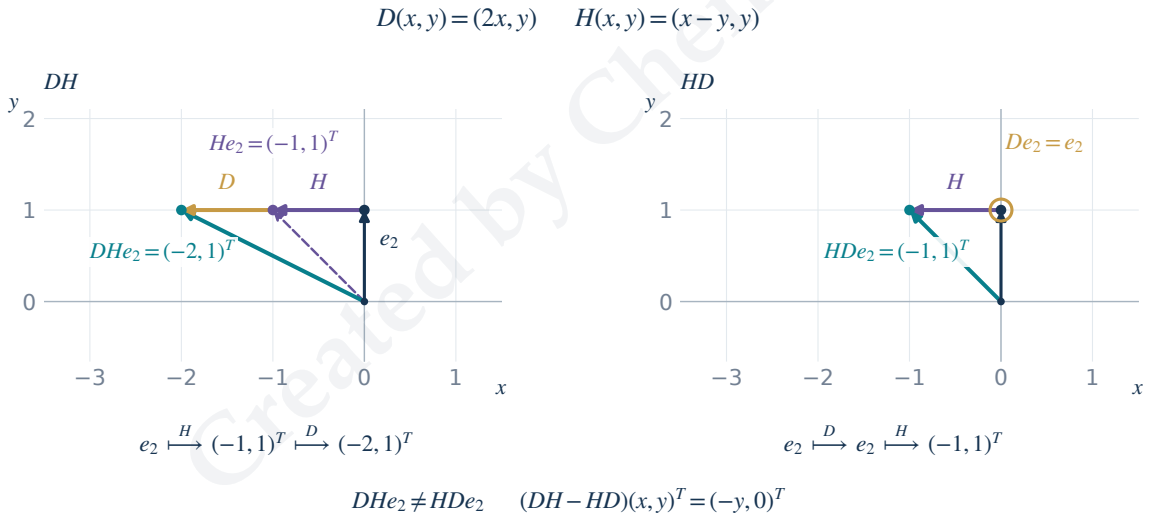


Figure 5.3. Operation order on the same vector: DH applies H first, whereas HD applies D first. The ring marks the fixed point $De_2 = e_2$, but the final results differ: $DHe_2 = (-2, 1)^T \neq (-1, 1)^T = HDe_2$.

Additional practice

EXERCISE 5.13 FOUNDATION

$$A = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 0 & 3 \end{pmatrix}.$$

Please determine the dimensions of AB, BA and please compute both. Please use $e_2 \in \mathbb{R}^2$ to verify $(AB)e_2 = A(Be_2)$. Can $AB = BA$?

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EXERCISE 5.13 FOUNDATION CONTINUED

WORKED SOLUTION

The dimensions are 2×2 and 3×3, respectively.

$$AB = \begin{pmatrix} 1\cdot 2 + 0(-1) - 1\cdot 0 & 1\cdot 1 + 0\cdot 0 - 1\cdot 3 \\ 0\cdot 2 + 2(-1) + 1\cdot 0 & 0\cdot 1 + 2\cdot 0 + 1\cdot 3 \end{pmatrix} = \begin{pmatrix} 2 & -2 \\ -2 & 3 \end{pmatrix},$$

$$BA = \begin{pmatrix} 2\cdot 1 + 1\cdot 0 & 2\cdot 0 + 1\cdot 2 & 2(-1) + 1\cdot 1 \\ (-1)1 + 0\cdot 0 & (-1)0 + 0\cdot 2 & (-1)(-1) + 0\cdot 1 \\ 0\cdot 1 + 3\cdot 0 & 0\cdot 0 + 3\cdot 2 & 0(-1) + 3\cdot 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 & -1 \\ -1 & 0 & 1 \\ 0 & 6 & 3 \end{pmatrix}.$$

Here $Be_2 = (1, 0, 3)^T$, so $A(Be_2) = (-2, 3)^T$. Also $(AB)e_2 = (-2, 3)^T$, the second column of AB. The dimensions differ, so AB and BA cannot be equal.

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EXERCISE 5.14 CHALLENGE

Let $n \geq 2$ and $A \in M_n(\mathbb{F})$ over an arbitrary field. Please prove:

1. A commutes with every diagonal matrix if and only if A is diagonal.
2. A commutes with every matrix in $M_n(\mathbb{F})$ if and only if $A = \lambda I_n$ for some $\lambda \in \mathbb{F}$.

You may use the matrix units E_{ij}, whose entry (i, j) is 1 and all other entries are zero.

WORKED SOLUTION

Suppose A commutes with every diagonal matrix. In particular, $AE_{kk} = E_{kk}A$ for every k. In entry (i, k) with $i \neq k$, the left side has value a_{ik} and the right side has value zero. Hence $a_{ik} = 0$ for every $i \neq k$, so A is diagonal. Conversely, two diagonal matrices commute because their product is diagonal and its diagonal entries are products of commuting scalars.
If A commutes with every matrix, part 1 gives $A = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. For $i \neq j$,

$$AE_{ij} = \lambda_i E_{ij}, \qquad E_{ij}A = \lambda_j E_{ij}.$$

Comparing entry (i, j) gives $\lambda_i = \lambda_j$. All diagonal entries are therefore equal, so $A = \lambda I_n$. Conversely, for every B, $(\lambda I_n)B = \lambda B = B(\lambda I_n)$. The proof uses only the scalars $0, 1$ and field multiplication, so it applies to finite fields as well.

NOTATION AND CONVENTIONS

For $1 \leq i \leq m$ and $1 \leq j \leq n$, the matrix $E_{ij} \in M_{m\times n}(F)$ has entry 1 at (i, j) and zero elsewhere. Every matrix $A = (a_{ij})$ has a unique representation $A = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij} E_{ij}$. Matrix addition and scalar multiplication therefore act componentwise, just as for columns in F^{mn}.

EXERCISE 5.15

Please determine whether A lies in the span of B_1, B_2 over $\mathbb{R}$.

$$A = \begin{pmatrix} -2 & 4 \\ 3 & -3 \end{pmatrix}, \qquad B_1 = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}, \qquad B_2 = \begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix}.$$

WORKED SOLUTION

$$aB_1 + bB_2 = \begin{pmatrix} a & -a \\ b & -b \end{pmatrix}.$$

Equality with A would require $a = -2$ and $-a = 4$, hence $a = -4$. This contradiction proves that A is not in the span. The choice $b = 3$ matches only the second row.

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EXERCISE 5.16

Let $A = \begin{pmatrix} 1 & -2 \\ 0 & 0 \end{pmatrix}$ over $\mathbb{R}$. Please characterize all matrices B satisfying $AB = BA$, and please show that they are precisely the linear combinations of A and $I_2 - A$.

WORKED SOLUTION

Write $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Computing both products gives

$$AB = \begin{pmatrix} a - 2c & b - 2d \\ 0 & 0 \end{pmatrix}, \qquad BA = \begin{pmatrix} a & -2a \\ c & -2c \end{pmatrix}.$$

Equality requires $c = 0$ and $b = -2(a - d)$. Hence

$$B = \begin{pmatrix} a & -2(a-d) \\ 0 & d \end{pmatrix} = aA + d(I_2 - A).$$

Substitution into both products shows conversely that every matrix of this form commutes with A.

EXERCISE 5.17

Work in $M_3(\mathbb{R})$. The matrix unit E_{ij} has entry 1 at (i, j) and zero elsewhere. Define

$$D_i = E_{ii} \quad (1 \le i \le 3), \qquad S_{ij} = E_{ij} + E_{ji}, \quad K_{ij} = E_{ij} - E_{ji} \quad (1 \le i < j \le 3).$$

$$\mathcal{B}_s = \{D_1, D_2, D_3, S_{12}, S_{13}, S_{23}\}, \qquad \mathcal{B}_a = \{K_{12}, K_{13}, K_{23}\}.$$

Please prove that $\mathcal{B}_s$ is a basis of the symmetric matrices and $\mathcal{B}_a$ is a basis of the skew-symmetric matrices. Please prove that their union is a basis of $M_3(\mathbb{R})$ and please compute the three dimensions.

WORKED SOLUTION

We have $D_i^T = D_i$, $S_{ij}^T = S_{ij}$, and $K_{ij}^T = -K_{ij}$. Every symmetric $A = (a_{ij})$ has the representation

$$A = \sum_{i=1}^{3} a_{ii} D_i + \sum_{1 \le i < j \le 3} a_{ij} S_{ij}.$$

The diagonal and upper-diagonal entries determine all coefficients uniquely. Thus the family spans and is independent, giving a basis of size $3 + 3 = 6$.
For skew-symmetric A, the diagonal is zero and $a_{ji} = -a_{ij}$. Hence

$$A = \sum_{1 \le i < j \le 3} a_{ij} K_{ij}.$$

Again, the entries above the diagonal uniquely determine the coefficients, so $\mathcal{B}_a$ is a basis of size 3.
Every $A \in M_3(\mathbb{R})$ satisfies

$$A = \frac{A + A^T}{2} + \frac{A - A^T}{2}.$$

The first summand is symmetric and the second skew-symmetric, so the union spans all

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EXERCISE 5.17 CONTINUED

matrices. The intersection of the two subspaces is $\{0\}$: $A^T = A = -A$ gives $2A = 0$, hence $A = 0$. Any zero combination of the union therefore splits into a zero combination in each basis, forcing every coefficient to vanish. The union is a basis, with dimensions

$$\dim \mathrm{Sym}_3(\mathbb{R}) = 6, \quad \dim \mathrm{ASym}_3(\mathbb{R}) = 3, \quad \dim M_3(\mathbb{R}) = 9 = 6 + 3.$$

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EXERCISE 5.18 CHALLENGE

Let $a_1, \ldots, a_{n+2} \in \mathbb{R}^n$, where $n \geq 1$. For a nonempty index set I, define the convex hull by

$$\operatorname{conv}\{a_i : i \in I\} = \left\{ \sum_{i \in I} \lambda_i a_i : \lambda_i \geq 0,\ \sum_{i \in I} \lambda_i = 1 \right\}.$$

Please prove that the indices can be partitioned into two nonempty sets I, J whose convex hulls intersect.

WORKED SOLUTION

Consider real coefficients $\alpha_1, \ldots, \alpha_{n+2}$ satisfying

$$\sum_{i=1}^{n+2} \alpha_i a_i = 0, \qquad \sum_{i=1}^{n+2} \alpha_i = 0.$$

This is a homogeneous system of $n+1$ equations in $n+2$ unknowns, so it has a nonzero solution. Since the coefficients sum to zero, at least one is positive and at least one is negative. Put

$$I = \{i : \alpha_i \geq 0\}, \qquad J = \{i : \alpha_i < 0\}, \qquad s = \sum_{i \in I} \alpha_i = -\sum_{j \in J} \alpha_j > 0.$$

These sets are nonempty and partition all indices. The vector equation yields

$$\sum_{i \in I} \frac{\alpha_i}{s} a_i = \sum_{j \in J} \frac{-\alpha_j}{s} a_j.$$

On each side the coefficients are nonnegative and sum to 1. The common point therefore lies in both convex hulls.

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TUTORIAL 6

Invertible Matrices and the Determinant

MOTIVATION AND GUIDING IDEA

Motivation. A basis turns the construction of a vector into the determination of unique coefficients. An invertible matrix guarantees that every input can be reconstructed from its output. Elementary row operations are reversible steps, so row reduction provides both a reconstruction algorithm and an algebraic explanation of why it works.

MOTIVATION AND GUIDING IDEA

Tutorial objectives. Develop methods for choosing and extending bases, recognize invertible matrices and calculate their inverses, and connect row reduction with products of elementary matrices. Check each solution through its meaning for a system of equations.

MOTIVATION AND GUIDING IDEA

From a basis to unique reconstruction. The columns of a square matrix form a basis if and only if every right-hand-side vector can be expressed uniquely using those columns. This is the meaning of invertibility and explains why an inverse solves a system for every right-hand side.

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6.1 Invertible Matrices

DEFINITION 6.1

A matrix $A \in M_n(\mathbb{F})$, with $n \geq 1$, is **invertible** if there is $B \in M_n(\mathbb{F})$ satisfying $AB = BA = I_n$.

The unique inverse B is denoted A^{-1}.

Basic Properties

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RECALL AND TOOLS

- If A is invertible, so is A^{-1}.
- If A, B are invertible, so is AB, and $(AB)^{-1} = B^{-1}A^{-1}$.
- If A is invertible, so is A^T, and $(A^T)^{-1} = (A^{-1})^T$.
- The identity is invertible, with $I^{-1} = I$.
- A square zero matrix of positive order is not invertible.

6.1.1 Exercises on the Algebra of Invertible Matrices

EXERCISE 6.1

Let A, B be square matrices, with A invertible, such that $A = B^2$ and $B = A^2$. Please prove that B is the inverse of A, that is, $B = A^{-1}$.

WORKED SOLUTION

We have

$$\begin{aligned} AB &= AA^2 = A^3, \\ A^3 &= A^3(AA^{-1}) = A^4A^{-1} \\ &= (A^2A^2)A^{-1} = B^2A^{-1} = AA^{-1} = I. \end{aligned}$$

Since $B = A^2$, also $BA = A^2A = A^3 = AB = I$. Therefore $AB = BA = I$, and $B = A^{-1}$.

6.1.2 An Algorithm for Finding the Inverse

PROPOSITION 6.1

A square matrix A is invertible if and only if it is row-equivalent to the identity matrix.

Algorithm

1. Form $(A \mid I)$ and row-reduce, aiming to put the left block into **reduced row-echelon form**.
2. Inspect the resulting left block:
   - If it is I, then A is invertible, and the right block is its inverse: $(I \mid A^{-1})$.
   - If it is not I, then A is not invertible.

EXERCISE 6.2

Please determine which matrices are invertible. For each invertible matrix, please find its inverse by row reduction and please verify the result.

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 6 \\ -2 & -4 \end{pmatrix}.$$

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EXERCISE 6.2 CONTINUED

WORKED SOLUTION

$$\left(\begin{array}{ccc|ccc} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & -2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right).$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 + 2R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & 1 & 2 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right).$$

$$A^{-1} = \begin{pmatrix} -2 & 1 & 2 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}, \qquad AA^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -2+2 & 1 & 2-2 \\ 1-1 & 0 & 1 \end{pmatrix} = I_3.$$

In B, the second row is $-2/3$ times the first. The operation $R_2 \leftarrow R_2 + (2/3)R_1$ creates a zero row, so B is not invertible.

6.2 Elementary Matrices

DEFINITION 6.2

- Elementary matrices are square matrices representing the elementary row operations used in row reduction.
- Applying an elementary row operation to a matrix is equivalent to multiplying it **on the left** by the corresponding elementary matrix.
- Each elementary matrix is obtained by applying its row operation to the identity matrix.

Consequence. Matrices A, B are row-equivalent if and only if there is a finite sequence of elementary matrices such that $B = E_1 E_2 \cdots E_k A$.

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1. **Interchanging rows** i and j is represented by $E^{(1)}_{i,j}$:

$$E^{(1)}_{i,j} = \begin{pmatrix} 1 & & & & \\ & 0 & & 1 & \\ & & 1 & & \\ & 1 & & 0 & \\ & & & & 1 \end{pmatrix}.$$

   - This matrix is its own inverse: interchange the same two rows again.

2. **Multiplying row** i by a nonzero scalar λ is represented by $E^{(2)}_{i,\lambda}$:

$$E^{(2)}_{i,\lambda} = \begin{pmatrix} 1 & & & & \\ & \ddots & & & \\ & & \lambda & & \\ & & & \ddots & \\ & & & & 1 \end{pmatrix}.$$

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- Its inverse is $E^{(2)}_{i,\lambda^{-1}}$: multiply the same row by the reciprocal scalar.

3. **Adding λ times row j** to row i, with $i \neq j$, is represented by $E^{(3)}_{i,j,\lambda}$. The entry λ is in row i, column j:

$$E^{(3)}_{i,j,\lambda} = \begin{pmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & \lambda & \\ & & & \ddots & & \\ & & & & 1 & \\ & & & & & 1 \end{pmatrix}.$$

- Its inverse is $E^{(3)}_{i,j,-\lambda}$: add $-\lambda$ times row j to row i.

EXERCISE 6.3

Please use the preceding reduction to write $A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix}$ as a product of elementary matrices.

WORKED SOLUTION

The reduction operations are $R_1 \leftrightarrow R_2$, then $R_3 \leftarrow R_3 - R_2$, then $R_1 \leftarrow R_1 + 2R_3$. Their matrices are

$$E_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}, \quad E_3 = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

$$E_3E_2E_1A = I_3, \qquad A = E_1^{-1}E_2^{-1}E_3^{-1}$$
$$= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The last two factors multiply to $\begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$. The first factor swaps the first two rows and recovers A. Each factor is elementary.

EXERCISE 6.4

Let $A, B, C \in \mathbb{R}^{n\times n}$ be invertible, and let $\alpha \in \mathbb{R}$ with $\alpha \neq 0$. Define

$$D := \alpha A^3 B^T C^2 A.$$

1. Please prove that D is invertible.
2. Please find an explicit expression for D^{-1} in terms of A, B, C, α.

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EXERCISE 6.4 CONTINUED

WORKED SOLUTION

1. Each of $\alpha I_n, A, A^3, B^T, C^2$ is invertible:
   - $\alpha \neq 0$, so αI_n is invertible.
   - A is invertible by assumption, so A^3 is invertible.
   - Since B is invertible, B^T is invertible with $(B^T)^{-1} = (B^{-1})^T$.
   - Since C is invertible, C^2 is invertible with $(C^2)^{-1} = (C^{-1})^2$.

   A product of invertible matrices is invertible, so D is invertible.

2. Write the ordered product $D = \alpha A^3 B^T C^2 A$. The inverse of a product is the product of the inverses in reverse order:

$$(XYZ)^{-1} = Z^{-1}Y^{-1}X^{-1}.$$

   Thus

$$D^{-1} = A^{-1}(C^2)^{-1}(B^T)^{-1}(A^3)^{-1}(\alpha I_n)^{-1}.$$

   Compute each factor:

$$(C^2)^{-1} = (C^{-1})^2, \quad (B^T)^{-1} = (B^{-1})^T, \quad (A^3)^{-1} = (A^{-1})^3, \quad (\alpha I_n)^{-1} = \frac{1}{\alpha} I_n.$$

   Substitution gives

$$D^{-1} = A^{-1}(C^{-1})^2(B^{-1})^T(A^{-1})^3 \cdot \frac{1}{\alpha} I_n = \frac{1}{\alpha} A^{-1}(C^{-1})^2(B^{-1})^T(A^{-1})^3.$$

   This is the required expression for D^{-1}.

EXERCISE 6.5

Let

$$A = \operatorname{diag}(\lambda_1, \ldots, \lambda_n) = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} \in \mathbb{R}^{n \times n}.$$

Please prove that A is invertible if and only if every $\lambda_i \neq 0$, and please find an explicit formula for A^{-1}.

WORKED SOLUTION

If A is invertible, then every $\lambda_i \neq 0$. Suppose A is invertible and, for a contradiction, that

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EXERCISE 6.5 CONTINUED

$\lambda_k = 0$ for some index k. Consider the standard vector

$$\vec{e}_k = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix},$$

with 1 in position k and zero in every other coordinate. Then

$$A\vec{e}_k = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ \lambda_k \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \vec{0},$$

since $\lambda_k = 0$. Thus $A\vec{e}_k = \vec{0}$ with $\vec{e}_k \neq \vec{0}$, a nontrivial solution of $Ax = 0$. If A were invertible, left multiplication by A^{-1} would give $e_k = A^{-1}0 = 0$, a contradiction. Therefore every $\lambda_i \neq 0$. **If every $\lambda_i \neq 0$, then A is invertible; we find its inverse.** Assume all these scalars are nonzero. Define

$$B := \operatorname{diag}\left(\frac{1}{\lambda_1}, \ldots, \frac{1}{\lambda_n}\right) = \begin{pmatrix} \frac{1}{\lambda_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{\lambda_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{\lambda_n} \end{pmatrix}.$$

Calculate

$$\begin{aligned} AB &= \operatorname{diag}(\lambda_1, \ldots, \lambda_n) \operatorname{diag}\left(\frac{1}{\lambda_1}, \ldots, \frac{1}{\lambda_n}\right) \\ &= \operatorname{diag}\left(\lambda_1 \frac{1}{\lambda_1}, \ldots, \lambda_n \frac{1}{\lambda_n}\right) \\ &= \operatorname{diag}(1, \ldots, 1) = I_n. \end{aligned}$$

The same calculation in reverse order gives $BA = I_n$. Hence B is the inverse of A, so

$$A^{-1} = B = \operatorname{diag}\left(\frac{1}{\lambda_1}, \ldots, \frac{1}{\lambda_n}\right).$$

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6.2.1 Equivalent Conditions

PROPOSITION 6.2

For a square matrix $A \in M_n(\mathbb{F})$, the following conditions are equivalent:

1. A is invertible.
2. A is row-equivalent to I_n.
3. $\operatorname{rank}(A) = n$.
4. For every column vector $\vec{b} \in \mathbb{F}^n$, the system $A\vec{x} = \vec{b}$ has a solution.
5. For every column vector $\vec{b} \in \mathbb{F}^n$, the system $A\vec{x} = \vec{b}$ has a unique solution $\vec{x} \in \mathbb{F}^n$.

6.2.2 The Connection with the Homogeneous System

In the preceding assertions, take $\vec{b} = \vec{0}$. There is then a unique solution, the trivial solution. Thus:

PROPOSITION 6.3

Another equivalent condition: $A \in M_n(\mathbb{R})$ is invertible if and only if its homogeneous system has only the trivial solution. In other words, the only solution of $A\vec{x} = \vec{0}$ is $\vec{x} = \vec{0}$.

EXERCISE 6.6

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix}, \qquad b = (\alpha, 2\gamma, 3\phi)^T, \qquad \alpha, \gamma, \phi \in \mathbb{R}.$$

Please solve $Ax = b$ in terms of the parameters and please prove that the system is homogeneous if and only if its unique solution is zero.

WORKED SOLUTION

$$x = A^{-1}b = \begin{pmatrix} -2 & 1 & 2 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha \\ 2\gamma \\ 3\phi \end{pmatrix} = \begin{pmatrix} -2\alpha + 2\gamma + 6\phi \\ \alpha \\ -\alpha + 3\phi \end{pmatrix}.$$

Indeed, $x_2 = \alpha$, $x_1 - 2x_3 = 2\gamma$, and $x_2 + x_3 = 3\phi$. The solution is unique because A is invertible. The system is homogeneous exactly when $b = 0$, equivalently $\alpha = \gamma = \phi = 0$, which gives $x = 0$. Conversely, if $x = 0$ is a solution, then $b = A0 = 0$.

EXERCISE 6.7

$$A = I_3 + N, \qquad N = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{pmatrix}.$$

Please compute N^2, N^3, please find A^{-1} without row reduction, and please solve $Ax = (3, 8, 3)^T$.

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EXERCISE 6.7 CONTINUED

WORKED SOLUTION

$$N^2 = \begin{pmatrix} 0 & 0 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad N^3 = 0.$$

$$(I+N)(I-N+N^2) = I - N + N^2 + N - N^2 + N^3 = I,$$

$$(I-N+N^2)(I+N) = I + N - N - N^2 + N^2 + N^3 = I.$$

$$A^{-1} = I - N + N^2 = \begin{pmatrix} 1 & 1 & -3 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{pmatrix}.$$

$$x = A^{-1}\begin{pmatrix} 3 \\ 8 \\ 3 \end{pmatrix} = \begin{pmatrix} 3+8-3\cdot 3 \\ 8-3\cdot 3 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}.$$

The only nonzero entry of N^2 is $(-1)\cdot 3 = -3$. Verification gives $Ax = (2-(-1), -1+3\cdot 3, 3)^T = (3,8,3)^T$.

Additional practice

EXERCISE 6.8 CHALLENGE

Let $u, v \in \mathbb{R}^n$, $n \geq 1$, and define $A = I_n + uv^T$.

1. Please prove that A is invertible if and only if $1 + v^T u \neq 0$, and please find its inverse in this case without using a determinant.
2. Please apply the formula to $u = (1,-1,0)^T$, $v = (2,0,1)^T$, and please solve $Ax = (5,0,-1)^T$.

WORKED SOLUTION

Write $P = uv^T$ and $c = v^T u \in \mathbb{R}$. Associativity gives

$$P^2 = (uv^T)(uv^T) = u(v^T u)v^T = cP.$$

If $1 + c \neq 0$, define $B = I_n - P/(1+c)$. Then

$$AB = (I_n + P)\left(I_n - \frac{P}{1+c}\right) = I_n + \left(1 - \frac{1}{1+c} - \frac{c}{1+c}\right)P = I_n.$$

Multiplication in the opposite order gives exactly the same expansion, so $BA = I_n$. Therefore

$$A^{-1} = I_n - \frac{uv^T}{1+v^T u}.$$

If $1 + c = 0$, then $c = -1$, so $u \neq 0$, whereas

$$Au = u + u(v^T u) = (1+c)u = 0.$$

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EXERCISE 6.8 CHALLENGE CONTINUED

An inverse would imply $u = 0$, a contradiction. This proves both directions.
For the given example,

$$v^T u = 2, \quad A = \begin{pmatrix} 3 & 0 & 1 \\ -2 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad A^{-1} = \begin{pmatrix} \frac{1}{3} & 0 & -\frac{1}{3} \\ \frac{2}{3} & 1 & \frac{1}{3} \\ 0 & 0 & 1 \end{pmatrix}.$$

$$x = A^{-1} \begin{pmatrix} 5 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} (5+1)/3 \\ 10/3 - 1/3 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}.$$

Verification: $Ax = (3 \cdot 2 - 1, -2 \cdot 2 + 3 + 1, -1)^T = (5, 0, -1)^T$.

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EXERCISE 6.9 CHALLENGE

Let $A \in M_n(\mathbb{R})$ satisfy $A^m = I_n$ for an integer $m \geq 2$, and define

$$P = \frac{1}{m} \sum_{k=0}^{m-1} A^k.$$

Please prove that $AP = PA = P$ and $P^2 = P$. Please find the inverse of $I_n + P$.

WORKED SOLUTION

Multiplication by A shifts the powers, so

$$AP = \frac{1}{m}(A + A^2 + \cdots + A^m) = P.$$

Since powers of A commute with A, we also have $PA = P$. Induction gives $A^k P = P$ for every $k \geq 0$. Thus

$$P^2 = \frac{1}{m} \sum_{k=0}^{m-1} A^k P = P.$$

Consequently

$$(I_n + P)(I_n - P/2) = I_n + P/2 - P^2/2 = I_n.$$

The reverse product is also I_n, since both factors are polynomials in P. Therefore $(I_n+P)^{-1} = I_n - P/2$.

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EXERCISE 6.10 CHALLENGE

Let $A, B \in M_n(\mathbb{R})$ and define the block matrix

$$C = \begin{pmatrix} A & B \\ B & A \end{pmatrix}.$$

Please prove that C is invertible if and only if $A+B$ and $A-B$ are invertible. In that case, please find C^{-1} in terms of $(A+B)^{-1}$ and $(A-B)^{-1}$.

WORKED SOLUTION

Define

$$J = \begin{pmatrix} I_n & I_n \\ I_n & -I_n \end{pmatrix}.$$

Since $J^2 = 2I_{2n}$, we have $J^{-1} = J/2$. Block multiplication gives

$$CJ = \begin{pmatrix} A+B & A-B \\ A+B & -(A-B) \end{pmatrix},$$

and therefore

$$J^{-1}CJ = \begin{pmatrix} A+B & 0 \\ 0 & A-B \end{pmatrix}.$$

Multiplication on either side by invertible matrices preserves invertibility. A block diagonal matrix is invertible exactly when both its diagonal blocks are invertible. This proves the equivalence. Writing $X = (A+B)^{-1}$ and $Y = (A-B)^{-1}$, we obtain

$$\begin{aligned} C^{-1} &= J \begin{pmatrix} X & 0 \\ 0 & Y \end{pmatrix} J^{-1} \\ &= \frac{1}{2} \begin{pmatrix} X+Y & X-Y \\ X-Y & X+Y \end{pmatrix}. \end{aligned}$$

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6.3 Uniqueness of Reduced Row Echelon Form

PROPOSITION 6.4

Every $A \in M_{m\times n}(F)$ has a unique reduced row echelon form.

Proof. Gaussian elimination proves existence. Suppose R, S are reduced row echelon matrices row-equivalent to A. Every row operation and its inverse express new rows as linear combinations of old rows. Thus exactly the same vectors can be obtained as combinations of the rows of R and of S.

Let the nonzero rows of R be $r_1, \dots, r_t$, with pivot positions $p_1 < \cdots < p_t$. In a nonzero combination, if c_k is the first nonzero coefficient, then the first nonzero entry of $\sum_i c_i r_i$ is in position p_k and equals c_k. Therefore the possible first nonzero positions of all such combinations are exactly $p_1, \dots, p_t$. The same set is obtained from S, so both matrices have the same pivot positions and the same number of nonzero rows.

Write row i of S as $s_i = \sum_j c_j r_j$. Entry p_j gives $c_j = (s_i)_{p_j}$, since a pivot column of R has one 1 and zeros elsewhere. Since S is also reduced, $(s_i)_{p_j}$ is 1 if $i = j$ and zero otherwise. Hence $s_i = r_i$. The zero rows also agree, so $S = R$.

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EXERCISE 6.11

Let $A \in M_{m\times n}(F)$ and $B \in M_{k\times n}(F)$. Suppose $Ax = 0$ and $Bx = 0$ have the same solutions. Please prove that the nonzero rows of the reduced row echelon forms of A and B agree.
Hint. Please check when a solution can satisfy $x_j = 1$ and $x_{j+1} = \cdots = x_n = 0$.

WORKED SOLUTION

Let the reduced forms be R, S. Row operations preserve solutions, so $Rx = 0$ and $Sx = 0$ also have the same solution set.
In a reduced system, x_j is free if and only if there is a solution with $x_j = 1$ and all later variables zero. If j is free, set it to 1 and every other free variable to zero. Every later pivot variable is zero, since its equation involves only free variables to its right. If j is a pivot, its equation is

$$x_j + \sum_{\ell>j} r_{i\ell}x_\ell = 0,$$

and the requested conditions would give $1 = 0$.
The solution set therefore determines the free positions and hence the pivot positions. Now set one free variable $x_j = 1$ and all other free variables to zero. The equation with pivot p_i gives

$$x_{p_i} + r_{ij} = 0, \qquad r_{ij} = -x_{p_i}.$$

Thus every entry in a free column is determined by the solutions. The pivot columns are already fixed by reduced form. Hence the nonzero rows of R, S agree. If there are no free variables, both forms have the same rows of I_n, so the conclusion still holds.

EXERCISE 6.12

Let p be prime and let $m, n \geq 1$. Please find the number of $m \times n$ reduced row echelon matrices over $\mathbb{Z}_p$ with exactly one pivot.

WORKED SOLUTION

There is one nonzero row, which is the first row. If its pivot is in column j, the preceding entries are zero, the pivot is 1, and each of the $n-j$ later entries has p choices. This gives p^{n-j} matrices with a pivot in column j.
The pivot position is unique, so the cases are disjoint. Summing gives

$$\sum_{j=1}^{n} p^{n-j} = 1 + p + \cdots + p^{n-1} = \frac{p^n - 1}{p - 1}.$$

The division here is ordinary integer division for counting, not an operation in the field. For example, $p = 3, n = 3$ gives $9 + 3 + 1 = 13$ forms, for every $m \geq 1$.

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6.4 The Determinant over a Field

DEFINITION 6.3

Let F be a field and let $n \geq 1$. A determinant is a function $D : M_n(F) \to F$ satisfying:

1. Linearity in each row separately, with the other rows fixed.
2. $D(A) = 0$ whenever two rows of A are equal. This is called **alternation**.
3. $D(I_n) = 1$.

We shall prove existence and uniqueness, and then write det for this function.

NOTATION AND CONVENTIONS

A permutation of $1, \ldots, n$ is an ordering of these numbers without repetitions. Their set is denoted by S_n. For $\sigma \in S_n$, let $\mathrm{inv}(\sigma)$ count the pairs $i < j$ with $\sigma(i) > \sigma(j)$, and define

$$\mathrm{sgn}(\sigma) = (-1)^{\mathrm{inv}(\sigma)}.$$

Swapping two positions changes the sign. Indeed, an adjacent swap changes the parity of the inversion count, and a swap of nonadjacent positions is a composition of an odd number of adjacent swaps.

PROPOSITION 6.5

The determinant exists and is unique over every field.
Existence. For $A = (a_{ij})$, define

$$D(A) = \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)}.$$

Every product uses exactly one entry from each row, so the expression is linear in each row. If rows r, s are equal, pair each permutation with the one obtained by swapping $\sigma(r), \sigma(s)$. Their products agree and their signs are opposite, so each pair cancels. At I_n, only the identity permutation contributes, giving 1. All three properties hold.
Uniqueness. Suppose D has these properties. Let e_j^T be the row with 1 in position j and zeros elsewhere. Alternation and linearity first give

$$0 = D(\ldots, u+v, \ldots, u+v, \ldots) = D(\ldots, u, \ldots, v, \ldots) + D(\ldots, v, \ldots, u, \ldots).$$

The two terms with equal rows vanish. Thus swapping rows changes the sign, and hence

$$D(e_{\sigma(1)}^T, \ldots, e_{\sigma(n)}^T) = \mathrm{sgn}(\sigma).$$

Write each row as $\sum_j a_{ij} e_j^T$ and expand using linearity in all rows. Any term with two repeated choices vanishes. The remaining terms correspond to permutations, so necessarily

$$D(A) = \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)}.$$

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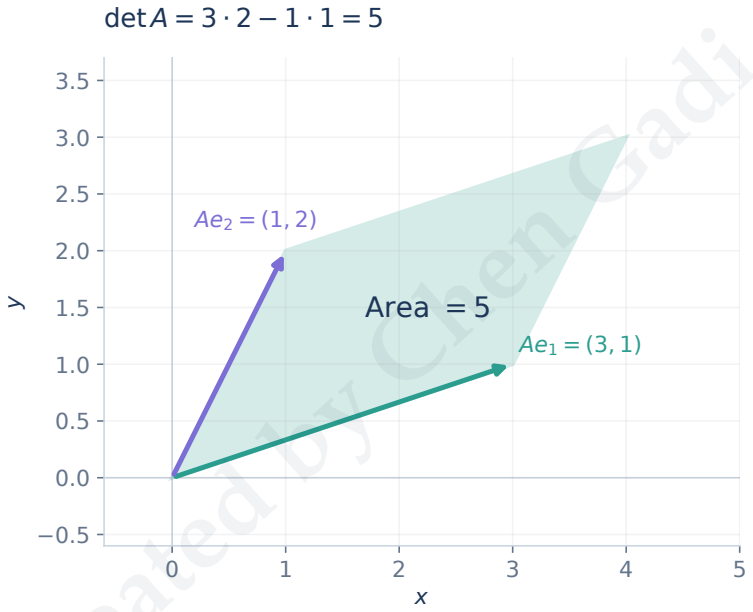
This is precisely the constructed function. The proof also works over $\mathbb{Z}_2$: cancelling opposite terms does not require division by 2.

NOTE

The geometric interpretation requires a real matrix. In $\mathbb{R}^2$, $|\det A|$ is the area scale factor; in $\mathbb{R}^3$, it is the volume scale factor. If $\det A = 0$, the image has smaller dimension. If $\det A \neq 0$, the sign indicates whether orientation is preserved or reversed.

6.5 The Determinant

The determinant is defined for square matrices. In the real plane, its absolute value is the area scale factor; its sign indicates whether orientation is preserved or reversed.



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Figure 6.1. A linear map sends the unit square to a parallelogram. The parallelogram's area is the absolute value of the determinant.

DEFINITION **6.4**

Let $n \in \mathbb{N}$. A function

$$\det : M_n(\mathbb{R}) \longrightarrow \mathbb{R}$$

is called the **determinant** if, for every $A \in M_n(\mathbb{R})$, it has the following properties:

1. **Interchanging rows (change of sign):** For $1 \leq i < j \leq n$, if $E^{(1)}_{i,j} \in M_n(\mathbb{R})$ is the elementary matrix interchanging rows i and j, then

$$\det\left(E^{(1)}_{i,j} A\right) = -\det(A).$$

2. **Multiplying a row by a scalar (homogeneity):** For $1 \leq i \leq n$ and $\lambda \in \mathbb{R} \setminus \{0\}$, if $E^{(2)}_{i,\lambda} \in M_n(\mathbb{R})$ is the elementary matrix multiplying row i by λ, then

$$\det\left(E^{(2)}_{i,\lambda} A\right) = \lambda \det(A).$$

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3. **Adding a multiple of another row (invariance):** For $1 \leq i \neq j \leq n$ and $\lambda \in \mathbb{R}$, if $E^{(3)}_{i,j,\lambda} \in M_n(\mathbb{R})$ is the elementary matrix performing $R_i \leftarrow R_i + \lambda R_j$, then

$$\det\left(E^{(3)}_{i,j,\lambda} A\right) = \det(A).$$

4. **Normalization:**

$$\det(I_n) = 1.$$

RECALL AND TOOLS

Two useful basic properties:

- If $A = (a_{ij}) \in M_n(\mathbb{R})$ is upper triangular, meaning that $a_{ij} = 0$ for every $i > j$, then

$$\det(A) = \prod_{k=1}^{n} a_{kk}.$$

- If rows i and j are equal for some $i \neq j$, or if A has a zero row, then $\det(A) = 0$. Similarly, two equal columns or a zero column imply $\det(A) = 0$.

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TUTORIAL 7

Determinant Properties, Cofactor Expansion, and Cramer's Rule

MOTIVATION AND GUIDING IDEA

By the end of this tutorial, you will be able to choose an efficient determinant calculation, track the effect of row operations, identify conditions for invertibility, and connect algebraic structure with changes of basis and area.

NOTATION AND CONVENTIONS

The set $F[X]$ consists of polynomials in X with coefficients in F. The degree of a nonzero polynomial is the largest power with a nonzero coefficient. If $p(a) = 0$, division by $X - a$ has constant remainder $p(a) = 0$, so $p(X) = (X - a)q(X)$. Induction on the degree therefore shows that a nonzero polynomial of degree d has at most d distinct roots. These are the facts used in the Vandermonde and Cauchy calculations below. For a matrix A with n columns, we also write $\ker A = \{x \in F^n : Ax = 0\}$ for its homogeneous solution set.

EXERCISE 7.1

Please compute the determinants using row operations, stating the effect of each operation on the determinant.

1. For $t \in \mathbb{R}$,
$$A(t) = \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 2 & 1 & t & 9 \end{pmatrix}.$$
2.
$$B = \begin{pmatrix} 2 & 1 & 0 & 3 & -1 \\ 0 & 1 & 2 & -1 & 1 \\ 2 & -2 & -6 & 6 & -4 \\ 1 & 0 & 1 & 0 & 2 \\ 0 & 2 & 0 & 1 & 1 \end{pmatrix}.$$

WORKED SOLUTION

1. Add multiples of other rows to the fourth row; these operations preserve the determi-

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EXERCISE 7.1 CONTINUED

nant:

$$A(t) \xrightarrow{R_4 \leftarrow R_4 - 2R_1} \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & -1 & t & 11 \end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 + R_2} \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & t+2 & 12 \end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 - (t+2)R_3} \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 8-2t \end{pmatrix}.$$

Multiplying the diagonal entries gives $\det A(t) = 1 \cdot 1 \cdot 1 \cdot (8 - 2t) = 8 - 2t$.

2. The rows of B satisfy $r_3 = r_1 - 3r_2$. Thus the two operations $R_3 \leftarrow R_3 - R_1$ and $R_3 \leftarrow R_3 + 3R_2$ produce a zero row without changing the determinant. A matrix with a zero row has determinant zero, so $\det B = 0$.

RECALL AND TOOLS

Basic properties (from the lectures):

- **Multiplicativity:** For every $A, B \in M_n(\mathbb{R})$,

$$\det(AB) = \det(A)\det(B).$$

- **Failure of additivity (in general):** In general, the following equality does not hold:

$$\det(A + B) = \det(A) + \det(B).$$

- **Invariance under transposition:** For every $A \in M_n(\mathbb{R})$,

$$\det(A^T) = \det(A).$$

- **Scaling the entire matrix:** For every $c \in \mathbb{R}$ and $A \in M_n(\mathbb{R})$,

$$\det(cA) = c^n \det(A),$$

where n is the size of the matrix.

Fundamental theorem:

PROPOSITION 7.1

For every $A \in M_n(\mathbb{R})$, A is invertible if and only if $\det(A) \neq 0$.

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EXERCISE 7.2

Let $A, B \in M_n(\mathbb{R})$, with A singular and B invertible. Please prove or disprove that the matrix in each part is singular:

1. $A^T B$.
2. $B^{-1}AB$.
3. $A + A^2 + A^3 + A^4$.
4. $(A + I)^2 - (A - I)^2$.
5. $(A + B)^2 + (A - B)^2$.

WORKED SOLUTION

Use $\det A = 0$, $\det B \neq 0$, and multiplicativity of the determinant.

1. $\det(A^T B) = \det(A^T)\det B = \det A \det B = 0$.
2. $\det(B^{-1}AB) = \det(B^{-1})\det A \det B = 0$.
3. Factor out A on the left:

$$A + A^2 + A^3 + A^4 = A(I + A + A^2 + A^3),$$

Therefore the determinant is zero.

4. Since $AI = IA = A$, we obtain

$$(A + I)^2 - (A - I)^2 = (A^2 + 2A + I) - (A^2 - 2A + I) = 4A.$$

Consequently, $\det(4A) = 4^n \det A = 0$.

5. Expand while preserving the order of multiplication:

$$(A + B)^2 + (A - B)^2 = (A^2 + AB + BA + B^2) + (A^2 - AB - BA + B^2) = 2(A^2 + B^2).$$

The determinant is therefore $2^n \det(A^2 + B^2)$, but it need not vanish. A counterexample is $A = 0$, $B = 2I_n$: the resulting matrix is $8I_n$, which is invertible and has determinant $8^n \neq 0$.

7.1 Further Determinant Expansions and Properties

7.1.1 The Determinant Formula

RECALL AND TOOLS

Let $A = (a_{ij}) \in M_n(\mathbb{R})$ be a square matrix of size n. For $1 \leq i, j \leq n$, the **minor matrix** corresponding to entry (i, j), denoted M_{ij}, is the square matrix of size $n - 1$ obtained by deleting row i and column j from A.

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DEFINITION 7.1

For every $n \geq 1$, the determinant is the unique function $\det : M_n(\mathbb{F}) \to \mathbb{F}$ that is linear in each row separately with the other rows fixed, vanishes when two rows are equal, and satisfies $\det I_n = 1$. Row operation rules and cofactor expansions follow. Expansion along row i gives $\det A = \sum_{j=1}^{n} a_{ij} C_{ij}$; the result is independent of the selected row. With the convention $\det(\varnothing) = 1$ for the empty matrix of size 0, the formula also holds for $n = 1$.

NOTE

For square matrices over any field, $\det(AB) = \det A \det B$, $\det(A^T) = \det A$, and $\det(cA) = c^n \det A$. Also, A is invertible exactly when $\det A \neq 0$; then $\det(A^{-1}) = (\det A)^{-1}$. The determinant is not generally additive: $\det(A + B) \neq \det A + \det B$ in general.

EXAMPLE 7.1

Let $A = \begin{pmatrix} 4 & -1 & 3 \\ 2 & 5 & 0 \\ -2 & 1 & 6 \end{pmatrix}$. Then $M_{11} = \begin{pmatrix} 5 & 0 \\ 1 & 6 \end{pmatrix}$ and $M_{23} = \begin{pmatrix} 4 & -1 \\ -2 & 1 \end{pmatrix}$.

DEFINITION 7.2

Let $A = (a_{ij}) \in M_n(\mathbb{R})$ be square of size n. Its **determinant**, denoted $\det(A)$, can be computed recursively in n:

1. If $n = 1$, then $A = (a_{11})$ and we define
$$\det(A) = a_{11}.$$
2. If $n = 2$, then
$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \qquad \det(A) = a_{11}a_{22} - a_{12}a_{21}.$$
3. If $n \geq 3$, then for every $1 \leq i \leq n$, expansion along row i gives
$$\det(A) = \sum_{j=1}^{n} a_{ij} \, (-1)^{i+j} \, \det(M_{ij}),$$
where M_{ij} is obtained by deleting row i and column j.
Equivalently, for every $1 \leq j \leq n$, expansion along column j gives
$$\det(A) = \sum_{i=1}^{n} a_{ij} \, (-1)^{i+j} \, \det(M_{ij}).$$

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EXERCISE 7.3

Please compute the determinants again, this time using cofactor expansion and its consequences.

1. For $t \in \mathbb{R}$,
$$A(t) = \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 2 & 1 & t & 9 \end{pmatrix}.$$
2.
$$B = \begin{pmatrix} 2 & 1 & 0 & 3 & -1 \\ 0 & 1 & 2 & -1 & 1 \\ 2 & -2 & -6 & 6 & -4 \\ 1 & 0 & 1 & 0 & 2 \\ 0 & 2 & 0 & 1 & 1 \end{pmatrix}.$$

WORKED SOLUTION

1. Expand along the first column. Its two nonzero entries are in positions $(1,1)$ and $(4,1)$, so
$$\det A(t) = \det M_{11} - 2\det M_{41}.$$
The minor matrices are
$$M_{11} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 1 & t & 9 \end{pmatrix}, \qquad M_{41} = \begin{pmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$
Expand the first along its first column and the second along its first row:
$$\det M_{11} = (1 \cdot 9 - 2t) + (2 \cdot 2 - 1 \cdot 1) = 12 - 2t,$$
$$\det M_{41} = 1(2 \cdot 2 - 1 \cdot 1) + (-1)(1 \cdot 1 - 2 \cdot 0) = 3 - 1 = 2.$$
Thus $\det A(t) = 12 - 2t - 2 \cdot 2 = 8 - 2t$, as obtained by the previous method.
2. Regard the determinant as a function of the rows. Since $r_3 = r_1 - 3r_2$, linearity in the third row gives
$$\begin{aligned}\det(r_1, r_2, r_1 - 3r_2, r_4, r_5) = {} & \det(r_1, r_2, r_1, r_4, r_5) \\ & - 3\det(r_1, r_2, r_2, r_4, r_5) = 0.\end{aligned}$$
Each term has two equal rows. Linearity in a row and vanishing for equal rows follow from the determinant expansion, so again $\det B = 0$.

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EXERCISE 7.4

Please find all $k \in \mathbb{R}$ for which

$$A(k) = \begin{pmatrix} 2 & k & 4 \\ 3 & k+1 & 6 \\ -1 & -1 & k+3 \end{pmatrix}$$

is singular, using its determinant.

WORKED SOLUTION

Expanding along the first row gives

$$\begin{aligned} \det A(k) &= 2\big((k+1)(k+3)+6\big) - k\big(3(k+3)+6\big) + 4(k-2) \\ &= 2k^2 + 8k + 18 - 3k^2 - 15k + 4k - 8 \\ &= -k^2 - 3k + 10 = -(k+5)(k-2). \end{aligned}$$

Thus the matrix is singular exactly when $k = -5$ or $k = 2$.

EXERCISE 7.5

1. Let $n \geq 2$ and let $x_1, \ldots, x_n$ be real (or complex) numbers. Define the Vandermonde matrix

$$V(x_1, \ldots, x_n) = \left(x_i^{j-1}\right)_{i,j=1}^{n} = \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{pmatrix}.$$

Please prove that

$$\det\big(V(x_1, \ldots, x_n)\big) = \prod_{1 \leq i < j \leq n} (x_j - x_i).$$

WORKED SOLUTION

1. Write

$$\Delta_n(x_1, \ldots, x_n) = \det\big(V(x_1, \ldots, x_n)\big).$$

We prove by induction on n that

$$\Delta_n(x_1, \ldots, x_n) = \prod_{1 \leq i < j \leq n} (x_j - x_i).$$

Base case, $n = 2$:

$$\Delta_2(x_1, x_2) = \det \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \end{pmatrix} = x_2 - x_1,$$

as required.

Inductive step: assume the statement holds for $n-1$, and prove it for n. Regard Δ_n as a polynomial in x_n, with $x_1, \ldots, x_{n-1}$ fixed. Expanding the determinant along row n gives

$$\Delta_n(x_1, \ldots, x_n) = \sum_{j=1}^{n} x_n^{j-1} \, (-1)^{n+j} \, \det(M_{nj}),$$

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EXERCISE 7.5 CONTINUED

Thus Δ_n is a polynomial in x_n of degree at most $n-1$. The only term containing x_n^{n-1} comes from column n, so its coefficient is

$$(-1)^{n+n}\det(M_{nn}) = \det\big(V(x_1,\dots,x_{n-1})\big) = \Delta_{n-1}(x_1,\dots,x_{n-1}).$$

If two of $x_1,\dots,x_{n-1}$ are equal, the matrix has two equal rows, and both sides of the desired formula are zero. Hence we may assume these numbers are pairwise distinct. For every $k \in \{1,\dots,n-1\}$, substituting $x_n = x_k$ makes rows k and n of V equal, so

$$\Delta_n(x_1,\dots,x_{n-1},x_k) = 0.$$

Therefore $(x_n - x_k)$ divides Δ_n for every $k = 1,\dots,n-1$ (as a polynomial in x_n). Since these roots are distinct and the degree is at most $n-1$, there is a polynomial P such that

$$\Delta_n(x_1,\dots,x_n) = \Big(\prod_{k=1}^{n-1}(x_n - x_k)\Big)\,P(x_1,\dots,x_{n-1}),$$

where P is independent of x_n.

Compare the coefficients of x_n^{n-1} on both sides. On the left the coefficient is $\Delta_{n-1}(x_1,\dots,x_{n-1})$; on the right the coefficient of x_n^{n-1} in $\prod_{k=1}^{n-1}(x_n - x_k)$ is 1. Hence

$$P(x_1,\dots,x_{n-1}) = \Delta_{n-1}(x_1,\dots,x_{n-1}).$$

It follows that

$$\Delta_n(x_1,\dots,x_n) = \Delta_{n-1}(x_1,\dots,x_{n-1})\prod_{k=1}^{n-1}(x_n - x_k).$$

Finally, by the induction hypothesis,

$$\Delta_{n-1}(x_1,\dots,x_{n-1}) = \prod_{1\le i<j\le n-1}(x_j - x_i),$$

and therefore

$$\Delta_n(x_1,\dots,x_n) = \Big(\prod_{1\le i<j\le n-1}(x_j - x_i)\Big)\Big(\prod_{k=1}^{n-1}(x_n - x_k)\Big) = \prod_{1\le i<j\le n}(x_j - x_i).$$

This proves the formula.

Additional practice

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EXERCISE 7.6 CHALLENGE

Let $A, B \in M_n(\mathbb{R})$ be invertible and satisfy $AB = -BA$. Please prove that n is even. For every $m \geq 1$, please construct such a pair of size $2m$ with $A^2 = B^2 = I_{2m}$.

WORKED SOLUTION

Determinant rules give

$$\det A \det B = \det(AB) = \det(-BA) = (-1)^n \det B \det A.$$

Both determinants are nonzero, so cancellation gives $1 = (-1)^n$. Over $\mathbb{R}$, n must be even. For every $m \geq 1$, take

$$A = \begin{pmatrix} I_m & 0 \\ 0 & -I_m \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & I_m \\ I_m & 0 \end{pmatrix}.$$

Direct block multiplication gives

$$AB = \begin{pmatrix} 0 & I_m \\ -I_m & 0 \end{pmatrix}, \qquad BA = \begin{pmatrix} 0 & -I_m \\ I_m & 0 \end{pmatrix} = -AB,$$

and $A^2 = B^2 = I_{2m}$. Thus both are invertible.

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EXERCISE 7.7 CHALLENGE

Let $n \geq 1$, and let $x_1, \ldots, x_n$ be distinct elements of a field F, and let $y_1, \ldots, y_n$ be distinct. Assume $x_i + y_j \neq 0$ for every i, j, and define

$$C = \left(\frac{1}{x_i + y_j} \right)_{i,j=1}^{n}.$$

Please prove that every square submatrix of C is invertible. Please deduce that the real matrix $H_n = (1/(i+j-1))_{i,j=1}^{n}$ is invertible for every $n \geq 1$.

WORKED SOLUTION

First suppose $Cc = 0$, and define

$$q(t) = \sum_{j=1}^{n} c_j \prod_{k \neq j} (t + y_k).$$

This polynomial has degree at most $n-1$. Row i of the equation gives

$$q(x_i) = \left(\prod_{k=1}^{n} (x_i + y_k) \right) \sum_{j=1}^{n} \frac{c_j}{x_i + y_j} = 0.$$

It has n distinct roots and must therefore be the zero polynomial. At $t = -y_j$, every term except term j vanishes, so

$$0 = q(-y_j) = c_j \prod_{k \neq j} (y_k - y_j).$$

The product is nonzero, forcing $c_j = 0$ for every j. Thus C is invertible. A square submatrix selects equally many row and column indices. The selected elements remain distinct within each list, and their sums are still nonzero, so the same proof applies. For the final assertion, take $x_i = i - 1$ and $y_j = j$ over $\mathbb{R}$. Every denominator $i + j - 1$ is positive, and $C = H_n$.

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EXERCISE 7.8 EXAM QUESTION

Please work in $\mathbb{Z}_7$. Please solve the system in terms of $\omega \in \mathbb{Z}_7$:

$$\begin{cases} x + y + z = 2, \\ x + \omega y + z = 5, \\ x + y + (\omega + 1) z = 2. \end{cases}$$

Please write the solution set and number of solutions in every case.

WORKED SOLUTION

Subtract the first equation from each of the other two:

$$\begin{cases} x + y + z = 2, \\ (\omega - 1) y = 3, \\ \omega z = 0. \end{cases}$$

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EXERCISE 7.8 EXAM QUESTION CONTINUED

1. If $\omega = 1$, the equation $0 = 3$ gives no solution.
2. If $\omega = 0$, then $-y = 3$, so $y = 4$. Choosing $z = t$ gives
$$S_0 = \{(5 - t, 4, t) : t \in \mathbb{Z}_7\}.$$
There are exactly seven solutions, since different values of t give different third coordinates.
3. If $\omega \in \{2, 3, 4, 5, 6\}$, both ω and $\omega - 1$ are invertible. Thus
$$z = 0, \qquad y = 3(\omega - 1)^{-1}, \qquad x = 2 - y.$$
Each case has a unique solution:

ω	2	3	4	5	6
x	6	4	1	3	0
y	3	5	1	6	2
z	0	0	0	0	0

As a check, the coefficient determinant is $\omega(\omega - 1)$, which vanishes only at $\omega = 0, 1$.

7.2 Cramer's Rule

NOTATION AND CONVENTIONS

Write $A = (a_1 \ \cdots \ a_n)$, where a_j is column j of A. For $b \in F^n$, let $A_i(b)$ be the matrix obtained by replacing column i of A with b.

PROPOSITION 7.2

Let $A \in M_n(F)$ with $\det A \neq 0$. The unique solution of $Ax = b$ is
$$x_i = \frac{\det A_i(b)}{\det A}, \qquad i = 1, \ldots, n.$$

Proof. Invertibility gives a unique solution. Matrix-vector multiplication gives
$$b = Ax = \sum_{j=1}^{n} x_j a_j.$$
By linearity of the determinant in column i,
$$\det A_i(b) = \sum_{j=1}^{n} x_j \det A_i(a_j).$$
When $j \neq i$, two columns are equal, so that term vanishes. When $j = i$, the matrix is A. Thus $\det A_i(b) = x_i \det A$. Dividing by $\det A \neq 0$ proves the formula.

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EXAMPLE 7.2

We solve $Ax = b$ over $\mathbb{R}$, where

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 0 \\ 2 \\ 7 \end{pmatrix}.$$

Expansion in the first row gives $\det A = 1 - 2(0 - 2) = 5$. Replacing the columns gives

$$\det A_1(b) = \det \begin{pmatrix} 0 & 2 & 0 \\ 2 & 1 & 1 \\ 7 & 0 & 1 \end{pmatrix} = -2(2 - 7) = 10,$$

$$\det A_2(b) = \det \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 2 & 7 & 1 \end{pmatrix} = 2 - 7 = -5,$$

$$\det A_3(b) = \det \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 2 & 0 & 7 \end{pmatrix} = 7 - 2(0 - 4) = 15.$$

Hence $x = (10/5, -5/5, 15/5)^T = (2, -1, 3)^T$. Direct substitution gives

$$Ax = \begin{pmatrix} 2 - 2 \\ -1 + 3 \\ 4 + 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 7 \end{pmatrix} = b.$$

EXERCISE 7.9

Let $a_0, \ldots, a_n \in F^n$. Let D_j be the determinant obtained from columns $a_0, \ldots, a_n$ by omitting a_j and keeping the order of the other columns. Please prove that

$$\sum_{j=0}^{n} (-1)^j D_j a_j = 0.$$

WORKED SOLUTION

Let a_{ij} be entry i of a_j. For each $i = 1, \ldots, n$, form

$$C_i = \begin{pmatrix} a_{i0} & a_{i1} & \cdots & a_{in} \\ a_{10} & a_{11} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{n0} & a_{n1} & \cdots & a_{nn} \end{pmatrix},$$

where the bottom n rows are the matrix $(a_0 \ \cdots \ a_n)$. The top row equals row i of the lower block, so $\det C_i = 0$. Expanding in the top row gives

$$0 = \sum_{j=0}^{n} (-1)^{1+(j+1)} a_{ij} D_j = \sum_{j=0}^{n} (-1)^j D_j a_{ij}.$$

This is entry i of the required sum. Every entry is zero, so the sum is zero.

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EXERCISE 7.9 CONTINUED

For example, over $\mathbb{R}$, take

$$a_0 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad a_1 = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad a_2 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

Then $D_0 = 11, D_1 = -2, D_2 = -5$, and

$$11a_0 + 2a_1 - 5a_2 = \begin{pmatrix} 11 + 4 - 15 \\ 22 - 2 - 20 \end{pmatrix} = 0.$$

If at least one minor is nonzero, the identity gives an explicit nontrivial dependence among the columns.

EXERCISE 7.10

Let A be an invertible real $n \times n$ matrix with integer entries. Please prove that the following conditions are equivalent:

1. For every $b \in \mathbb{Z}^n$, the solution of $Ax = b$ belongs to $\mathbb{Z}^n$.
2. $\det A \in \{1, -1\}$.

Here $\mathbb{Z}^n$ is the set of columns with integer entries.

WORKED SOLUTION

If $\det A = 1$ or -1, then for every $b \in \mathbb{Z}^n$ all entries of $A_i(b)$ are integers. Their determinants are integers, and Cramer's rule gives

$$x_i = \frac{\det A_i(b)}{\det A} \in \mathbb{Z}.$$

Conversely, choose $b = e_j$ in turn, where e_j has entry 1 in position j and zeros elsewhere. The solution is column j of A^{-1}, so every entry of A^{-1} is an integer. Thus both $\det A$ and $\det A^{-1}$ are integers. Multiplicativity gives

$$(\det A)(\det A^{-1}) = \det I_n = 1.$$

Two integers with product 1 are both 1 or both -1, so $\det A \in \{1, -1\}$.

For example, $\begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$ has determinant 1, so every integer right-hand side gives an integer solution. In contrast, for $A = \operatorname{diag}(2, 1)$ and $b = (1, 0)^T$, the solution is $(1/2, 0)^T$.

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TUTORIAL 8

Adjugates and Vector Spaces

NOTATION AND CONVENTIONS

The set $F[X]$ consists of polynomials in X with coefficients in F, and $F[X]_{\leq d}$ consists of those of degree at most d, including zero. Equality means equality of coefficients at every power. Addition and scalar multiplication act on coefficients. The formal derivative of $p(X) = \sum_{j=0}^{d} a_j X^j$ is $p'(X) = \sum_{j=1}^{d} j a_j X^{j-1}$; over a finite field, j is interpreted as a field element. The supplementary appendix contains further polynomial material.

MOTIVATION AND GUIDING IDEA

Motivation. A homogeneous linear constraint allows two admissible states to be added, or an admissible state to be scaled, without leaving the admissible set. Span describes which states can be constructed; independence identifies components that are not redundant. Together, these ideas describe a system using a small collection of independent components.

NOTATION AND CONVENTIONS

Notation guide. $\mathbb{F}$ is the scalar field, V a vector space over it, 0_V the zero vector, and $\alpha \in \mathbb{F}$ a scalar. $v \in S$ denotes membership and $S \subseteq V$ inclusion. $\operatorname{span}_{\mathbb{F}} S$ is the span over the indicated field. $U \cup W$ denotes union, $U \cap W$ intersection, $U + W$ a sum of subspaces, and $U \oplus W$ a direct sum. The symbol $\setminus$ denotes set difference and $\varnothing$ the empty set.

DEFINITION 8.1

Vector space axioms and solution tools. A vector space over a field F is a set V equipped with operations $V \times V \to V$ and $F \times V \to V$ satisfying, for all $u, v, w \in V$ and $\alpha, \beta \in F$,

$$u + v = v + u, \qquad (u + v) + w = u + (v + w).$$

There is $0_V \in V$ such that $v + 0_V = v$, and every $v \in V$ has an additive inverse $-v \in V$ satisfying $v + (-v) = 0_V$. In addition,

$$\alpha(u + v) = \alpha u + \alpha v, \qquad (\alpha + \beta)v = \alpha v + \beta v,$$

$$(\alpha\beta)v = \alpha(\beta v), \qquad 1_F v = v.$$

For $A \in M_{m\times n}(F)$, the system $Ax = b$ is consistent if and only if $\operatorname{rank} A = \operatorname{rank}[A \mid b]$. If x_0 is a particular solution, then its full solution set is $x_0 + \ker A$: indeed, $Ax = b$ is equivalent to $A(x - x_0) = 0$. This solution set is a subspace of F^n if and only if $b = 0$, since containing the zero vector forces $A0 = b = 0$.

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NOTE

A vector is an element of the vector space, so a matrix or polynomial can also be a vector. In an independence test, the unknowns are the *coefficients of the combination*; in a span-membership test, the target vector is given. For a nonhomogeneous condition such as $x + 2y + z = 3$, first check whether the zero vector belongs to the set.

MOTIVATION AND GUIDING IDEA

Objectives. We will formulate problems as linear systems, perform row reduction while preserving the solution set, and distinguish free variables from pivot variables. We will connect homogeneous systems with vector spaces and test closure systematically.

MOTIVATION AND GUIDING IDEA

Tutorial objectives. Test linear dependence and independence, find spanning sets, and distinguish the union, intersection, and sum of subspaces. State the scalar field in every test and connect the equations with their geometric interpretation.

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8.1 The Classical Adjugate

DEFINITION 8.2

Let $A = (a_{ij}) \in M_n(\mathbb{F})$ with $n \geq 2$. Write $A_{\widehat{i,j}}$ for the matrix obtained by deleting row i and column j. The scalar

$$C_{ij} = (-1)^{i+j} \det A_{\widehat{i,j}}$$

is its (i, j) cofactor. The classical adjugate is

$$\operatorname{adj}(A) = (C_{ij})^T.$$

For $n = 1$, set $\operatorname{adj}((a)) = (1)$. The notation $\operatorname{adj}(A)$ denotes the classical adjugate. The Hermitian adjoint will be denoted by $A^* = \overline{A}^T$.

PROPOSITION 8.1

Every square matrix over a field satisfies

$$A \operatorname{adj}(A) = \operatorname{adj}(A)A = (\det A)I_n.$$

Indeed, entry (i, j) of the first product is

$$(A \operatorname{adj}(A))_{ij} = \sum_{k=1}^{n} a_{ik}C_{jk}.$$

For $i = j$, this is expansion along row j. For $i \neq j$, it is the determinant expansion of the matrix obtained by replacing row j with row i. That matrix has two identical rows and determinant zero. Expansion along columns proves the identity for the second product. In

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particular, if $\det A \neq 0$, then

$$A^{-1} = \frac{1}{\det A}\,\mathrm{adj}(A).$$

EXAMPLE 8.1

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$(C_{ij}) = \begin{pmatrix} d & -c \\ -b & a \end{pmatrix}, \qquad \mathrm{adj}(A) = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Both products can be computed directly:

$$A\,\mathrm{adj}(A) = \begin{pmatrix} ad - bc & -ab + ba \\ cd - dc & -cb + da \end{pmatrix} = (ad - bc)I_2,$$

$$\mathrm{adj}(A)A = \begin{pmatrix} da - bc & db - bd \\ -ca + ac & -cb + ad \end{pmatrix} = (ad - bc)I_2.$$

When $ad - bc \neq 0$, division by that scalar gives the usual inverse formula.

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EXERCISE 8.1

Consider the real matrix

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \end{pmatrix}.$$

Please compute $\det A$, every cofactor, $\operatorname{adj}(A)$, and A^{-1}. Please verify both adjugate product identities.

WORKED SOLUTION

The matrix is triangular, so $\det A = 1 \cdot 1 \cdot 2 = 2$. The minors and signs give

$$\begin{aligned} C_{11} &= 1 \cdot 2 - 3 \cdot 0 = 2, & C_{12} &= -(0 \cdot 2 - 3 \cdot 0) = 0, & C_{13} &= 0 \cdot 0 - 1 \cdot 0 = 0, \\ C_{21} &= -(2 \cdot 2 - 0 \cdot 0) = -4, & C_{22} &= 1 \cdot 2 - 0 \cdot 0 = 2, & C_{23} &= -(1 \cdot 0 - 2 \cdot 0) = 0, \\ C_{31} &= 2 \cdot 3 - 0 \cdot 1 = 6, & C_{32} &= -(1 \cdot 3 - 0 \cdot 0) = -3, & C_{33} &= 1 \cdot 1 - 2 \cdot 0 = 1. \end{aligned}$$

Thus

$$\operatorname{adj}(A) = \begin{pmatrix} 2 & -4 & 6 \\ 0 & 2 & -3 \\ 0 & 0 & 1 \end{pmatrix}, \qquad A^{-1} = \begin{pmatrix} 1 & -2 & 3 \\ 0 & 1 & -3/2 \\ 0 & 0 & 1/2 \end{pmatrix}.$$

Direct multiplication verifies

$$A \operatorname{adj}(A) = \begin{pmatrix} 2 & -4+4 & 6-6 \\ 0 & 2 & -3+3 \\ 0 & 0 & 2 \end{pmatrix} = 2I_3,$$

$$\operatorname{adj}(A) A = \begin{pmatrix} 2 & 4-4 & -12+12 \\ 0 & 2 & 6-6 \\ 0 & 0 & 2 \end{pmatrix} = 2I_3.$$

Dividing by 2 also verifies $AA^{-1} = A^{-1}A = I_3$.

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8.2 Vector spaces

RECALL AND TOOLS

1. To define a vector space V over a field $\mathbb{F}$, we specify an addition of vectors in V and a multiplication of a vector in V by a scalar in $\mathbb{F}$, satisfying the vector space axioms. These are **in addition** to addition and multiplication in $\mathbb{F}$. A product of two vectors is not part of the vector space axioms; if needed later, it will be defined separately.
2. Every field is a vector space over itself, with the field's addition and multiplication.
3. A vector space V is always defined over a field $\mathbb{F}$. Always specify the field of scalars. For example, as discussed in the lecture, $\mathbb{C}$ can be regarded as a vector space over $\mathbb{C}$ or over $\mathbb{R}$. These are two different vector spaces.

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EXAMPLE 8.2

Let $\mathbb{F}$ be a field.

(a) $\mathbb{F}^n$: the set of n-tuples over $\mathbb{F}$, where $n \in \mathbb{N}$.

(b) $M_{m\times n}(\mathbb{F})$: the set of $m \times n$ matrices over $\mathbb{F}$.

(c) $\mathbb{F}[X]$: the set of all polynomials with coefficients in $\mathbb{F}$.

EXERCISE 8.2

Please determine whether the following sets are subspaces of $\mathbb{R}[X]$ over $\mathbb{R}$:

1. $V = \{p(X) \in \mathbb{R}[X] : p(-2) = 0,\ p(3) \neq 0\}$.
2. $V = \{p(X) \in \mathbb{R}[X] : \exists c \in \mathbb{R} \setminus \{0\},\ p(c) = 0\}$.

WORKED SOLUTION

1. No. The zero polynomial is absent. Addition also fails: for
$$p(X) = (X+2)(X-8), \qquad q(X) = (X+2)^2,$$
we have $p(-2) = q(-2) = 0$ and $p(3) = -25$, $q(3) = 25$, but
$$(p+q)(X) = (X+2)(2X-6), \qquad (p+q)(3) = 0.$$
2. No. The polynomials $(X-2)^2$ and $(X+2)^2$ belong to the set, with roots $2, -2$, respectively. Their sum is $2X^2 + 8 > 0$ for all real X, so it does not belong.

8.3 Subspaces

DEFINITION 8.3

Let V be a vector space over a field $\mathbb{F}$, and let $U \subseteq V$ be a subset that is itself a vector space over $\mathbb{F}$ under the addition and scalar multiplication inherited from V. Then U is a vector subspace of V.

PROPOSITION 8.2

Let V be a vector space over a field $\mathbb{F}$, and let $U \subseteq V$ be a subset. Then U is a vector subspace if and only if all of the following conditions hold:

1. $0 \in U$, where this is the zero vector of V.
2. U is closed under addition.
3. U is closed under scalar multiplication.

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EXAMPLE 8.3

1. The set $V = \{p(X) \in F[X] : p(-3) = 0\}$ is a subspace of $F[X]$. It contains the zero polynomial. For $p, q \in V$ and $\alpha \in F$,

$$(p+q)(-3) = p(-3) + q(-3) = 0, \qquad (\alpha p)(-3) = \alpha p(-3) = 0.$$

   Thus it is closed under addition and scalar multiplication.

2. For $n \geq 3$, the set $W = \{x \in \mathbb{R}^n : x_1 + x_3 = 0\}$ is a subspace. It contains zero. If $x, y \in W$, then

$$(x_1 + y_1) + (x_3 + y_3) = (x_1 + x_3) + (y_1 + y_3) = 0.$$

   Also $(\alpha x)_1 + (\alpha x)_3 = \alpha(x_1 + x_3) = 0$ for every $\alpha \in \mathbb{R}$.

EXERCISE 8.3

Please prove or please disprove:

1. The set $U = \{(a, 3a^2) : a \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^2$.
2. The symmetric matrices form a subspace of $M_n(F)$.

WORKED SOLUTION

1. This is not a subspace: $(1, 3) \in U$, but

$$(1,3) + (1,3) = (2,6) \notin U,$$

   because a first coordinate of 2 requires second coordinate $3 \cdot 2^2 = 12$.

2. The zero matrix is symmetric. If $A^T = A$, $B^T = B$ and $\lambda \in F$, then

$$(A+B)^T = A^T + B^T = A + B, \qquad (\lambda A)^T = \lambda A^T = \lambda A.$$

   Thus all three subspace-test conditions hold.

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8.4 Vector Spaces

RECALL AND TOOLS

1. To define a vector space V over a field $\mathbb{F}$, specify addition of vectors in V and multiplication of a vector in V by a scalar in $\mathbb{F}$, satisfying the vector-space axioms. These operations are specified **in addition** to the addition and multiplication in $\mathbb{F}$. Multiplication of two vectors is not a vector-space axiom; any such operation used later must be defined separately.
2. Every field is a vector space over itself, using its own addition and multiplication.
3. A vector space V is always defined over a field $\mathbb{F}$, which must be stated. For example, as discussed in the lecture, $\mathbb{C}$ can be viewed as a vector space over $\mathbb{C}$ or over $\mathbb{R}$. These

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are two different vector-space structures.

EXAMPLE 8.4

Let $\mathbb{F}$ be a field.

(a) $\mathbb{F}^n$: the set of n-tuples over $\mathbb{F}$ ($n \in \mathbb{N}$).

(b) $M_{m\times n}(\mathbb{F})$: the set of $m \times n$ matrices over $\mathbb{F}$.

(c) $\mathbb{F}[X]$: the set of all polynomials with coefficients in $\mathbb{F}$.

EXERCISE 8.4

Please determine whether the following sets are subspaces of $\mathbb{R}[X]$ over $\mathbb{R}$ with the usual operations.

1. $V = \{p(X) \in \mathbb{R}[X] : p(-3) = 0,\ p(2) \neq 0\}$.
2. $W = \{p(X) \in \mathbb{R}[X] : \exists c \in [0, 1],\ p(c) = 0\}$.

WORKED SOLUTION

1. No. The zero polynomial does not belong to V. Addition also fails: for
$$p(X) = (X + 3)(X - 1), \qquad q(X) = -(X + 3),$$
we have $p(-3) = q(-3) = 0$, $p(2) = 5$, and $q(2) = -5$. Thus $p, q \in V$, but
$$(p + q)(X) = (X + 3)(X - 2), \qquad (p + q)(2) = 0,$$
so $p + q \notin V$.
2. No. Both $p(X) = X$ and $q(X) = 1 - X$ belong to W, with roots 0 and 1. Their sum is the constant polynomial 1, which has no root in the interval. Thus W is not closed under addition.

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8.4.1 Subspaces

DEFINITION 8.4

Let V be a vector space over a field $\mathbb{F}$. A subset $U \subseteq V$ is a **vector subspace** of V if it is itself a vector space over $\mathbb{F}$ under the addition and scalar multiplication inherited from V.

PROPOSITION 8.3

Let V be a vector space over a field $\mathbb{F}$ and $U \subseteq V$. Then U is a subspace if and only if all the following conditions hold:

1. $0 \in U$, where 0 is the zero vector of V.
2. U is closed under addition.

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3. U is closed under scalar multiplication.

EXAMPLE 8.5

1. $V = \{p(X) \in \mathbb{F}[X] : p(-3) = 0\}$.
2. $U = \{x \in \mathbb{R}^n : x_1 = 2x_2\}$, where $n \geq 2$.

Both sets are subspaces. The zero polynomial lies in V, and for $p, q \in V$ and $\alpha \in \mathbb{F}$,

$$(p+q)(-3) = p(-3) + q(-3) = 0, \qquad (\alpha p)(-3) = \alpha p(-3) = 0.$$

The zero vector also lies in U. If $x, y \in U$ and $\alpha \in \mathbb{R}$, then

$$(x+y)_1 = x_1 + y_1 = 2x_2 + 2y_2 = 2(x+y)_2,$$

$$(\alpha x)_1 = \alpha x_1 = 2\alpha x_2 = 2(\alpha x)_2.$$

Thus both sets are closed under addition and scalar multiplication.

EXERCISE 8.5

Please prove or disprove.

1. $U = \{(a, 5a^2)^T : a \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^2$.
2. The symmetric matrices form a subspace of $M_n(\mathbb{F})$.

WORKED SOLUTION

1. No.
$$(1,5)^T \in U, \qquad (1,5)^T + (1,5)^T = (2,10)^T \notin U,$$
because $5 \cdot 2^2 = 20 \neq 10$, so addition is not closed.
2. Yes. $0^T = 0$, and if $A^T = A$, $B^T = B$, and $\lambda \in \mathbb{F}$, then
$$(A+B)^T = A^T + B^T = A + B, \qquad (\lambda A)^T = \lambda A^T = \lambda A.$$
All three subspace conditions hold.

8.5 Linear Combinations

DEFINITION 8.5

Let V be a vector space over a field $\mathbb{F}$ and let $v_1, \ldots, v_k \in V$. A vector $v \in V$ is a **linear combination** of $v_1, \ldots, v_k$ if there are $a_1, \ldots, a_k \in \mathbb{F}$ such that $v = a_1 v_1 + \cdots + a_k v_k$.

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DEFINITION 8.6

Let V be a vector space over a field $\mathbb{F}$. For a set $S \subseteq V$, not necessarily finite, define $\operatorname{span}_{\mathbb{F}} S$ as the set of all finite linear combinations of elements of S:

$$\operatorname{span}_{\mathbb{F}} S = \left\{ \sum_{j=1}^{k} \alpha_j v_j \;\middle|\; k \geq 0,\ v_j \in S,\ \alpha_j \in \mathbb{F} \right\}.$$

The empty sum ($k = 0$) is allowed and equals the zero vector. In particular, $\operatorname{span}_{\mathbb{F}} \varnothing = \{0\}$.

NOTE

If V is a vector space over $\mathbb{F}$ and $S \subseteq V$, then $\operatorname{span} S$ is a subspace of V.

DEFINITION 8.7

If V is a vector space over $\mathbb{F}$ and $S \subseteq V$ satisfies $V = \operatorname{span} S$, then S **spans** V and is called a **spanning set**. If S is finite, V is **finitely generated**.

NOTE

Pay attention to the scalar field. A set $S \subseteq V$ can span over one field and fail to span over another. When more than one field is possible, write $\operatorname{span}_{\mathbb{F}} S$ to specify the field.

EXERCISE 8.6

Let V be a vector space over $\mathbb{F}$, let $\varnothing \neq S \subseteq V$ be a finite set of size k, and let $v \in V$. Please prove that

$$\operatorname{span}(S \cup \{v\}) = \operatorname{span} S \quad \Longleftrightarrow \quad v \in \operatorname{span} S.$$

WORKED SOLUTION

$\Longrightarrow$: Assume $\operatorname{span}(S \cup \{v\}) = \operatorname{span} S$. Write $v = 0 \cdot v_1 + \cdots + 0 \cdot v_k + 1 \cdot v$. Thus v is a linear combination of elements of $S \cup \{v\}$, so $v \in \operatorname{span}(S \cup \{v\})$. The assumed equality gives $v \in \operatorname{span} S$.

$\Longleftarrow$: The inclusion $\operatorname{span}(S \cup \{v\}) \supseteq \operatorname{span} S$ always holds: any linear combination of elements of S is also one of elements of $S \cup \{v\}$, with coefficient zero on v. For the opposite inclusion, write $S = \{s_1, \ldots, s_k\}$ and take $x \in \operatorname{span}(S \cup \{v\})$. There are $a_1, \ldots, a_{k+1} \in \mathbb{F}$ with

$$a_1 s_1 + \cdots + a_k s_k + a_{k+1} v = x.$$

Since $v \in \operatorname{span} S$, there are $b_1, \ldots, b_k \in \mathbb{F}$ with $b_1 s_1 + \cdots + b_k s_k = v$. Substituting gives

$$(a_1 + a_{k+1} b_1) s_1 + \cdots + (a_k + a_{k+1} b_k) s_k = x.$$

Hence $x \in \operatorname{span} S$, as required.

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DEFINITION 8.8

Let V be a vector space over $\mathbb{F}$ and $S \subseteq V$.

1. The set S is **linearly dependent** if there are pairwise distinct vectors $v_1, \ldots, v_k \in S$ and scalars $\alpha_1, \ldots, \alpha_k \in \mathbb{F}$, not all zero, such that
$$\alpha_1 v_1 + \cdots + \alpha_k v_k = 0.$$
2. The set S is **linearly independent** if, for every finite choice of pairwise distinct vectors $v_1, \ldots, v_k \in S$, this equality implies $\alpha_1 = \cdots = \alpha_k = 0$.

The vectors need not be tuples: they may also be polynomials, matrices, or elements of another vector space.

8.6 Bases and Dimension

DEFINITION 8.9

Let V be a vector space over a field $\mathbb{F}$. A set $\{u_1, \ldots, u_n\} \subseteq V$ is a **basis** of V if it is linearly independent and spans V.

A finite-dimensional vector space can have many bases, but all its bases have the same number of elements. This number is the **dimension** of the space.

EXAMPLE 8.6

$$\dim(\mathbb{F}^n) = n, \qquad \dim M_{m\times n}(\mathbb{F}) = mn, \qquad \dim \mathbb{F}_d[x] = d+1.$$

For example, the standard basis of $\mathbb{R}^3$ is

$$e_1 = (1,0,0)^T;\ e_2 = (0,1,0)^T;\ e_3 = (0,0,1)^T.$$

The standard basis of $\mathbb{R}_3[x]$ is $(1, x, x^2, x^3)$.

RECALL AND TOOLS

In a vector space of known finite dimension n, any two of the following three properties suffice to establish that a set is a basis:

1. Linear independence.
2. Spanning the space.
3. Having exactly n elements, the dimension of the space.

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8.7 Bases and Dimension

DEFINITION 8.10

Let V be a vector space over a field $\mathbb{F}$. A set $\{u_1, \ldots, u_n\} \subseteq V$ is a **basis** of V if it is linearly independent and spans V.

A finite-dimensional vector space can have many bases, but all its bases have the same number of elements. This number is the **dimension** of the space.

EXAMPLE 8.7

$$\dim(\mathbb{F}^n) = n, \qquad \dim M_{m\times n}(\mathbb{F}) = mn, \qquad \dim \mathbb{F}_d[x] = d+1.$$

For example, the standard basis of $\mathbb{R}^4$ is

$$e_1 = (1,0,0,0)^T;\ e_2 = (0,1,0,0)^T;\ e_3 = (0,0,1,0)^T;\ e_4 = (0,0,0,1)^T.$$

The standard basis of $\mathbb{R}_1[x]$ is $(1, x)$.

RECALL AND TOOLS

Let V be a vector space over a field $\mathbb{F}$ with finite dimension, $\dim V < \infty$.

1. If B_1, B_2 are bases of V, then $|B_1| = |B_2|$. Thus dimension is well-defined: $\dim V$ is the cardinality of any basis of V.
2. If $W = \{w_1, \ldots, w_m\} \subseteq V$ is linearly independent, then
$$|W| = m \leq \dim V.$$
One cannot choose more independent vectors than the number of vectors in a basis.
3. If $W = \{w_1, \ldots, w_m\} \subseteq V$ spans V, that is, $V = \operatorname{span}(W)$, then
$$|W| = m \geq \dim V.$$
Every spanning set therefore has at least $\dim V$ vectors.
4. Suppose $W = \{w_1, \ldots, w_m\} \subseteq V$ satisfies:
   - $\dim V = |W| = m$;
   - W is either linearly independent or spans V.

   Then W is a basis of V. If W is independent and has exactly $\dim V$ vectors, no further vector can be added while preserving independence, so W already spans. If W spans and has exactly $\dim V$ vectors, there can be no nontrivial zero linear combination of its elements, so they are independent.

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PROPOSITION 8.4

An additional important fact is the following. Let V have dimension n, and let $W = \{w_1, \ldots, w_k\} \subseteq V$ be linearly independent, with $k \leq n$. There is a set

$$U = \{u_{k+1}, \ldots, u_n\} \subseteq V$$

such that

$$B = W \cup U = \{w_1, \ldots, w_k, u_{k+1}, \ldots, u_n\}$$

is a basis of V. Thus **every linearly independent set in a finite-dimensional space can be extended to a basis**. This is a central tool for constructing bases and calculating dimensions.

RECALL AND TOOLS

To establish that a set is a basis of a space of known finite dimension n, it suffices to verify any two of:

1. Linear independence.
2. Spanning the space.
3. Having exactly n elements, the dimension of the space.

8.8 The Dimension Formula

PROPOSITION 8.5

Let V be a finitely generated vector space over a field $\mathbb{F}$, and let U be a subspace of V.

1. U is finitely generated and $\dim U \leq \dim V$.
2. $\dim U = \dim V$ if and only if $U = V$.

8.9 The Dimension Formula

PROPOSITION 8.6

Let V be a finitely generated vector space over a field $\mathbb{F}$, and let U be a subspace of V.

1. U is finitely generated and $\dim U \leq \dim V$.
2. $\dim U = \dim V$ if and only if $U = V$.

EXERCISE 8.7

Let $p(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$ be a real polynomial.

1. Please find all polynomials satisfying $p(0) = 0$, $p(2) = 4$, $p'(0) = 0$, $p'(1) = 0$. Please determine whether the solution set is a subspace and whether $t = 1$ is a local maximum or minimum.

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EXERCISE 8.7 CONTINUED

2. Please repeat with $p(0) = 2$, $p(2) = 6$, $p'(0) = 0$, $p'(1) = 0$.

3. Add $p'(2) = 5$ to the second part. Please determine whether the system is consistent and whether its solution set is a subspace.

WORKED SOLUTION

We have $p'(t) = a_1 + 2a_2t + 3a_3t^2$. In part one, $a_0 = a_1 = 0$, while the remaining conditions give

$$4a_2 + 8a_3 = 4, \qquad 2a_2 + 3a_3 = 0.$$

Subtracting twice the second equation from the first gives $2a_3 = 4$, so $a_3 = 2$ and $a_2 = -3$. The unique solution is

$$p_1(t) = -3t^2 + 2t^3.$$

Check: $p_1(2) = -12 + 16 = 4$ and $p_1'(1) = -6 + 6 = 0$. The zero polynomial fails $p(2) = 4$, so the solution set is not a subspace. Also $p_1''(t) = -6 + 12t$, hence $p_1''(1) = 6 > 0$, giving a local minimum.
In part two, $a_0 = 2$, $a_1 = 0$, and the other equations are unchanged. Thus

$$p_2(t) = 2 - 3t^2 + 2t^3.$$

Again the solution is unique, its set is not a subspace, and $t = 1$ is a local minimum. In part three, this unique polynomial gives

$$p_2'(2) = -12 + 24 = 12 \neq 5.$$

The system is inconsistent, and the empty solution set is not a subspace.

EXERCISE 8.8

Let

$$p(t) = a_0 + a_1t + a_2t^2 + a_3t^3 \in \mathbb{R}_3[t].$$

1. Please find all polynomials satisfying $p(0) = 0$, $p(1) = -1$, and $p'(0) = p'(1/2) = 0$. Please determine whether the solution set is a subspace and whether $t = 1/2$ is a local maximum or minimum.

2. Please repeat for $p(0) = 2$, $p(1) = 1$, and $p'(0) = p'(1/2) = 0$.

3. Please add $p'(1) = 1$ to the preceding conditions and please determine whether a solution exists.

WORKED SOLUTION

$$p'(t) = a_1 + 2a_2t + 3a_3t^2.$$

In part 1, $a_0 = a_1 = 0$. The remaining conditions give

$$a_2 + a_3 = -1, \qquad a_2 + \tfrac{3}{4}a_3 = 0.$$

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EXERCISE 8.8 CONTINUED

Subtracting gives $a_3/4 = -1$, so $a_3 = -4$ and $a_2 = 3$. The unique solution is

$$p_1(t) = 3t^2 - 4t^3.$$

Indeed, $p_1(1) = -1$ and $p_1'(1/2) = 3 - 3 = 0$. The solution set is not a subspace because the zero polynomial fails $p(1) = -1$. Also,

$$p_1''(t) = 6 - 24t, \qquad p_1''(1/2) = -6 < 0,$$

so this is a local maximum.
In part 2, $a_0 = 2$, $a_1 = 0$, and $a_2 + a_3 = 1 - 2 = -1$. The same system determines a_2, a_3, giving

$$p_2(t) = 2 + 3t^2 - 4t^3.$$

The solution is again unique, and its set is not a subspace because it does not contain the zero polynomial. The derivatives coincide, so $t = 1/2$ is again a local maximum.
In part 3, $p_2'(1) = 6 - 12 = -6 \neq 1$. Thus the additional condition contradicts the preceding ones. The solution set is empty and is not a subspace.

DEFINITION 8.11

Hypotheses and tools. Let V have dimension $n < \infty$ over $\mathbb{F}$. Every independent set in V can be extended to a basis; relative to an ordered basis, every $v \in V$ has a unique expression. If $A \in M_n(\mathbb{F})$, $n \geq 1$, is invertible, then $A^{-1} \in M_n(\mathbb{F})$ satisfies $AA^{-1} = A^{-1}A = I_n$. The inverse is unique: if B, C are inverses, then $B = B(AC) = (BA)C = C$.
The following are equivalent: A is invertible; its reduced row-echelon form is I_n; rank $A = n$; $Ax = b$ has a unique solution for every $b \in \mathbb{F}^n$; $Ax = 0$ has only the zero solution; the columns of A form a basis of $\mathbb{F}^n$. For square matrices of the same order, $(AB)^{-1} = B^{-1}A^{-1}$ when both factors are invertible.
If row operations are applied in the order $E_1, E_2, \ldots, E_k$, the result is $E_k \cdots E_2 E_1 A$. If this equals I_n, then $A^{-1} = E_k \cdots E_1$, whereas $A = E_1^{-1} \cdots E_k^{-1}$. The factor order is part of the assertion.

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Additional practice

8.10 Supplementary Exercises: Span, Redundancy, and the Scalar Field

EXERCISE 8.9

$$U = \{(x, y, z) \in \mathbb{R}^3 : x + 2y + z = 0\}, \qquad u = (2, -1, 0), \quad v = (0, 1, -2), \quad w = (2, 0, -2).$$

Please prove $U = \operatorname{span}\{u, v\}$. Please test independence and the effect of adding w. Please determine whether $x + 2y + z = 3$ defines a subspace.

WORKED SOLUTION

$$au + bv = (2a, -a + b, -2b), \qquad 2a + 2(-a + b) - 2b = 0.$$

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EXERCISE 8.9 CONTINUED

Thus span$\{u, v\} \subseteq U$. Conversely, $(x, y, z) \in U$ implies $y = -(x + z)/2$, so

$$(x, y, z) = \frac{x}{2}u - \frac{z}{2}v.$$

Hence $U = \text{span}\{u, v\}$. In $au + bv = 0$, the first and third coordinates give $a = b = 0$, proving independence. Since $w = u + v$, adding w preserves the span but introduces the dependence $u + v - w = 0$. The set defined by $x + 2y + z = 3$ is not a subspace because it excludes the zero vector.

EXERCISE 8.10

For $S = \{1, 2 + i\} \subseteq \mathbb{C}$, please test linear dependence and spanning over $\mathbb{R}$ and over $\mathbb{C}$.

WORKED SOLUTION

Over $\mathbb{R}$, $a + b(2 + i) = 0$ gives $b = 0$ from the imaginary part, then $a = 0$. Thus the pair is independent. Every $x + iy \in \mathbb{C}$ satisfies

$$x + iy = (x - 2y) \cdot 1 + y(2 + i),$$

so the pair spans $\mathbb{C}$ over $\mathbb{R}$. Over $\mathbb{C}$, the nontrivial relation

$$-(2 + i) \cdot 1 + 1 \cdot (2 + i) = 0$$

shows dependence. The pair still spans because 1 alone spans $\mathbb{C}$ over itself.

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EXERCISE 8.11 CHALLENGE

Let $S \subseteq \mathbb{R}^n$ be nonempty and assume $(1-t)x + ty \in S$ for all $x, y \in S$ and $t \in \mathbb{R}$.

1. Please choose $a \in S$ and prove that $U = \{x - a : x \in S\}$ is a subspace and $S = a + U$, where $a + U = \{a + u : u \in U\}$.
2. Is it enough to require only $(x + y)/2 \in S$ for all $x, y \in S$? Please prove or disprove.

WORKED SOLUTION

We have $0 = a - a \in U$. For $u \in U$ and $\lambda \in \mathbb{R}$, the hypothesis gives

$$(1 - \lambda)a + \lambda(a + u) = a + \lambda u \in S,$$

so $\lambda u \in U$. For $u, v \in U$, taking the midpoint of $a + u$ and $a + v$ gives $(u + v)/2 \in U$. Scalar closure with scalar 2 then gives $u + v \in U$. Hence U is a subspace. Its definition gives both inclusions in $S = a + U$. Midpoint closure alone is insufficient. The set $\mathbb{Q} \subseteq \mathbb{R}$ is closed under midpoints. If it were $a + U$ for a real subspace U, the fact that $0 \in \mathbb{Q}$ would imply $\mathbb{Q} = U$. But $1 \in U$ would then imply $\sqrt{3} \cdot 1 \in U$, a contradiction.

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EXERCISE 8.12 CHALLENGE

Let $U_1, \ldots, U_m$ be proper subspaces of a real vector space V, meaning $U_j \neq V$ for every j.

1. Please prove that $U_1 \cup \cdots \cup U_m \neq V$.
2. Please construct four proper subspaces of $\mathbb{Z}_3^2$ whose union is the whole space.

WORKED SOLUTION

We induct on m. The case $m = 1$ is the assumption. By the induction hypothesis, choose

$$a \notin U_1 \cup \cdots \cup U_{m-1}, \qquad b \notin U_m.$$

Consider $a + tb$ for $t \in \mathbb{R}$. For each $j < m$, at most one value of t gives a point in U_j. Two distinct values would imply $b \in U_j$ by subtraction and division by their difference, and then $a \in U_j$, a contradiction. The same line meets U_m in at most one point, since two would imply $b \in U_m$. Only finitely many parameters are excluded, whereas $\mathbb{R}$ is infinite. This proves the claim. Over $\mathbb{Z}_3$, take

$$U_1 = \operatorname{span}\{(1,0)\}, \quad U_2 = \operatorname{span}\{(0,1)\},$$
$$U_3 = \operatorname{span}\{(1,1)\}, \quad U_4 = \operatorname{span}\{(1,2)\}.$$

Each subspace has three vectors, so each is proper. A vector (x, y) with $x = 0$ belongs to U_2. If $x \neq 0$, then $(x, y) = x(1, y/x)$, where y/x is 0, 1, or 2, so the vector lies in U_1, U_3, or U_4. The union covers all nine vectors.

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TUTORIAL 9

Dimension Formulas, Row and Column Spaces, and Rank

NOTATION AND CONVENTIONS

For $A \in M_{m\times n}(F)$, the row space $\operatorname{Row} A \leq F^n$ is the span of its rows, and the column space $\operatorname{Col} A \leq F^m$ is the span of its columns. Rows are viewed as vectors of F^n. The symbol $\dim U$ denotes the dimension of a subspace U. The rank $\operatorname{rank} A$ is the number of pivots in reduced row echelon form; below we show that it equals both the row-space and column-space dimensions.

DEFINITION 9.1

A precise formulation with hypotheses. For $S \subseteq V$, its span consists of all finite sums $\sum_{j=1}^{k} \alpha_j v_j$ with $v_j \in S$ and $\alpha_j \in \mathbb{F}$, including the empty sum, which equals 0_V. It is the smallest subspace containing S. Independence means that every finite zero linear combination of distinct members of S has only zero coefficients. In particular, the empty set is independent and its span is $\{0_V\}$.
For subspaces $U, W \leq V$ over the same field, $U+W$ is the smallest subspace containing both. The condition $U \cap W = \{0_V\}$ is equivalent to every $x \in U+W$ having a unique expression $x = u + w$ with $u \in U, w \in W$. Indeed, two expressions give $u - u' = w' - w \in U \cap W$. Conversely, if $z \in U \cap W$, then $z = z + 0_V = 0_V + z$, so uniqueness forces $z = 0_V$.

MOTIVATION AND GUIDING IDEA

Motivation. Matrix multiplication describes composition of linear operations: in ABx, apply B first and then A. Checking dimensions reveals in advance whether the composition is possible. A basis gives a unique description of every vector, and dimension counts the coefficients required for that description.

NOTATION AND CONVENTIONS

Notation guide. $M_{m\times n}(\mathbb{F})$ is the space of matrices with m rows and n columns; a_{ij} is the entry in row i, column j. $I_n = (\delta_{ij})$, where $\delta_{ij} = 1$ if $i = j$ and is zero otherwise. e_j is the jth standard basis vector, and Ae_j is column j of A. A^T denotes transpose, $\operatorname{diag}(d_1, \ldots, d_n)$ a diagonal matrix, $|B|$ the cardinality of a finite set, and $\dim_{\mathbb{F}} V$ the dimension over $\mathbb{F}$.

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DEFINITION 9.2

Hypotheses to check. If $A \in M_{k\times m}(\mathbb{F})$ and $B \in M_{m\times n}(\mathbb{F})$, then $AB \in M_{k\times n}(\mathbb{F})$ with $(AB)_{ij} = \sum_{r=1}^{m} a_{ir}b_{rj}$. Also $(AB)^T = B^T A^T$. Over $\mathbb{C}$, the transpose A^T does not include complex conjugation.
For square matrices, define $\mathrm{Sym}_n(\mathbb{F}) = \{A : A^T = A\}$ and $\mathrm{ASym}_n(\mathbb{F}) = \{A : A^T = -A\}$. Both sets are subspaces over any field. If the characteristic is different from 2, a skew-symmetric matrix has zero diagonal and every matrix has a unique symmetric/skew-symmetric decomposition using division by 2. In characteristic 2 the two conditions coincide and do not imply a zero diagonal.
If $\dim_{\mathbb{F}} V = n < \infty$, a set of n vectors is a basis if and only if it is independent, and if and only if it spans. For $U, W \le V$, $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$; subtracting the intersection dimension prevents shared directions from being counted twice.

MOTIVATION AND GUIDING IDEA

Tutorial objectives. Compute with matrices while checking dimensions and multiplication order, recognize symmetric structure, and choose bases. Connect algebraic calculations with sums and intersections of subspaces.

9.1 Intersections and Sums of Subspaces

MOTIVATION AND GUIDING IDEA

Distinguish union from sum. For example, the two coordinate axes in $\mathbb{R}^2$ are subspaces, but their union is not closed under addition: $(2, 0)$ and $(0, -3)$ belong to the union, whereas their sum $(2, -3)$ does not. The sum of the two axes is the whole plane.

RECALL AND TOOLS

If $U, W \subseteq V$ are subspaces of a vector space V, their **intersection** is

$$U \cap W = \{v \in V : v \in U \text{ and } v \in W\}.$$

DEFINITION 9.3

For subspaces $U, W \subseteq V$, define their **sum** by

$$U + W = \{u + w : u \in U, w \in W\}.$$

Then $U + W$ is a subspace of V containing both U and W. If also $U \cap W = \{0\}$, the sum is called a **direct sum** and is denoted $U \oplus W$.

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EXERCISE 9.1

Please write

$$\text{Sym}(\mathbb{F}) = \{A \in M_n(\mathbb{F}) : A = A^T\}, \qquad \text{aSym}(\mathbb{F}) = \{A \in M_n(\mathbb{F}) : A = -A^T\}$$

for the symmetric and skew-symmetric matrices. Please prove that both sets are subspaces of $M_n(\mathbb{F})$. In addition, please prove that if $\text{char}(\mathbb{F}) \neq 2$, that is, $1 + 1 \neq 0$ in the field, then

$$M_n(\mathbb{F}) = \text{Sym}(\mathbb{F}) \oplus \text{aSym}(\mathbb{F}).$$

WORKED SOLUTION

We verify that $\text{Sym}(\mathbb{F})$ is a subspace. For $\text{aSym}(\mathbb{F})$, the zero matrix belongs to the set, and if $A^T = -A$, $B^T = -B$, and $\lambda \in \mathbb{F}$, then

$$(A + B)^T = A^T + B^T = -A - B = -(A + B), \qquad (\lambda A)^T = \lambda A^T = -\lambda A.$$

Thus skew-symmetric matrices are also closed under addition and scalar multiplication and form a subspace.
Now let $A, B \in \text{Sym}(\mathbb{F})$ and $\lambda \in \mathbb{F}$. First, $0 \in \text{Sym}(\mathbb{F})$. For addition, $(A + B)^T = A^T + B^T = A + B$, so $A + B$ is symmetric. For scalar multiplication, $(\lambda A)^T = \lambda A^T = \lambda A$, so λA is symmetric. To prove the direct-sum assertion, first prove that the sum is the whole matrix space and then prove that the intersection is zero. For the equality $M_n(\mathbb{F}) = \text{Sym}(\mathbb{F}) + \text{aSym}(\mathbb{F})$, check both inclusions:

1. $M_n(\mathbb{F}) \supseteq \text{Sym}(\mathbb{F}) + \text{aSym}(\mathbb{F})$ holds because both summands are subspaces of $M_n(\mathbb{F})$.
2. For the reverse inclusion, take $A \in M_n(\mathbb{F})$ and write

$$A = \frac{A + A^T}{2} + \frac{A - A^T}{2}.$$

The first summand is symmetric, since

$$\left(\frac{A + A^T}{2}\right)^T = \frac{A^T + (A^T)^T}{2} = \frac{A + A^T}{2},$$

and the second is skew-symmetric, since

$$\left(\frac{A - A^T}{2}\right)^T = \frac{A^T - (A^T)^T}{2} = \frac{A^T - A}{2} = -\frac{A - A^T}{2}.$$

Hence $M_n(\mathbb{F}) \subseteq \text{Sym}(\mathbb{F}) + \text{aSym}(\mathbb{F})$.

Finally, suppose $A \in \text{Sym}(\mathbb{F}) \cap \text{aSym}(\mathbb{F})$. Symmetry gives $A = A^T$, while skew-symmetry gives $-A = A^T$. Thus $A = -A$, and $a_{ij} = -a_{ij}$ for every $1 \leq i, j \leq n$. Consequently $a_{ij} + a_{ij} = 2a_{ij} = 0$. Because $\text{char}(\mathbb{F}) \neq 2$, the scalar $2 = 1 + 1$ is nonzero and invertible in the field. Therefore every $a_{ij} = 0$, so $A = 0$, as required.

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RECALL AND TOOLS

In characteristic 2, $-A = A$, so the conditions $A^T = A$ and $A^T = -A$ coincide. This is exactly why the symmetric/skew-symmetric decomposition using division by 2 requires characteristic different from 2.

9.2 Intersections and Sums of Subspaces

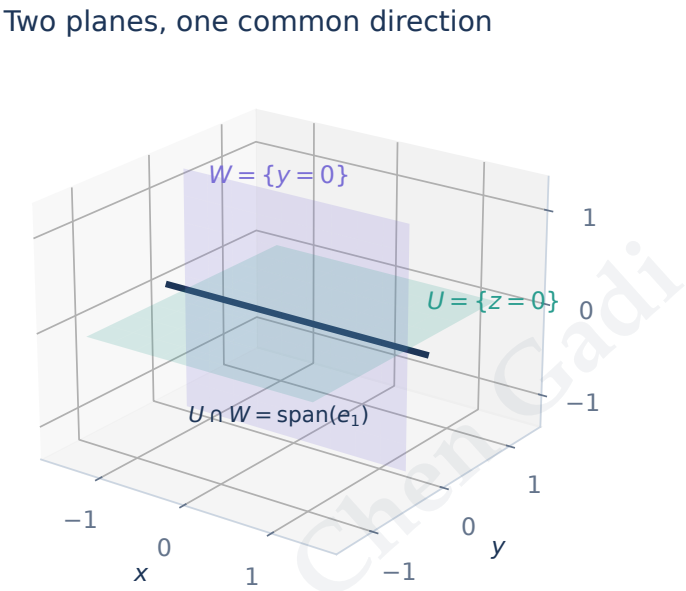


Figure 9.1. Two distinct planes through the origin in $\mathbb{R}^3$ may intersect in a line. Here $U = \{z = 0\}$ and $W = \{y = 0\}$, so $U \cap W = \text{span}\{(1, 0, 0)\}$ and $U + W = \mathbb{R}^3$.

RECALL AND TOOLS

For subspaces $U, W \subseteq V$ of a vector space V, their intersection is

$$U \cap W = \{v \in V : v \in U \text{ and } v \in W\}.$$

DEFINITION 9.4

For subspaces $U, W \subseteq V$, define

$$U + W = \{u + w : u \in U, w \in W\}.$$

This is a subspace of V containing U and W. If also $U \cap W = \{0\}$, the sum is a **direct sum**, denoted $U \oplus W$.

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EXERCISE 9.2

Let

$$\mathrm{Sym}(\mathbb{F}) = \{A \in M_n(\mathbb{F}) : A = A^T\}, \qquad \mathrm{aSym}(\mathbb{F}) = \{A \in M_n(\mathbb{F}) : A = -A^T\}.$$

Please prove that the symmetric and skew-symmetric matrices are subspaces of $M_n(\mathbb{F})$, and that whenever $\mathrm{char}(\mathbb{F}) \neq 2$,

$$M_n(\mathbb{F}) = \mathrm{Sym}(\mathbb{F}) \oplus \mathrm{aSym}(\mathbb{F}).$$

WORKED SOLUTION

We show that $\mathrm{Sym}(\mathbb{F})$ is a subspace. For the skew-symmetric set, the corresponding identities are

$$0^T = -0, \quad (A+B)^T = -A - B = -(A+B), \quad (\lambda A)^T = -\lambda A,$$

so this set is also a subspace. For the symmetric set, take $A, B \in \mathrm{Sym}(\mathbb{F})$ and $\lambda \in \mathbb{F}$. Clearly $0 \in \mathrm{Sym}(\mathbb{F})$. Closure under addition follows from $(A+B)^T = A^T + B^T = A + B$. Closure under scalar multiplication follows from $(\lambda A)^T = \lambda A^T = \lambda A$. To establish the direct sum, first prove equality with the sum and then show that the intersection is zero. Check both inclusions for $M_n(\mathbb{F}) = \mathrm{Sym}(\mathbb{F}) + \mathrm{aSym}(\mathbb{F})$:

1. $M_n(\mathbb{F}) \supseteq \mathrm{Sym}(\mathbb{F}) + \mathrm{aSym}(\mathbb{F})$ follows because both summands are subspaces of $M_n(\mathbb{F})$.
2. Take $A \in M_n(\mathbb{F})$. Then
$$A = \frac{A + A^T}{2} + \frac{A - A^T}{2}.$$
The first summand is symmetric because
$$\left(\frac{A + A^T}{2}\right)^T = \frac{A^T + (A^T)^T}{2} = \frac{A + A^T}{2}.$$
The second is skew-symmetric because
$$\left(\frac{A - A^T}{2}\right)^T = \frac{A^T - (A^T)^T}{2} = \frac{A^T - A}{2} = -\frac{A - A^T}{2}.$$
This proves the reverse inclusion.

Now take $A \in \mathrm{Sym}(\mathbb{F}) \cap \mathrm{aSym}(\mathbb{F})$. Symmetry gives $A = A^T$, and skew-symmetry gives $-A = A^T$. Therefore $A = -A$, so $a_{ij} = -a_{ij}$ for every $1 \leq i, j \leq n$. Thus $a_{ij} + a_{ij} = 2a_{ij} = 0$. As the field has characteristic different from 2, the scalar 2 is nonzero, so every $a_{ij} = 0$. Hence $A = 0$, as required.

The dimension formula:

PROPOSITION 9.1

If V is finite-dimensional over $\mathbb{F}$ and $U, W \subseteq V$ are subspaces, then

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

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EXERCISE 9.3

1. Let V have dimension n, and let U, W be subspaces with $\dim U = n-1$ and $W \not\subseteq U$. Please prove $V = U + W$ and $\dim(U \cap W) = \dim W - 1$.
2. $V = \mathbb{R}_2[X]$, $U = \operatorname{span}\{X, X^2+4\}$, $W = \{p \in \mathbb{R}_2[X] : p'(2) = 0\}$. Please compute the dimensions of the sum and intersection, and please determine whether the sum equals V and is direct.

WORKED SOLUTION

1. Choose $w \in W \setminus U$. Then $U \subsetneq U + W \subseteq V$, so

$$n - 1 = \dim U < \dim(U + W) \leq n.$$

Thus $\dim(U + W) = n$ and $U + W = V$. The dimension formula gives

$$n = (n-1) + \dim W - \dim(U \cap W),$$

hence $\dim(U \cap W) = \dim W - 1$.

2. The generators of U are independent, so $\dim U = 2$. Also $1 \in W \setminus U$. For W,

$$p(X) = aX^2 + bX + c \in W \iff 4a + b = 0 \iff b = -4a.$$

$$W = \operatorname{span}\{1, X^2 - 4X\}, \qquad \dim W = 2.$$

Part 1 gives $U + W = V$. The sum is not direct, since

$$(X-2)^2 = (X^2+4) - 4X \in U, \qquad \frac{d}{dX}(X-2)^2\Big|_{X=2} = 0.$$

$$U \cap W = \operatorname{span}\{(X-2)^2\}, \quad \dim(U \cap W) = 1, \quad \dim(U + W) = 2 + 2 - 1 = 3.$$

The dimension formula:

PROPOSITION 9.2

If V is finite-dimensional over $\mathbb{F}$ and $U, W \subseteq V$ are subspaces, then

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

EXERCISE 9.4

1. Let V have dimension n, and let U, W be subspaces with $\dim U = n-1$ and $W \not\subseteq U$. Please prove $V = U + W$ and $\dim(U \cap W) = \dim W - 1$.
2. $V = \mathbb{R}_2[X]$, $U = \operatorname{span}\{X, X^2+9\}$, $W = \{p \in \mathbb{R}_2[X] : p'(-3) = 0\}$. Please compute the dimensions of the sum and intersection, and please determine whether the sum equals V and is direct.

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EXERCISE 9.4 CONTINUED

WORKED SOLUTION

1. Choose $w \in W \setminus U$. Then $U \subsetneq U + W \subseteq V$, so

$$n - 1 = \dim U < \dim(U + W) \leq n.$$

Thus $\dim(U + W) = n$ and $U + W = V$. The dimension formula gives

$$n = (n - 1) + \dim W - \dim(U \cap W),$$

hence $\dim(U \cap W) = \dim W - 1$.

2. The generators of U are independent, so $\dim U = 2$. Also $1 \in W \setminus U$. For W,

$$p(X) = aX^2 + bX + c \in W \iff -6a + b = 0 \iff b = 6a.$$

$$W = \text{span}\{1, X^2 + 6X\}, \qquad \dim W = 2.$$

Part 1 gives $U + W = V$. The sum is not direct, since

$$(X + 3)^2 = (X^2 + 9) + 6X \in U, \qquad \frac{d}{dX}(X + 3)^2\Big|_{X=-3} = 0.$$

$$U \cap W = \text{span}\{(X + 3)^2\}, \quad \dim(U \cap W) = 1, \quad \dim(U + W) = 2 + 2 - 1 = 3.$$

9.3 Row and column spaces and the rank theorem

RECALL AND TOOLS

The **row space** $\mathcal{R}(A)$ of $A \in \mathbb{F}^{m\times n}$ is the subspace of $\mathbb{F}^n$ spanned by its rows. .

- Elementary row operations preserve the row space of a matrix.

PROPOSITION 9.3

- **Claim 1.** If $A, B \in M_{m\times n}(\mathbb{F})$ are row equivalent, then $\mathcal{R}(A) = \mathcal{R}(B)$.
- **Claim 2:** If A is in row echelon form, its nonzero rows form a basis of $\mathcal{R}(A)$, (and are therefore linearly independent in $\mathbb{F}^n$).

RECALL AND TOOLS

The **column space** $\mathcal{C}(A)$ of $A \in \mathbb{F}^{m\times n}$ is the subspace of $\mathbb{F}^m$ spanned by its columns. .

- Row reduction **does not in general preserve the column space**; only its **dimension** is preserved.
- $\mathcal{C}(A) = \mathcal{R}(A^T)$ when a row vector is identified with its transposed column vector.

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PROPOSITION 9.4

The rank theorem: Every matrix A satisfies $\dim \mathcal{C}(A) = \dim \mathcal{R}(A)$. This common dimension is called the **rank of the matrix**.

RECALL AND TOOLS

- The row and column spaces need not be the same subspace. For a rectangular matrix they even lie in different ambient spaces, $\mathbb{F}^n$ and $\mathbb{F}^m$, respectively; their dimensions are nevertheless equal.
- Equivalently, rank$A$ is the number of pivots in the reduced row echelon form.
- For $A \in \mathbb{R}^{m\times n}$, $\text{rank}(A) \leq \min(m, n)$.
- A square matrix $A \in \mathbb{R}^{n\times n}$ is invertible if and only if $rank(A) = n$
- $rank(A \cdot B) \leq min\{rank(A), rank(B)\}$
- $rank(A + B) \leq rank(A) + rank(B)$

EXERCISE 9.5

$$A = \begin{pmatrix} 2 & 6 & -4 \\ 1 & 3 & -2 \end{pmatrix}.$$

Please find a basis and dimension for the row space and column space of A.

WORKED SOLUTION

Swap the rows and subtract twice the first row from the second. The column-space basis is taken from the original matrix.

$$A \sim \begin{pmatrix} 1 & 3 & -2 \\ 2 & 6 & -4 \end{pmatrix} \sim \begin{pmatrix} 1 & 3 & -2 \\ 0 & 0 & 0 \end{pmatrix}.$$

$$\text{Row}(A) = \text{span}\{(1, 3, -2)\}, \qquad \text{Col}(A) = \text{span}\{(2, 1)^T\}.$$

$$\text{Col}_2(A) = 3\,\text{Col}_1(A), \qquad \text{Col}_3(A) = -2\,\text{Col}_1(A), \qquad \dim \text{Row}(A) = \dim \text{Col}(A) = 1.$$

EXERCISE 9.6

Let $A \in \mathbb{R}^{m\times n}, B \in \mathbb{R}^{n\times m}$ satisfy $AB = I_m$. Please find the ranks of A and B.
What can be concluded about m and n?

WORKED SOLUTION

The rank inequalities give $\text{rank}(AB) = \text{rank}(I_m) = m \leq \text{rank}(A) \leq m$.
Thus $\text{rank}(A) = m$. For B, $m = \text{rank}(AB) \leq \text{rank}(B) \leq m$, so $\text{rank}(B) = m$ as well.
Since $\text{rank}(A) = m \leq \min(m, n)$, we obtain $m \leq n$.
The matrices are not assumed square, so one cannot conclude that B is an inverse of A.

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9.3.1 Reducing spanning sets and extending independent sets to bases in $\mathbb{F}^n$

Every spanning set contains a basis, and every linearly independent set extends to a basis.

Reducing a spanning set to a basis

MOTIVATION AND GUIDING IDEA

1. **Method A:** We want a basis consisting of vectors from the original set (as in the preceding exercise).
   - Form a matrix whose **columns** are the vectors of S and reduce it to row echelon form.
   - For each pivot column, choose the corresponding **original column** as a basis vector. These columns are linearly independent.
   - The column space is not preserved by row operations; therefore the columns must be taken from the original matrix.
2. **Method B:** Find a **new** linearly independent set spanning the same space.
   - Form a matrix whose **rows** are the vectors of S, and reduce it to row echelon form.
   - Its nonzero echelon rows form a basis.
   - Row operations preserve the row space, so the basis may be taken after reduction.

Extending a linearly independent set to a basis

MOTIVATION AND GUIDING IDEA

- Form a matrix whose **rows** are the vectors of S, and reduce it to row echelon form.
- For each nonpivot column j, adjoin the standard basis vector e_j to S. To justify this construction, first replace the original rows by the echelon rows, preserving their span. At the pivot indices, the added standard basis vectors are zero, whereas the echelon rows give a triangular matrix with nonzero diagonal. Hence a vanishing linear combination first forces the coefficients of the echelon rows to be zero, and then those of the added standard basis vectors. Since the original rows are independent, replacing the echelon rows back by the originals preserves the basis property.

EXERCISE **9.7**

$$S = \{(1,2,0,1,-1)^T, (2,4,1,3,0)^T, (1,2,1,2,2)^T\} \subseteq \mathbb{R}^5.$$

Please prove that S is linearly independent and please extend it to a basis of $\mathbb{R}^5$.

WORKED SOLUTION

The three nonzero echelon rows prove independence. Pivot columns are $1, 3, 5$. Adding e_2, e_4 gives five independent rows: the pivot columns first force the three echelon-row coefficients

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EXERCISE 9.7 CONTINUED

to vanish, and columns $2, 4$ then force the remaining two coefficients to vanish.

$$\begin{pmatrix} 1 & 2 & 0 & 1 & -1 \\ 2 & 4 & 1 & 3 & 0 \\ 1 & 2 & 1 & 2 & 2 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 2 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

$$B = S \cup \{e_2, e_4\}.$$

EXERCISE 9.8

$$V = \operatorname{span}\{(1,3,2,-1)^T, (2,7,5,0)^T, (0,1,1,2)^T\} \subseteq \mathbb{R}^4.$$

Please find a basis and dimension for V and please extend that basis to a basis of $\mathbb{R}^4$.

WORKED SOLUTION

The first two coordinates show that the two vectors in B are independent. They span all three given vectors. The final determinant verifies the extension.

$$(2,7,5,0)^T = 2(1,3,2,-1)^T + (0,1,1,2)^T.$$

$$B = \{(1,3,2,-1)^T, (0,1,1,2)^T\}, \qquad \dim V = 2.$$

$$\det \begin{pmatrix} 1 & 3 & 2 & -1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = 1, \qquad D = B \cup \{e_3, e_4\}.$$

EXERCISE 9.9 CHALLENGE

Let U, W be subspaces of a vector space V over an arbitrary field $\mathbb{F}$. Please prove that the following are equivalent:

1. $U \cup W$ is a subspace of V.
2. $U \subseteq W$ or $W \subseteq U$.
3. $U + W = U \cup W$.

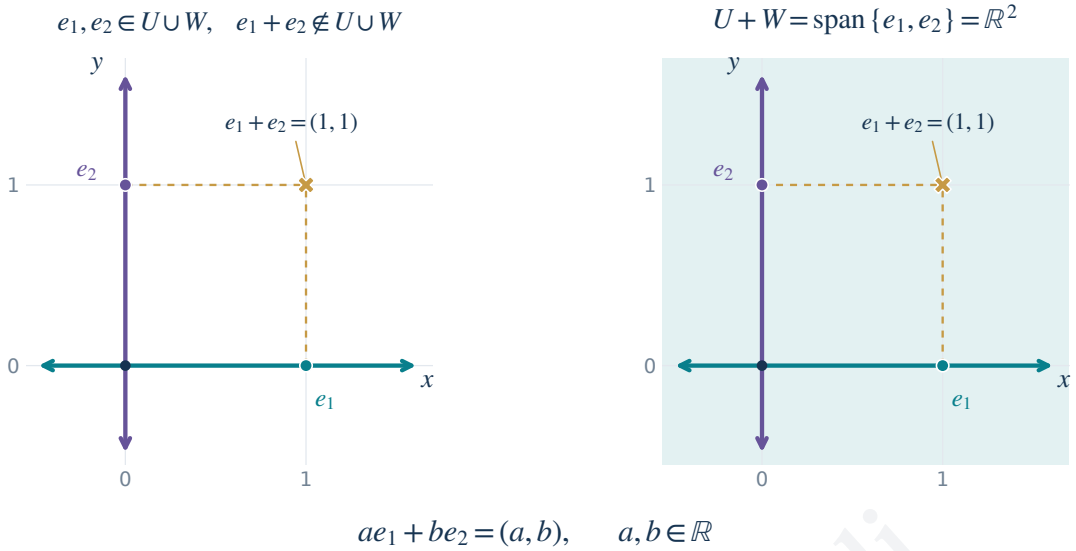
WORKED SOLUTION

(1) $\Rightarrow$ (2): Suppose neither inclusion holds. Choose $u \in U \setminus W$ and $w \in W \setminus U$. Both belong to the union, so closure under addition gives $u + w \in U \cup W$. If $u + w \in U$, then $w = (u + w) - u \in U$, a contradiction. If $u + w \in W$, then $u = (u + w) - w \in W$, again a contradiction. Hence one inclusion must hold.
(2) $\Rightarrow$ (3): If $U \subseteq W$, then $U \cup W = W$ and $U + W = W$: every such sum belongs to W, and each $w \in W$ equals $0_V + w$. The other case follows by interchanging the subspaces.
(3) $\Rightarrow$ (1): The sum $U + W$ is a subspace, so the equal set $U \cup W$ is also a subspace. The proof divides by no scalar and therefore applies to finite fields and characteristic 2 as well.

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$$U = \mathbb{R}e_1, \qquad W = \mathbb{R}e_2, \qquad e_1 = (1,0), \quad e_2 = (0,1)$$



$$ae_1 + be_2 = (a,b), \qquad a,b \in \mathbb{R}$$

Figure 9.2. An illustration over $\mathbb{R}$: the coordinate axes $U = \mathbb{R}e_1$ and $W = \mathbb{R}e_2$ are subspaces. Their union contains e_1 and e_2 but excludes their sum $(1,1)$, so it is not closed under addition. In contrast, every $(a,b) \in \mathbb{R}^2$ equals $ae_1 + be_2$, giving $U + W = \mathbb{R}^2$. Shading in the right panel represents the full plane within the viewing window.

9.4 Exercises

EXERCISE 9.10

Let V be a vector space over a field $\mathbb{F}$, and let $S, R \subseteq V$ be arbitrary subsets, not necessarily finite.
Please write

$$\mathrm{span}(S) = \left\{ \sum_{i=1}^{k} \alpha_i s_i \;\middle|\; k \in \mathbb{Z}_{\geq 0},\ \alpha_i \in \mathbb{F},\ s_i \in S \right\},$$

The empty sum is the zero vector, so $\mathrm{span}(\varnothing) = \{0\}$. Define $\mathrm{span}(R)$ and $\mathrm{span}(S \cup R)$ similarly.
Please show that, for all $S, R \subseteq V$,

$$\mathrm{span}(S) + \mathrm{span}(R) = \mathrm{span}(S \cup R).$$

WORKED SOLUTION

Write

$$U := \mathrm{span}(S), \qquad W := \mathrm{span}(R).$$

Recall that the sum $U + W$ is defined by

$$U + W := \{u + w \mid u \in U,\ w \in W\}.$$

We prove

$$\mathrm{span}(S) + \mathrm{span}(R) = \mathrm{span}(S \cup R)$$

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EXERCISE 9.10 CONTINUED

by establishing both inclusions:

$$\text{span}(S) + \text{span}(R) \subseteq \text{span}(S \cup R) \quad \text{and} \quad \text{span}(S \cup R) \subseteq \text{span}(S) + \text{span}(R).$$

1. First prove
$$\text{span}(S) + \text{span}(R) \subseteq \text{span}(S \cup R).$$
Let $v \in \text{span}(S) + \text{span}(R)$. Then $v = u + w$ for some $u \in \text{span}(S)$ and $w \in \text{span}(R)$:
$$v = u + w.$$
By the definition of $\text{span}(S)$, there are vectors $s_1, \ldots, s_k \in S$ and scalars $\alpha_1, \ldots, \alpha_k \in \mathbb{F}$ such that
$$u = \sum_{i=1}^{k} \alpha_i s_i.$$
Similarly, there are vectors $r_1, \ldots, r_\ell \in R$ and scalars $\beta_1, \ldots, \beta_\ell \in \mathbb{F}$ such that
$$w = \sum_{j=1}^{\ell} \beta_j r_j.$$
Therefore,
$$v = u + w = \sum_{i=1}^{k} \alpha_i s_i + \sum_{j=1}^{\ell} \beta_j r_j.$$
Every s_i and every r_j lies in $S \cup R$. Thus v is a finite linear combination of vectors in $S \cup R$.
By the definition of span,
$$v \in \text{span}(S \cup R).$$
Hence
$$\text{span}(S) + \text{span}(R) \subseteq \text{span}(S \cup R).$$
2. For the reverse inclusion, prove
$$\text{span}(S \cup R) \subseteq \text{span}(S) + \text{span}(R).$$
Let $v \in \text{span}(S \cup R)$. There are $x_1, \ldots, x_t \in S \cup R$ and $\gamma_1, \ldots, \gamma_t \in \mathbb{F}$ such that
$$v = \sum_{i=1}^{t} \gamma_i x_i.$$
Partition the indices into disjoint sets: assign to S every vector belonging to S, and assign the rest to R. A vector in the intersection is counted only in the first set.
Define
$$I_S := \{i \in \{1, \ldots, t\} \mid x_i \in S\}, \qquad I_R := \{1, \ldots, t\} \setminus I_S.$$
Then v can be written as
$$v = \sum_{i \in I_S} \gamma_i x_i + \sum_{i \in I_R} \gamma_i x_i.$$

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EXERCISE 9.10 CONTINUED

Set

$$u := \sum_{i \in I_S} \gamma_i x_i, \qquad w := \sum_{i \in I_R} \gamma_i x_i.$$

Observe that

- for $i \in I_S$, $x_i \in S$, so u is a linear combination of vectors in S; thus

$$u \in \text{span}(S).$$

- for $i \in I_R$, $x_i \in R$, so w is a linear combination of vectors in R; thus

$$w \in \text{span}(R).$$

We have obtained

$$v = u + w$$

with $u \in \text{span}(S)$ and $w \in \text{span}(R)$, hence

$$v \in \text{span}(S) + \text{span}(R).$$

This proves

$$\text{span}(S \cup R) \subseteq \text{span}(S) + \text{span}(R).$$

The two inclusions give

$$\text{span}(S) + \text{span}(R) = \text{span}(S \cup R),$$

as required.

EXERCISE 9.11

$$U = \text{span}\{1, x, 2 + x\}, \qquad W = \{p \in \mathbb{R}_3[x] : p(-1) = p(2) = 0\}.$$

Please find bases and dimensions for $U + W$ and $U \cap W$.

WORKED SOLUTION

A polynomial of degree at most one with two distinct roots is zero. Thus the intersection is zero and the sum is direct. The four listed polynomials are independent by their degrees and span the ambient space. The intersection has the empty basis.

$$U = \text{span}\{1, x\}, \qquad p \in W \iff p = (x + 1)(x - 2)(a + bx).$$

$$W = \text{span}\{x^2 - x - 2,\ x^3 - x^2 - 2x\}, \qquad \dim W = 2.$$

$$U \cap W = \{0\}, \quad \dim(U \cap W) = 0, \qquad U + W = \mathbb{R}_3[x], \quad \dim(U + W) = 4.$$

$$B_{U+W} = (1, x, x^2 - x - 2, x^3 - x^2 - 2x).$$

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EXERCISE 9.12

$$M = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 1 & 1 & 1 \\ 2 & -1 & 5 \end{pmatrix}, \qquad y = (4, -1, 3, 10)^T.$$

Please find row-space and column-space bases and the number of dependent rows. Please decide whether $Mx = y$ is solvable and whether an existing solution is unique. Please replace the fourth entry of y by 9 and please find every solution.

WORKED SOLUTION

The first two rows are independent and the other two depend on them. The third column is twice the first minus the second, so the listed columns form a basis. The row relations give necessary right-hand-side conditions; they are sufficient because the first two equations can be solved for any two right-hand sides. The original right-hand side is inconsistent. After correction, setting $x_3 = t$ gives the displayed solutions. The one-dimensional kernel prevents uniqueness.

$$R_3 = R_1 + R_2, \qquad R_4 = 2R_1 - R_2, \qquad \operatorname{rank} M = 2.$$

$$B_{\text{row}} = \{(1, 0, 2), (0, 1, -1)\}, \quad B_{\text{col}} = \{(1, 0, 1, 2)^T, (0, 1, 1, -1)^T\}.$$

$$y \in \operatorname{Col} M \iff y_3 = y_1 + y_2, \quad y_4 = 2y_1 - y_2.$$

$$2 \cdot 4 - (-1) = 9 \neq 10, \qquad \ker M = \operatorname{span}\{(-2, 1, 1)^T\}.$$

$$x = (4, -1, 0)^T + t(-2, 1, 1)^T, \qquad t \in \mathbb{R}.$$

Additional practice

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EXERCISE 9.13 CHALLENGE

Let $A \in M_n(F)$ over any field. Please prove

$$\operatorname{rank} A + \operatorname{rank}(I_n - A) \geq n,$$

with equality if and only if $A^2 = A$.

WORKED SOLUTION

Put $U = \operatorname{Col} A$ and $W = \operatorname{Col}(I_n - A)$. Since $x = Ax + (I_n - A)x$, we have $U + W = F^n$. The dimension formula gives

$$\operatorname{rank} A + \operatorname{rank}(I_n - A) = n + \dim(U \cap W) \geq n.$$

If equality holds, then $U \cap W = \{0\}$. For every x, the vector $A(I_n - A)x = (I_n - A)Ax$ belongs to both spaces, so it is zero. Hence $A^2 = A$. Conversely, assume $A^2 = A$ and let $y = Au = (I_n - A)v$. The first representation gives $Ay = y$, while the second gives $Ay = (A - A^2)v = 0$. Therefore $y = 0$, the intersection is trivial, and equality follows.

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EXERCISE 9.14 CHALLENGE

Let $m, n \geq 2$ and let $W \neq \{0\}$ be a subspace of $M_{m\times n}(F)$ in which every matrix has rank at most 1. Please prove that at least one of the following holds:

1. There is a fixed $u \in F^m \setminus \{0\}$ such that every $A \in W$ has the form uv^T for some $v \in F^n$.
2. There is a fixed $v \in F^n \setminus \{0\}$ such that every $A \in W$ has the form uv^T for some $u \in F^m$.

WORKED SOLUTION

Every nonzero rank-one matrix is uv^T with $u, v \neq 0$: choose a nonzero column u and collect the scalar multiples expressing the other columns into v^T. First consider uv^T and rs^T. If both u, r and v, s are independent pairs, the matrix with rows v^T, s^T has rank 2 and maps onto F^2. Thus there are x, y with

$$v^T x = 1,\ s^T x = 0, \qquad v^T y = 0,\ s^T y = 1.$$

The sum sends x to u and y to r, so its rank is at least 2. Consequently a sum of rank at most 1 forces dependence of at least one pair of factors. Choose $A_0 = u_0 v_0^T \in W$ nonzero. If every column space lies in $\text{span}\{u_0\}$, the first alternative holds. Otherwise choose B whose column direction is independent of u_0. Since $A_0 + B \in W$, the preceding argument forces the row factor of B to be a multiple of v_0. Absorbing that nonzero scalar into its column factor gives $B = rv_0^T$ with r, u_0 independent. For any nonzero $C = cd^T \in W$, suppose d, v_0 were independent. Applying the same argument to $A_0 + C$ forces $c \in \text{span}\{u_0\}$; applying it to $B + C$ forces $c \in \text{span}\{r\}$. This gives $c = 0$, a contradiction. Every row factor is therefore a multiple of v_0, proving the second alternative.

EXERCISE 9.15

Let A, B be square matrices such that AB is invertible. Please prove that both A and B are invertible. **This is the converse of the product assertion from the lecture.**

WORKED SOLUTION

Use the fact that for square matrices of the same order, a one-sided inverse is also a two-sided inverse. Indeed, $AX = I$ implies $n = \text{rank}(AX) \leq \text{rank}(A) \leq n$, so A is invertible. From $(AB)(AB)^{-1} = I$, obtain $A[B(AB)^{-1}] = I$. Thus A is invertible, with inverse $B(AB)^{-1}$. Similarly, $(AB)^{-1}(AB) = I$ gives $[(AB)^{-1}A]B = I$. Thus B is invertible, with inverse $(AB)^{-1}A$.

EXERCISE 9.16 EXAM QUESTION

Work over $\mathbb{Z}_3 = \{0, 1, 2\}$, with all arithmetic modulo 3. For each $\omega \in \mathbb{Z}_3$, define

$$A(\omega) = \begin{pmatrix} 1 & \omega & 1 \\ \omega & 1 & 1 \\ 1 & 1 & \omega \end{pmatrix}, \qquad b = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \qquad c = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}.$$

1. Please determine the values of ω for which $b \in \text{Col}(A(\omega))$, and please find all solutions of $A(\omega)x = b$ in those cases.
2. Please determine the values of ω for which both $b \in \text{Col}(A(\omega))$ and $c \in \text{Col}(A(\omega))$ hold.

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EXERCISE 9.16 EXAM QUESTION CONTINUED

WORKED SOLUTION

All operations below are modulo 3. For $\omega = 0, 2$ the matrix is invertible, so the displayed solutions for b are unique. For $\omega = 1$ its column space consists of equal-entry vectors, so the system with b is inconsistent. Both parts therefore have the answer $\omega \in \{0, 2\}$. The last products also give c in each of these two cases.

$$A(0) = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \quad A(1) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad A(2) = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

$$\det A(\omega) = -\omega^3 + 3\omega - 2, \qquad \det A(0) = 1, \quad \det A(2) = 2 \quad \text{in } \mathbb{Z}_3.$$

$$A(0)\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} = b, \qquad A(2)\begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} = b.$$

$$\mathrm{Col}(A(1)) = \{(a, a, a)^T : a \in \mathbb{Z}_3\}, \qquad b \notin \mathrm{Col}(A(1)).$$

$$A(0)\begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} = c, \qquad A(2)\begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} = c.$$

EXERCISE 9.17

$$m = Cq, \qquad C = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 1 \end{pmatrix}.$$

Please find a column-space basis. Please prove that C is invertible and please calculate C^{-1} and q in terms of m.

WORKED SOLUTION

The matrix is triangular with diagonal entries 1, so it is invertible and its columns form a basis. Solving the equations in order gives the inverse formula.

$$\det C = 1, \qquad B_{\text{col}} = \{(1, 2, -1)^T, (0, 1, 3)^T, (0, 0, 1)^T\}.$$

$$m_1 = q_1, \quad m_2 = 2q_1 + q_2, \quad m_3 = -q_1 + 3q_2 + q_3.$$

$$q_1 = m_1, \quad q_2 = m_2 - 2m_1, \quad q_3 = m_3 - 3m_2 + 7m_1.$$

$$C^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 7 & -3 & 1 \end{pmatrix}, \qquad CC^{-1} = C^{-1}C = I_3.$$

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EXERCISE 9.18 CHALLENGE

Let $n \geq 2$. Let W consist of all real $n \times n$ matrices whose row sums and column sums all equal the same number. Please prove that W is a subspace and find a basis and its dimension. You may use E_{ij}, the matrix with a single 1 in position (i, j).

WORKED SOLUTION

Addition adds the common sums, scalar multiplication scales them, and zero belongs to W. Hence W is a subspace. Let W_0 be the matrices with all row and column sums zero. For $i, j < n$, define

$$F_{ij} = E_{ij} - E_{in} - E_{nj} + E_{nn}.$$

Each lies in W_0. Every $A \in W_0$ satisfies

$$A = \sum_{i=1}^{n-1} \sum_{j=1}^{n-1} a_{ij} F_{ij}.$$

The upper left $(n-1) \times (n-1)$ blocks agree. Zero row sums then determine the last column, zero column sums determine the last row, and the final entry is determined as well. These matrices are independent: entry (i, j) with $i, j < n$ reads off the coefficient of F_{ij}. If the common sum of $A \in W$ is c, then $A - cI_n \in W_0$. Since $I_n \notin W_0$, a basis is

$$\{I_n\} \cup \{F_{ij} : 1 \leq i, j < n\}, \qquad \dim W = 1 + (n-1)^2.$$

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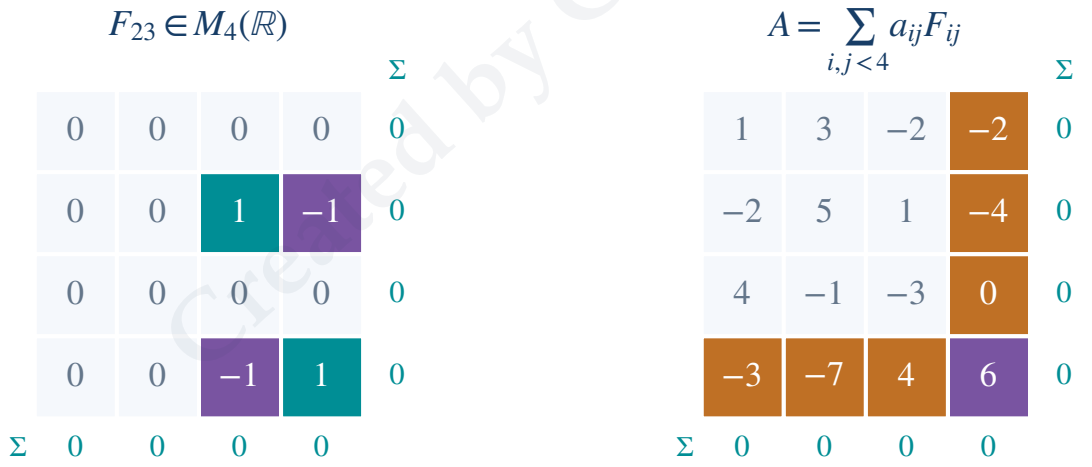


Figure 9.3. Left: the matrix F_{23} for $n = 4$. Right: an interior block completed so that every row and column sum is zero.

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EXERCISE 9.19 CHALLENGE

A permutation matrix has exactly one entry equal to 1 in each row and each column, with all other entries zero. Let $n \geq 2$. Please prove that $A \in M_n(\mathbb{R})$ is a linear combination of permutation matrices if and only if all its row sums and column sums equal the same number.

WORKED SOLUTION

Every permutation matrix has row and column sums equal to 1. Thus every linear combination has all those sums equal to the sum of its coefficients.
Conversely, let c be the common sum of A and put $B = A - cI_n$. All row and column sums of B are zero. For $i, j < n$, define

$$F_{ij} = E_{ij} - E_{in} - E_{nj} + E_{nn},$$

where E_{ij} has one 1 in position (i, j) and zeros elsewhere. Then

$$B = \sum_{i=1}^{n-1} \sum_{j=1}^{n-1} b_{ij} F_{ij}.$$

The upper left blocks agree, and zero row and column sums determine the remaining entries. Each F_{ij} is the difference of two permutation matrices. Choose a permutation matrix P with ones in positions (i, j) and (n, n), matching the remaining $n - 2$ rows and columns in any order. Swap rows i, n to obtain a permutation matrix Q. Then

$$P - Q = E_{ij} + E_{nn} - E_{in} - E_{nj} = F_{ij}.$$

Thus every F_{ij} is a linear combination of permutation matrices. The identity is itself a permutation matrix, so $A = B + cI_n$ is a linear combination of them as well.

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TUTORIAL 10

Null Spaces and Linear Maps

NOTATION AND CONVENTIONS

For $A \in M_{m\times n}(F)$, the homogeneous solution set is denoted by

$$\ker A = \{x \in F^n : Ax = 0\}.$$

Here F^n is the set of columns of length n over the field F, and 0 on the right is the zero column of length m. The rank $\operatorname{rank} A$ is the number of leading entries in its reduced row echelon form. A variable whose column has no leading entry is a free variable.

NOTATION AND CONVENTIONS

If $A \in M_{m\times n}(F)$, then $Ax = b$ is a system of m equations in n variables, with $x \in F^n$ and $b \in F^m$. The notation $[A \mid b]$ denotes its augmented matrix, and $A \sim B$ denotes row equivalence. The rank $\operatorname{rank} A$ is the number of pivots. Define $\ker A = \{x \in F^n : Ax = 0\}$. The expression $\operatorname{span}\{v_1, \ldots, v_k\}$ denotes all combinations $\alpha_1 v_1 + \cdots + \alpha_k v_k$ with $\alpha_j \in F$. The notation $x_0 + U = \{x_0 + u : u \in U\}$ denotes a translate of a subspace U. Distinguish the empty set $\varnothing$, with no elements, from $\{0\}$, which has one element.

MOTIVATION AND GUIDING IDEA

Motivation. In $Ax = b$, the column space determines attainable measurements, while the kernel determines undetectable changes in the state. Distinguishing these roles allows existence and uniqueness to be tested before solving explicitly.

NOTATION AND CONVENTIONS

- $\mathbb{F}^{m\times n}$ is the space of matrices with m rows and n columns over the field $\mathbb{F}$; A^T denotes the transpose.
- $\mathcal{R}(A) = \operatorname{Row}(A) \subseteq \mathbb{F}^n$ is the row space; $\mathcal{C}(A) = \operatorname{Col}(A) \subseteq \mathbb{F}^m$ is the column space. Identify a row with its transposed column when matrix multiplication requires it.
- $N(A) = \ker A = \{x \in \mathbb{F}^n : Ax = 0\}$ is the kernel; $\operatorname{nullity}(A) = \dim \ker A$ is the nullity; $\operatorname{rank}(A) = \dim \operatorname{Col}(A) = \dim \operatorname{Row}(A)$ is the rank.
- $\operatorname{span} S$ consists of all finite linear combinations of vectors in S; e_j is the standard basis vector with entry 1 at index j and 0 elsewhere.
- $x_0 + \ker A = \{x_0 + v : v \in \ker A\}$ is an affine space, a translate of the kernel. Its affine dimension is $\dim \ker A$; it need not be a vector subspace.

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DEFINITION 10.1

Let $A \in \mathbb{F}^{m\times n}$, and define $T_A : \mathbb{F}^n \to \mathbb{F}^m$ by $T_A(x) = Ax$. Then $\operatorname{Im} T_A = \operatorname{Col}(A)$ and $\ker T_A = N(A)$, so

$$\operatorname{rank}(A) + \dim \ker A = n.$$

The right-hand side is the dimension of the **domain**. The system $Ax = b$ is solvable if and only if $b \in \operatorname{Col}(A)$. If x_0 is one solution, **all** solutions form $x_0 + \ker A$. An existing solution is therefore unique if and only if $\ker A = \{0\}$. The nonzero echelon rows form a basis of the row space, but a column-space basis must use the **original** columns at the pivot indices.

MOTIVATION AND GUIDING IDEA

Tutorial objectives. Distinguish the row space, column space and null space, and use rank and nullity to quantify information and freedom in a system. The measurement examples apply to engineering systems in many disciplines.

10.1 The null space and nullity of a matrix

DEFINITION 10.2

Let $\mathbb{F}$ be a field and $A \in \mathbb{F}^{m\times n}$.
The **null space of** A, also called its *kernel*, is the set of column vectors

$$N(A) := \{x \in \mathbb{F}^n \mid Ax = 0\},$$

that solve the homogeneous system $Ax = 0$.

- $N(A)$ is a vector subspace of $\mathbb{F}^n$.
- Its dimension is the **nullity** of A, denoted

$$\dim N(A) = \operatorname{null}(A).$$

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PROPOSITION 10.1

For $A \in \mathbb{F}^{m\times n}$,

$$\operatorname{rank}(A) + \operatorname{null}(A) = n.$$

- A has n columns, each a vector in $\mathbb{F}^m$.
- Reduce A to reduced row echelon form.
- The number of pivot columns equals $\operatorname{rank}(A)$. The columns of the *original matrix* at these pivot indices form a basis of $\operatorname{Col}(A)$; row reduction itself need not preserve the column space.
- There are $n - \operatorname{rank}(A)$ nonpivot columns. These correspond to free variables in $Ax = 0$, each contributing one independent parameter to its solution space. Consequently,

$$\dim N(A) = \operatorname{null}(A) = n - \operatorname{rank}(A).$$

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- Therefore,

$$\operatorname{rank}(A) + \operatorname{null}(A) = \operatorname{rank}(A) + \big(n - \operatorname{rank}(A)\big) = n.$$

EXERCISE 10.1

$$A = \begin{pmatrix} 1 & -3 & 0 \\ 0 & 1 & 2 \end{pmatrix}.$$

Please find a basis and dimension for the kernel, please give every solution of $Ax = 0$, and please check rank-nullity.

WORKED SOLUTION

The first two columns are pivot columns. Every solution arises from $t \in \mathbb{R}$; the nonzero spanning vector is a kernel basis.

$$x_1 - 3x_2 = 0, \quad x_2 + 2x_3 = 0, \quad x_3 = t \implies x_2 = -2t,\ x_1 = -6t.$$

$$\ker A = \operatorname{span}\{(-6, -2, 1)^T\}, \qquad \operatorname{rank} A + \dim \ker A = 2 + 1 = 3.$$

MOTIVATION AND GUIDING IDEA

Different questions require different spaces. The column space describes attainable right-hand sides. The null space describes changes in the input that leave the output unchanged. Rank and nullity measure these two kinds of freedom.

10.2 Interpreting matrix-vector multiplication

EXERCISE 10.2

Let $\mathbb{F}$ be a field and $B \in \mathbb{F}^{n \times m}$. Define

$$J := \{Bv \mid v \in \mathbb{F}^m\} \subseteq \mathbb{F}^n.$$

Please show that

$$J = \operatorname{Col}(B).$$

WORKED SOLUTION

Denote the columns of B by

$$B = \begin{pmatrix} | & & | \\ b_1 & \dots & b_m \\ | & & | \end{pmatrix},$$

so that $b_1, \dots, b_m \in \mathbb{F}^n$ are its column vectors.
Now let

$$v = \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_m \end{pmatrix} \in \mathbb{F}^m.$$

By the definition of matrix-vector multiplication,

$$Bv = \lambda_1 b_1 + \cdots + \lambda_m b_m.$$

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EXERCISE 10.2 CONTINUED

We prove both inclusions.

1. First, $J \subseteq \mathrm{Col}(B)$.

   Let $y \in J$. Then $y = Bv$ for some $v \in \mathbb{F}^m$.

   Every such vector is a linear combination of $b_1, \ldots, b_m$:

   $$y = Bv = \lambda_1 b_1 + \cdots + \lambda_m b_m$$

   for suitable scalars $\lambda_1, \ldots, \lambda_m \in \mathbb{F}$.

   Thus y belongs to their span, namely

   $$y \in \mathrm{span}\{b_1, \ldots, b_m\} = \mathrm{Col}(B).$$

   This proves $J \subseteq \mathrm{Col}(B)$.

2. Conversely, $\mathrm{Col}(B) \subseteq J$.

   Let $w \in \mathrm{Col}(B)$. There are scalars $\eta_1, \ldots, \eta_m \in \mathbb{F}$ such that

   $$w = \eta_1 b_1 + \cdots + \eta_m b_m.$$

   Define

   $$v := \begin{pmatrix} \eta_1 \\ \vdots \\ \eta_m \end{pmatrix} \in \mathbb{F}^m.$$

   The preceding calculation gives

   $$Bv = \eta_1 b_1 + \cdots + \eta_m b_m = w.$$

   Hence $w = Bv$ for some $v \in \mathbb{F}^m$, so

   $$w \in J.$$

   This proves $\mathrm{Col}(B) \subseteq J$.

The two inclusions yield

$$J = \mathrm{Col}(B),$$

as required.

EXERCISE 10.3

Let $\mathbb{F}$ be a field and let $A, B \in \mathbb{F}^{n \times n}$ satisfy

$$AB = 0.$$

Please show that

$$\mathrm{Col}(B) \subseteq N(A),$$

Every column of B, and every linear combination of its columns, is therefore annihilated by A.

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EXERCISE 10.3 CONTINUED

WORKED SOLUTION

Use the definitions of column space and null space.
It suffices to prove that $Av = 0$ for every $v \in \mathrm{Col}(B)$.
By the definition of column space, $v \in \mathrm{Col}(B)$ implies that some $w \in \mathbb{F}^n$ satisfies

$$v = Bw.$$

Compute

$$Av = A(Bw) = (AB)w.$$

Since $AB = 0$,

$$(AB)w = 0w = 0,$$

and therefore

$$Av = 0.$$

We have proved that $Av = 0$ for every $v \in \mathrm{Col}(B)$, or equivalently,

$$v \in N(A).$$

Consequently,

$$\mathrm{Col}(B) \subseteq N(A),$$

as required.

EXERCISE 10.4 FOUNDATION

$$A = \begin{pmatrix} 1 & -1 & 0 & 2 \\ 0 & 0 & 1 & 3 \\ 1 & -1 & 1 & 5 \end{pmatrix}, \qquad b = (2, 4, 6)^T.$$

Please find row-space, column-space and kernel bases and please check rank-nullity. Please give a solvability condition for a general right-hand side and please find all solutions for the given b.

WORKED SOLUTION

The first two rows are independent and pivot columns are $1, 3$. The kernel vectors have independent free entries in positions $2, 4$. The row relation gives the solvability condition, and the first two equations prove sufficiency. The displayed solutions are a translate of the kernel.

$$R_3 = R_1 + R_2, \quad B_{\text{row}} = \{(1, -1, 0, 2), (0, 0, 1, 3)\}, \quad \text{rank } A = 2,$$

$$B_{\text{col}} = \{(1, 0, 1)^T, (0, 1, 1)^T\},$$

$$x_2 = s, \quad x_4 = t \quad \Longrightarrow \quad x_1 = s - 2t, \quad x_3 = -3t,$$

$$B_{\text{ker}} = \{(1, 1, 0, 0)^T, (-2, 0, -3, 1)^T\}, \qquad 2 + 2 = 4.$$

$$\exists x : \; Ax = b \iff b_3 = b_1 + b_2,$$

$$x = (2, 0, 4, 0)^T + s(1, 1, 0, 0)^T + t(-2, 0, -3, 1)^T, \qquad s, t \in \mathbb{R}.$$

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MOTIVATION AND GUIDING IDEA

Motivation. The same information may appear as a polynomial, a matrix or a coordinate vector. A linear map compares such descriptions while preserving linear combinations. The kernel asks what is lost; the image asks what can be attained.

NOTATION AND CONVENTIONS

- $T : V \to W$ denotes a map with domain V and codomain W; $T(v) \in W$ is the image of v. The vectors $0_V, 0_W$ are the zero vectors in the corresponding spaces.
- $\ker T = \{v \in V : T(v) = 0_W\}$ is the kernel; $\operatorname{Im} T = T(V) = \{T(v) : v \in V\}$ is the image. Do not automatically identify the image with the codomain.
- $\operatorname{rank} T = \dim \operatorname{Im} T$ is the rank; $\dim \ker T$ is the nullity. The notation $V \cong W$ means that a linear isomorphism between them exists.
- $B = (v_1, \ldots, v_n)$ is an **ordered** basis. If $v = \sum_{j=1}^{n} a_j v_j$, then $[v]_B = (a_1, \ldots, a_n)^T$. The order of basis vectors determines the order of entries.
- $\mathbb{F}_{\leq n}[X]$ is the space of polynomials of degree at most n, including the zero polynomial. For $T : V \to V$, $T^2 = T \circ T$ denotes composition, not entrywise squaring.

DEFINITION **10.3**

Let V, W be vector spaces over the same field $\mathbb{F}$. A map $T : V \to W$ is linear if it is well defined and satisfies

$$T(av + bw) = aT(v) + bT(w) \qquad (v, w \in V,\ a, b \in \mathbb{F}).$$

It is injective if $T(v) = T(w)$ implies $v = w$, and surjective if every $y \in W$ equals $T(v)$ for some $v \in V$. For a linear map, injectivity is equivalent to $\ker T = \{0_V\}$, and surjectivity to $\operatorname{Im} T = W$. An isomorphism is a bijective linear map; its inverse is linear. If $\dim V < \infty$, then

$$\dim V = \dim \ker T + \dim \operatorname{Im} T.$$

The coordinate map $v \mapsto [v]_B$ is an isomorphism $V \to \mathbb{F}^n$. A vector and its coordinate column are related by a choice of basis; they should not be equated without an appropriate identification.

MOTIVATION AND GUIDING IDEA

Tutorial objectives. Recognize linear maps, compute kernels and images, and use coordinates to pass between polynomials, matrices and vectors. Distinguish a mathematical object from its representation in a chosen basis.

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10.3 Linear maps

DEFINITION 10.4

Let V, W be vector spaces over the same field $\mathbb{F}$.
A map $T : V \to W$ is **linear** if, for all $v_1, v_2 \in V$ and $\lambda \in \mathbb{F}$,

1. $T(v_1 + v_2) = T(v_1) + T(v_2)$.
2. $T(\lambda v_1) = \lambda T(v_1)$.

Interpretation A linear map sends vectors from one vector space to another while preserving their linear structure. It respects vector addition and scalar multiplication.

The first condition says that adding and then applying the map gives the same result as applying the map and then adding. Thus addition commutes with applying the map.

The second condition says that multiplying by a scalar before applying the map gives the same result as multiplying the image by that scalar. In a real geometric setting, scaling an input by λ scales its image by the same factor.

NOTE

Together these properties imply that, for all $\alpha, \beta \in \mathbb{F}$ and $v_1, v_2 \in V$,

$$T(\alpha v_1 + \beta v_2) = \alpha T(v_1) + \beta T(v_2),$$

so a linear map preserves every linear combination.
Geometrically, real linear maps include scalings, rotations and reflections. They send a line to a line or a point and fix the zero vector. In particular, $T(0_V) = 0_W$ always holds.
The main consequences are:

- More generally, $T(\sum_{i=1}^{n} \lambda_i v_i) = \sum_{i=1}^{n} \lambda_i T(v_i)$.
- A necessary condition for $T : V \to W$ to be linear is $T(0_V) = 0_W$. (A map with $T(0_V) \neq 0_W$ is not linear.)

EXERCISE 10.5

$$V = \mathbb{R}_{\leq n}[X], \qquad n \geq 1.$$
$$T_1(P) = P(X) + 2P(-1), \qquad T_2(P) = X^2 P(X),$$
$$T_3(P) = P(X + 2) - P(2).$$

Please decide which formulas define linear maps $V \to V$.

WORKED SOLUTION

T_1 and T_3 do not increase polynomial degree, so they are well-defined and linear. The second formula does not define a self-map of V, since it sends X^n outside V.

$$T_1(\alpha P + \beta Q) = \alpha P + \beta Q + 2\alpha P(-1) + 2\beta Q(-1) = \alpha T_1(P) + \beta T_1(Q).$$

$$T_2(X^n) = X^{n+2} \notin V.$$

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EXERCISE 10.5 CONTINUED

$$T_3(\alpha P + \beta Q) = \alpha P(X+2) + \beta Q(X+2) - \alpha P(2) - \beta Q(2) = \alpha T_3(P) + \beta T_3(Q).$$

MOTIVATION AND GUIDING IDEA

Two guiding questions. The kernel records information lost by a map; the image records attainable outputs. A trivial kernel means that distinct inputs remain distinguishable. An image equal to the codomain means that every output in that codomain is attainable.

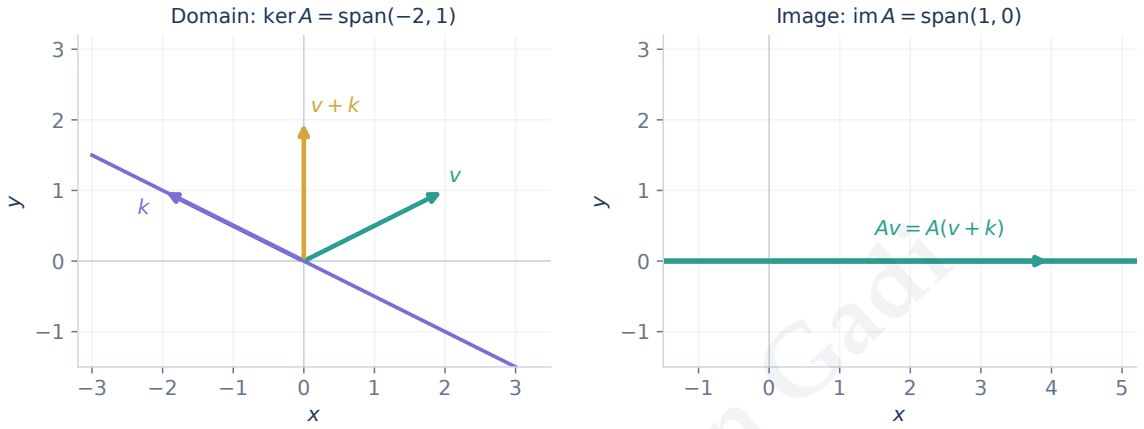


Figure 10.1. The kernel lies in the domain and the image in the codomain. Several inputs can have the same output when the kernel is nontrivial.

10.4 The kernel and image of a linear map

RECALL AND TOOLS

Let $T : V \to W$ be linear.
The **kernel of** T consists of all vectors sent to 0_W:

$$\ker(T) = \{v \in V | T(v) = 0\}$$

The kernel is a subspace of V.
The **image of** T is the set of all outputs attained by the map:

$$\mathrm{Im}(T) = \{T(v) | v \in V\} = \{w \in W | \exists v \in V, T(v) = w\}$$

The image is a subspace of W. It consists of outputs actually attained, whereas the codomain W is specified as part of the definition of the map. These need not be equal.
The sets $\ker T$ and $\mathrm{Im}\, T$ are nonempty, since $T(0_V) = 0_W$.

- Always $\mathrm{Im}\, T \subseteq W$. If $\mathrm{Im}\, T = W$, the map is **surjective**, or **onto**.
- T is **injective**, or **one-to-one**, if, for all $v_1, v_2 \in V$,

  $$T(v_1) = T(v_2) \implies v_1 = v_2$$

  Equivalently, no two distinct inputs have the same image.

  $$v_1 \neq v_2 \implies T(v_1) \neq T(v_2)$$

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- A linear map T is an **isomorphism** if it is injective and surjective.
- Vector spaces V, W are **isomorphic** if there **exists** an isomorphism $T : V \to W$. We then write $V \cong W$.

PROPOSITION 10.2

A linear map $T : V \to W$ is **injective** if and only if $\ker T = \{0_V\}$.
For a homogeneous linear system, this means that the trivial solution is the only solution.

PROPOSITION 10.3

Let $T : V \to W$ be a linear map between finite-dimensional vector spaces. Then

1. If T is injective, $\dim V \leq \dim W$.
2. If T is surjective, $\dim W \leq \dim V$.
3. If $\dim V = \dim W$, then T is injective **if and only if** it is surjective.

EXERCISE 10.6

Let $n \geq 2$ and $V = M_n(\mathbb{F})$. Define $T : V \to \mathbb{F}$ by

$$T(A) = \mathrm{Tr}(A).$$

1. Please prove that T is linear.
2. Please find $\ker T$.
3. Please find $\mathrm{Im}\, T$.
4. Please compute $\dim \ker T$ and $\dim \mathrm{Im}\, T$, and please verify rank-nullity.

WORKED SOLUTION

Write $A = (a_{ij}), B = (b_{ij})$.

1. Linearity:

$$T(A+B) = \mathrm{Tr}(A+B) = \sum_{i=1}^{n}(a_{ii}+b_{ii}) = \sum_{i=1}^{n} a_{ii} + \sum_{i=1}^{n} b_{ii} = T(A) + T(B),$$

For every $\lambda \in \mathbb{F}$,

$$T(\lambda A) = \mathrm{Tr}(\lambda A) = \sum_{i=1}^{n} \lambda a_{ii} = \lambda \sum_{i=1}^{n} a_{ii} = \lambda T(A).$$

Thus T is linear.

2. Kernel:

$$\ker(T) = \{A \in M_n(\mathbb{F}) \ : \ \mathrm{Tr}(A) = 0\}.$$

This is the subspace of matrices with trace zero.

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EXERCISE 10.6 CONTINUED

3. Image: for any $\alpha \in \mathbb{F}$, take

$$A = \operatorname{diag}(\alpha, 0, \ldots, 0) \in M_n(\mathbb{F}),$$

Then $\operatorname{Tr}(A) = \alpha$, so

$$\operatorname{Im}(T) = \mathbb{F}.$$

4. Dimensions: $\dim M_n(\mathbb{F}) = n^2$. Since $\operatorname{Im} T = \mathbb{F}$, its dimension is 1. Rank-nullity therefore gives

$$\dim \ker(T) = n^2 - 1.$$

Alternatively, $n^2 - 1$ entries may be chosen freely, and the last diagonal entry is fixed by requiring trace zero.

EXERCISE 10.7

$$V = C[-1, 2], \qquad S : V \to \mathbb{R}, \qquad S(f) = \int_{-1}^{2} f(x)\, dx.$$

Please prove that S is linear. Please find its kernel and image and please give a nonzero kernel element.

WORKED SOLUTION

Linearity of integration proves linearity. Every real value is obtained from a suitable constant function. The function $x - \frac{1}{2}$ is nonzero and lies in the kernel.

$$S(\alpha f + \beta g) = \int_{-1}^{2} (\alpha f + \beta g) = \alpha \int_{-1}^{2} f + \beta \int_{-1}^{2} g = \alpha S(f) + \beta S(g).$$

$$\ker S = \left\{ f \in C[-1, 2] : \int_{-1}^{2} f(x)\, dx = 0 \right\}, \qquad \operatorname{Im} S = \mathbb{R}.$$

$$S(c/3) = \frac{c}{3}(2 - (-1)) = c, \qquad S(x - \tfrac{1}{2}) = \left[\tfrac{x^2}{2} - \tfrac{x}{2} \right]_{-1}^{2} = 1 - 1 = 0.$$

EXERCISE 10.8

Let V be a vector space over a field F, and let $T : V \to V$ be linear. Please prove:
(a) $\ker T \subseteq \ker T^2$; (b) $\operatorname{Im} T^2 \subseteq \operatorname{Im} T$.

WORKED SOLUTION

(a) We must show that every vector in the set on the left also belongs to the set on the right. Let $v \in \ker T$. Then $T^2(v) = T(T(v)) = T(0) = 0$, so $v \in \ker T^2$.
(b) Let $w \in \operatorname{Im} T^2$. Some $v \in V$ satisfies $w = T^2(v) = T(T(v))$.
Set $u = T(v)$. Then $w = T(u)$, so $w \in \operatorname{Im} T$.

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DEFINITION 10.5

If $T : V \to W$ is an **isomorphism**, each $w \in W$ has a unique $v \in V$ with $T(v) = w$. Thus the **inverse map** $T^{-1} : W \to V$ is defined by $T^{-1}(w) = v$ whenever $T(v) = w$. Applying T to v gives w; applying T^{-1} to w recovers v.

EXERCISE 10.9

$$V = M_2(\mathbb{R}), \qquad \Phi\begin{pmatrix} a & b \\ c & d \end{pmatrix} = (a, b, c, d)^T.$$

$$U = \{x \in \mathbb{R}^4 : x_2 = x_3\}, \qquad W = \{x \in \mathbb{R}^4 : x_1 = x_4 = 0,\ x_2 = -x_3\}.$$

Please prove that Φ is an isomorphism. Please find $\Phi^{-1}(U)$ and $\Phi^{-1}(W)$ and their dimensions.

WORKED SOLUTION

Listing matrix entries preserves addition and scalar multiplication; the inverse formula proves bijectivity. The preimages are the symmetric and skew-symmetric matrices. Their displayed free parameters give the independent spanning matrices and the dimensions.

$$\Phi^{-1}((a, b, c, d)^T) = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

$$\Phi^{-1}(U) = \left\{\begin{pmatrix} a & b \\ b & d \end{pmatrix} : a, b, d \in \mathbb{R}\right\} = \operatorname{span}\left\{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\right\}.$$

$$\Phi^{-1}(W) = \left\{\begin{pmatrix} 0 & b \\ -b & 0 \end{pmatrix} : b \in \mathbb{R}\right\} = \operatorname{span}\left\{\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\right\}.$$

$$\dim U = 3, \qquad \dim W = 1.$$

EXERCISE 10.10 FOUNDATION

Please solve over $\mathbb{R}$ the system

$$\begin{cases} x + y + z = 6, \\ x - y + z = 2. \end{cases}$$

Please write its solution set as a particular solution plus multiples of a fixed vector. Please determine whether it is a subspace and please describe all differences between pairs of solutions.

WORKED SOLUTION

Subtracting the second equation from the first gives $2y = 4$, so $y = 2$ and $x + z = 4$. Put $z = t$ to get

$$S = \{(4 - t, 2, t) : t \in \mathbb{R}\} = (4, 2, 0) + \operatorname{span}\{(-1, 0, 1)\}.$$

It is not a subspace, because zero does not satisfy the equations. The difference of two solutions is

$$(4 - t, 2, t) - (4 - s, 2, s) = (t - s)(-1, 0, 1).$$

Every real multiple occurs by choosing $s = 0$. Thus the differences form exactly

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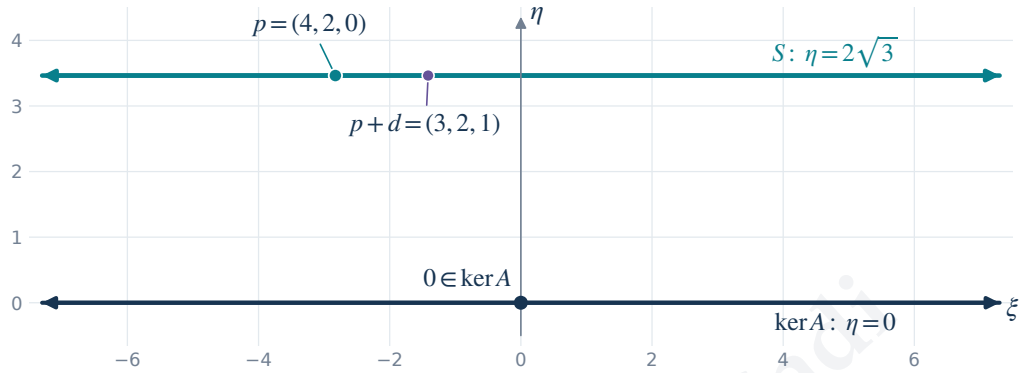
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EXERCISE 10.10 FOUNDATION CONTINUED

$\operatorname{span}\{(-1,0,1)\}$, the solution set of the homogeneous system.

$$S=\{(4-t,2,t):t\in\mathbb{R}\}=p+\ker A$$

$$\Phi(\xi,\eta)=\xi u+\eta v,\qquad u=\frac{(-1,0,1)}{\sqrt{2}},\quad v=\frac{(1,1,1)}{\sqrt{3}}$$



$$\ker A=\operatorname{span}\{d\},\quad d=(-1,0,1),\qquad 0\notin S$$

Figure 10.2. The real system $x+y+z=6$, $x-y+z=2$ has solution set $S=p+\ker A$, where $p=(4,2,0)$ and $\ker A=\operatorname{span}\{(-1,0,1)\}$. The ξ,η axes describe the plane $\operatorname{span}\{u,v\}\subseteq\mathbb{R}^3$ through the map Φ displayed in the figure. The vectors u,v are perpendicular unit vectors, so the chart preserves lengths and angles. The translated line excludes the origin, whereas the kernel contains it.

EXERCISE 10.11 CHALLENGE

Let $a,b\in\mathbb{R}$. Please classify the solutions of

$$\begin{cases}x+y+z=5,\\ x+ay+z=b,\\ x+y+az=b.\end{cases}$$

In each case, please find the coefficient matrix rank, the number of free variables when consistent, and the solution set. Also find the kernel of the coefficient matrix, and please determine whether the solution set is a subspace.

WORKED SOLUTION

Subtract the first equation from both the second and the third. The equivalent system is

$$\begin{cases}x+y+z=5,\\ (a-1)y=b-5,\\ (a-1)z=b-5.\end{cases}$$

1. If $a\neq 1$, division by $a-1$ is permitted. Put $c=(b-5)/(a-1)$. The unique solution is

$$S=\{(5-2c,c,c)\}.$$

There are three pivots, so the rank is 3 and there are no free variables. In the homoge-

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EXERCISE 10.11 CHALLENGE CONTINUED

neous system, $y = z = 0$ and then $x = 0$, so $\ker A = \{0\}$.

2. If $a = 1$ and $b = 5$, the last two equations after subtraction are $0 = 0$. The rank is 1, and there are two free variables. Choosing $y = s, z = t$ gives

$$S = \{(5 - s - t, s, t) : s, t \in \mathbb{R}\} = (5, 0, 0) + \text{span}\{(-1, 1, 0), (-1, 0, 1)\}.$$

The kernel is $\text{span}\{(-1, 1, 0), (-1, 0, 1)\}$.

3. If $a = 1$ and $b \neq 5$, the equation $0 = b - 5$ is a contradiction, so $S = \varnothing$. The coefficient matrix still has rank 1, while the augmented matrix has rank 2. The kernel depends only on A, so it is the same kernel as in the preceding case.

The solution set is never a subspace: in the consistent cases the zero vector fails to satisfy $x + y + z = 5$, and in the inconsistent case the set is empty. A number of degrees of freedom of a solution set cannot be defined through a choice of solution parameters when no solutions exist.

DEFINITION 10.6

Let $\mathbb{F}$ be a field and $A \in \mathbb{F}^{m \times n}$. Write its columns as

$$A = \begin{pmatrix} | & & | \\ a_1 & \dots & a_n \\ | & & | \end{pmatrix}, \qquad a_j \in \mathbb{F}^m.$$

Let $v = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \in \mathbb{F}^n$.

1. **A linear combination of columns.** Matrix-vector multiplication is defined by

$$Av = v_1 a_1 + \cdots + v_n a_n.$$

Thus Av is the linear combination of the columns of A with coefficients $v_1, \dots, v_n$. It follows that

$$\{Av \mid v \in \mathbb{F}^n\} = \text{span}\{a_1, \dots, a_n\} = \text{Col}(A).$$

2. **A linear map.** The function

$$T_A : \mathbb{F}^n \to \mathbb{F}^m, \qquad T_A(v) = Av,$$

is linear: for all $u, v \in \mathbb{F}^n$ and $\lambda \in \mathbb{F}$,

$$T_A(u + v) = A(u + v) = Au + Av,$$

$$T_A(\lambda v) = A(\lambda v) = \lambda(Av).$$

Thus multiplication by A is a linear map, and its columns describe its action on the standard basis of $\mathbb{F}^n$:

$$T_A(e_j) = a_j.$$

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10.5 Determining a map from vector images

PROPOSITION 10.4

Let V, W be vector spaces over the same field $\mathbb{F}$, let $B = (v_1, \dots, v_n)$ be an ordered basis of V, and let $w_1, \dots, w_n \in W$ be arbitrary vectors, not necessarily independent. There is a unique linear map $T : V \to W$ with $T(v_i) = w_i$ for every i. Explicitly, if $v = \sum_{i=1}^{n} a_i v_i$, then

$$T(v) = \sum_{i=1}^{n} a_i w_i.$$

The formula is well defined because basis expansions are unique. It is linear because coordinates of sums and scalar multiples are computed entry by entry. Any linear map with the prescribed images must obey this formula, proving uniqueness. In particular, between any two vector spaces over the same field there is at least one linear map: the zero map.

EXERCISE 10.12

(a) $T(1,0,0)^T = (2,0,1)^T$, $T(1,1,0)^T = (1,1,0)^T$, $T(1,0,1)^T = (0,2,1)^T$;
(b) $T(2,1,4)^T = (1,2,0)^T$, $T(1,0,2)^T = (1,-1,1)^T$, $T(0,1,0)^T = (-1,4,0)^T$.

Please decide in each case whether a linear map $T : \mathbb{R}^3 \to \mathbb{R}^3$ satisfies the requirements. If it exists, please find an explicit formula.

WORKED SOLUTION

In part (a), the three inputs form a basis, so the map exists uniquely. Computing the images of the standard basis gives the formula. In part (b), the proposed outputs do not preserve the displayed input relation, so no such map exists.

(a) $T(e_1) = (2,0,1)^T$, $T(e_2) = (1,1,0)^T - T(e_1) = (-1,1,-1)^T$,

$T(e_3) = (0,2,1)^T - T(e_1) = (-2,2,0)^T$, $T(x,y,z)^T = (2x - y - 2z, y + 2z, x - y)^T$.

(b) $(2,1,4)^T = 2(1,0,2)^T + (0,1,0)^T$,

$2(1,-1,1)^T + (-1,4,0)^T = (1,2,2)^T \neq (1,2,0)^T$.

MOTIVATION AND GUIDING IDEA

Existence criterion for prescribed images. The requirements $T(u_i) = w_i$ can be satisfied by a linear map if and only if, for every choice of coefficients a_i,

$$\sum_i a_i u_i = 0 \quad \Longrightarrow \quad \sum_i a_i w_i = 0.$$

Under this condition, define a map on $U = \operatorname{span}\{u_i\}$ by $T(\sum_i a_i u_i) = \sum_i a_i w_i$. The difference of two representations is a dependence relation, so both yield the same image. Linearity follows from the definition. If the domain is finite-dimensional, extend a basis of U to a basis of V, and freely prescribe images of the extra basis vectors. The map is unique if $U = V$. If $U \neq V$ and $W \neq \{0\}$, two different images may be chosen for an extra basis vector, so the extension is not unique. If the u_i already form a basis, there are no nontrivial dependence

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relations, and every choice of images defines a unique linear map.

EXERCISE 10.13

$$T:\mathbb{R}^2\to\mathbb{R}_3[X],\qquad T((1,0)^T)=1+2X^2,\quad T((1,1)^T)=4+X+2X^2.$$

Please find the unique linear map satisfying these requirements.

WORKED SOLUTION

The two given inputs form a basis. Linearity forces the formula, and substitution reproduces both prescribed outputs.

$$T((0,1)^T)=T((1,1)^T)-T((1,0)^T)=3+X,$$

$$T((x,y)^T)=x(1+2X^2)+y(3+X)=(x+3y)+yX+2xX^2.$$

EXERCISE 10.14

$$P:\mathbb{R}^4\to\mathbb{R}^4,\qquad P(x)=x-\frac{x_1+x_2+x_3+x_4}{4}(1,1,1,1)^T.$$

Please prove that P is linear, please find its kernel and image, and please prove $P^2=P$.

WORKED SOLUTION

The image lies in the zero-sum subspace. Every vector in that subspace is fixed by P, proving equality with the image. The kernel and image dimensions are $1,3$, as rank-nullity requires.

$$\mu(x)=\frac{x_1+x_2+x_3+x_4}{4},$$

$$\begin{aligned}P(\alpha x+\beta y)&=\alpha x+\beta y-(\alpha\mu(x)+\beta\mu(y))(1,1,1,1)^T\\&=\alpha P(x)+\beta P(y).\end{aligned}$$

$$P(x)=0\iff x=\mu(x)(1,1,1,1)^T,\qquad \ker P=\operatorname{span}\{(1,1,1,1)^T\}.$$

$$\sum_{j=1}^{4}(Px)_j=\sum_{j=1}^{4}x_j-4\mu(x)=0.$$

$$\operatorname{Im}P=\{y\in\mathbb{R}^4:y_1+y_2+y_3+y_4=0\}=\operatorname{span}\{e_1-e_4,e_2-e_4,e_3-e_4\}.$$

$$\mu(Px)=0\quad\Longrightarrow\quad P(Px)=Px.$$

Additional practice

EXERCISE 10.15 CHALLENGE

$$P:V\to V,\quad P^2=P.$$

$$V=\ker P\oplus\operatorname{Im}P,\quad \ker(I-P)=\operatorname{Im}P,\quad \operatorname{Im}(I-P)=\ker P.$$

$$P_0(x,y,z)^T=(x+2z,y-z,0)^T\quad\text{on }\mathbb{R}^3.$$

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EXERCISE 10.15 CHALLENGE CONTINUED

Please prove the three identities and the existence and uniqueness of the projection with kernel U and image W when $V = U \oplus W$. Please compute the explicit decomposition for P_0.

WORKED SOLUTION

The first decomposition splits a vector into a kernel part and an image part. The map acts as the identity on its image, so the intersection is zero. The displayed calculations prove all three identities. For a given $U \oplus W$, uniqueness of decomposition makes $P(u+w) = w$ well-defined and linear. It is the only possibility because it must vanish on the kernel and fix the image. In the example, the two parts lie in the stated subspaces.

$$v = (v - Pv) + Pv, \quad P(v - Pv) = 0, \quad P(Pv) = Pv.$$

$$w \in \ker P \cap \operatorname{Im} P \implies Pw = 0 = w \implies w = 0.$$

$$(I - P)v = 0 \iff Pv = v \iff v \in \operatorname{Im} P, \quad P(I - P) = 0, \quad u \in \ker P \implies (I - P)u = u.$$

$$P(u + w) = w \quad (u \in U, w \in W),$$

$$\ker P_0 = \operatorname{span}\{(-2, 1, 1)^T\}, \quad \operatorname{Im} P_0 = \{(a, b, 0)^T : a, b \in \mathbb{R}\},$$

$$(x, y, z)^T = z(-2, 1, 1)^T + (x + 2z, y - z, 0)^T.$$

10.6 From a Spanning Description to Linear Equations

RECALL AND TOOLS

Let $U \leq \mathbb{F}^n$. If the columns of C form a basis of U, then

$$x \in U \iff Ct = x \text{ is consistent.}$$

Apply row operations to $[C \mid x]$, leaving the entries of x as variables. If $\dim U = d$, the reduced form of C is $\begin{pmatrix} I_d \\ 0 \end{pmatrix}$. The bottom rows of the augmented matrix give $n - d$ homogeneous equations in the entries of x. Place their coefficients in a matrix A to obtain $U = \ker A$.
Conversely, row reduction of A expresses all solutions of $Ax = 0$ in terms of the free variables. Setting one free variable to 1 and the others to 0 gives a basis of the kernel. Here $\ker A = \{x : Ax = 0\}$.
If $d = n$, there are no equations; the $1 \times n$ zero matrix is a possible choice. If $d = 0$, then $U = \{0\} = \ker I_n$.

EXAMPLE 10.1

Over a field $\mathbb{F}$, let

$$U = \operatorname{span}\left\{ \begin{pmatrix} 1 \\ 0 \\ 2 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \\ 3 \end{pmatrix} \right\}.$$

The two columns are independent, as their first two entries show. Reduce the augmented

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matrix:

$$\left(\begin{array}{cc|c} 1 & 0 & x_1 \\ 0 & 1 & x_2 \\ 2 & 1 & x_3 \\ -1 & 3 & x_4 \end{array}\right) \longrightarrow \left(\begin{array}{cc|c} 1 & 0 & x_1 \\ 0 & 1 & x_2 \\ 0 & 0 & x_3 - 2x_1 - x_2 \\ 0 & 0 & x_4 + x_1 - 3x_2 \end{array}\right).$$

The operations are $R_3 \leftarrow R_3 - 2R_1 - R_2$ and $R_4 \leftarrow R_4 + R_1 - 3R_2$. Hence

$$U = \left\{ x \in \mathbb{F}^4 : \begin{array}{rcl} -2x_1 - x_2 + x_3 & = & 0, \\ x_1 - 3x_2 + x_4 & = & 0 \end{array} \right\} = \ker \begin{pmatrix} -2 & -1 & 1 & 0 \\ 1 & -3 & 0 & 1 \end{pmatrix}.$$

To recover the spanning description, set $x_1 = s$ and $x_2 = t$. Then $x_3 = 2s+t$ and $x_4 = -s+3t$, so

$$x = s \begin{pmatrix} 1 \\ 0 \\ 2 \\ -1 \end{pmatrix} + t \begin{pmatrix} 0 \\ 1 \\ 1 \\ 3 \end{pmatrix}.$$

The same calculation works over finite fields, interpreting the numbers and operations in the field.

EXERCISE 10.16

Let $A \in M_{r\times n}(\mathbb{F})$ have rank $r \geq 1$, and let $B \in M_{k\times n}(\mathbb{F})$. Please prove that

$$\ker A \subseteq \ker B \quad \Longleftrightarrow \quad \text{there is a unique } C \in M_{k\times r}(\mathbb{F}) \text{ with } B = CA.$$

WORKED SOLUTION

If $B = CA$ and $Ax = 0$, then $Bx = CAx = C0 = 0$. Thus $\ker A \subseteq \ker B$.
Now assume the inclusion. Since $\operatorname{rank} A = r$, each standard column $e_j \in \mathbb{F}^r$ has a preimage $d_j \in \mathbb{F}^n$ satisfying $Ad_j = e_j$. Place these solutions in the columns of

$$D = (d_1 \ \cdots \ d_r) \in M_{n\times r}(\mathbb{F}), \qquad AD = I_r.$$

For every $x \in \mathbb{F}^n$,

$$A(x - DAx) = Ax - (AD)Ax = Ax - Ax = 0.$$

The inclusion gives $B(x - DAx) = 0$, hence $Bx = BDAx$. This identity holds for every x, so

$$B = BDA.$$

Therefore $C = BD$ works. If also $B = C'A$, then

$$BD = C'AD = C'I_r = C'.$$

Every admissible matrix must therefore equal BD, proving uniqueness.

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EXERCISE 10.17

Let $U \leq \mathbb{F}^n$ have dimension $1 \leq d \leq n$. Please prove that d standard coordinates can be selected so that their values determine each vector in U uniquely, and every choice of d values occurs. Please prove that fewer than d coordinates cannot suffice.

WORKED SOLUTION

Choose a basis $u_1, \ldots, u_d$ of U and place it in the columns of

$$C = (u_1 \ \cdots \ u_d) \in M_{n\times d}(\mathbb{F}).$$

The rank of C is d. Thus d independent rows can be selected. Write their indices as $j_1 < \cdots < j_d$, and let R be the square matrix consisting of these rows. It has rank d and is invertible. Every $x \in U$ has a unique expression $x = Ct$. The selected coordinates satisfy

$$\begin{pmatrix} x_{j_1} \\ \vdots \\ x_{j_d} \end{pmatrix} = Rt.$$

If their prescribed values form a column $y \in \mathbb{F}^d$, then

$$t = R^{-1}y, \qquad x = CR^{-1}y.$$

This is a unique vector of U, and every y is possible.
If only $k < d$ coordinates are selected, the corresponding rows of C form a $k \times d$ matrix. Its homogeneous system has a solution $t \neq 0$. The columns of C are independent, so $Ct \neq 0$, but all k selected coordinates of Ct vanish. Those coordinates do not distinguish Ct from the zero vector.

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EXERCISE 10.18 CHALLENGE

Let V have finite dimension $n \geq 2$ over a field F. Let $T : V \to V$ be linear and satisfy $T(v) \in \text{span}\{v\}$ for every $v \in V$. Please prove that one scalar λ satisfies $T(v) = \lambda v$ for all v. Please check whether imposing the condition only on a fixed basis is sufficient.

WORKED SOLUTION

Choose a basis $e_1, \ldots, e_n$ and write $T(e_j) = \lambda_j e_j$. For $i \neq j$, the hypothesis for $e_i + e_j$ gives

$$\lambda_i e_i + \lambda_j e_j = T(e_i + e_j) = \mu(e_i + e_j)$$

for some μ. Independence implies $\lambda_i = \lambda_j = \mu$. All basis scalars therefore equal one λ, and linearity gives

$$T\left(\sum_j a_j e_j\right) = \sum_j a_j \lambda e_j = \lambda \sum_j a_j e_j.$$

A basis condition alone is insufficient. On $\mathbb{R}^2$, $T(x, y) = (x, 2y)$ preserves the lines of the two standard basis vectors, but $T(1, 1) = (1, 2)$ is not a multiple of $(1, 1)$.

EXERCISE 10.19 EXAM QUESTION

1. Let $n \geq 2$ and $A, B \in M_n(\mathbb{R})$ satisfy $B \neq 0$ and $\text{rank}(A) = n - 1$. Please prove that $AB = 0$ implies $\text{rank}(B) = 1$.

2. Let $A, B \in M_{2031}(\mathbb{R})$ satisfy $AB = 0$. Please prove that

$$\text{rank}(A) + \text{rank}(B) \leq 2031.$$

WORKED SOLUTION

The key implication in both parts is

$$AB = 0 \quad \Longrightarrow \quad \text{Im}(B) \subseteq \ker A.$$

Indeed, every $y \in \text{Im}\, B$ has the form $y = Bx$. Thus $Ay = A(Bx) = (AB)x = 0$, so $y \in \ker A$.

1. The inclusion and rank-nullity give

$$\begin{aligned} \text{rank}(B) &= \dim \text{Im}(B) \\ &\leq \dim \ker A \\ &= n - \text{rank}(A) \\ &= n - (n - 1) = 1. \end{aligned}$$

   Since $B \neq 0$, at least one column of B is nonzero, so $\text{rank}(B) \geq 1$. The two bounds yield $\text{rank}(B) = 1$.

2. Apply the same inclusion in $\mathbb{R}^{2031}$:

$$\begin{aligned} \text{rank}(B) &= \dim \text{Im}(B) \\ &\leq \dim \ker A \\ &= 2031 - \text{rank}(A). \end{aligned}$$

   Rearranging gives

$$\text{rank}(A) + \text{rank}(B) \leq 2031.$$

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EXERCISE 10.19 EXAM QUESTION CONTINUED

If every output of B is annihilated by A, the entire image of B must lie in the kernel of A. The dimension of that kernel therefore bounds the rank of B.

EXERCISE 10.20

$$A = \begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \end{pmatrix}, \qquad z = ax_1 + bx_2 + cx_3.$$

Please find a necessary and sufficient condition for uniquely recovering $x \in \mathbb{R}^3$ from Ax and z. Please give one useful extra reading and one reading determined by the original two.

WORKED SOLUTION

Unique recovery means that the extra reading does not vanish on the kernel direction. Equivalently, the augmented square matrix is invertible. The first example satisfies the condition; the second is the sum of the two original readings.

$$\ker A = \operatorname{span}\{(3,-2,1)^T\}, \qquad (a,b,c)(3,-2,1)^T = 3a - 2b + c.$$

$$\det \begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \\ a & b & c \end{pmatrix} = 3a - 2b + c.$$

$$3a - 2b + c \neq 0.$$

$$z = x_3 : \quad (a,b,c) = (0,0,1), \qquad z = x_1 + x_2 - x_3 : \quad (a,b,c) = (1,1,-1).$$

EXERCISE 10.21 FOUNDATION

For $t \in \mathbb{R}$, consider

$$A_t = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 4 & 1 \\ 3 & 6 & t \end{pmatrix}.$$

1. Please compute $\det A_t$ using one row operation, and please determine when A_t is invertible.
2. At $t = 0$, please find a basis of $\ker A_0$ and please find $\operatorname{rank} A_0$.
3. Please construct B_t from A_t by, in this order, interchanging the first two rows, multiplying the third row by 3, and adding 5 times the third row to the first row. Please compute $\det B_t$ without writing B_t explicitly.

WORKED SOLUTION

1. The operation $R_3 \leftarrow R_3 - 3R_1$ preserves the determinant and gives

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 4 & 1 \\ 0 & 0 & t \end{pmatrix}.$$

Hence $\det A_t = 1 \cdot 4 \cdot t = 4t$, and A_t is invertible exactly when $t \neq 0$.

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EXERCISE 10.21 FOUNDATION CONTINUED

2. At $t = 0$, the kernel equations are $x_1 + 2x_2 = 0$ and $4x_2 + x_3 = 0$. Setting $x_2 = s$ gives

$$\ker A_0 = \{s(-2, 1, -4)^T : s \in \mathbb{R}\} = \text{span}\{(-2, 1, -4)^T\}.$$

The kernel has dimension 1, so rank-nullity gives $\text{rank } A_0 = 3 - 1 = 2$.

3. The interchange multiplies the determinant by -1, scaling the row multiplies it by 3, and the final operation preserves it. Thus

$$\det B_t = (-1) \cdot 3 \cdot \det A_t = -12t.$$

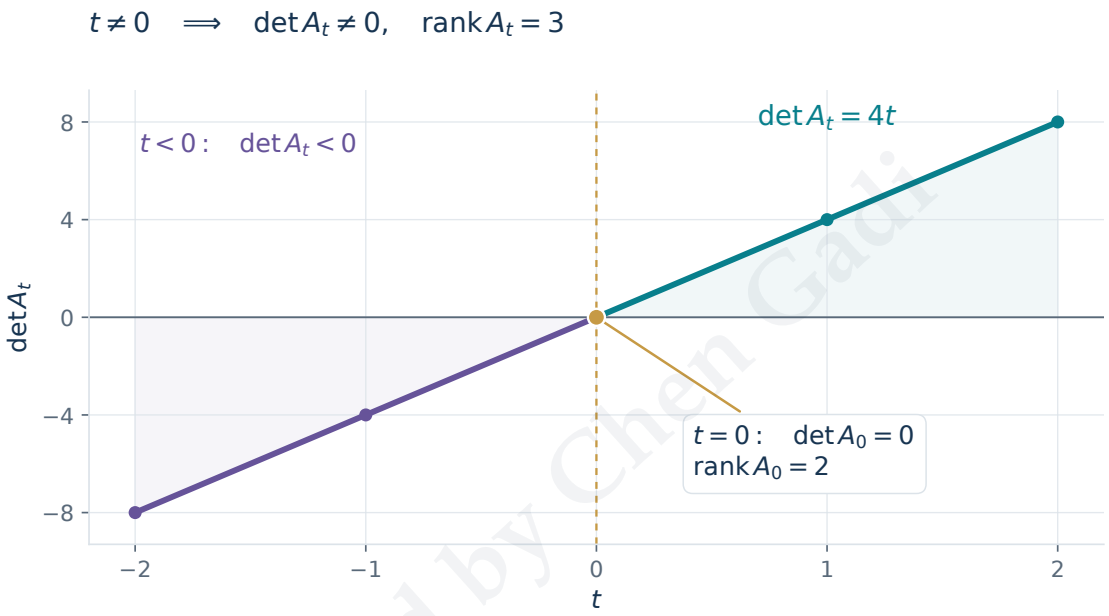


Figure 10.3. For the matrix A_t in the exercise, the graph $\det A_t = 4t$ crosses zero only at $t = 0$. For every $t \neq 0$, the matrix is invertible and has rank 3; at the gold point, $\text{rank } A_0 = 2$, with kernel $\text{span}\{(-2, 1, -4)^T\}$.

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EXERCISE 10.22 CHALLENGE

Let $A \in M_n(\mathbb{F})$ be invertible, $U \in \mathbb{F}^{n \times k}$, and $V \in \mathbb{F}^{k \times n}$, with $n, k \geq 1$.

1. Please use determinants of block matrices to prove

$$\det(A + UV) = \det A \, \det(I_k + VA^{-1}U).$$

Do not assume that U or V is square or invertible.

2. For $s \in \mathbb{R}$, please use the identity to compute the determinant of

$$C_s = \begin{pmatrix} s+2 & 0 & 1 \\ 0 & s+2 & -1 \\ s+1 & s+1 & 1 \end{pmatrix}$$

and please determine when it is invertible. At the singular parameter value, please find its kernel and rank.

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EXERCISE 10.22 CHALLENGE CONTINUED

WORKED SOLUTION

1. First prove $\det(I_n + XY) = \det(I_k + YX)$ for $X \in \mathbb{F}^{n\times k}$ and $Y \in \mathbb{F}^{k\times n}$. Set

$$K = \begin{pmatrix} I_n & X \\ -Y & I_k \end{pmatrix}, \quad L = \begin{pmatrix} I_n & 0 \\ Y & I_k \end{pmatrix}, \quad H = \begin{pmatrix} I_n & -X \\ 0 & I_k \end{pmatrix}.$$

Both L and H are block triangular with determinant 1. We have

$$LK = \begin{pmatrix} I_n & X \\ 0 & I_k + YX \end{pmatrix}, \qquad HK = \begin{pmatrix} I_n + XY & 0 \\ -Y & I_k \end{pmatrix}.$$

Multiplicativity and the determinant formula for block triangular matrices give

$$\det K = \det(I_k + YX) = \det(I_n + XY).$$

Now $A + UV = A(I_n + A^{-1}UV)$. Substitute $X = A^{-1}U$ and $Y = V$ to obtain the required identity.

2. Take $A = I_3$ and write $C_s = I_3 + UV$, where

$$U = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad V = \begin{pmatrix} s+1 & 0 & 1 \\ 0 & s+1 & -1 \end{pmatrix}.$$

Direct multiplication verifies this factorization and gives

$$I_2 + VU = \begin{pmatrix} s+3 & 1 \\ -1 & s+1 \end{pmatrix}.$$

Therefore

$$\det C_s = \det(I_2 + VU) = (s+1)(s+3) + 1 = (s+2)^2.$$

Thus C_s is invertible exactly when $s \neq -2$. At $s = -2$, its kernel equations are $x_3 = 0$ and $-x_1 - x_2 + x_3 = 0$, so

$$\ker C_{-2} = \operatorname{span}\{(1, -1, 0)^T\}, \qquad \operatorname{rank} C_{-2} = 3 - 1 = 2.$$

The multiplicity of a root of the determinant as a polynomial in s is not the kernel dimension: this root is double, while the kernel has dimension 1.

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EXERCISE 10.23

Let $A \in M_n(\mathbb{F})$ with $n \geq 2$.

1. Please prove that $\operatorname{adj}(A)$ has rank n, 1, or 0 according as rank $A = n$, rank $A = n-1$, or rank $A \leq n-2$.
2. Please deduce that $\det(\operatorname{adj}(A)) = (\det A)^{n-1}$.

WORKED SOLUTION

1. We use the rank characterization by minors. If a matrix has rank r, select r independent columns. The resulting matrix has row rank r, so selecting r independent rows gives an invertible $r \times r$ minor. Every larger minor has dependent columns and determinant zero. If A is invertible, then $\operatorname{adj}(A) = (\det A)A^{-1}$ is invertible and has rank n. If rank $A = n-1$, some minor of order $n-1$ is nonzero. At least one cofactor is therefore nonzero, so $\operatorname{adj}(A) \neq 0$. On the other hand,

$$A \operatorname{adj}(A) = 0,$$

so every column of the adjugate belongs to $\ker A$. Rank-nullity gives

$$\dim \ker A = n - (n-1) = 1.$$

Hence $1 \leq \operatorname{rank}(\operatorname{adj}(A)) \leq 1$, and the rank is 1. If rank $A \leq n-2$, every minor of order $n-1$ vanishes. Every cofactor is zero, so $\operatorname{adj}(A) = 0$.

2. For invertible A, take determinants in $A \operatorname{adj}(A) = (\det A)I_n$:

$$\det A \cdot \det(\operatorname{adj}(A)) = (\det A)^n.$$

Division by $\det A \neq 0$ gives the result. If A is singular, part 1 gives $\operatorname{rank}(\operatorname{adj}(A)) \leq 1 < n$, so both sides of the required identity are zero.

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TUTORIAL 11

Coordinates, Matrix Representations, and Change of Basis

11.1 Rank-nullity for linear maps

PROPOSITION 11.1

Let $T : V \to W$ be linear, with V finite-dimensional.
Then

$$\dim V = \dim \ker T + \dim \operatorname{Im} T.$$

EXERCISE 11.1

(a) $T : \mathbb{R}^3 \to \mathbb{R}^3$, $\ker T = \operatorname{span}\{(1,-1,2)^T\}$, $\operatorname{Im} T = \operatorname{span}\{(2,0,1)^T\}$;

(b) $T : M_2(\mathbb{R}) \to \mathbb{R}^2$, $\ker T = \operatorname{span}\left\{\begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}\right\}$.

Please decide whether a linear map exists in each case.

WORKED SOLUTION

Neither map exists. Part (a) contradicts rank-nullity. In part (b), the image would have to be a three-dimensional subspace of a two-dimensional space.

(a) $\dim \ker T + \dim \operatorname{Im} T = 1 + 1 = 2 \neq 3.$

(b) $\dim \operatorname{Im} T = \dim M_2(\mathbb{R}) - \dim \ker T = 4 - 1 = 3 > 2.$

MOTIVATION AND GUIDING IDEA

Motivation. A representing matrix is a computational tool that depends on the ordered bases of the domain and codomain. Changing coordinates leaves the map unchanged but can greatly simplify repeated calculations. Always identify the input and output coordinate systems to avoid reversing a change of basis.

NOTATION AND CONVENTIONS

- For ordered bases $B = (u_1, \dots, u_n)$ of U and $C = (v_1, \dots, v_m)$ of V, the notation $[T]_C^B$ denotes an $m \times n$ matrix taking B-coordinates as input and returning C-coordinates.
- $S \circ T$ applies T first and S second; I_V is the identity map on V.

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- $P_{B\leftarrow B'} = [I_V]_B^{B'}$ takes $[v]_{B'}$ to $[v]_B$. The arrow indicates the direction of coordinate conversion.
- If B is an ordered basis of $\mathbb{F}^n$ and E is the standard basis, $S_B = [\, b_1 \ \cdots \ b_n \,] = [I]_E^B$ is the basis-column matrix. Hence $[v]_E = S_B[v]_B$.
- $[T]_B$ abbreviates $[T]_B^B$ for an operator $T : V \to V$. Square matrices A, D are similar if $D = P^{-1}AP$ for an invertible P.

DEFINITION 11.1

Let U, V be finite-dimensional vector spaces over the same field, with ordered bases B, C, and let $T : U \to V$ be linear. Its representing matrix is defined by its columns:

$$[T]_C^B = \big[\, [T(u_1)]_C \ \cdots \ [T(u_n)]_C \,\big], \qquad [T(u)]_C = [T]_C^B[u]_B.$$

For another ordered basis B' of the domain, the columns of $P_{B\leftarrow B'}$ are the B-coordinate vectors of the vectors of B'. Its inverse is $P_{B'\leftarrow B}$. If $R : V \to W$ is linear and D is an ordered basis of W, then

$$[R \circ T]_D^B = [R]_D^C[T]_C^B.$$

If T is an isomorphism, $[T^{-1}]_B^C = ([T]_C^B)^{-1}$. For an operator with standard matrix A, the matrix in the ordered basis B is $S_B^{-1}AS_B$. Domain and codomain bases must match at every matrix product.

MOTIVATION AND GUIDING IDEA

Construct a map from prescribed vector images, check compatibility with linear dependence relations, compute a representing matrix and change bases. The emphasis is on the connection between an abstract map, coordinates and matrix calculations.

11.2 Composition of maps

RECALL AND TOOLS

- Let U, V, W be vector spaces over a field $\mathbb{F}$.
- Let $T : U \to V$ and $S : V \to W$ be linear maps.
- The map $S \circ T : U \to W$, defined for $u \in U$ by $(S \circ T)(u) = S(T(u))$, is their **composition**.
- $S \circ T$ is linear over $\mathbb{F}$.
- If both maps are surjective, their composition is surjective. If both are injective, their composition is injective. In particular, a composition of isomorphisms is an isomorphism.

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EXERCISE 11.2

$$T : \mathbb{R}^2 \to M_2(\mathbb{R}), \qquad T(a,b)^T = \begin{pmatrix} 2a & a-b \\ a+b & 3b \end{pmatrix},$$

$$S : M_2(\mathbb{R}) \to \mathbb{R}_3[X], \qquad S\begin{pmatrix} q & r \\ s & t \end{pmatrix} = qX^2 + (r-2s)X + t.$$

Please compute $S \circ T$.

WORKED SOLUTION

Apply T first and substitute its four matrix entries into the formula for S.

$$(S \circ T)(a,b)^T = (2a)X^2 + \big((a-b) - 2(a+b)\big)X + 3b = 2aX^2 - (a+3b)X + 3b.$$

NOTATION AND CONVENTIONS

Notation guide. For an ordered basis $B = (b_1, \ldots, b_n)$, the notation $[v]_B = (c_1, \ldots, c_n)^T$ means $v = \sum_{j=1}^{n} c_j b_j$. A^{-1} denotes the inverse, A^T the transpose, and $(A \mid I_n)$ an augmented matrix. $R_i \leftrightarrow R_j$ interchanges rows; $R_i \leftarrow \lambda R_i$ requires $\lambda \neq 0$; $R_i \leftarrow R_i + \lambda R_j$ requires $i \neq j$. rank A is the number of pivots in a row-echelon form. Distinguish a matrix unit E_{ij} from an entire elementary matrix E.

11.3 Coordinates

DEFINITION 11.2

Let V be a vector space over a field $\mathbb{F}$ with ordered basis $B = (v_1, \ldots, v_n)$.
Every $v \in V$ has a unique representation as a linear combination of the basis vectors. Thus there are unique $\alpha_1, \ldots, \alpha_n \in \mathbb{F}$ such that

$$v = \alpha_1 v_1 + \cdots + \alpha_n v_n$$

The coefficients $\alpha_1, \ldots, \alpha_n$ are the **coordinates of** v **relative to** B.

The column vector $\begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \in \mathbb{F}^n$ is the **coordinate vector of** v **relative to** B, denoted $[v]_B$.

EXERCISE 11.3

$$B = ((1,1,0)^T, (0,1,1)^T, (1,0,1)^T), \qquad v = (5,3,6)^T.$$

Please find $[v]_B$ and please check it by reconstructing v.

WORKED SOLUTION

The basis-column matrix has determinant 2, so it is a basis. Equating the components of the

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EXERCISE 11.3 CONTINUED

linear combination gives the three equations.

$$a(1,1,0)^T + b(0,1,1)^T + c(1,0,1)^T = (a+c, a+b, b+c)^T.$$

$$a+c=5, \quad a+b=3, \quad b+c=6 \implies a=1, \quad b=2, \quad c=4.$$

$$[v]_B = (1,2,4)^T, \qquad (1,1,0)^T + 2(0,1,1)^T + 4(1,0,1)^T = (5,3,6)^T.$$

Conversely, knowing a vector's coordinates in a chosen basis allows us to reconstruct the vector.

With the notation above, $v = \alpha_1 v_1 + \cdots + \alpha_n v_n$ **if and only if** $[v]_B = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix}$.

EXERCISE 11.4

$$B = ((2,1)^T, (-1,1)^T), \qquad [v]_B = (3,-2)^T.$$

Please find v in standard coordinates.

WORKED SOLUTION

The coordinate entries are the coefficients of the ordered basis vectors.

$$v = 3(2,1)^T - 2(-1,1)^T = (6,3)^T + (2,-2)^T = (8,1)^T.$$

PROPOSITION 11.2

Let V have dimension n over a field $\mathbb{F}$, and let B be an ordered basis of V.
The coordinate map $T_B : V \to \mathbb{F}^n$, defined by $T_B(v) = [v]_B$, is an isomorphism. In particular, $V \cong \mathbb{F}^n$.

RECALL AND TOOLS

Let V be a vector space over $\mathbb{F}$, let B be an ordered basis with n vectors, and let $u_1, \ldots, u_m$ be vectors in V. Then:

1. $\{u_1, \ldots, u_m\}$ is linearly independent (in V) **if and only if** $\{[u_1]_B, \ldots, [u_m]_B\}$ is linearly independent (in $\mathbb{F}^n$).

2. $\{u_1, \ldots, u_m\}$ spans V **if and only if** $\{[u_1]_B, \ldots, [u_m]_B\}$ spans $\mathbb{F}^n$.

3. $\{u_1, \ldots, u_m\}$ is a basis of V **if and only if** $\{[u_1]_B, \ldots, [u_m]_B\}$ is a basis of $\mathbb{F}^n$.

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EXERCISE 11.5

$$p_1 = 1 + 2X, \qquad p_2 = X + 3X^2, \qquad p_3 = 1 + X + 2X^2.$$

Please prove that these polynomials are independent in $\mathbb{R}_2[X]$ using coordinates in $(1, X, X^2)$.

WORKED SOLUTION

The coordinate columns are therefore independent. The coordinate map is an isomorphism, so the polynomials are independent as well.

$$[p_1] = (1, 2, 0)^T, \quad [p_2] = (0, 1, 3)^T, \quad [p_3] = (1, 1, 2)^T.$$

$$\det \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 0 & 3 & 2 \end{pmatrix} = (2 - 3) + (6 - 0) = 5 \neq 0.$$

EXERCISE 11.6

$$U = \operatorname{span}\{U_1, U_2, U_3\}, \qquad V = \operatorname{span}\{V_1, V_2\} \subseteq M_2(\mathbb{R}),$$

$$U_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad U_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad U_3 = \begin{pmatrix} 0 & 0 \\ 1 & 2 \end{pmatrix}, \quad V_1 = \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}, \quad V_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Please find bases and dimensions for U, V, and $U \cap V$.

WORKED SOLUTION

In each list, the entries uniquely determine the coefficients, proving independence. Membership in U requires a relation between the second-row entries. Substituting a general element of V determines the intersection.

$$U = \left\{ \begin{pmatrix} a & b \\ c & 2c \end{pmatrix} : a, b, c \in \mathbb{R} \right\}, \qquad V = \left\{ \begin{pmatrix} r & s \\ r + s & 2r \end{pmatrix} : r, s \in \mathbb{R} \right\}.$$

$$\begin{pmatrix} r & s \\ r + s & 2r \end{pmatrix} \in U \iff 2r = 2(r + s) \iff s = 0.$$

$$B_U = (U_1, U_2, U_3), \quad \dim U = 3, \qquad B_V = (V_1, V_2), \quad \dim V = 2,$$

$$U \cap V = \operatorname{span}\{V_1\}, \qquad \dim(U \cap V) = 1.$$

EXERCISE 11.7 FOUNDATION

Let $w_1 = (1, 2, 0)^T$, $w_2 = (0, 1, -1)^T$, $e_3 = (0, 0, 1)^T$. Please prove that $B = (w_1, w_2, e_3)$ is a basis of $\mathbb{R}^3$. Please find the inverse of its column matrix by row reduction and please compute $[b]_B$ for $b = (3, 8, 5)^T$.

WORKED SOLUTION

The equation $aw_1 + bw_2 + ce_3 = 0$ gives $a = 0$, $2a + b = 0$, and $-b + c = 0$, so $a = b = c = 0$.

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EXERCISE 11.7 FOUNDATION CONTINUED

Three independent vectors in $\mathbb{R}^3$ form a basis.

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}.$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_3 \leftarrow R_3 + R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & -2 & 1 & 1 \end{array}\right).$$

$$A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -2 & 1 & 1 \end{pmatrix}, \qquad [b]_B = A^{-1} \begin{pmatrix} 3 \\ 8 \\ 5 \end{pmatrix} = \begin{pmatrix} 3 \\ -6+8 \\ -6+8+5 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 7 \end{pmatrix}.$$

Indeed, $3w_1 + 2w_2 + 7e_3 = (3, 6+2, -2+7)^T = (3, 8, 5)^T$. The representation is unique because B is a basis.

EXERCISE 11.8

Please determine whether $B = (x^2 - x,\ x + 2,\ 1)$ is a basis of $\mathbb{R}_2[x]$ and of $\mathbb{R}_3[x]$. In the first case, please find the coordinates of $p(x) = ax^2 + bx + c$.

WORKED SOLUTION

Write p as a combination with coefficients α, β, γ and compare coefficients:

$$ax^2 + bx + c = \alpha(x^2 - x) + \beta(x + 2) + \gamma.$$

$$\alpha = a, \quad -\alpha + \beta = b, \quad 2\beta + \gamma = c.$$

$$[p]_B = (a,\ a + b,\ c - 2a - 2b)^T.$$

For each a, b, c there is a unique coordinate triple, so B is a basis of $\mathbb{R}_2[x]$. Every combination has degree at most 2 and cannot equal x^3, so this is not a basis of $\mathbb{R}_3[x]$.

EXERCISE 11.9

Please determine whether $B = (x^2 + 2x,\ x - 3,\ 2)$ is a basis of $\mathbb{R}_2[x]$ and of $\mathbb{R}_3[x]$. In the first case, please find the coordinates of $p(x) = ax^2 + bx + c$.

WORKED SOLUTION

Write p as a combination with coefficients α, β, γ and compare coefficients:

$$ax^2 + bx + c = \alpha(x^2 + 2x) + \beta(x - 3) + 2\gamma.$$

$$\alpha = a, \quad 2\alpha + \beta = b, \quad -3\beta + 2\gamma = c.$$

$$[p]_B = (a,\ b - 2a,\ (c + 3b - 6a)/2)^T.$$

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EXERCISE 11.9 CONTINUED

For each a, b, c there is a unique coordinate triple, so B is a basis of $\mathbb{R}_2[x]$. Every combination has degree at most 2 and cannot equal x^3, so this is not a basis of $\mathbb{R}_3[x]$.

11.4 Representing a linear map by a matrix

DEFINITION 11.3

Let U be an n-dimensional vector space over $\mathbb{F}$ with ordered basis $B = (u_1, \ldots, u_n)$.
Let V be an m-dimensional vector space over the same field with ordered basis $C = (v_1, \ldots, v_m)$.
For a linear map $T : U \to V$, define the matrix

$$[T]_C^B = \begin{pmatrix} | & | & | & | \\ [T(u_1)]_C & [T(u_2)]_C & \cdots & [T(u_n)]_C \\ | & | & | & | \end{pmatrix} \in \mathbb{F}^{m \times n}$$

$[T]_C^B$ is the **matrix of T relative to the ordered bases B and C.**

Construction

First compute $T(u_i)$, the images of the basis vectors of the domain.

Then express each image in coordinates relative to C, and use these coordinate columns in the given order.

PROPOSITION 11.3

Let U, V be vector spaces over $\mathbb{F}$, with ordered bases B of U and C of V. For a linear map $T : U \to V$ and every $u \in U$,

$$[T(u)]_C = [T]_C^B \cdot [u]_B$$

Compute this matrix once for the chosen ordered bases. It then converts the coordinate vector of any new input into the coordinate vector of its image relative to C.

EXERCISE 11.10

$$V = M_3(\mathbb{R}), \qquad T(A) = 2(A - A^T),$$
$$U = \{A : A = A^T\}, \qquad W = \{A : A = -A^T\}.$$

Please prove linearity, please find the kernel and image and their dimensions, and please give a representing matrix in a basis formed by a basis of U followed by a basis of W.

WORKED SOLUTION

Here E_{ij} has a single 1 in position (i, j) and zeros elsewhere. The free matrix entries give the displayed bases. The map vanishes on symmetric matrices and multiplies skew-symmetric matrices by 4, giving the diagonal representation.

$$T(\alpha A + \beta B) = 2\big(\alpha A + \beta B - \alpha A^T - \beta B^T\big) = \alpha T(A) + \beta T(B).$$

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EXERCISE 11.10 CONTINUED

$$T(A)=0 \iff A=A^T, \qquad \ker T=U.$$

$$T(A)^T=-T(A), \qquad K\in W \implies T(K/4)=K.$$

$$\operatorname{Im} T=W, \qquad B_U=(E_{11},E_{22},E_{33},E_{12}+E_{21},E_{13}+E_{31},E_{23}+E_{32}),$$

$$B_W=(E_{12}-E_{21},E_{13}-E_{31},E_{23}-E_{32}), \qquad \dim U=6, \quad \dim W=3, \quad 6+3=9.$$

$$A=\frac{A+A^T}{2}+\frac{A-A^T}{2}\in U+W, \quad U\cap W=\{0\}, \qquad [T]_{(B_U,B_W)}=\operatorname{diag}(0,0,0,0,0,0,4,4,4).$$

EXERCISE 11.11

$$[T]_E=A=\begin{pmatrix}0&2&-1\\0&3&1\\0&0&4\end{pmatrix}, \qquad V=\mathbb{R}^3.$$

Please find nonzero vectors v,u satisfying $T(v)=0$ and $T(u)=4u$. Then please find w satisfying $T(w)=w+u$.

WORKED SOLUTION

The first column is zero, so e_1 lies in the kernel. Solving the second system gives an eigenvector for 4. Scaling that eigenvector by $\frac{1}{3}$ yields the last equation.

$$v=(1,0,0)^T, \qquad Av=0.$$

$$(A-4I)u=0 \iff -4u_1+2u_2-u_3=0, \quad -u_2+u_3=0.$$

$$u=(1,4,4)^T, \qquad Au=(4,16,16)^T=4u.$$

$$w=\tfrac{1}{3}u=(\tfrac{1}{3},\tfrac{4}{3},\tfrac{4}{3})^T, \qquad T(w)=\tfrac{4}{3}u=w+u.$$

EXERCISE 11.12

Let $V=\mathbb{C}[x]$, the vector space of all complex polynomials.
Define the linear operator $N:V\to V$ by

$$(Np)(x):=-(7-x)\,p'(x).$$

For each $\theta\in\mathbb{R}$, define $U_\theta:V\to V$ by

$$U_\theta:=\exp(-i\theta N^2)=\sum_{k=0}^{\infty}\frac{(-i\theta)^k}{k!}N^{2k}.$$

Please use the following two ordered bases:

$$E=(1,x,x^2,x^3,\dots), \qquad B=(7,\ (7-x),\ (7-x)^2,\ (7-x)^3,\dots).$$

1. Please prove that N and U_θ are linear.
2. For $n\geq 1$, set $b_n(x)=(7-x)^n$, and set $b_0(x)=7$. Please compute $N(b_n)$ and $N^2(b_n)$.
3. Please compute $U_\theta(b_n)$ for all $n\geq 0$ and please deduce the form of $[U_\theta]_B$.

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EXERCISE 11.12 CONTINUED

4. Please find the matrix of U_θ restricted to $P_3 = \text{span}\{1, x, x^2, x^3\}$ in the ordered basis $E_3 = (1, x, x^2, x^3)$.

WORKED SOLUTION

1. Differentiation is linear, so

$$N(\alpha p + \beta r) = -(7-x)(\alpha p' + \beta r') = \alpha N(p) + \beta N(r).$$

To justify the series, restrict to P_d, the polynomials of degree at most d. This space is invariant under N, so on P_d the series is the exponential series of a finite matrix, converging for every θ. These definitions agree as d increases and therefore define a map on $\mathbb{C}[x]$. Linearity follows by taking limits of the partial sums in each coordinate.

2. For $n \geq 1$,

$$b_n'(x) = \frac{d}{dx}(7-x)^n = n(7-x)^{n-1} \cdot (-1) = -n(7-x)^{n-1}.$$

so

$$N(b_n) = -(7-x)\, b_n'(x) = -(7-x)\big(-n(7-x)^{n-1}\big) = n(7-x)^n = nb_n.$$

and hence

$$N^2(b_n) = N(nb_n) = nN(b_n) = n \cdot nb_n = n^2 b_n.$$

Also $N(b_0) = N(7) = 0$, so $N^2(b_0) = 0$.

3. Since $N^{2k}(b_n) = n^{2k} b_n$ for $n \geq 1$, the series gives

$$U_\theta(b_n) = \sum_{k=0}^{\infty} \frac{(-i\theta)^k}{k!}\, n^{2k}\, b_n = \exp(-i\theta n^2)\, b_n.$$

For $n = 0$, $U_\theta(b_0) = b_0$.

Thus the matrix in B is diagonal:

$$[U_\theta]_B = \text{diag}\big(1,\ e^{-i\theta},\ e^{-i4\theta},\ e^{-i9\theta}, \dots\big).$$

4. Write $q_1 = e^{-i\theta}$, $q_4 = e^{-i4\theta}$ and $q_9 = e^{-i9\theta}$. On P_3, use b_0, b_1, b_2, b_3. Their images are $b_0, q_1 b_1, q_4 b_2, q_9 b_3$, respectively.

Expand these four polynomials in E_3:

$$b_0 = 7, \qquad b_1 = 7 - x, \qquad b_2 = (7-x)^2 = x^2 - 14x + 49,$$

$$b_3 = (7-x)^3 = -x^3 + 21x^2 - 147x + 343.$$

Let C be the matrix with coordinate columns $[b_0]_{E_3}, [b_1]_{E_3}, [b_2]_{E_3}, [b_3]_{E_3}$:

$$C = \begin{pmatrix} 7 & 7 & 49 & 343 \\ 0 & -1 & -14 & -147 \\ 0 & 0 & 1 & 21 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

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EXERCISE 11.12 CONTINUED

This matrix is invertible. The formula $[U_\theta]_{E_3} = C\,\mathrm{diag}(1, q_1, q_4, q_9)C^{-1}$ gives the following matrix:

$$[U_\theta]_{E_3} = \begin{pmatrix} 1 & 7(1-q_1) & 49(1-2q_1+q_4) & 343(1-3q_1+3q_4-q_9) \\ 0 & q_1 & 14(q_1-q_4) & 147(q_1-2q_4+q_9) \\ 0 & 0 & q_4 & 21(q_4-q_9) \\ 0 & 0 & 0 & q_9 \end{pmatrix}.$$

EXERCISE 11.13

$$T : \mathbb{R}^3 \to \mathbb{R}^2, \qquad T(x,y,z)^T = (3x+2y-z, 2x-y+4z)^T.$$

Please compute the standard-basis matrix using the images of the basis vectors.

WORKED SOLUTION

The basis-vector images are the matrix columns. The final multiplication verifies the formula for every input.

$$T(e_1) = (3,2)^T, \quad T(e_2) = (2,-1)^T, \quad T(e_3) = (-1,4)^T,$$

$$[T] = \begin{pmatrix} 3 & 2 & -1 \\ 2 & -1 & 4 \end{pmatrix}, \qquad [T]\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3x+2y-z \\ 2x-y+4z \end{pmatrix}.$$

EXERCISE 11.14

$$T : \mathbb{R}_2[X] \to \mathbb{R}^2, \qquad T(p) = (p(0), p(-1))^T,$$

$$B = (1, 2+X, 1+X+2X^2), \qquad C = ((1,-1)^T, (0,1)^T).$$

Please compute $[T]_C^B$ and please verify $[T(p)]_C = [T]_C^B[p]_B$ for $p = a + bX + cX^2$.

WORKED SOLUTION

Compute each image and convert it to C-coordinates. To verify the identity, first equate coefficients in the B-expansion and then multiply by the matrix.

$$[(r,s)^T]_C = (r, r+s)^T,$$

$$T(1) = (1,1)^T, \quad T(2+X) = (2,1)^T, \quad T(1+X+2X^2) = (1,2)^T,$$

$$M = [T]_C^B = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 3 \end{pmatrix}.$$

$$p = \alpha + \beta(2+X) + \gamma(1+X+2X^2) \implies \gamma = c/2, \quad \beta = b - c/2, \quad \alpha = a - 2b + c/2.$$

$$[p]_B = \begin{pmatrix} a - 2b + c/2 \\ b - c/2 \\ c/2 \end{pmatrix}, \qquad M[p]_B = \begin{pmatrix} a \\ 2a - b + c \end{pmatrix}.$$

$$T(p) = (a, a-b+c)^T, \qquad [T(p)]_C = (a, 2a-b+c)^T.$$

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11.4.1 Change-of-coordinate matrices

RECALL AND TOOLS

Definition: Let V be a finite-dimensional vector space over $\mathbb{F}$, with ordered bases B, B'. The **change-of-coordinate matrix** $[I_V]_B^{B'}$ from B' to B is the representing matrix $[T]_B^{B'}$ for the **identity map** $T : V \to V, T(v) = v$ for every $v \in V$.

- The fundamental identity is $[v]_B = [I_V]_B^{B'}[v]_{B'}$.
- The identity map is denoted I_V, with $I_V(v) = v$.
- Its columns are the coordinate vectors of the vectors of B' relative to the ordered basis B.

EXERCISE **11.15**

$$B = (1 + X, 2 + 3X), \qquad B' = (3 + 4X, 1 + 2X), \qquad p = a + bX.$$

Please compute the transition matrix $[I]_B^{B'}$ and please verify $[p]_B = [I]_B^{B'}[p]_{B'}$.

WORKED SOLUTION

The basis matrices contain polynomial coefficients in $(1, X)$. The transition first converts to standard coordinates and then to B. The coordinate formulas and final product explicitly verify the identity.

$$S_B = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}, \quad S_{B'} = \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}, \quad S_B^{-1} = \begin{pmatrix} 3 & -2 \\ -1 & 1 \end{pmatrix},$$

$$[I]_B^{B'} = S_B^{-1} S_{B'} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

$$[p]_B = \begin{pmatrix} 3a - 2b \\ -a + b \end{pmatrix}, \qquad [p]_{B'} = \begin{pmatrix} a - b/2 \\ -2a + 3b/2 \end{pmatrix},$$

$$\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} a - b/2 \\ -2a + 3b/2 \end{pmatrix} = \begin{pmatrix} 3a - 2b \\ -a + b \end{pmatrix}.$$

11.4.2 Matrices of compositions and inverse maps

PROPOSITION **11.4**

Let U, V, W be finite-dimensional vector spaces over $\mathbb{F}$, with ordered bases A, B, C, respectively.
Let $S : U \to V$ and $T : V \to W$ be linear. Then

$$[T \circ S]_C^A = [T]_C^B [S]_B^A$$

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Corollary

If $T : V \to W$ is an **isomorphism** and B, C are ordered bases of V, W, respectively, then

$$[T^{-1}]_B^C = \left([T]_C^B\right)^{-1}$$

11.5 Five descriptions of a linear map

Define $T : \mathbb{R}^2 \to \mathbb{R}^2$ so that, for every

$$(x, y) \in \mathbb{R}^2$$

we have

$$T(x, y) = (2x + 2y,\ 2x - 2y).$$

We now describe this same map in several ways.

Method 1: A formula for a general vector

This is a **general formula**: it specifies $T(v)$ for arbitrary $v = (x, y) \in \mathbb{R}^2$, rather than just particular vectors:

$$T(x, y) = (2x + 2y,\ 2x - 2y).$$

It defines the map directly on its entire domain.

Method 2: Images of a basis

Choose the standard basis:

$$e_1 = (1, 0), \qquad e_2 = (0, 1).$$

Then

$$T(e_1) = T(1, 0) = (2, 2), \qquad T(e_2) = T(0, 1) = (2, -2).$$

For $(x, y) = xe_1 + ye_2$, extend linearly:

$$T(x, y) = xT(e_1) + yT(e_2) = x(2, 2) + y(2, -2) = (2x + 2y,\ 2x - 2y).$$

Thus specifying the images of a basis defines the same map.

Method 3: A matrix

Write the input as a column:

$$\begin{pmatrix} x \\ y \end{pmatrix}.$$

Define

$$A = \begin{pmatrix} 2 & 2 \\ 2 & -2 \end{pmatrix}.$$

Then

$$T(x, y) = A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x + 2y \\ 2x - 2y \end{pmatrix}.$$

This matrix represents the same map in the standard basis.

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Method 4: Familiar linear operations

View T as performing the following operations:

- First component: twice the sum of the coordinates.
- Second component: twice the difference of the coordinates.

Thus

$$T(x, y) = \big(2\text{sum}(x, y), 2\text{diff}(x, y)\big),$$

These operations are linear, so T is linear.

Method 5: A structural decomposition

Specify that T satisfies

$$T(e_1 + e_2) = (4, 0), \qquad T(e_1 - e_2) = (0, 4),$$

and that T is linear.

Since $\{e_1+e_2, e_1-e_2\}$ is a basis of $\mathbb{R}^2$, these requirements determine T uniquely. Calculation gives the same formula:

$$T(x, y) = (2x + 2y,\ 2x - 2y).$$

Remarks

- A **general formula** defines $T(v)$ for an arbitrary vector v.
- A linear map may be specified by a formula, basis images, a matrix or a structural description.
- The descriptions differ, but the map itself is the same.

11.5.1 Summary of the methods

RECALL AND TOOLS

1. **An explicit formula for a general vector**
   Specify $T(v)$ for every $v \in V$.
   Check: the formula is well defined and satisfies both linearity conditions.

2. **Basis images and linear extension**
   If $B = \{v_1, \dots, v_n\}$ is a basis of V, freely choose

   $$T(v_1), \dots, T(v_n) \in W$$

   and define, for $v = \sum_{i=1}^{n} \alpha_i v_i$,

   $$T(v) := \sum_{i=1}^{n} \alpha_i\, T(v_i).$$

   Each choice of basis images determines exactly one linear map.

3. **A matrix**
   For $V = \mathbb{F}^n$ and $W = \mathbb{F}^m$, define

   $$T(x) = Ax$$

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where A is an $m \times n$ matrix.
Every matrix defines a linear map. After choosing ordered bases, every linear map between finite-dimensional spaces has a matrix representation.
Remark: the matrix depends on the chosen bases; the map itself does not.

4. **A familiar linear operation**
Use an operation such as differentiation, integration or multiplication by a fixed scalar.
Examples:
$$T(P) = P', \qquad T(f) = \int_{-1}^{2} f(x)\,dx.$$
Specify a domain and codomain, check well-definedness and verify linearity in each case.

5. **A direct-sum decomposition**
A map can be specified by its action on the components of a direct sum.
For example, if $V = U \oplus W$, define T by
$$T(u) = 0 \ \ (\forall u \in U), \qquad T(w) = w \ \ (\forall w \in W).$$
Linearity follows from the direct-sum decomposition and the linear action on each component.

Common errors

- Adding a nonzero constant vector destroys linearity: $T(v) = v + b$ with $b \neq 0$ is not linear.
- Always specify the domain and codomain: $T : V \to W$. The image $\operatorname{Im} T$ may be a proper subspace of the codomain.
- Distinguish a vector, its coordinate vector in an ordered basis, and the matrix representing a map.

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Additional practice

EXERCISE 11.16 FOUNDATION

$$T(x, y) = (2x - y, x + y), \quad B = ((1, 0)^T, (1, 1)^T), \quad C = ((1, 1)^T, (0, 1)^T), \quad v = (3, 2)^T.$$

Please compute $[T]_C^B$ and $[T^{-1}]_B^C$. Please verify the coordinate identity and please recover v using the inverse.

WORKED SOLUTION

Convert the images of the B-vectors to C-coordinates and place them in the columns. The inverse takes C-coordinates and returns B-coordinates. Reconstruction in B recovers $(3, 2)^T$.

$$[(r, s)^T]_C = (r, s - r)^T, \quad T(1, 0)^T = (2, 1)^T, \quad T(1, 1)^T = (1, 2)^T.$$

$$M = [T]_C^B = \begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix}, \quad \det M = 3, \quad [T^{-1}]_B^C = M^{-1} = \frac{1}{3}\begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}.$$

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EXERCISE 11.16 FOUNDATION CONTINUED

$$v = (1,0)^T + 2(1,1)^T, \quad [v]_B = (1,2)^T, \quad T(v) = (4,5)^T,$$

$$[T(v)]_C = (4,1)^T = M(1,2)^T, \qquad M^{-1}(4,1)^T = \frac{1}{3}(3,6)^T = (1,2)^T.$$

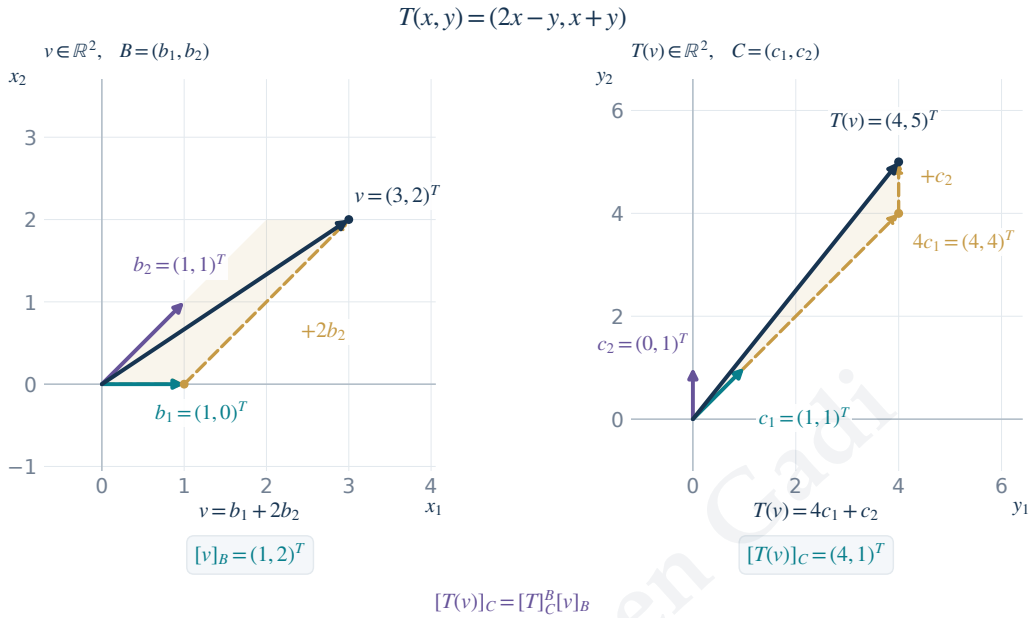


Figure 11.1. The arrows show vectors in standard coordinates; the gold paths show the decompositions $v = b_1 + 2b_2$ and $T(v) = 4c_1 + c_2$. Coordinate columns appear separately: $[v]_B = (1,2)^T$ and $[T(v)]_C = (4,1)^T$. The vector $T(v)$ itself is $(4,5)^T$.

11.6 Determinants and Change of Basis

RECALL AND TOOLS

We also use determinants over a field $\mathbb{F}$, such as $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{Z}_p$ with p prime. Arithmetic in the formulas is performed in that field. In particular,

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

If $\mathcal{B}, \mathcal{C}$ are bases of the same space and the columns of S are the coordinates of the vectors of $\mathcal{C}$ in $\mathcal{B}$, then

$$[T]_{\mathcal{C}} = S^{-1}[T]_{\mathcal{B}} S.$$

Multiplicativity gives

$$\det[T]_{\mathcal{C}} = \det(S^{-1}) \det[T]_{\mathcal{B}} \det S = \det[T]_{\mathcal{B}}.$$

Here $T : V \to V$ is linear and $\dim V < \infty$. Thus we may define $\det T = \det[T]_{\mathcal{B}}$; this scalar does not depend on the choice of basis.

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EXERCISE 11.17

Let V be a real vector space with basis $\mathcal{B} = (b_1, b_2)$, and let $T : V \to V$ satisfy

$$T(b_1) = 4b_1 + b_2, \qquad T(b_2) = b_1 + 2b_2.$$

For $\mathcal{C} = (b_1, b_1 + b_2)$, please compute $[T]_{\mathcal{B}}$ and $[T]_{\mathcal{C}}$, and please verify that their determinants agree.

WORKED SOLUTION

The images of the basis vectors form the columns, so

$$A = [T]_{\mathcal{B}} = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \det A = 4 \cdot 2 - 1 \cdot 1 = 7.$$

Write $c_1 = b_1$, $c_2 = b_1 + b_2$. Then $b_2 = c_2 - c_1$, and hence

$$T(c_1) = 4c_1 + (c_2 - c_1) = 3c_1 + c_2,$$

$$T(c_2) = T(b_1) + T(b_2) = 5b_1 + 3b_2 = 2c_1 + 3c_2.$$

Therefore

$$[T]_{\mathcal{C}} = \begin{pmatrix} 3 & 2 \\ 1 & 3 \end{pmatrix}, \qquad \det[T]_{\mathcal{C}} = 3 \cdot 3 - 2 \cdot 1 = 7.$$

The change-of-basis calculation gives the same result:

$$S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad S^{-1}AS = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 4 & 5 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 1 & 3 \end{pmatrix}.$$

EXERCISE 11.18

$$p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 \in \mathbb{F}_{\le 3}[x], \qquad \ell_1(p) = p(0), \quad \ell_2(p) = p(-1), \quad \ell_3(p) = p'(0).$$

Please find the matrix of the three maps in the basis $(1, x, x^2, x^3)$, row-space and column-space bases, and the rank. Please find all p satisfying $p(0) = 2$, $p(-1) = 3$, $p'(0) = 1$, and the dimension of the solution set.

WORKED SOLUTION

Expanding the evaluations and derivative gives the matrix. The displayed minor is invertible over every field, so the rank is 3 and the listed rows and columns are bases. The solution set is an affine line of dimension 1. Integer constants denote the corresponding field elements, so the calculation also holds in positive characteristic.

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & -1 & 1 & -1 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad \det \begin{pmatrix} 1 & 0 & 0 \\ 1 & -1 & 1 \\ 0 & 1 & 0 \end{pmatrix} = -1.$$

$$B_{\text{row}} = \{(1,0,0,0), (1,-1,1,-1), (0,1,0,0)\}, \qquad B_{\text{col}} = \{(1,1,0)^T, (0,-1,1)^T, (0,1,0)^T\}.$$

$$a_0 = 2, \quad a_1 = 1, \quad 2 - 1 + a_2 - a_3 = 3 \implies a_2 = 2 + a_3.$$

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EXERCISE 11.18 CONTINUED

$$p(x) = 2 + x + 2x^2 + t(x^2 + x^3), \qquad t \in \mathbb{F}.$$

EXERCISE 11.19 FOUNDATION

$$T : \mathbb{R}_{\leq 2}[X] \to \mathbb{R}^2, \qquad T(p) = (p(2), p(-2))^T.$$

Please prove linearity, please find the kernel and image, and please check rank-nullity. Please construct a linear map S in the opposite direction with $T \circ S = I$. Please decide whether $S \circ T = I$.

WORKED SOLUTION

Evaluation preserves addition and scalar multiplication. The formula for S is linear and takes the required values at the two points, so T is onto. The reverse composition preserves the evaluations but loses the quadratic coefficient, so it is not the identity.

$$p = a + bX + cX^2 \in \ker T \iff a + 2b + 4c = 0, \quad a - 2b + 4c = 0 \iff b = 0,\ a = -4c.$$

$$\ker T = \operatorname{span}\{X^2 - 4\}, \quad \dim \ker T = 1, \qquad S((u,v)^T) = \frac{u+v}{2} + \frac{u-v}{4}X.$$

$$(T \circ S)(u,v)^T = (u,v)^T, \qquad \operatorname{Im} T = \mathbb{R}^2, \quad 1 + 2 = 3.$$

$$(S \circ T)(a + bX + cX^2) = a + 4c + bX, \qquad (S \circ T)(X^2) = 4 \neq X^2.$$

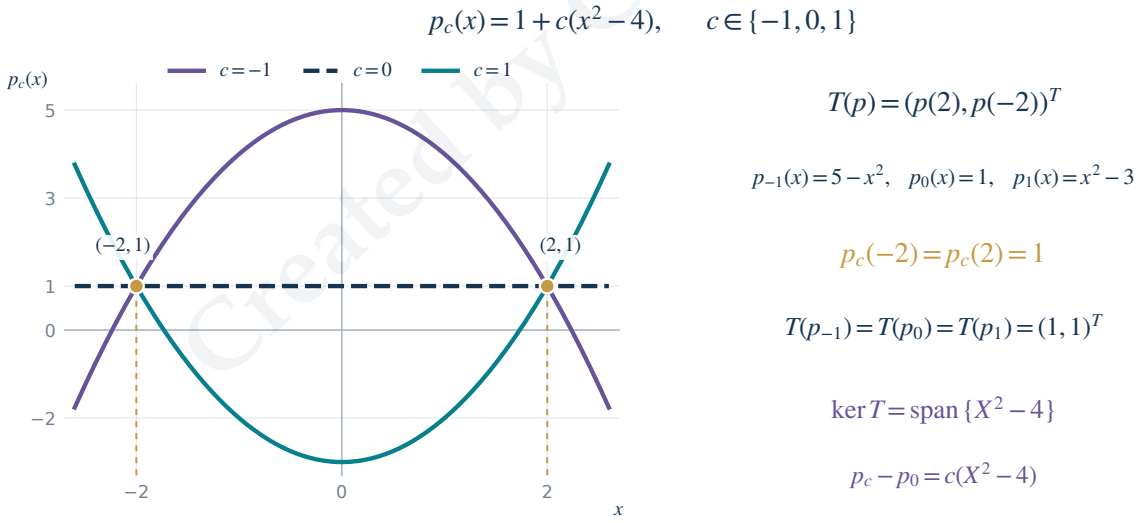


Figure 11.2. Three distinct polynomials give the same two measurements: $p_c(2) = p_c(-2) = 1$ for $c = -1, 0, 1$. Their differences $p_c - p_0 = c(X^2 - 4)$ lie in the kernel of $T(p) = (p(2), p(-2))^T$, so these measurements do not distinguish the polynomials.

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EXERCISE 11.20 CHALLENGE

Let V, W, U be finite-dimensional spaces over a field F, and let $S : V \to W$ and $T : V \to U$ be linear. Please prove that $\ker S \subseteq \ker T$ if and only if there is a linear map $R : W \to U$ with $T = R \circ S$. When S is onto, please prove that R is unique.

WORKED SOLUTION

If $T = R \circ S$ and $x \in \ker S$, then $T(x) = R(0) = 0$, so $x \in \ker T$. Conversely, assume the kernel inclusion and define on $\operatorname{Im} S$

$$R_0(Sx) = Tx.$$

This is well-defined: if $Sx = Sy$, then $x - y \in \ker S \subseteq \ker T$, so $Tx = Ty$. Also

$$R_0(aSx + bSy) = R_0(S(ax + by)) = T(ax + by) = aTx + bTy,$$

which proves linearity. Choose a basis of $\operatorname{Im} S$ and extend it to a basis of W. Define R by R_0 on the first basis vectors and by zero on the added vectors, then extend linearly. This gives $R(Sx) = Tx$ for every x. If S is onto, every $w \in W$ is Sx for some x. The equality forces $R(w) = Tx$, so there is only one possible map R.

EXERCISE 11.21 CHALLENGE

$$A \in \mathbb{F}^{m\times n}, \quad B \in \mathbb{F}^{n\times p}.$$

$$\operatorname{rank}(AB) = \operatorname{rank} B - \dim(\operatorname{Col} B \cap \ker A).$$

$$A_0 = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix}, \qquad B_0 = \begin{pmatrix} 2 & 0 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{over } \mathbb{R}.$$

Please prove the identity and please deduce $\operatorname{rank}(AB) \geq \operatorname{rank} A + \operatorname{rank} B - n$. Please verify the identity for A_0, B_0.

WORKED SOLUTION

Apply rank-nullity to the restriction L. Its image and kernel are exactly the spaces in the identity. In the example, the kernel of A_0 lies in the column space of B_0, so their intersection has dimension 1.

$$L : \operatorname{Col} B \to \mathbb{F}^m, \qquad L(y) = Ay, \quad \operatorname{Im} L = \operatorname{Col}(AB), \quad \ker L = \operatorname{Col} B \cap \ker A.$$

$$\operatorname{rank} B = \dim \operatorname{Im} L + \dim \ker L, \qquad \dim \ker L \leq \dim \ker A = n - \operatorname{rank} A.$$

$$\ker A_0 = \operatorname{span}\{(2, -1, 1)^T\}, \quad (2, -1, 1)^T = B_0 e_1 - B_0 e_2, \quad \operatorname{rank} B_0 = 2,$$

$$A_0 B_0 = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}, \qquad \operatorname{rank}(A_0 B_0) = 1 = 2 - 1.$$

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TUTORIAL 12

Similarity, Eigenvalues, and the Characteristic Polynomial

MOTIVATION AND GUIDING IDEA

A change of basis preserves the linear map but can greatly simplify its matrix. The determinant tests invertibility without computing an inverse, detects dependence among rows, and measures changes in area or volume. Before a long calculation, look for zeros, linear dependence, or block structure.

NOTATION AND CONVENTIONS

- $M_n(\mathbb{F})$ is the space of square matrices of size n over $\mathbb{F}$; I_n is the identity matrix, and A^T is the transpose.
- $A \sim B$ means that $B = P^{-1}AP$ for some invertible P. The columns of P are the new basis vectors expressed in the old basis.
- $\det A$ is the determinant. The alternative notation $|A|$ sometimes means the same scalar; it is not an absolute value of a matrix.
- M_{ij} is obtained by deleting row i and column j; $C_{ij} = (-1)^{i+j} \det(M_{ij})$ is the corresponding cofactor.
- $R_i \leftarrow R_i + cR_j$ means replacing row i by itself plus c times row j. The determinant preservation rule requires $i \neq j$.

NOTE

Distinguish two transformations. Similarity, $B = P^{-1}AP$, describes an operator after the same basis change in domain and codomain; it preserves the determinant. Congruence, $B = P^TAP$, describes a basis change for a bilinear form; here $\det B = (\det P)^2 \det A$. The formulas cannot be interchanged.

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12.1 Similar matrices

RECALL AND TOOLS

Matrices $A, B \in \mathbb{F}^{n\times n}$ are similar if there is an invertible $P \in \mathbb{F}^{n\times n}$ such that

$$A = P^{-1}BP$$

If A, B are similar, then

1. $$\det(A) = \det(B)$$
2. $$\text{trace}(A) = \text{trace}(B)$$
3. $$\text{rank}(A) = \text{rank}(B)$$

EXERCISE 12.1

$$A, B \in \mathbb{F}^{n\times n}.$$

Please prove or disprove: if A, B are similar, then A^k, B^k are similar for every $k \geq 1$; if A^k, B^k are similar for some $k \geq 1$, then A, B are similar.

WORKED SOLUTION

The first claim is true: cancelling adjacent inverse factors gives similarity through the same P. The second claim is false over every field. The displayed cubes agree, but a matrix similar to zero must be zero, whereas B is nonzero.

$$B = P^{-1}AP \quad \Longrightarrow \quad B^k = (P^{-1}AP)^k = P^{-1}A^kP.$$

$$A = 0_{3\times 3}, \qquad B = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix},$$

$$A^3 = B^3 = 0, \qquad B^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad B \neq 0.$$

EXERCISE 12.2

$$T : \mathbb{R}^2 \to \mathbb{R}^2, \quad T(x, y)^T = (2x + y, x - y)^T, \qquad B = ((1, 1)^T, (1, -2)^T).$$

Please compute $[T]_E^B$, please find v with $T(v) = (9, -6)^T$, and please check the result in both coordinate systems.

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EXERCISE 12.2 CONTINUED

WORKED SOLUTION

The columns are the images in standard output coordinates. Dividing by 3 gives the input B-coordinates, and rebuilding in B gives the vector itself.

$$T(1,1)^T = (3,0)^T, \qquad T(1,-2)^T = (0,3)^T, \qquad [T]_E^B = 3I_2.$$

$$[v]_B = (3,-2)^T, \qquad v = 3(1,1)^T - 2(1,-2)^T = (1,7)^T.$$

$$T(1,7)^T = (2+7, 1-7)^T = (9,-6)^T, \qquad 3I_2(3,-2)^T = (9,-6)^T.$$

12.2 Similar Matrices

RECALL AND TOOLS

Matrices $A, B \in \mathbb{F}^{n\times n}$ are called similar if there is an invertible matrix $P \in \mathbb{F}^{n\times n}$ such that

$$A = P^{-1}BP$$

If A, B are similar, then

1. $$\det(A) = \det(B)$$
2. $$\operatorname{trace}(A) = \operatorname{trace}(B)$$
3. $$\operatorname{rank}(A) = \operatorname{rank}(B)$$

EXERCISE 12.3

Let $A, B \in \mathbb{F}^{n\times n}$. Please prove or disprove:

1. If A, B are similar, then A^k, B^k are similar for every integer $k \geq 1$.
2. If A^k, B^k are similar for some integer $k \geq 1$, then A, B are similar.

WORKED SOLUTION

1. If $B = P^{-1}AP$ with P invertible, cancellation using $PP^{-1} = I$ gives

   $$B^k = (P^{-1}AP)^k = P^{-1}A^kP.$$

   Thus the powers are similar via the same matrix P.

2. The statement is false. Over any field, take

   $$A = 0_{2\times 2}, \qquad B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

   We have $A^2 = B^2 = 0$, but A, B are not similar: $P^{-1}AP = 0$ for every invertible P,

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EXERCISE 12.3 CONTINUED

whereas $B \neq 0$. Over $\mathbb{R}$ or $\mathbb{C}$, the example $A = -2I_2$, $B = 2I_2$ also works: their squares are equal, and a scalar matrix is similar only to itself. This example gives no counterexample in characteristic 2, where $-2I_2 = 2I_2$.

Common pitfall: The equality $A^k = (Q^{-1}BQ)^k$ does not imply $A = Q^{-1}BQ$. Taking a power is not an injective operation on matrices, as these examples show.

12.3 Similar Matrices

DEFINITION 12.1

Let $A, B \in M_n(\mathbb{F})$.
We say that A is **similar** to B (and write $A \sim B$) if there is an invertible $P \in M_n(\mathbb{F})$ such that

$$A = P^{-1}BP$$

PROPOSITION 12.1

Let $A, B \in M_n(\mathbb{F})$. They are similar if and only if there exist an n-dimensional vector space V, a linear operator $T : V \to V$, and two ordered bases $\mathcal{E}, \mathcal{F}$ such that

$$A = [T]_{\mathcal{E}}, \qquad B = [T]_{\mathcal{F}}.$$

If P converts coordinates from $\mathcal{E}$ to $\mathcal{F}$, so that $[v]_{\mathcal{F}} = P[v]_{\mathcal{E}}$, then

$$A = P^{-1}BP.$$

Thus similarity describes the same operation in two coordinate systems.

EXERCISE 12.4 EXAM QUESTION

Please prove or disprove the following statements.

1. There do not exist fields $\mathbb{F}_1 \subseteq \mathbb{F}_2$ and a vector space V over $\mathbb{F}_2$, also regarded as an $\mathbb{F}_1$-vector space by restricting scalars, such that

$$\dim_{\mathbb{F}_1} V = 2 \dim_{\mathbb{F}_2} V.$$

2. If $A \in M_n(\mathbb{F})$ is invertible, then every $B \in M_{n\times m}(\mathbb{F})$ satisfies

$$N(AB) = N(B).$$

3. Let $n \geq 1$. For every integer k with $0 \leq k \leq n^2$, there exists $A \in M_n(\mathbb{R})$ with $\mathrm{tr}(A) = k$ that is similar only to itself.

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EXERCISE 12.4 EXAM QUESTION CONTINUED

WORKED SOLUTION

1. The claim is false. Take $\mathbb{F}_1 = \mathbb{R}$, $\mathbb{F}_2 = \mathbb{C}$ and $V = \mathbb{C}^2$. Appropriate bases are

$$B_{\mathbb{C}} = ((1,0),(0,1)), \qquad B_{\mathbb{R}} = ((1,0),(i,0),(0,1),(0,i)).$$

$$\dim_{\mathbb{C}} V = 2, \qquad \dim_{\mathbb{R}} V = 4 = 2\dim_{\mathbb{C}} V.$$

2. True. Prove both inclusions. If $x \in N(B)$, then $Bx = 0$, and

$$ABx = A(Bx) = A0 = 0.$$

so $x \in N(AB)$. Hence $N(B) \subseteq N(AB)$. Conversely, if $x \in N(AB)$, then $ABx = 0$. Multiplication on the left by A^{-1} gives

$$Bx = (A^{-1}A)Bx = A^{-1}0 = 0.$$

so $x \in N(B)$, proving $N(AB) \subseteq N(B)$. Together the inclusions give the required equality.

3. True. Define

$$A = \frac{k}{n} I_n.$$

Since $n \geq 1$ and the field is $\mathbb{R}$, the quotient is defined, and

$$\operatorname{tr}(A) = \sum_{j=1}^{n} \frac{k}{n} = n\frac{k}{n} = k.$$

For every invertible $P \in M_n(\mathbb{R})$,

$$P^{-1}AP = P^{-1}\left(\frac{k}{n} I_n\right) P = \frac{k}{n} P^{-1}P = \frac{k}{n} I_n = A.$$

Thus every matrix similar to A equals A. In fact this construction works for every real k, including all the values in the statement.

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12.4 Eigenvalues and Eigenvectors

12.4.1 Motivation and Background

MOTIVATION AND GUIDING IDEA

Eigenvalues and eigenvectors help simplify complex systems. Linear operators (and matrices) generally rotate vectors and mix their coordinates. We seek nonzero vectors on which the action is multiplication by a scalar. Such vectors are **eigenvectors** , and the scalar is an **eigenvalue** . Over $\mathbb{R}$, a positive eigenvalue stretches or contracts along the same direction, a negative one also reverses direction, and a zero eigenvalue sends the vector to zero.

Eigenvalues describe models throughout engineering. The following examples show how one algebraic equation appears in different settings:

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- **Google's ranking algorithm (PageRank):** In a random browsing model, a transition matrix M sends a probability distribution to the next one. A stationary ranking distribution $\vec{r}$ satisfies

$$M\vec{r} = \vec{r}, \qquad r_i \geq 0, \qquad \sum_i r_i = 1.$$

  This is an eigenvalue equation with $\lambda = 1$. The full PageRank model adds a random jump mechanism to address the link structure; the exercise below studies a simplified model.

- **Quantum mechanics:** In a finite-dimensional quantum model, a pure state is a normalized vector and a Hermitian Hamiltonian H represents energy. The eigenvalue equation

$$H\vec{\psi} = E\vec{\psi}$$

  identifies states of definite energy and possible energy values. It is an application of the same linear equation; a general description of mixed states requires the additional concept of a density matrix.

- **Systems of differential equations:** If A is diagonalizable and $A = PDP^{-1}$, substituting $\vec{y} = P\vec{z}$ into $\vec{y}' = A\vec{y}$ gives

$$P\vec{z}' = AP\vec{z} = PD\vec{z} \quad \Longrightarrow \quad \vec{z}' = D\vec{z}.$$

  Each component satisfies $z_i' = \lambda_i z_i$, so $z_i(t) = z_i(0)e^{\lambda_i t}$. For a continuous linear system, all solutions decay to zero if every eigenvalue has negative real part. If an eigenvalue has positive real part, some solutions grow. A zero real part requires further analysis; its sign alone does not imply decay.

- **Classifying critical points in calculus:** For a twice continuously differentiable real function, consider the symmetric Hessian at a critical point. If all its eigenvalues are positive, the point is a strict local minimum; if all are negative, it is a strict local maximum. Positive and negative eigenvalues imply a saddle point. When zero eigenvalues occur without both signs, the second derivative test may be inconclusive.

DEFINITION 12.2

Let V be a vector space over a field $\mathbb{F}$, and let $T : V \to V$ be a linear map (a linear operator). A scalar $\lambda \in \mathbb{F}$ is an **eigenvalue** of T if there is a **nonzero** vector $\vec{v} \in V$ (that is, $\vec{v} \neq \vec{0}_V$) such that

$$T(\vec{v}) = \lambda\vec{v}$$

This is the **eigenvalue equation** . The vector $\vec{v}$ is an **eigenvector** of T associated with λ.

DEFINITION 12.3

Let $A \in \mathbb{F}^{n\times n}$ be a square matrix of size n.
A scalar $\lambda \in \mathbb{F}$ is an **eigenvalue** of A if there is a **nonzero** column vector $\vec{x} \in \mathbb{F}^n$ (that is, $\vec{x} \neq \vec{0}$) such that

$$A\vec{x} = \lambda\vec{x}$$

This is the **eigenvalue equation** . The vector $\vec{x}$ is an **eigenvector** of A associated with λ.

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NOTE

Over $\mathbb{R}$, the line span$\{\vec{x}\}$ is invariant. If $\lambda < 0$, direction is reversed; if $\lambda = 0$, the image is the zero vector. The algebraic concept also applies over $\mathbb{C}$, but a complex eigenvalue cannot always be interpreted as stretching a real line.

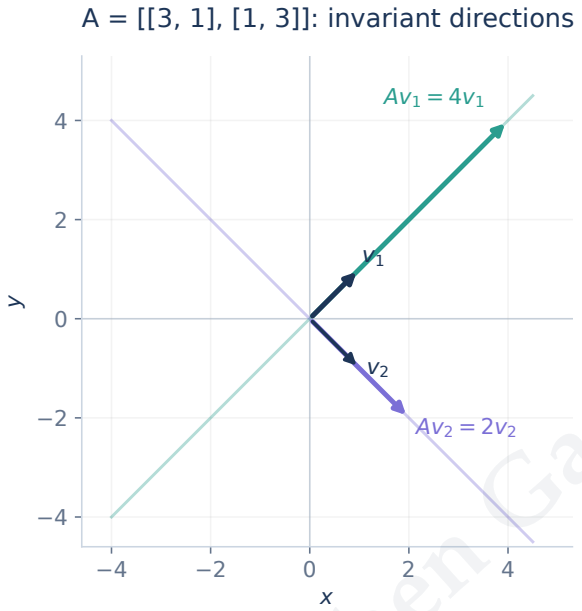


Figure 12.1. Vectors in eigendirections remain on the lines they span. A general vector may change direction because its components are stretched by different factors.

12.5 The Characteristic Polynomial

DEFINITION **12.4**

For $A \in M_n(F)$, the characteristic polynomial is $p_A(t) = \det(tI_n - A)$, where t is a polynomial variable and I_n the identity matrix. There is an eigenvector for $\lambda \in F$ if and only if $\lambda I_n - A$ is singular, equivalently $p_A(\lambda) = 0$. The solution space of $Av = \lambda v$ is the eigenspace $E_\lambda = \ker(A - \lambda I_n)$. It contains zero, whereas an eigenvector is required to be nonzero.

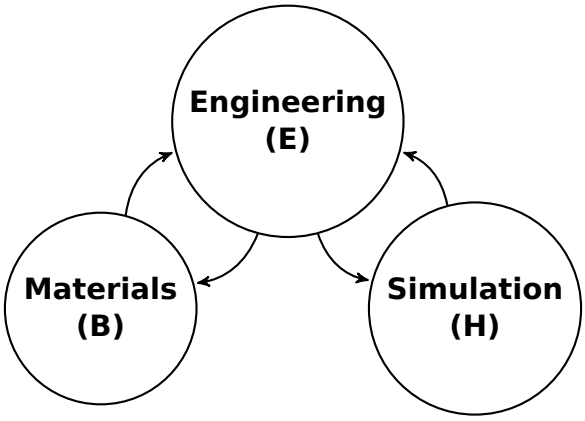
EXERCISE **12.5**

Three sites B, E, H are linked as follows. From either B or H, the next site is always E. From E, the next site is B with probability $1/3$ and H with probability $2/3$.

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EXERCISE 12.5 CONTINUED



Please find a stationary ranking distribution whose entries sum to 1, and please compare the three rankings.

WORKED SOLUTION

In the order (B, E, H), the transition matrix is

$$M = \begin{pmatrix} 0 & 1/3 & 0 \\ 1 & 0 & 1 \\ 0 & 2/3 & 0 \end{pmatrix}.$$

The equation $Mr = r$ gives

$$r_B = \tfrac{1}{3} r_E, \qquad r_H = \tfrac{2}{3} r_E, \qquad r_B + r_H = r_E.$$

Thus $1 = r_B + r_E + r_H = 2r_E$, so

$$r = (1/6, 1/2, 1/3)^T.$$

Site E has three times the rank of B and $3/2$ times the rank of H. A stationary distribution does not guarantee convergence from every start. Here

$$(0,1,0)^T \longmapsto (1/3, 0, 2/3)^T \longmapsto (0,1,0)^T,$$

so this initial distribution produces a cycle of length 2.

EXERCISE 12.6

On the space V of all real sequences, define

$$T(x_1, x_2, x_3, \ldots) = (x_1, 3x_2, 5x_3, \ldots), \qquad (Tx)_n = (2n-1)x_n.$$

Please find all eigenvalues and all eigenvectors of T.

WORKED SOLUTION

In each coordinate, the eigenvalue equation is

$$(2n-1)x_n = \lambda x_n, \qquad (2n - 1 - \lambda)x_n = 0.$$

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EXERCISE 12.6 CONTINUED

If λ is not a positive odd integer, every coefficient is nonzero, so $x = 0$. For $\lambda = 2m - 1$, only coordinate m can be nonzero. Therefore

$$\lambda_m = 2m - 1, \qquad E_{\lambda_m} = \text{span}\{e_m\}, \qquad m = 1, 2, \ldots.$$

All eigenvectors are ce_m with $c \neq 0$. For example, $Te_1 = e_1$ and $Te_2 = 3e_2$.

NOTE

The exercise concerns all sequences, with no summability or norm assumption. The unit sequences $\{e_n\}$ are not an algebraic basis of this space: finite linear combinations give only finitely supported sequences, not, for example, $(1, 1, 1, \ldots)$. An algebraically similar model can use indices $n = 0, 1, 2, \ldots$ and the rule $Ne_n = ne_n$. This resembles a number operator. The exercise alone does not imply that all physical energies are discrete or that every physical operator acts on the space of all sequences.

EXERCISE 12.7

Let $n \geq 2$, let $V = M_n(\mathbb{C})$ be the space of square matrices, and let $p, q \in \mathbb{C}$ with $q \neq 0$. Define the linear operator $T : V \to V$ by

$$T(A) = pA + qA^T$$

Please find the eigenvalues and eigenvectors of T directly from the eigenvalue equation.

WORKED SOLUTION

Let $A \neq 0$ be an eigenvector of T with eigenvalue λ. Then

$$pA + qA^T = \lambda A \tag{12.1}$$

Isolate the transpose term:

$$qA^T = (\lambda - p)A$$

Divide by q (which is nonzero) to express A^T in terms of A:

$$A^T = \frac{\lambda - p}{q} A \tag{12.2}$$

Transpose both sides of (12.2):

$$(A^T)^T = \left(\frac{\lambda - p}{q} A\right)^T$$

Using the properties of transposition,

$$A = \frac{\lambda - p}{q} A^T$$

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EXERCISE 12.7 CONTINUED

Substitute the expression for A^T from (12.2) into the last equation:

$$A = \frac{\lambda - p}{q} \cdot \underbrace{\left(\frac{\lambda - p}{q} A\right)}_{A^T}$$

$$A = \left(\frac{\lambda - p}{q}\right)^2 A$$

Rearrange and factor out A on the right:

$$\left[1 - \left(\frac{\lambda - p}{q}\right)^2\right] A = 0$$

Since $A \neq 0$ (an eigenvector), the scalar in brackets must vanish, so $\left(\frac{\lambda-p}{q}\right)^2 = 1$. There are two cases:

1. **First case:** $\frac{\lambda-p}{q} = 1$. Solving for λ gives the first eigenvalue:

   $$\lambda_1 = p + q$$

   To identify its eigenvectors, substitute $\frac{\lambda-p}{q} = 1$ directly into (12.2):

   $$A^T = 1 \cdot A \implies A^T = A$$

   Thus the eigenvectors for λ_1 are precisely the nonzero symmetric matrices.

2. **Second case:** $\frac{\lambda-p}{q} = -1$. Solving for λ gives the second eigenvalue:

   $$\lambda_2 = p - q$$

   Substitute $\frac{\lambda-p}{q} = -1$ into (12.2):

   $$A^T = -1 \cdot A \implies A^T = -A$$

   Thus the eigenvectors for λ_2 are precisely the nonzero skew-symmetric matrices.

DEFINITION 12.5

Definition: the eigenspace V_λ

Let λ be an eigenvalue. The eigenspace V_λ is the set of all eigenvectors for λ together with the zero vector. The zero vector belongs to this subspace but is not an eigenvector. It follows directly from the eigenvalue equation:

1. **For a linear operator T:** Starting from $T(\vec{v}) = \lambda\vec{v}$, rearrange:

   $$T(\vec{v}) - \lambda\vec{v} = \vec{0} \implies T(\vec{v}) - \lambda I_V(\vec{v}) = \vec{0}$$

   Factor:

   $$(T - \lambda I_V)\vec{v} = \vec{0}$$

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Thus the eigenspace is the kernel of the operator

$$V_\lambda = \ker(T - \lambda I_V)$$

2. **For a matrix A:** Starting from $A\vec{v} = \lambda\vec{v}$, rearrange:

$$\lambda\vec{v} - A\vec{v} = \vec{0} \implies \lambda I_n\vec{v} - A\vec{v} = \vec{0}$$

Factor out $\vec{v}$ on the right:

$$(\lambda I_n - A)\vec{v} = \vec{0}$$

Thus the eigenspace is the nullspace of the matrix:

$$V_\lambda = N(\lambda I_n - A)$$

12.6 An Introduction to Diagonalization

DEFINITION 12.6

A matrix $A \in M_n(\mathbb{F})$ is diagonalizable over $\mathbb{F}$ if there are an invertible matrix $P \in M_n(\mathbb{F})$ and a diagonal matrix $D \in M_n(\mathbb{F})$ such that $A = PDP^{-1}$. If p_j is column j of P and d_j is diagonal entry j of D, then $AP = PD$ means precisely

$$Ap_j = d_j p_j \qquad (1 \le j \le n).$$

Invertibility of P means its columns form a basis. Thus diagonalization amounts to choosing a basis of eigenvectors. In those coordinates, each component is multiplied by its corresponding scalar.

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EXAMPLE 12.1

Over $\mathbb{R}$, consider

$$A = \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix}.$$

We have

$$A\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 10 \\ 5 \end{pmatrix} = 5\begin{pmatrix} 2 \\ 1 \end{pmatrix}, \qquad A\begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

The vectors are independent because their column matrix has determinant -5. Hence

$$P = \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}, \quad D = \begin{pmatrix} 5 & 0 \\ 0 & 0 \end{pmatrix}, \quad P^{-1} = \frac{1}{5}\begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}.$$

The inverse can be checked using $P^2 = 5I_2$. Also,

$$AP = \begin{pmatrix} 10 & 0 \\ 5 & 0 \end{pmatrix} = PD, \qquad PDP^{-1} = \begin{pmatrix} 2 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix} = A.$$

Each vector has a unique decomposition $x = \alpha(2,1)^T + \beta(1,-2)^T$, and

$$Ax = 5\alpha(2,1)^T.$$

Thus the action in the new coordinates is $(\alpha, \beta)^T \mapsto (5\alpha, 0)^T$.

EXAMPLE 12.2

Over $\mathbb{R}$, the matrix

$$J = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

has no basis of eigenvectors. Its eigenvalue equation is

$$(J - \lambda I)\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} (2-\lambda)x + y \\ (2-\lambda)y \end{pmatrix} = 0.$$

If $\lambda \neq 2$, the second equation gives $y = 0$, and the first then gives $x = 0$. Thus the only eigenvalue is 2. For that value, $y = 0$, so all eigenvectors lie on $\text{span}\{(1,0)^T\}$. Two independent vectors cannot be chosen from that line. Hence J is not diagonalizable.

EXERCISE 12.8 CHALLENGE

$$A = \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix}, \quad B = ((1,1)^T, (1,-1)^T), \quad x_{k+1} = Ax_k, \quad x_0 = (2,1)^T.$$

Please compute $D = [T]_B$ for $T(x) = Ax$ and $P = [I]_E^B$. Please prove $A^k = PD^kP^{-1}$ for every $k \geq 0$ and please solve the recurrence.

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EXERCISE 12.8 CHALLENGE CONTINUED

WORKED SOLUTION

The columns of P are the new basis vectors in standard coordinates. Cancelling adjacent $P^{-1}P$ factors proves the power identity; for $k = 0$ both sides are the identity. Induction on the recurrence gives $x_k = A^k x_0$. In the basis B, the two components are multiplied separately by 6 and 2.

$$A(1,1)^T = 6(1,1)^T, \quad A(1,-1)^T = 2(1,-1)^T,$$

$$D = \begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix}, \quad P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad P^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad A = PDP^{-1},$$

$$A^k = PD^kP^{-1} = \frac{1}{2}\begin{pmatrix} 6^k + 2^k & 6^k - 2^k \\ 6^k - 2^k & 6^k + 2^k \end{pmatrix},$$

$$x_k = A^k x_0 = \frac{1}{2}\begin{pmatrix} 3\cdot 6^k + 2^k \\ 3\cdot 6^k - 2^k \end{pmatrix}, \quad x_0 = \frac{1}{2}(4,2)^T = (2,1)^T.$$

EXERCISE 12.9

1. Let $n \geq 2$ and let $w \in \mathbb{C}$ be a primitive nth root of unity: $w^n = 1$ and $w^k \neq 1$ for every $1 \leq k \leq n-1$. Define the matrix

$$F = \left(w^{(i-1)(j-1)}\right)_{i,j=1}^{n} \in M_n(\mathbb{C}).$$

   Please compute $\det(F)$ explicitly.

2. Let $n \geq 3$, let $1 \leq p \leq n-1$, and please write $q = n - p$. For $\alpha, \beta \in \mathbb{R}$, define $A(\alpha,\beta) \in M_n(\mathbb{R})$ by

$$a_{ii} = 1 \quad (1 \leq i \leq n),$$

   and, for every $i \neq j$,

$$a_{ij} = \begin{cases} \alpha, & 1 \leq i,j \leq p \text{ or } p+1 \leq i,j \leq n, \\ \beta, & \text{otherwise.} \end{cases}$$

   Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be the linear map $T(v) = A(\alpha,\beta)v$. Please determine for which α, β the map T is an isomorphism, and please compute $\det(A(\alpha,\beta))$.

WORKED SOLUTION

First display the structure of the matrices:

$$F = \left(w^{(i-1)(j-1)}\right)_{i,j=1}^{n} = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & w & w^2 & \cdots & w^{n-1} \\ 1 & w^2 & w^4 & \cdots & w^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & w^{n-1} & w^{2(n-1)} & \cdots & w^{(n-1)(n-1)} \end{pmatrix}.$$

$$A(\alpha,\beta) = \left(\begin{array}{c|c} A_{11} & A_{12} \\ \hline A_{21} & A_{22} \end{array}\right),.$$

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EXERCISE 12.9 CONTINUED

The block sizes are

$$A_{11} \in M_p(\mathbb{R}), \qquad A_{22} \in M_q(\mathbb{R}),$$
$$A_{12} \in M_{p\times q}(\mathbb{R}), \qquad A_{21} \in M_{q\times p}(\mathbb{R}).$$

$$A_{11} = \begin{pmatrix} 1 & \alpha & \alpha & \cdots & \alpha \\ \alpha & 1 & \alpha & \cdots & \alpha \\ \alpha & \alpha & 1 & \cdots & \alpha \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \alpha & \alpha & \alpha & \cdots & 1 \end{pmatrix}_{p\times p}, \qquad A_{22} = \begin{pmatrix} 1 & \alpha & \alpha & \cdots & \alpha \\ \alpha & 1 & \alpha & \cdots & \alpha \\ \alpha & \alpha & 1 & \cdots & \alpha \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \alpha & \alpha & \alpha & \cdots & 1 \end{pmatrix}_{q\times q}.$$

$$A_{12} = \begin{pmatrix} \beta & \beta & \cdots & \beta \\ \beta & \beta & \cdots & \beta \\ \vdots & \vdots & \ddots & \vdots \\ \beta & \beta & \cdots & \beta \end{pmatrix}_{p\times q}, \qquad A_{21} = \begin{pmatrix} \beta & \beta & \cdots & \beta \\ \beta & \beta & \cdots & \beta \\ \vdots & \vdots & \ddots & \vdots \\ \beta & \beta & \cdots & \beta \end{pmatrix}_{q\times p}.$$

$$P^{-1}A(\alpha,\beta)P = (1-\alpha)I_{n-2} \oplus B = \left(\begin{array}{c|c} (1-\alpha)I_{n-2} & 0 \\ \hline 0 & B \end{array}\right),$$

where the lower-right block has size 2:

$$B = \begin{pmatrix} 1+(p-1)\alpha & q\beta \\ p\beta & 1+(q-1)\alpha \end{pmatrix}.$$

1. **Fact used below (Vandermonde determinant).** For $x_1, \ldots, x_n \in \mathbb{C}$, define

$$V(x_1, \ldots, x_n) = \left(x_i^{j-1}\right)_{i,j=1}^{n}.$$

Then

$$\det\big(V(x_1, \ldots, x_n)\big) = \prod_{1\le i<j\le n} (x_j - x_i).$$

Set

$$x_i = w^{i-1} \qquad (i = 1, \ldots, n).$$

For every i, j,

$$F_{ij} = w^{(i-1)(j-1)} = (w^{i-1})^{j-1} = x_i^{j-1}.$$

Therefore

$$F = V(x_1, \ldots, x_n).$$

By the Vandermonde determinant formula,

$$\det(F) = \prod_{1\le i<j\le n} (x_j - x_i) = \prod_{0\le i<j\le n-1} \left(w^j - w^i\right).$$

For every $0 \le i < j \le n-1$,

$$w^j - w^i = w^i\left(w^{j-i} - 1\right),$$

so

$$\det(F) = \left(\prod_{0\le i<j\le n-1} w^i\right)\left(\prod_{0\le i<j\le n-1} (w^{j-i} - 1)\right).$$

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EXERCISE 12.9 CONTINUED

Write

$$S = \sum_{0 \le i < j \le n-1} i.$$

Then

$$\prod_{0 \le i < j \le n-1} w^i = w^S.$$

We now compute S: For each $i \in \{0, \dots, n-2\}$, there are exactly $(n-1-i)$ values of j with $i < j \le n-1$, so

$$S = \sum_{i=0}^{n-2} i(n-1-i) = \frac{n(n-1)(n-2)}{6}.$$

Consequently,

$$\prod_{0 \le i < j \le n-1} w^i = w^{\frac{n(n-1)(n-2)}{6}}.$$

Also group the factors by $m = j - i \in \{1, \dots, n-1\}$. For each m, there are exactly $(n-m)$ pairs (i, j) with $j - i = m$, so

$$\prod_{0 \le i < j \le n-1} (w^{j-i} - 1) = \prod_{m=1}^{n-1} (w^m - 1)^{n-m}.$$

Therefore

$$\boxed{\det(F) = w^{\frac{n(n-1)(n-2)}{6}} \prod_{m=1}^{n-1} (w^m - 1)^{n-m}}.$$

Because w is a primitive nth root of unity, $w^m \neq 1$ for $1 \le m \le n-1$, so $w^m - 1 \neq 0$. Every factor in the product is therefore nonzero, and $\det(F) \neq 0$.

2. **Theorems used below (from class).**

(a) Isomorphisms and representing matrices. Let V be a finite-dimensional real vector space, let $T : V \to V$ be linear, and let $[T]_{\mathcal{B}}$ represent T in a basis $\mathcal{B}$. Then

$$T \text{ is an isomorphism} \iff [T]_{\mathcal{B}} \text{ is invertible.}$$

(b) Invertibility and the determinant. For every $A \in M_n(\mathbb{R})$,

$$A \text{ is invertible} \iff \det(A) \neq 0.$$

(This is the standard determinant criterion for invertibility.)

(c) Invariance of the determinant under change of basis. If P is invertible, then

$$\det(P^{-1}AP) = \det(A),$$

because

$$\det(P^{-1}AP) = \det(P^{-1}) \det(A) \det(P) = \det(A).$$

We can now solve the problem.

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EXERCISE 12.9 CONTINUED

Write $A = A(\alpha, \beta)$. Define

$$U = \Big\{ x \in \mathbb{R}^n : x_{p+1} = \cdots = x_n = 0, \quad \sum_{i=1}^{p} x_i = 0 \Big\},$$

$$W = \Big\{ x \in \mathbb{R}^n : x_1 = \cdots = x_p = 0, \quad \sum_{i=p+1}^{n} x_i = 0 \Big\}.$$

Clearly U, W are subspaces, since they are defined by homogeneous linear equations in the coordinates.

We first prove that $Ax = (1-\alpha)x$ for every $x \in U$. Let $x \in U$. Then $x_{p+1} = \cdots = x_n = 0$ and $\sum_{j=1}^{p} x_j = 0$.

If $1 \le i \le p$, the structure of row i of A (diagonal entry 1, entries α elsewhere in its block, and entries β between blocks) gives

$$(Ax)_i = 1 \cdot x_i + \sum_{\substack{1 \le j \le p \\ j \ne i}} \alpha x_j + \sum_{j=p+1}^{n} \beta x_j.$$

The last sum vanishes because $x_{p+1} = \cdots = x_n = 0$, so

$$(Ax)_i = x_i + \alpha \sum_{\substack{1 \le j \le p \\ j \ne i}} x_j.$$

Now

$$\sum_{\substack{1 \le j \le p \\ j \ne i}} x_j = \sum_{j=1}^{p} x_j - x_i = 0 - x_i = -x_i,$$

and hence

$$(Ax)_i = x_i + \alpha(-x_i) = (1-\alpha)x_i.$$

If $p+1 \le i \le n$, the coefficients of coordinates $1, \ldots, p$ in this row are all β (the second block of x is zero). Thus

$$(Ax)_i = \sum_{j=1}^{p} \beta x_j = \beta \sum_{j=1}^{p} x_j = 0 = (1-\alpha)x_i.$$

Therefore $Ax = (1-\alpha)x \in U$. In particular, U is A-invariant, and the restriction $A|_U$ is multiplication by $(1-\alpha)$. The same argument gives $Ay = (1-\alpha)y$ for every $y \in W$. Thus W is also A-invariant and $A|_W = (1-\alpha)\mathrm{Id}_W$.

Define

$$u = (\underbrace{1, \ldots, 1}_{p}, \underbrace{0, \ldots, 0}_{q}), \qquad v = (\underbrace{0, \ldots, 0}_{p}, \underbrace{1, \ldots, 1}_{q}), \qquad S = \mathrm{span}\{u, v\}.$$

We next compute Au and Av.

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EXERCISE 12.9 CONTINUED

For every $1 \le i \le p$,

$$(Au)_i = 1 \cdot 1 + \sum_{\substack{1 \le j \le p \\ j \ne i}} \alpha \cdot 1 + \sum_{j=p+1}^{n} \beta \cdot 0 = 1 + (p-1)\alpha.$$

and, for every $p+1 \le i \le n$,

$$(Au)_i = \sum_{j=1}^{p} \beta \cdot 1 + \Big(1 \cdot 0 + \sum_{\substack{p+1 \le j \le n \\ j \ne i}} \alpha \cdot 0\Big) = p\beta.$$

Therefore

$$Au = \big(1 + (p-1)\alpha\big)u + p\beta\, v.$$

Similarly,

$$Av = q\beta\, u + \big(1 + (q-1)\alpha\big)v.$$

Thus S is A-invariant, and its representing matrix in the basis (u, v) is

$$B = \begin{pmatrix} 1 + (p-1)\alpha & q\beta \\ p\beta & 1 + (q-1)\alpha \end{pmatrix}.$$

We claim that

$$\mathbb{R}^n = (U \oplus W) \oplus S.$$

First, $U \cap W = \{0\}$, since a vector in both subspaces must have every coordinate zero. Also, if $au + bv \in U \oplus W$, the sum of its first-block coordinates is $ap = 0$, so $a = 0$; the sum of its second-block coordinates is $bq = 0$, so $b = 0$. Hence $S \cap (U \oplus W) = \{0\}$.

Next, every $x \in \mathbb{R}^n$ admits a decomposition $x = x_{UW} + x_S$ with $x_{UW} \in U \oplus W$ and $x_S \in S$. Write

$$s_1 = \sum_{i=1}^{p} x_i, \qquad s_2 = \sum_{i=p+1}^{n} x_i,$$

and define

$$x_S = \frac{s_1}{p}u + \frac{s_2}{q}v \in S, \qquad x_{UW} = x - x_S.$$

The sum of the first-block coordinates of x_{UW} is $s_1 - s_1 = 0$, and the second-block sum is $s_2 - s_2 = 0$, so $x_{UW} \in U \oplus W$. This proves the direct-sum decomposition.

Choose a basis $\mathcal{B}$ by concatenating bases of U and W and the basis (u, v) of S. In this basis, the matrix of A has the block diagonal form

$$(1-\alpha)I_{n-2} \oplus B.$$

Let P be the change-of-coordinates matrix from $\mathcal{B}$ to the standard basis. Then

$$P^{-1}AP = (1-\alpha)I_{n-2} \oplus B.$$

Combining facts (b) and (c),

$$\det(A) = \det(P^{-1}AP) = \det\big((1-\alpha)I_{n-2} \oplus B\big).$$

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EXERCISE 12.9 CONTINUED

The determinant of a block diagonal matrix is the product of the determinants of its blocks (this follows by cofactor expansion, or by a suitable factorization and multiplicativity $\det(AB) = \det(A)\det(B)$). Therefore

$$\det(A) = \det\big((1-\alpha)I_{n-2}\big) \cdot \det(B) = (1-\alpha)^{n-2}\det(B).$$

Compute $\det(B)$:

$$\det(B) = \big(1+(p-1)\alpha\big)\big(1+(q-1)\alpha\big) - pq\,\beta^2.$$

and hence

$$\boxed{\det(A) = (1-\alpha)^{n-2}\Big(\big(1+(p-1)\alpha\big)\big(1+(q-1)\alpha\big) - pq\,\beta^2\Big)}.$$

By fact (a), T is an isomorphism if and only if its representing matrix A in the standard basis is invertible. By fact (b), this is equivalent to $\det(A) \neq 0$. Thus

$$T \text{ is an isomorphism} \iff \det(A) \neq 0 \iff (1-\alpha)^{n-2} \neq 0 \text{ and } \det(B) \neq 0.$$

Since $\alpha, \beta \in \mathbb{R}$ and $n \geq 3$, $(1-\alpha)^{n-2} \neq 0$ holds if and only if $\alpha \neq 1$. The exact condition is therefore

$$\boxed{\alpha \neq 1} \quad \text{and} \quad \boxed{\big(1+(p-1)\alpha\big)\big(1+(q-1)\alpha\big) \neq pq\,\beta^2}.$$

EXERCISE 12.10

For $s \in \mathbb{R}$, define $T_s : \mathbb{R}^2 \to \mathbb{R}^2$ by

$$T_s(v) = A_s v, \qquad A_s = \begin{pmatrix} 3 & 1 \\ s & 2 \end{pmatrix}.$$

The triangle Δ has vertices $(0,0)$, $(1,0)$, $(0,1)$. Please find the vertices and area of its image, please determine when T_s is invertible, and please describe the image when $s = 6$.

WORKED SOLUTION

The images of the vertices are $(0,0)$, $(3,s)$, $(1,2)$. The original triangle has area $1/2$, and the area scale factor is $|\det A_s| = |3\cdot 2 - 1\cdot s| = |6-s|$. Hence the image has area $|6-s|/2$. Also, T_s is invertible exactly when $6 - s \neq 0$, namely when $s \neq 6$. When $s = 6$, the columns satisfy $(3,6)^T = 3(1,2)^T$. Every image is therefore a multiple of $(1,2)^T$, and the triangle collapses to a line segment on $y = 2x$, with zero area. For $s < 6$, orientation is preserved; for $s > 6$, it is reversed. For example, at $s = 1$ the image has area $5/2$.

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EXERCISE 12.11 EXAM QUESTION

1. Let $A \in M_n(\mathbb{R})$ satisfy $\text{rank}(A^2) < \text{rank}(A)$. Please prove that there exists $v \in \mathbb{R}^n$ such that $A^2v = 0$ and $Av \neq 0$.

2. Let $A, B \in M_n(\mathbb{R})$. Please prove that

$$\text{rank}(A) + \text{rank}(B) \leq n + \text{rank}(AB).$$

Please deduce that if $\text{rank}(A) + \text{rank}(B) > n$, then $AB \neq 0$.

3. Let $N \in M_n(\mathbb{R})$ satisfy $N^2 = 0$ and $\text{rank}(N) = r$. Please find $\det(I_n + tN)$ and $\text{rank}(I_n + tN)$ for every $t \in \mathbb{R}$.

WORKED SOLUTION

1. If $Av = 0$, then $A^2v = A(Av) = 0$. Hence $\ker A \subseteq \ker A^2$. By the rank-nullity theorem and the given inequality,

$$\begin{aligned}\dim \ker A^2 &= n - \text{rank}(A^2) \\ &> n - \text{rank}(A) = \dim \ker A.\end{aligned}$$

Thus the inclusion is strict: $\ker A \subsetneq \ker A^2$. Choose $v \in \ker A^2 \setminus \ker A$. Membership in the first kernel gives $A^2v = 0$, while exclusion from the second gives $Av \neq 0$. This vector remains nonzero after one application of A and becomes zero after two applications.

2. Consider the restricted map

$$T : \text{Im}\, B \longrightarrow \mathbb{R}^n, \qquad T(u) = Au.$$

Its image is exactly $\text{Im}(AB)$: every $u \in \text{Im}\, B$ has the form $u = Bv$, so $T(u) = ABv$; conversely, every vector ABv arises in this way. Its kernel is $\text{Im}\, B \cap \ker A$. Applying rank-nullity to T gives

$$\text{rank}(B) = \dim(\text{Im}\, B \cap \ker A) + \text{rank}(AB).$$

However,

$$\dim(\text{Im}\, B \cap \ker A) \leq \dim \ker A = n - \text{rank}(A).$$

Substitute and rearrange:

$$\begin{aligned}\text{rank}(B) &\leq n - \text{rank}(A) + \text{rank}(AB), \\ \text{rank}(A) + \text{rank}(B) &\leq n + \text{rank}(AB).\end{aligned}$$

If the ranks sum to more than n, then $\text{rank}(AB) > 0$, so AB is not the zero matrix.

3. First compute the two products:

$$\begin{aligned}(I_n + tN)(I_n - tN) &= I_n - tN + tN - t^2N^2 = I_n, \\ (I_n - tN)(I_n + tN) &= I_n + tN - tN - t^2N^2 = I_n.\end{aligned}$$

Thus $I_n + tN$ is invertible for every $t \in \mathbb{R}$, with inverse $I_n - tN$. Immediately,

$$\text{rank}(I_n + tN) = n.$$

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EXERCISE 12.11 EXAM QUESTION CONTINUED

To compute the determinant, note that $N^2 = 0$ implies $\operatorname{Im} N \subseteq \ker N$. Hence $r \leq n - r$, or $2r \leq n$. Choose a basis $(u_1, \ldots, u_r)$ of $\operatorname{Im} N$ and extend it to a basis $(u_1, \ldots, u_r, w_1, \ldots, w_{n-2r})$ of $\ker N$. For each j, choose v_j with $Nv_j = u_j$.

The list

$$\mathcal{B} = (u_1, \ldots, u_r, w_1, \ldots, w_{n-2r}, v_1, \ldots, v_r)$$

is a basis. Indeed, applying N to a vanishing linear combination gives a combination of the u_j, so every coefficient of a v_j is zero. The remaining coefficients vanish because they multiply a basis of $\ker N$. Thus the list contains n linearly independent vectors. In this basis,

$$[N]_{\mathcal{B}} = \begin{pmatrix} 0 & 0 & I_r \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad [I_n + tN]_{\mathcal{B}} = \begin{pmatrix} I_r & 0 & tI_r \\ 0 & I_{n-2r} & 0 \\ 0 & 0 & I_r \end{pmatrix}.$$

A block of size zero is simply omitted. This is upper triangular with every diagonal entry equal to 1. Since the determinant is preserved under change of basis,

$$\det(I_n + tN) = 1 \qquad \text{for every } t \in \mathbb{R}.$$

Neither answer depends on r: the determinant is 1 and the rank is n.

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TUTORIAL 13

Diagonalization, Cayley-Hamilton, Jordan Form, and Inner Products

MOTIVATION AND GUIDING IDEA

Diagonalization replaces coordinates whose components interact by coordinates in which each component is multiplied by its own scalar. This helps compute powers, understand update processes, and identify preserved or decaying components. Finding eigenvalues is only the first step: enough linearly independent eigenvectors must exist over the field being used.

NOTATION AND CONVENTIONS

- $A \in M_n(\mathbb{F})$ is square; $\lambda \in \mathbb{F}$ is a scalar, and $v \in \mathbb{F}^n \setminus \{0\}$ is an eigenvector when $Av = \lambda v$.
- $E_\lambda(A) = \ker(A - \lambda I_n) = \ker(\lambda I_n - A)$ is the eigenspace. The notations V_λ and $N(\lambda I_n - A)$ used in the notes describe the same space.
- $p_A(t) = \det(tI_n - A)$ is the characteristic polynomial; t is neither a vector nor a matrix parameter. The notations $P_A(x)$ and $\chi_A(t)$ express the same convention with a different variable name.
- $r_a(\lambda)$ is the algebraic multiplicity, and $r_g(\lambda) = \dim E_\lambda(A)$ is the geometric multiplicity.
- $P = (v_1 \ \cdots \ v_n)$ diagonalizes A when its columns form an eigenbasis. If $Av_i = \lambda_i v_i$, then $AP = PD$, with $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$.
- For $f(t) = \sum_{j=0}^{m} c_j t^j$, define $f(A) = \sum_{j=0}^{m} c_j A^j$, where $A^0 = I_n$.

DEFINITION 13.1

A matrix $A \in M_n(\mathbb{F})$ is diagonalizable over $\mathbb{F}$ if there are an invertible $P \in M_n(\mathbb{F})$ and a diagonal $D \in M_n(\mathbb{F})$ with $P^{-1}AP = D$. Equivalently, $\mathbb{F}^n$ has a basis of eigenvectors. The precise criterion is that p_A split into linear factors over $\mathbb{F}$ and that $r_g(\lambda) = r_a(\lambda)$ for every eigenvalue. Splitting is automatic over $\mathbb{C}$; over $\mathbb{R}$, it must be checked.

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NOTE

If $A = PDP^{-1}$, then $f(A) = Pf(D)P^{-1}$ for every polynomial f. In particular, $A^k = PD^kP^{-1}$ for every integer $k \geq 0$; negative powers require invertibility. If $Av = \lambda v$ and $v \neq 0$, then $f(A)v = f(\lambda)v$, but distinct eigenvalues of A can give the same value of f. Diagonalizability of a power or polynomial of A therefore does not imply diagonalizability of A.

NOTE

Sum eigenspace dimensions over distinct eigenvalues. Eigenspaces for distinct eigenvalues form a direct sum. Their dimensions sum to n exactly when there are enough eigenvectors for a basis; a smaller sum means that an eigenbasis is missing.

MOTIVATION AND GUIDING IDEA

By the end of this tutorial, you will be able to identify directions preserved by a linear map, compute eigenspaces and multiplicities, test diagonalizability, and use an eigenbasis to calculate powers and understand an iterative process. We distinguish statements that hold generally from those requiring a particular field or additional hypotheses.

13.1 Eigenspaces, the Characteristic Polynomial, and Multiplicities

DEFINITION 13.2

Definition: characteristic polynomial and multiplicities

1. **Characteristic polynomial $P_A(x)$:** It is defined by
$$P_A(x) = \det(xI_n - A)$$
2. **Algebraic multiplicity $r_a(\lambda)$:** The multiplicity (exponent) of λ as a root of the characteristic polynomial $P_A(x)$.
3. **Geometric multiplicity $r_g(\lambda)$:** The dimension of the eigenspace for λ:
$$r_g(\lambda) = \dim(V_\lambda) = \dim(N(\lambda I_n - A))$$

EXERCISE 13.1

Let
$$A = \begin{pmatrix} -2 & 1 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$
Please find its characteristic polynomial, eigenspaces, and multiplicities. Please determine whether A is diagonalizable.

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EXERCISE 13.1 CONTINUED

WORKED SOLUTION

Triangularity gives

$$p_A(\lambda) = (\lambda + 2)^2(\lambda - 3)^2.$$

Each eigenvalue $-2, 3$ has algebraic multiplicity 2. The equation $(A + 2I)v = 0$ gives $v_2 = 0$ and $5v_3 = 5v_4 = 0$, so

$$E_{-2} = \text{span}\{e_1\}, \qquad \dim E_{-2} = 1.$$

The equation $(A - 3I)v = 0$ gives $-5v_1 + v_2 = 0$ and $-5v_2 = 0$, so

$$E_3 = \text{span}\{e_3, e_4\}, \qquad \dim E_3 = 2.$$

The eigenspace dimensions sum to $3 < 4$, so A is not diagonalizable.

DEFINITION 13.3

For $\lambda \in \mathbb{F}$ and $A \in M_n(\mathbb{F})$, λ is an eigenvalue exactly when $\ker(A - \lambda I_n) \neq \{0\}$, equivalently $p_A(\lambda) = 0$. The eigenspace contains zero, but zero itself is not an eigenvector. For an eigenvalue, the algebraic multiplicity is the integer $r \geq 1$ such that $p_A(t) = (t - \lambda)^r q(t)$ and $q(\lambda) \neq 0$; the geometric multiplicity satisfies $1 \leq r_g(\lambda) \leq r_a(\lambda)$.

13.2 Diagonalization and Eigenbases

MOTIVATION AND GUIDING IDEA

Main objective: find a basis of eigenvectors

DEFINITION 13.4

Definition: diagonalizable matrix and diagonalizing basis

Let $A \in M_n(\mathbb{F})$.

1. A is **diagonalizable over** $\mathbb{F}$ if it is similar over $\mathbb{F}$ to a diagonal matrix. That is, there are an invertible matrix P and a diagonal matrix D, both over $\mathbb{F}$, such that

   $$D = P^{-1}AP \iff A = PDP^{-1}$$

   In this case, the diagonal entries of D are the eigenvalues of A:

   $$D = \text{diag}(\lambda_1, \ldots, \lambda_n)$$

2. The columns of the diagonalizing matrix P form a basis of **eigenvectors** of A, in the order of the diagonal entries of D. Since P is invertible, its columns form a basis of $\mathbb{F}^n$. This is a **diagonalizing basis** , or **eigenbasis** .

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PROPOSITION 13.1

Diagonalization theorem (criterion for diagonalizability)
A matrix $A \in M_n(\mathbb{F})$ is diagonalizable over $\mathbb{F}$ **if and only if** both conditions hold:

1. **Complete splitting:** The characteristic polynomial $P_A(x)$ splits into linear factors over $\mathbb{F}$, meaning that all its roots (counted with multiplicity) lie in $\mathbb{F}$.
2. **Equal multiplicities:** For each eigenvalue λ, the geometric multiplicity equals the algebraic multiplicity:
$$r_g(\lambda) = r_a(\lambda)$$

Useful consequence: If a matrix A of size n has n **distinct** eigenvalues in $\mathbb{F}$, then $r_a = 1$ for each. Consequently $r_g = 1$ (since $1 \le r_g \le r_a$), and A is diagonalizable over $\mathbb{F}$.

NOTE

Why does $1 \le r_g(\lambda) \le r_a(\lambda)$ always hold? Write $r = \dim V_\lambda$. Since λ is an eigenvalue, a nonzero eigenvector exists, so $r \ge 1$. Extend a basis of V_λ to a basis of the full space. In this basis, the matrix has the form
$$\begin{pmatrix} \lambda I_r & C \\ 0 & D \end{pmatrix}.$$
The lower-left block is zero because each of the first r basis vectors maps to a multiple of itself. Thus the characteristic polynomial is $(x-\lambda)^r \det(xI - D)$, and $r_g(\lambda) = r \le r_a(\lambda)$.

DEFINITION 13.5

Definition: a diagonalizable linear operator
Let $T : V \to V$ be a linear operator on a finite-dimensional vector space V of dimension n over $\mathbb{F}$. The operator T is **diagonalizable over** $\mathbb{F}$ if there is a basis $\mathcal{B}$ of V in which $[T]_{\mathcal{B}}$ is diagonal.
Such a basis necessarily consists of eigenvectors of T and is called a **diagonalizing basis** .

PROPOSITION 13.2

Let $v_1, \dots, v_k$ be eigenvectors of a matrix A (or an operator T), associated with **distinct** eigenvalues $\lambda_1, \dots, \lambda_k$.
Then $\{v_1, \dots, v_k\}$ is **linearly independent** .

MOTIVATION AND GUIDING IDEA

An algorithm for diagonalizing a linear operator
To diagonalize T (or find a diagonalizing basis), proceed as follows:

1. Choose a convenient basis $\mathcal{C}$ of V (often the standard basis).
2. Form its representing matrix $A = [T]_{\mathcal{C}}$.
3. Solve the matrix problem:
   - Compute the characteristic polynomial $P_A(x)$.
   - Find the eigenvalues λ in $\mathbb{F}$.

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- For each λ, find a basis of the eigenspace (the kernel) V_λ.

4. Test diagonalizability: add the geometric multiplicities ($\sum r_g(\lambda)$). If the sum is less than n, the operator is not diagonalizable over $\mathbb{F}$. If it equals n, continue.

5. Recover the diagonalizing basis: The vectors found in step 3 are coordinate vectors (in $\mathbb{F}^n$). Convert them back to vectors of V using the initial basis $\mathcal{C}$. The combined lists form the diagonalizing basis $\mathcal{B}$.

6. The representing matrix in the new basis is

$$[T]_{\mathcal{B}} = D = \text{diag}(\lambda_1, \ldots, \lambda_n)$$

13.2.1 Exercises on Diagonalization

EXAMPLE 13.1

A polynomial that splits into linear factors over $\mathbb{R}$: $p(x) = (x+2)(x-1)(x-4)$.

EXAMPLE 13.2

A polynomial that **does not** split into linear factors over $\mathbb{R}$: $q(x) = x^2 + 4$.
Observe that $q(x)$ **does** split into linear factors over $\mathbb{C}$: $q(x) = (x-2i)(x+2i)$

EXERCISE 13.2

For each matrix, please determine whether it is diagonalizable over $\mathbb{R}$ and over $\mathbb{C}$. Whenever it is diagonalizable, please give suitable P, D over the relevant field:

$$A_1 = \begin{pmatrix} 5 & 3 \\ 0 & 5 \end{pmatrix}, \quad A_2 = \begin{pmatrix} -2 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 0 & -4 \\ 1 & 0 \end{pmatrix}.$$

WORKED SOLUTION

1. For A_1, the characteristic polynomial is $(\lambda - 5)^2$. The equation $(A_1 - 5I)v = 0$ forces $3v_2 = 0$, so the eigenspace is $\text{span}\{(1,0)^T\}$ and has dimension 1. Its algebraic multiplicity 2 exceeds its geometric multiplicity, so it is not diagonalizable over either field.

2. A_2 has distinct eigenvalues $-2, 1, 4$. For -2, choose e_1. For 1, the equation $-3v_1 + 3v_2 = 0$ gives $(1,1,0)^T$. For 4, choose e_3. Hence

$$P = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad D = \text{diag}(-2, 1, 4).$$

   The columns are linearly independent, and their images give $A_2P = PD$. This is a diagonalization over $\mathbb{R}$, and hence also over $\mathbb{C}$.

3. The characteristic polynomial of A_3 is $\lambda^2 + 4$. It has no real root, so there is no real diagonalization. Over $\mathbb{C}$, the eigenvalues are $2i, -2i$, with respective eigenvectors

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EXERCISE 13.2 CONTINUED

$(2i, 1)^T, (-2i, 1)^T$:

$$A_3(\lambda, 1)^T = (-4, \lambda)^T = \lambda(\lambda, 1)^T \quad \text{when } \lambda^2 = -4.$$

Therefore

$$P_{\mathbb{C}} = \begin{pmatrix} 2i & -2i \\ 1 & 1 \end{pmatrix}, \quad D_{\mathbb{C}} = \operatorname{diag}(2i, -2i).$$

Since $\det P_{\mathbb{C}} = 4i \neq 0$, this is a valid complex diagonalization.

EXERCISE 13.3

As in the earlier transpose exercise, assume $n \geq 2$ and consider $T : M_n(\mathbb{R}) \to M_n(\mathbb{R})$ defined by

$$T(A) = pA + qA^T$$

where $p, q \in \mathbb{R}$ and $q \neq 0$.
Please find the characteristic polynomial $P_T(x)$ and the algebraic and geometric multiplicities (r_a, r_g) of each eigenvalue found earlier.

WORKED SOLUTION

Previously we decomposed the full space $V = M_n(\mathbb{R})$ into a direct sum of eigenspaces:

$$V = V_{p+q} \oplus V_{p-q} = S \oplus K$$

Here S is the space of symmetric matrices and K is the space of skew-symmetric matrices.

1. **For the eigenvalue $\lambda_1 = p + q$:** The eigenspace consists of the symmetric matrices:

   $$V_{\lambda_1} = S = \{A \in M_n(\mathbb{R}) \mid A^T = A\}$$

   Its geometric multiplicity is its dimension (the number of entries on or above the diagonal):

   $$r_g(p + q) = \dim(S) = \frac{n(n + 1)}{2}$$

2. **For the eigenvalue $\lambda_2 = p - q$:** The eigenspace consists of the skew-symmetric matrices:

   $$V_{\lambda_2} = K = \{A \in M_n(\mathbb{R}) \mid A^T = -A\}$$

   Its geometric multiplicity is its dimension (the number of entries strictly above the diagonal, since diagonal entries must be zero):

   $$r_g(p - q) = \dim(K) = \frac{n(n - 1)}{2}$$

3. **The characteristic polynomial $P_T(x)$:** The geometric multiplicities sum to the dimension of the full space (n^2), so the operator is diagonalizable. Thus each algebraic multiplicity equals its geometric multiplicity ($r_a = r_g$).

   The characteristic polynomial is the product of the linear factors, each raised to its al-

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EXERCISE 13.3 CONTINUED

gebraic multiplicity:

$$P_T(x) = (x - \lambda_1)^{r_a(\lambda_1)} \cdot (x - \lambda_2)^{r_a(\lambda_2)}$$

Substitute the values found above:

$$P_T(x) = \Big(x - (p+q)\Big)^{\frac{n(n+1)}{2}} \cdot \Big(x - (p-q)\Big)^{\frac{n(n-1)}{2}}$$

NOTE

The dimension of a solution subspace can be found by counting independent homogeneous linear constraints: If L is the linear constraint map on V, then

$$\dim \ker L = \dim V - \operatorname{rank} L$$

Here the ambient space is $M_n(\mathbb{R})$, of dimension n^2.

1. **For symmetric matrices (S):** The condition is $A_{ij} = A_{ji}$. This is a constraint only for $i < j$ (on the diagonal $A_{ii} = A_{ii}$ is an identity, and the equations below the diagonal repeat those above it). There are $\frac{n(n-1)}{2}$ independent constraints, one per entry strictly above the diagonal. Thus the dimension is

$$n^2 - \frac{n(n-1)}{2} = \frac{2n^2 - n^2 + n}{2} = \frac{n(n+1)}{2}$$

2. **For skew-symmetric matrices (K):** The condition is $A_{ji} = -A_{ij}$. There are two types of constraints:

   - For $i < j$, there are $\frac{n(n-1)}{2}$ constraints (as in the symmetric case).
   - For $i = j$ (on the diagonal), $A_{ii} = -A_{ii}$ implies $A_{ii} = 0$. There are n such constraints.

   The total number of independent constraints is $\frac{n(n-1)}{2} + n$. Thus the dimension is

$$n^2 - \left(\frac{n(n-1)}{2} + n\right) = \frac{n(n-1)}{2}$$

NOTE

A useful check: the sum of the exponents (degrees) in the characteristic polynomial must equal the dimension of the space on which the operator acts.

$$\frac{n(n+1)}{2} + \frac{n(n-1)}{2} = \frac{n^2 + n + n^2 - n}{2} = \frac{2n^2}{2} = n^2$$

This is indeed the dimension of $M_n(\mathbb{R})$.

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EXERCISE 13.4

Let $n \in \mathbb{N}$, and consider a general upper triangular matrix $A \in M_n(\mathbb{R})$:

$$A = \begin{pmatrix} \lambda_1 & * & \cdots & * \\ 0 & \lambda_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$$

Its diagonal entries $\lambda_1, \ldots, \lambda_n$ are **pairwise distinct** (that is, $\lambda_i \neq \lambda_j$ whenever $i \neq j$).
Define the polynomial $P(x) = 7x^7 - 4x^4$.
Please prove that there **exists** a matrix $X \in M_n(\mathbb{R})$ satisfying

$$P(X) = A$$

WORKED SOLUTION

Solution strategy: Although A has unspecified entries (asterisks) above the diagonal, the essential fact is that its diagonal entries are **distinct**. A matrix of size n with n distinct eigenvalues is diagonalizable. We can therefore work with the simpler diagonal matrix, solve the resulting scalar problem using calculus, and return to the original coordinates.

1. The characteristic polynomial of an upper triangular matrix is the product of its diagonal factors:
$$P_A(x) = (x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_n)$$
Its eigenvalues are therefore exactly $\lambda_1, \ldots, \lambda_n$. They are distinct, so A has n linearly independent eigenvectors. Thus A is diagonalizable: there are an invertible M and a diagonal matrix $D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ such that
$$A = MDM^{-1}$$

2. Seek a solution of the form $X = MEM^{-1}$, where $E = \mathrm{diag}(c_1, \ldots, c_n)$ is a diagonal matrix to be constructed. The requirement $P(X) = A$ becomes the following equation for diagonal matrices:
$$P(E) = D$$
Since both are diagonal, this is equivalent to n separate scalar equations:
$$P(c_i) = \lambda_i \quad \text{for every } i = 1, \ldots, n$$

3. Consider the real function $f(t) = 7t^7 - 4t^4$.
   - It is continuous.
   - Its leading term has odd degree and positive coefficient, so $\lim_{t \to \pm\infty} f(t) = \pm\infty$.

   The intermediate value theorem implies that f maps onto $\mathbb{R}$. Thus for each eigenvalue λ_i, regardless of the unspecified entries in A, there exists $c_i \in \mathbb{R}$ such that $f(c_i) = \lambda_i$.

4. **Construct the solution:** Choose these c_i as the diagonal entries of E. Define
$$X = MEM^{-1}$$

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EXERCISE 13.4 CONTINUED

and verify:

$$P(X) = MP(E)M^{-1} = M\,\mathrm{diag}(P(c_1),\dots,P(c_n))M^{-1}$$
$$= M\,\mathrm{diag}(\lambda_1,\dots,\lambda_n)M^{-1} = MDM^{-1} = A$$

This proves existence of the required matrix.

PROPOSITION 13.3

If $A \sim B$, the following quantities and properties are preserved:

1. **Determinant:** $\det(A) = \det(B)$.
2. **Trace:** $\mathrm{tr}(A) = \mathrm{tr}(B)$.
3. **Characteristic polynomial:** $P_A(x) = P_B(x)$.
4. **Eigenvalues:** A and B have exactly the same eigenvalues, with the same algebraic and geometric multiplicities.
5. **Rank:** $\mathrm{rank}(A) = \mathrm{rank}(B)$.
6. **Diagonalizability:** A is diagonalizable $\iff$ B is diagonalizable.

NOTE

Warning:
Equality of characteristic polynomials (and hence of eigenvalues, determinants, and traces) **does not guarantee** similarity.

A standard counterexample is $A = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$. Both have characteristic polynomial x^2, but they are not similar: the zero matrix is similar only to itself (prove that the same is true of every scalar matrix).

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13.3 The Cayley-Hamilton Theorem

DEFINITION 13.6

For $f(t) = \sum_{k=0}^{m} a_k t^k$ and $A \in M_n(\mathbb{F})$ with $n \geq 1$, define

$$f(A) = \sum_{k=0}^{m} a_k A^k, \qquad A^0 = I_n.$$

In particular, the constant term a_0 becomes $a_0 I_n$. Our convention is

$$\chi_A(t) = \det(tI_n - A) = t^n + c_{n-1}t^{n-1} + \cdots + c_1 t + c_0.$$

The convention $\det(A - tI_n)$ gives $(-1)^n \chi_A(t)$ and hence the same vanishing identity.

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PROPOSITION 13.4

Every square matrix satisfies its characteristic polynomial:

$$\chi_A(A) = A^n + c_{n-1}A^{n-1} + \cdots + c_1 A + c_0 I_n = 0.$$

We prove this using the adjugate. For $n = 1$, the identity is $A - aI_1 = 0$ with $A = (a)$. For $n \geq 2$, write

$$B(t) = \operatorname{adj}(tI_n - A) = \sum_{j=0}^{n-1} B_j t^j.$$

Every cofactor is a polynomial of degree at most $n-1$. The adjugate identity gives

$$(tI_n - A)B(t) = \chi_A(t) I_n.$$

Comparing coefficients of these polynomials in t yields

$$B_{n-1} = I_n, \qquad B_{j-1} - AB_j = c_j I_n \quad (1 \leq j \leq n-1), \qquad -AB_0 = c_0 I_n.$$

Starting at the top coefficient, use $B_{j-1} = AB_j + c_j I_n$. First $B_{n-2} = A + c_{n-1} I_n$, and continued substitution gives

$$B_0 = A^{n-1} + c_{n-1}A^{n-2} + \cdots + c_1 I_n.$$

Substituting this expression into $-AB_0 = c_0 I_n$ and rearranging proves the theorem.

RECALL AND TOOLS

Polynomial division by χ_A gives a computational tool. If

$$f(t) = q(t)\chi_A(t) + r(t), \qquad \deg r < n,$$

where $r = 0$ is also allowed, then

$$f(A) = q(A)\chi_A(A) + r(A) = r(A).$$

Evaluation at the same matrix preserves multiplication: in the expansion of $q(A)g(A)$, each product of powers is $A^i A^j = A^{i+j}$. Thus high powers reduce to combinations of $I, A, \ldots, A^{n-1}$. If A is invertible, then $c_0 = (-1)^n \det A \neq 0$. Put $c_n = 1$. Multiplying the Cayley-Hamilton identity by A^{-1} gives

$$A^{-1} = -\frac{1}{c_0} \sum_{k=1}^{n} c_k A^{k-1}.$$

The same statements hold for a linear map $T : V \to V$ on a finite-dimensional space, with I_V in place of I_n.

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EXERCISE 13.5

Consider the real matrix

$$A = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & 2 \end{pmatrix}.$$

Please find χ_A, and please use Cayley-Hamilton to compute A^{-1} and A^5.

WORKED SOLUTION

Expansion along the first row gives

$$\begin{aligned} \chi_A(t) &= \det \begin{pmatrix} t & 0 & -1 \\ -1 & t & 1 \\ 0 & -1 & t-2 \end{pmatrix} \\ &= t\big(t(t-2)+1\big) - 1 = t^3 - 2t^2 + t - 1. \end{aligned}$$

Hence

$$A^3 = 2A^2 - A + I.$$

It follows that $A(A^2 - 2A + I) = I$ and $(A^2 - 2A + I)A = I$, so $A^{-1} = A^2 - 2A + I$. Direct calculation yields

$$A^2 = \begin{pmatrix} 0 & 1 & 2 \\ 0 & -1 & -1 \\ 1 & 2 & 3 \end{pmatrix}, \qquad A^{-1} = \begin{pmatrix} 1 & 1 & 0 \\ -2 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

Reduce the next powers in turn:

$$\begin{aligned} A^4 &= 2A^3 - A^2 + A = 2(2A^2 - A + I) - A^2 + A = 3A^2 - A + 2I, \\ A^5 &= 3A^3 - A^2 + 2A = 3(2A^2 - A + I) - A^2 + 2A = 5A^2 - A + 3I. \end{aligned}$$

Finally,

$$A^5 = \begin{pmatrix} 3 & 5 & 9 \\ -1 & -2 & -4 \\ 5 & 9 & 16 \end{pmatrix}.$$

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EXERCISE 13.6

Consider the real matrix

$$A = \begin{pmatrix} 3 & 2 & 0 \\ 0 & 3 & 2 \\ 0 & 0 & 3 \end{pmatrix}.$$

Please find a formula for A^m for every integer $m \geq 0$, and please determine whether A is diagonalizable.

WORKED SOLUTION

Its characteristic polynomial is $(t-3)^3$. Put $N = A - 3I$. Cayley-Hamilton gives $N^3 = 0$. In this case this can also be checked directly:

$$N = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}, \qquad N^2 = \begin{pmatrix} 0 & 0 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad N^3 = 0.$$

The matrices $3I$ and N commute, so the binomial theorem applies to $(3I+N)^m$. Every term containing a power of N greater than 2 vanishes. With $\binom{m}{2} = m(m-1)/2$, we obtain

$$A^m = 3^m I + m3^{m-1}N + \binom{m}{2}3^{m-2}N^2 = \begin{pmatrix} 3^m & 2m3^{m-1} & 2m(m-1)3^{m-2} \\ 0 & 3^m & 2m3^{m-1} \\ 0 & 0 & 3^m \end{pmatrix}.$$

This also holds at $m = 0, 1$: coefficients of the terms absent from the binomial expansion are zero. The only eigenvalue is 3. If A were diagonalizable, its diagonal matrix would be $3I$, implying $A = P(3I)P^{-1} = 3I$. But entry $(1,2)$ of A is 2, so A is not diagonalizable.

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EXERCISE 13.7

Let $A \in M_4(\mathbb{R})$ have

$$\chi_A(t) = t^4 + 2t^3 - 3t^2 - 4t + 5.$$

Please prove that $A + I$ is invertible, and please express its inverse as a polynomial in A of degree at most 3.

WORKED SOLUTION

Division by $t + 1$ gives

$$t^4 + 2t^3 - 3t^2 - 4t + 5 = (t + 1)(t^3 + t^2 - 4t) + 5.$$

Expanding the product gives $t^4 + 2t^3 - 3t^2 - 4t$; the remainder is $5 = \chi_A(-1)$. By Cayley-Hamilton,

$$0 = (A + I)(A^3 + A^2 - 4A) + 5I.$$

Set

$$B = -\frac{1}{5}(A^3 + A^2 - 4A).$$

Then $(A+I)B = I$. Since both matrices are polynomials in A, they commute, so $B(A+I) = I$ as well. Therefore

$$(A + I)^{-1} = -\frac{1}{5}(A^3 + A^2 - 4A).$$

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EXERCISE 13.8

Let $A \in M_3(\mathbb{R})$ have $\chi_A(t) = t(t-2)(t+1)$. Define

$$P = \frac{A^2 + A}{6}, \qquad Q = \frac{A^2 - 2A}{3}, \qquad R = I - P - Q.$$

Please prove that P, Q, R are idempotent (each satisfies $X^2 = X$), and that the product of any two distinct members is zero. Please compute A^m using them for every $m \geq 0$, and please describe the resulting decomposition of each vector into three eigenspaces.

WORKED SOLUTION

Since $\chi_A(t) = t^3 - t^2 - 2t$, Cayley-Hamilton gives

$$A^3 = A^2 + 2A, \qquad A^4 = 3A^2 + 2A.$$

Thus

$$P^2 = \frac{A^4 + 2A^3 + A^2}{36} = \frac{6A^2 + 6A}{36} = P,$$

$$Q^2 = \frac{A^4 - 4A^3 + 4A^2}{9} = \frac{3A^2 - 6A}{9} = Q,$$

$$PQ = QP = \frac{A^4 - A^3 - 2A^2}{18} = 0.$$

Now $RP = PR = P - P^2 - PQ = 0$, and similarly $RQ = QR = 0$. Also,

$$R^2 = (I - P - Q)^2 = I - 2P - 2Q + P^2 + Q^2 + PQ + QP = R.$$

The formulas for P, Q give $A = 2P - Q$. For $m \geq 1$, all mixed products vanish, so

$$A^m = 2^m P + (-1)^m Q.$$

At $m = 0$, we have $A^0 = I = R + P + Q$. To identify the components, calculate

$$AP = \frac{A^3 + A^2}{6} = 2P, \qquad AQ = \frac{A^3 - 2A^2}{3} = -Q, \qquad AR = A - 2P + Q = 0.$$

Consequently, for every $x \in \mathbb{R}^3$,

$$x = Rx + Px + Qx, \qquad Rx \in \ker A, \quad Px \in \ker(A - 2I), \quad Qx \in \ker(A + I).$$

Evaluating the defining polynomials of P, Q, R at $0, 2, -1$ also gives

$$(Px, Qx, Rx) = \begin{cases} (0, 0, x), & Ax = 0, \\ (x, 0, 0), & Ax = 2x, \\ (0, x, 0), & Ax = -x. \end{cases}$$

The decomposition is unique: applying R, P, Q to such a decomposition recovers its three components, since mixed products are zero and each matrix is idempotent.

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EXERCISE 13.9

Let $A, B \in M_2(\mathbb{C})$ satisfy $\operatorname{tr} A = \operatorname{tr} B = 0$.

1. Please prove that $AB + BA = \operatorname{tr}(AB) I_2$.
2. If also $\det A = -2$, $\det B = 5$, and $\operatorname{tr}(AB) = 0$, please compute $(A + B)^8$.

WORKED SOLUTION

1. The characteristic polynomial of a 2×2 matrix is

$$\chi_X(t) = t^2 - (\operatorname{tr} X)t + \det X.$$

Since $A + B$ also has trace zero, Cayley-Hamilton gives

$$A^2 = -(\det A) I_2, \quad B^2 = -(\det B) I_2, \quad (A + B)^2 = -\det(A + B) I_2.$$

Expanding $(A + B)^2 = A^2 + AB + BA + B^2$ yields

$$AB + BA = cI_2, \qquad c = \det A + \det B - \det(A + B).$$

We have $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ because

$$\operatorname{tr}(AB) = \sum_{i,j=1}^{2} a_{ij} b_{ji} = \sum_{j,i=1}^{2} b_{ji} a_{ij} = \operatorname{tr}(BA).$$

Taking traces gives $2 \operatorname{tr}(AB) = 2c$. Division by 2 is valid over $\mathbb{C}$, so $c = \operatorname{tr}(AB)$.

2. We have $A^2 = 2I_2$, $B^2 = -5I_2$, and $AB + BA = 0$. Hence

$$(A + B)^2 = 2I_2 - 5I_2 = -3I_2, \qquad (A + B)^8 = \left((A + B)^2\right)^4 = (-3)^4 I_2 = 81 I_2.$$

13.4 Diagonalization, Powers, and Parameters

EXERCISE 13.10

Let $A \in M_{5\times 5}(\mathbb{R})$ have 1 on the superdiagonal and 0 elsewhere (that is, $A_{12} = A_{23} = A_{34} = A_{45} = 1$, with all other entries zero).

1. Please show that A is nilpotent, meaning that some positive power is zero, and please find its characteristic polynomial.
2. Please find $\dim \ker(A^2)$ and $\dim \ker(A)$.
3. Please prove that A is not diagonalizable.

WORKED SOLUTION

1. The matrix is upper triangular with zero diagonal, so $P_A(x) = x^5$. By the Cayley-Hamilton theorem, $A^5 = 0$, and hence A is nilpotent.

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EXERCISE 13.10 CONTINUED

2. For A, there is one free variable (the first column is zero and the remaining columns are linearly independent), so $\dim\ker(A) = 1$. In A^2, the entries equal to 1 lie on the second superdiagonal. There are two zero columns and three linearly independent nonzero columns, giving two free variables: $\dim\ker(A^2) = 2$.

3. The only eigenvalue is $\lambda = 0$, of algebraic multiplicity $r_a = 5$. The previous part gives geometric multiplicity $r_g = \dim\ker(A) = 1$. Since $1 \neq 5$, the matrix is not diagonalizable.

EXERCISE 13.11

Let $A \in M_{n\times n}(\mathbb{F})$, where $\mathbb{F}$ has characteristic different from 2 (so $1_\mathbb{F} + 1_\mathbb{F} \neq 0_\mathbb{F}$, and $2^{-1} = \frac{1}{2}$ exists). Suppose $A^2 = I$.

1. Please prove that every eigenvalue of A lies in $\{1, -1\}$.

2. Please prove that A is diagonalizable. (Hint: express every vector as a sum of vectors in the two eigenspaces.)

3. In this part, assume additionally that $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Given $\mathrm{tr}(A) = t$, please determine $\dim\ker(A - I)$ and $\dim\ker(A + I)$ in terms of n and t.

WORKED SOLUTION

1. Let $\lambda \in \mathbb{F}$ be an eigenvalue. By definition, there is a nonzero vector $v \in \mathbb{F}^n$, $v \neq 0$, with

$$Av = \lambda v$$

Multiply the equality on the left by A:

$$A(Av) = A(\lambda v)$$

Use associativity on the left and take the scalar outside on the right (linearity):

$$A^2 v = \lambda(Av)$$

Substitute $A^2 = I$ on the left and $Av = \lambda v$ on the right:

$$Iv = \lambda(\lambda v)$$

$$v = \lambda^2 v$$

Move all terms to one side:

$$\lambda^2 v - v = 0 \implies (\lambda^2 - 1)v = 0$$

Since $v \neq 0$ (an eigenvector is nonzero), its scalar coefficient must vanish:

$$\lambda^2 - 1 = 0 \implies \lambda^2 = 1$$

The solutions in the field are $\lambda = 1$ or $\lambda = -1$.

2. To prove diagonalizability, it suffices to find a basis of eigenvectors for the full space.

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EXERCISE 13.11 CONTINUED

Equivalently, the eigenspaces must span the space: $V = V_1 + V_{-1}$. For any $v \in \mathbb{F}^n$, use the identity below (division by 2 is permitted because the characteristic is not 2):

$$v = \frac{1}{2}(v + Av) + \frac{1}{2}(v - Av)$$

Denote the two summands by

$$u_1 = \frac{1}{2}(v + Av), \quad u_{-1} = \frac{1}{2}(v - Av)$$

Then $v = u_1 + u_{-1}$. Each summand lies in its corresponding eigenspace; a summand may be zero, in which case it is not itself an eigenvector: **Check u_1:** Apply A:

$$A(u_1) = A\left(\frac{1}{2}(v + Av)\right) = \frac{1}{2}(Av + A^2v)$$

Substitute $A^2 = I$:

$$= \frac{1}{2}(Av + v) = u_1$$

Thus $u_1 \in \ker(A - I)$ (and if $u_1 \neq 0$, it is an eigenvector for 1). **Check u_{-1}:** Apply A:

$$A(u_{-1}) = A\left(\frac{1}{2}(v - Av)\right) = \frac{1}{2}(Av - A^2v)$$

Substitute $A^2 = I$:

$$= \frac{1}{2}(Av - v) = -\frac{1}{2}(v - Av) = -u_{-1}$$

Thus $u_{-1} \in \ker(A + I)$ (and if $u_{-1} \neq 0$, it is an eigenvector for -1). **Conclusion:** Every vector lies in $\ker(A-I)+\ker(A+I)$. Their intersection is trivial: if $Az = z$ and $Az = -z$, then $2z = 0$, so $z = 0$. Consequently, this direct sum is the entire space. Concatenating bases of the two kernels gives an eigenbasis, so A is diagonalizable.

3. The matrix A is diagonalizable, so it is similar to a diagonal matrix D whose diagonal entries are 1 and -1. Write $k = \dim\ker(A - I)$ (the geometric multiplicity of 1, equal to its number of occurrences on the diagonal). $m = \dim\ker(A + I)$ (the geometric multiplicity of -1, equal to its number of occurrences on the diagonal).

   Two equations are available: 1. The matrix has size n, so the multiplicities sum to n:

$$k + m = n$$

   2. Trace is preserved under similarity, so it equals the sum of the diagonal entries of D:

$$k \cdot (1) + m \cdot (-1) = t \implies k - m = t$$

   This is a system of two equations in the two unknowns k, m. Add the equations:

$$(k + m) + (k - m) = n + t \implies 2k = n + t \implies k = \frac{n + t}{2}$$

   Subtract the second equation from the first:

$$(k + m) - (k - m) = n - t \implies 2m = n - t \implies m = \frac{n - t}{2}$$

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EXERCISE 13.11 CONTINUED

Answer:

$$\dim\ker(A-I)=\frac{n+t}{2},\quad \dim\ker(A+I)=\frac{n-t}{2}.$$

The field assumption in this part is essential for interpreting the trace as the difference of integer dimensions. In positive characteristic, the trace determines only their difference as a field element, which need not determine the integer dimensions.

EXERCISE 13.12

For $t\in\mathbb{R}$, let

$$A(t)=\begin{pmatrix}3 & t+1 & 0\\ 0 & 3 & 0\\ 0 & 0 & 7\end{pmatrix}.$$

Please find its eigenvalues and multiplicities, please determine when it is diagonalizable, and please prove a formula for $A(t)^n$ for every $n\geq 1$.

WORKED SOLUTION

Its characteristic polynomial is $p_A(\lambda)=(\lambda-3)^2(\lambda-7)$, with algebraic multiplicities $2,1$. The equation $(A-3I)v=0$ is $(t+1)v_2=0$, $4v_3=0$. Thus the eigenspace for 3 has dimension 2 when $t=-1$ and dimension 1 otherwise. The eigenspace for 7 has dimension 1. Hence $A(t)$ is diagonalizable exactly when $t=-1$. For every $n\geq 1$,

$$A(t)^n=\begin{pmatrix}3^n & n3^{n-1}(t+1) & 0\\ 0 & 3^n & 0\\ 0 & 0 & 7^n\end{pmatrix}.$$

The base case $n=1$ gives $A(t)$. On multiplication by $A(t)$ on the right, entry $(1,2)$ becomes

$$3^n(t+1)+n3^{n-1}(t+1)\cdot 3=(n+1)3^n(t+1),$$

and the diagonal becomes $3^{n+1},3^{n+1},7^{n+1}$. This proves the induction step. Separately, $A^0=I_3$.

EXERCISE 13.13

Let $A\in M_{n\times n}(\mathbb{F})$ be square over $\mathbb{F}$, and let $u,v\in\mathbb{F}^n$ be nonzero column vectors. Define $B=A+uv^T$ (this is a rank-one update).

1. Suppose $x\in\mathbb{F}^n$, $x\neq 0$, satisfies:
   - x is an eigenvector of A with eigenvalue λ (that is, $Ax=\lambda x$).
   - $v^Tx=0$.

   Please prove that x is also an eigenvector of B, and please identify its eigenvalue.
2. Now assume $A=\operatorname{diag}(\alpha_1,\dots,\alpha_n)$ and $u=v=(1,1,\dots,1)^T$. Please prove that for every scalar λ outside $\{\alpha_1,\dots,\alpha_n\}$,

$$\lambda \text{ is an eigenvalue of } B \iff 1+\sum_{i=1}^{n}\frac{1}{\alpha_i-\lambda}=0$$

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EXERCISE 13.13 CONTINUED

3. For $n = 3$ and $A = \text{diag}(-1, 2, 4)$, please compute the characteristic polynomial of B explicitly (with u, v as in the preceding part).

WORKED SOLUTION

1. Check directly by multiplying B by x:

$$Bx = (A + uv^T)x$$

Expand:

$$= Ax + u(v^T x)$$

Substitute $Ax = \lambda x$ (because x is an eigenvector of A) and $v^T x = 0$ (by assumption):

$$= \lambda x + u(0) = \lambda x$$

Conclusion: We obtained $Bx = \lambda x$. Since $x \neq 0$, it is an eigenvector of B with the same eigenvalue λ.

2. We seek the condition $\det(\lambda I - B) = 0$. Substitute B:

$$\det(\lambda I - (A + uv^T)) = \det((\lambda I - A) - uv^T)$$

Write $D = \lambda I - A$. This is diagonal, with entries $\lambda - \alpha_i$. **Why is D invertible?** By assumption, λ differs from every α_i, so all its diagonal entries are nonzero.

Factor out D inside the determinant:

$$\det(D - uv^T) = \det(D(I - D^{-1}uv^T))$$

By multiplicativity ($\det(XY) = \det(X)\det(Y)$),

$$= \det(D) \cdot \det(I - (D^{-1}u)v^T)$$

For λ to be an eigenvalue, the full expression must be zero. Since D is invertible, $\det(D) \neq 0$, so the second factor must vanish:

$$\det(I - (D^{-1}u)v^T) = 0$$

We use $\det(I + xy^T) = 1 + y^T x$, justifying it even when $y^T x = 0$. If $x = 0$, both sides equal 1. Otherwise, extend x to a basis. In this basis, the image of xy^T lies in $\text{span}\{x\}$ and its first column is $(y^T x, 0, \ldots, 0)^T$. Thus $I + xy^T$ has an upper triangular representing matrix with diagonal $(1 + y^T x, 1, \ldots, 1)$, giving determinant $1 + y^T x$. No nonzero eigenvalue of xy^T needs to be assumed. Here $x = -D^{-1}u$ and $y = v$. Hence the determinant is

$$1 - v^T(D^{-1}u) = 0$$

Compute $v^T D^{-1}u$: The diagonal entries of D^{-1} are $\frac{1}{\lambda - \alpha_i}$. Multiplication by the all-ones vectors adds them:

$$\sum_{i=1}^{n} \frac{1}{\lambda - \alpha_i} = 1$$

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EXERCISE 13.13 CONTINUED

Rearranging gives

$$1-\sum_{i=1}^{n}\frac{1}{\lambda-\alpha_i}=0 \implies 1+\sum_{i=1}^{n}\frac{1}{\alpha_i-\lambda}=0$$

3. Here $A=\operatorname{diag}(-1,2,4)$, so

$$B=\begin{pmatrix}0&1&1\\1&3&1\\1&1&5\end{pmatrix}.$$

The preceding identity gives

$$\begin{aligned}p_B(\lambda)&=(\lambda+1)(\lambda-2)(\lambda-4)\\&\quad-(\lambda-2)(\lambda-4)-(\lambda+1)(\lambda-4)-(\lambda+1)(\lambda-2)\\&=\lambda^3-8\lambda^2+12\lambda+6.\end{aligned}$$

After clearing denominators this is a polynomial identity, valid also at $\lambda=-1,2,4$.

EXERCISE 13.14

Two units exchange information at each step according to

$$\begin{pmatrix}x_{k+1}\\y_{k+1}\end{pmatrix}=A\begin{pmatrix}x_k\\y_k\end{pmatrix},\qquad A=\begin{pmatrix}2/3&1/3\\1/3&2/3\end{pmatrix}.$$

Please find an eigenbasis, an explicit formula for A^k and for the two readings, and their limits as $k\to\infty$.

WORKED SOLUTION

Compute the action of A on two directions:

$$A\begin{pmatrix}1\\1\end{pmatrix}=\begin{pmatrix}1\\1\end{pmatrix},\qquad A\begin{pmatrix}1\\-1\end{pmatrix}=\begin{pmatrix}1/3\\-1/3\end{pmatrix}=\frac{1}{3}\begin{pmatrix}1\\-1\end{pmatrix}.$$

These vectors are linearly independent, so

$$P=\begin{pmatrix}1&1\\1&-1\end{pmatrix},\qquad P^{-1}=\frac{1}{2}\begin{pmatrix}1&1\\1&-1\end{pmatrix},\qquad D=\operatorname{diag}(1,1/3).$$

For every integer $k\geq 0$,

$$A^k=PD^kP^{-1}=\frac{1}{2}\begin{pmatrix}1+3^{-k}&1-3^{-k}\\1-3^{-k}&1+3^{-k}\end{pmatrix}.$$

Consequently,

$$x_k=\frac{x_0+y_0}{2}+3^{-k}\frac{x_0-y_0}{2},\qquad y_k=\frac{x_0+y_0}{2}-3^{-k}\frac{x_0-y_0}{2}.$$

The mean is preserved, while the difference $x_k-y_k=3^{-k}(x_0-y_0)$ is divided by three at each step. Both readings therefore converge to the initial mean $(x_0+y_0)/2$. The eigenvalue 1

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EXERCISE 13.14 CONTINUED

describes the preserved component, and $1/3$ describes the decaying component.

EXERCISE 13.15 EXAM QUESTION

Let $\gamma \in \mathbb{Z}_{\geq 0}$. On $\mathbb{C}[x]_{\leq 3}$ define

$$T_\gamma(a_0 + a_1x + a_2x^2 + a_3x^3) = 2a_1 - 2a_0x + (\gamma a_2 - a_3)x^2 + 3a_3x^3.$$

Please find its matrix in $(1, x, x^2, x^3)$, please determine when it is diagonalizable, and, for the smallest admissible γ, please find an eigenbasis and the characteristic polynomial.

WORKED SOLUTION

The basis images are $-2x, 2, \gamma x^2, -x^2 + 3x^3$, so

$$A_\gamma = \begin{pmatrix} 0 & 2 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ 0 & 0 & \gamma & -1 \\ 0 & 0 & 0 & 3 \end{pmatrix}, \qquad p_{T_\gamma}(t) = (t^2 + 4)(t - \gamma)(t - 3).$$

If $\gamma \neq 3$, the four eigenvalues $2i, -2i, \gamma, 3$ are distinct, so the map is diagonalizable. At $\gamma = 3$, the eigenspace for 3 is $\operatorname{span}\{x^2\}$: the first block has no eigenvalue 3, and the second block forces $a_3 = 0$. The geometric multiplicity is 1 and the algebraic multiplicity is 2, so diagonalization fails. The smallest admissible parameter is $\gamma = 0$. An eigenbasis is

$$\mathcal{D} = (1 + ix, 1 - ix, x^2, 3x^3 - x^2), \qquad [T_0]_{\mathcal{D}} = \operatorname{diag}(2i, -2i, 0, 3).$$

For example, $T_0(3x^3 - x^2) = -3x^2 + 9x^3 = 3(3x^3 - x^2)$; the other checks are direct. The characteristic polynomial is $(t^2 + 4)t(t - 3)$.

EXERCISE 13.16 EXAM QUESTION

In $\mathbb{R}^4$, define

$$\mathcal{F} = (e_1, e_1 + e_2, e_2 + e_3, 2e_3 + e_4),$$

and let T send the standard basis $\mathcal{E}$ to this basis in order. Let S satisfy

$$[S]_{\mathcal{F}} = J = \begin{pmatrix} 0 & -2 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & -4 \\ 0 & 0 & 4 & 0 \end{pmatrix}, \qquad U = T^{-1}ST.$$

Please compute $[U]_{\mathcal{E}}$ and please determine whether U is diagonalizable over $\mathbb{R}$ and $\mathbb{C}$. Whenever possible, please find an eigenbasis. For $W = 3I$, please determine whether U, W can have the same matrix in one basis, or whether a change of basis can turn the matrix of U into $3I$.

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EXERCISE 13.16 EXAM QUESTION CONTINUED

WORKED SOLUTION

Set

$$P = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

We have $[T]_{\mathcal{E}} = P$, $[S]_{\mathcal{E}} = PJP^{-1}$, hence

$$[U]_{\mathcal{E}} = P^{-1}(PJP^{-1})P = J.$$

Its characteristic polynomial is $(t^2+4)(t^2+16)$, so it has no real eigenvalues and no real diagonalization. Over $\mathbb{C}$, an eigenbasis is

$$(e_1 - ie_2, e_1 + ie_2, e_3 - ie_4, e_3 + ie_4),$$

with diagonal matrix $\operatorname{diag}(2i, -2i, 4i, -4i)$. If two maps have the same matrix in the same basis, they are equal. Here $U \neq W$, since $Ue_1 = 2e_2 \neq 3e_1$. A change of basis cannot turn J into $3I$ either: a scalar matrix is similar only to itself.

EXERCISE 13.17 EXAM QUESTION

Let $A \in M_n(\mathbb{R})$ be diagonalizable, with $n \geq 3$, and suppose it has exactly two distinct eigenvalues. Assume

$$\operatorname{rank}(A+I) = n-2, \qquad \operatorname{tr} A = 3n-8, \qquad \det A = 3^{n-2}.$$

Please find the eigenvalues and multiplicities, please write the characteristic polynomial, and please compute $\operatorname{tr}(A^k)$ for every $k \geq 1$.

WORKED SOLUTION

Rank-nullity gives $\dim\ker(A+I) = 2$. Thus -1 has geometric multiplicity 2, and diagonalizability makes its algebraic multiplicity 2 as well. The other eigenvalue λ has multiplicity $n-2$. The trace condition gives

$$-2 + (n-2)\lambda = 3n-8, \qquad (n-2)\lambda = 3n-6 = 3(n-2).$$

Since $n \geq 3$, we obtain $\lambda = 3$. The determinant agrees: $(-1)^2 3^{n-2} = 3^{n-2}$. Consequently,

$$p_A(t) = (t+1)^2(t-3)^{n-2}, \qquad \operatorname{tr}(A^k) = 2(-1)^k + (n-2)3^k.$$

The last formula follows by powering a diagonal representation and using trace invariance under similarity.

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Additional practice

EXERCISE 13.18 FOUNDATION

Consider the real matrix

$$A = \begin{pmatrix} 4 & 2 & 0 \\ 2 & 4 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

1. Please find its characteristic polynomial and all eigenspaces.
2. Please give real matrices P, D satisfying $A = PDP^{-1}$.
3. Please find an explicit formula for A^k for every integer $k \geq 0$.

WORKED SOLUTION

1. The block structure gives

$$p_A(t) = (t+1)\det\begin{pmatrix} t-4 & -2 \\ -2 & t-4 \end{pmatrix} = (t+1)((t-4)^2 - 4) = (t+1)(t-2)(t-6).$$

The eigenvalues are distinct. Solving the kernel equations, or direct verification, gives

$$E_6 = \text{span}\{(1,1,0)^T\}, \quad E_2 = \text{span}\{(1,-1,0)^T\}, \quad E_{-1} = \text{span}\{(0,0,1)^T\}.$$

Each eigenspace has dimension 1, as also follows from its algebraic multiplicity 1.

2. Place the eigenbasis vectors in the columns:

$$P = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad P^{-1} = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad D = \text{diag}(6, 2, -1).$$

We have $\det P = -2 \neq 0$, and the eigenvalue equations give $AP = PD$. Thus $A = PDP^{-1}$.

3. Computing PD^kP^{-1} gives

$$A^k = \begin{pmatrix} (6^k + 2^k)/2 & (6^k - 2^k)/2 & 0 \\ (6^k - 2^k)/2 & (6^k + 2^k)/2 & 0 \\ 0 & 0 & (-1)^k \end{pmatrix}.$$

The formula also holds at $k = 0$, giving I_3. For $k \geq 1$, all entries are integers despite the division by 2 in this expression.

EXERCISE 13.19 CHALLENGE

Let $A \in M_n(\mathbb{F})$ have n distinct eigenvalues $\lambda_1, \ldots, \lambda_n$ in $\mathbb{F}$.

1. Please prove that $B \in M_n(\mathbb{F})$ satisfies $AB = BA$ if and only if there is a unique polynomial f of degree at most $n-1$ (including the zero polynomial) with $B = f(A)$.

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EXERCISE 13.19 CHALLENGE CONTINUED

2. Over $\mathbb{R}$, please find such a polynomial for

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}, \qquad B = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}.$$

WORKED SOLUTION

1. First, every power of A commutes with A, so every linear combination of its powers commutes with it. Thus $B = f(A)$ implies $AB = BA$. Conversely, assume $AB = BA$. Since A has n distinct eigenvalues in the field, it has an eigenbasis $(v_1, \ldots, v_n)$ with $Av_i = \lambda_i v_i$, and every eigenspace is one-dimensional. We have

$$A(Bv_i) = B(Av_i) = \lambda_i Bv_i.$$

Thus $Bv_i \in E_{\lambda_i}(A) = \text{span}\{v_i\}$, including the possibility $Bv_i = 0$. Hence $Bv_i = \mu_i v_i$ for some $\mu_i \in \mathbb{F}$. Define explicitly

$$f(t) = \sum_{i=1}^{n} \mu_i \prod_{\substack{1 \le j \le n \\ j \ne i}} \frac{t - \lambda_j}{\lambda_i - \lambda_j}.$$

All denominators are nonzero, so this polynomial is defined over $\mathbb{F}$. For $n = 1$, an empty product equals 1. Its degree is at most $n-1$, and at $t = \lambda_i$ only the ith summand remains, giving $f(\lambda_i) = \mu_i$. Therefore

$$f(A)v_i = f(\lambda_i)v_i = \mu_i v_i = Bv_i.$$

The two maps agree on a basis, so $f(A) = B$. If another polynomial g of degree at most $n - 1$ satisfies $g(A) = B$, then $(f - g)(\lambda_i) = 0$ for every i. A nonzero polynomial of degree at most $n - 1$ cannot have n distinct roots, so $f = g$. Uniqueness requires the degree bound; without it, a multiple of p_A may be added.

2. On $(1, 1)^T, (1, -1)^T$, the eigenvalues of A are $4, 2$, and those of B are $9, 5$. Writing $f(t) = at + b$ gives

$$4a + b = 9, \qquad 2a + b = 5.$$

Thus $2a = 4$, so $a = 2$, $b = 1$. Indeed,

$$B = 2A + I_2, \qquad f(t) = 2t + 1.$$

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EXERCISE 13.20 CHALLENGE

For $n \geq 2$, define $S(x_1, \ldots, x_n) = (x_n, x_1, \ldots, x_{n-1})$ on $\mathbb{C}^n$.

1. Please find an eigenbasis of S and its characteristic polynomial.
2. For a prime p, please define the same shift on $\mathbb{Z}_p^p$ and prove that it is not diagonalizable over $\mathbb{Z}_p$.

WORKED SOLUTION

Let $\omega = e^{2\pi i/n}$ and define

$$v_k = (1, \omega^{-k}, \omega^{-2k}, \ldots, \omega^{-(n-1)k})^T, \qquad 0 \leq k < n.$$

The first coordinate of Sv_k is $\omega^{-(n-1)k} = \omega^k$. Coordinate $j \geq 2$ is $\omega^{-(j-2)k} = \omega^k \omega^{-(j-1)k}$. Hence $Sv_k = \omega^k v_k$. The n eigenvalues are distinct, so these eigenvectors form a basis, and

$$p_S(t) = \prod_{k=0}^{n-1} (t - \omega^k) = t^n - 1.$$

Over $\mathbb{Z}_p$, $S^p = I$. Since S and I commute and $\binom{p}{k}$ is divisible by p for $1 \leq k < p$, the binomial formula gives

$$(S - I)^p = S^p - I = 0.$$

If S were diagonalizable, so would $S - I$ be. A diagonal matrix with a zero positive power has every diagonal entry zero. This would imply $S = I$, contradicting $S(e_1) = e_2 \neq e_1$ for $p \geq 2$.

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EXERCISE 13.21 CHALLENGE

Let $A \in M_n(\mathbb{C})$ and assume A^2 is diagonalizable. Please prove that A is diagonalizable if and only if $\ker A = \ker A^2$. Please give a 2×2 example in which A^2 is diagonalizable and A is not.

WORKED SOLUTION

If A is diagonalizable, an eigenbasis makes it diagonal. A diagonal entry is zero exactly when its square is zero, giving equality of the kernels. Conversely, suppose the kernels agree. Since A^2 is diagonalizable,

$$\mathbb{C}^n = \bigoplus_{\mu} E_\mu(A^2), \qquad E_\mu(A^2) = \{v : A^2 v = \mu v\}.$$

Each summand is preserved by A, since $A^2(Av) = A(A^2 v) = \mu Av$. On $E_0(A^2) = \ker A$, the map A is zero. For $\mu \neq 0$, choose $\lambda \in \mathbb{C}$ with $\lambda^2 = \mu$. For $v \in E_\mu(A^2)$, set

$$v_+ = \tfrac{1}{2}(v + \lambda^{-1} Av), \qquad v_- = \tfrac{1}{2}(v - \lambda^{-1} Av).$$

Then $v = v_+ + v_-$ and direct substitution gives $Av_+ = \lambda v_+$ and $Av_- = -\lambda v_-$. Their eigenspaces intersect trivially because $\lambda \neq 0$. Thus each summand has a basis of eigenvectors for A, and their union is an eigenbasis of the whole space. For a counterexample, $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ has $A^2 = 0$. Its only possible eigenvalue is zero. If A were diagonalizable, it would be zero, contradicting its displayed entry.

13.5 Jordan Form

DEFINITION 13.7

Let $k \geq 1$ be an integer and $\lambda \in F$. The Jordan block of size k with eigenvalue λ is

$$J_k(\lambda) = \begin{pmatrix} \lambda & 1 & & 0 \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{pmatrix} = \lambda I_k + N_k.$$

The matrix N_k has 1 immediately above the diagonal and zero elsewhere; set $N_1 = 0$. We have $N_k^k = 0$, while $N_k^{k-1} \neq 0$ for $k \geq 2$. A matrix with a positive power equal to zero is called **nilpotent**.

RECALL AND TOOLS

If the characteristic polynomial of $A \in M_n(F)$ splits into linear factors over F, there is an invertible P such that

$$P^{-1} A P = \operatorname{diag}\big(J_{k_1}(\lambda_1), \ldots, J_{k_s}(\lambda_s)\big).$$

Here diag denotes a block diagonal matrix: the displayed blocks are on the diagonal, with zero elsewhere. This is the Jordan form theorem. Over $\mathbb{C}$, the splitting condition always

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holds. Over $\mathbb{R}$, it holds exactly when all roots of the characteristic polynomial are real. Block sizes and eigenvalues are determined up to the order of the blocks.
Diagonalization occurs exactly when every block has size 1. For a given λ, the sum of its block sizes is its algebraic multiplicity, and the number of its blocks is its geometric multiplicity.

DEFINITION 13.8

A **Jordan chain** for λ is a list $v_1, \dots, v_k$ satisfying

$$v_1 \neq 0, \qquad (A - \lambda I)v_1 = 0, \qquad (A - \lambda I)v_j = v_{j-1} \quad (2 \leq j \leq k).$$

Thus v_1 is an eigenvector; the other chain vectors are called generalized eigenvectors. Every such chain is linearly independent: if $\sum_{j=1}^{k} a_j v_j = 0$ and a_ℓ is the last nonzero coefficient, applying $(A - \lambda I)^{\ell-1}$ gives $a_\ell v_1 = 0$, a contradiction. In this order,

$$A(v_1 \ \cdots \ v_k) = (v_1 \ \cdots \ v_k)J_k(\lambda).$$

If the columns of P form a basis made of these chains, then $AP = PJ$, so $P^{-1}AP = J$.

EXAMPLE 13.3

For

$$A = \begin{pmatrix} 5 & 4 \\ -1 & 1 \end{pmatrix}, \qquad N = A - 3I = \begin{pmatrix} 2 & 4 \\ -1 & -2 \end{pmatrix},$$

we have

$$\chi_A(t) = (t-5)(t-1) + 4 = (t-3)^2, \qquad N^2 = \begin{pmatrix} 4-4 & 8-8 \\ -2+2 & -4+4 \end{pmatrix} = 0.$$

The equation $Nx = 0$ is $x_1 + 2x_2 = 0$, so the eigenspace has dimension one. Choose

$$v_1 = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad Nv_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad Nv_2 = v_1.$$

This is a chain of length 2. With

$$P = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}, \qquad P^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix},$$

we obtain

$$AP = \begin{pmatrix} 6 & 5 \\ -3 & -1 \end{pmatrix}, \qquad P^{-1}AP = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix} = J_2(3).$$

The superdiagonal entry 1 records $Av_2 = 3v_2 + v_1$. There is no basis of two eigenvectors.

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EXAMPLE 13.4

Let

$$A = \begin{pmatrix} -1 & 3 & -2 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

Choose $v_1 = (1,1,0)^T$, $v_2 = (0,1,1)^T$, and $v_3 = (1,0,0)^T$. Direct calculation gives

$$Av_1 = (2,2,0)^T = 2v_1, \qquad Av_2 = (1,3,2)^T = 2v_2 + v_1, \qquad Av_3 = -v_3.$$

The first two vectors form a chain for 2, while the third is an eigenvector for -1. Set

$$P = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad P^{-1} = \begin{pmatrix} 0 & 1 & -1 \\ 0 & 0 & 1 \\ 1 & -1 & 1 \end{pmatrix}.$$

Then

$$AP = \begin{pmatrix} 2 & 1 & -1 \\ 2 & 3 & 0 \\ 0 & 2 & 0 \end{pmatrix}, \qquad P^{-1}AP = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Thus $\chi_A(t) = (t-2)^2(t+1)$, but the eigenspace for 2 has dimension one, so A is not diagonalizable.

EXERCISE 13.22

Let $N \in M_7(\mathbb{C})$ satisfy $N^3 = 0$, rank $N = 4$, and rank $N^2 = 2$. Please find its Jordan form and $\dim \ker N^j$ for every integer $j \geq 1$.

WORKED SOLUTION

Every eigenvalue satisfies $\lambda^3 = 0$, so $\lambda = 0$. Each block has size at most 3, since the cube of a larger nilpotent block is nonzero. Let b_k be the number of blocks of size k, for $k = 1, 2, 3$. For each $J_k(0)$,

$$\operatorname{rank} J_k(0) = k - 1, \qquad \operatorname{rank}(J_k(0)^2) = \max\{k-2, 0\}.$$

These are the numbers of nonzero columns: the first power moves one step along the chain, and the second moves two steps. Hence

$$\begin{aligned} b_1 + 2b_2 + 3b_3 &= 7, \\ b_2 + 2b_3 &= 4, \\ b_3 &= 2. \end{aligned}$$

The last equation gives $b_3 = 2$, the second gives $b_2 = 0$, and the first gives $b_1 = 1$. The Jordan form is therefore

$$\operatorname{diag}\big(J_3(0), J_3(0), J_1(0)\big).$$

Rank-nullity gives

$$\dim \ker N = 7 - 4 = 3, \qquad \dim \ker N^2 = 7 - 2 = 5.$$

For $j \geq 3$, $N^j = 0$, so $\dim \ker N^j = 7$.

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EXERCISE 13.23

Let $A \in M_n(\mathbb{C})$ satisfy $A^3 = A^2$ and $\operatorname{rank} A = \operatorname{rank} A^2$. Please prove that $A^2 = A$. Please construct an example with $A^3 = A^2$ but $A^2 \neq A$.

WORKED SOLUTION

Every eigenvalue satisfies $\lambda^3 = \lambda^2$, so $\lambda \in \{0, 1\}$. Pass to Jordan form. On a block for 0, write the block as N. The condition is

$$N^3 = N^2, \qquad N^2(N - I) = 0.$$

The matrix $N - I$ is invertible: if $N^k = 0$, then $(I - N)^{-1} = I + N + \cdots + N^{k-1}$. Thus $N^2 = 0$, and the block has size at most 2. On a block for 1, write $J = I + N$. Then

$$J^3 - J^2 = N + 2N^2 + N^3 = N(I + N)^2 = 0.$$

Also $I + N$ is invertible, with inverse $I - N + \cdots + (-N)^{k-1}$, so $N = 0$. Every block for 1 therefore has size 1. A block $J_2(0)$ contributes 1 to rank A and zero to rank A^2. Size-one blocks for $0, 1$ contribute the same rank to both. Equality of the ranks excludes every $J_2(0)$. Thus $A = PDP^{-1}$ for a diagonal matrix D with entries $0, 1$, so $A^2 = PD^2P^{-1} = PDP^{-1} = A$. For example, $B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ satisfies $B^3 = B^2 = 0$, but $B^2 \neq B$. Here $\operatorname{rank} B = 1$ and $\operatorname{rank} B^2 = 0$.

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EXERCISE 13.24 CHALLENGE

Let $\dim V = n$ over a field F, and let $T : V \to V$ be linear with $T^2 = 0$. Put $r = \operatorname{rank} T$. Please prove that $2r \le n$ and that some basis gives the matrix

$$\begin{pmatrix} 0 & I_r & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The three row and column block sizes are $r, r, n - 2r$. Blocks of size zero are omitted.

WORKED SOLUTION

The identity $T^2 = 0$ gives $\operatorname{Im} T \subseteq \ker T$. Rank-nullity yields $r \le n - r$, so $2r \le n$. Choose a basis $u_1, \dots, u_r$ of the image and extend it to a basis $u_1, \dots, u_r, z_1, \dots, z_{n-2r}$ of the kernel. For each i, choose w_i with $T(w_i) = u_i$. The list

$$\mathcal{B} = (u_1, \dots, u_r, w_1, \dots, w_r, z_1, \dots, z_{n-2r})$$

is independent. Applying T to a zero linear combination shows that all coefficients of the w_i vanish, because the u_i are independent. The remaining combination is in a basis of the kernel, so all its coefficients vanish too. The list has n vectors and is therefore a basis. In this basis $T(u_i) = T(z_j) = 0$ and $T(w_i) = u_i$. Writing these images as columns gives the displayed matrix. When $r = 0$, $T = 0$ and every basis works.

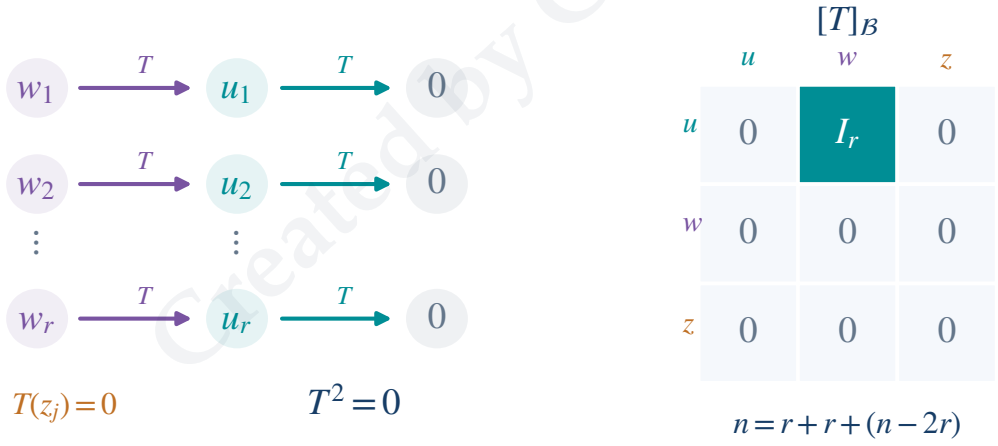


Figure 13.1. The schematic shows the case $r \ge 3$. Each w_i maps to u_i, and each u_i maps to zero. The vectors z_j complete $u_1, \dots, u_r$ to a basis of the kernel. The basis groups are ordered u, w, z.

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EXERCISE 13.25 CHALLENGE

Let $A \in M_n(F)$ have rank 1, where $n \geq 2$, and put $\alpha = \operatorname{tr} A$, the sum of the diagonal entries. Please prove that A is similar to

$$\begin{cases} \operatorname{diag}(\alpha, 0, \ldots, 0), & \alpha \neq 0, \\ E_{12}, & \alpha = 0, \end{cases}$$

where E_{12} has a single 1 in position $(1, 2)$. Please deduce that two rank-one matrices of the same size and trace are similar.

WORKED SOLUTION

Write $A = uv^T$ with $u, v \neq 0$ and define $\varphi(x) = v^T x$. Then

$$Ax = \varphi(x)u, \qquad \operatorname{tr} A = \sum_i u_i v_i = \varphi(u) = \alpha.$$

Since $\varphi \neq 0$, its kernel has dimension $n-1$. If $\alpha \neq 0$, then $u \notin \ker \varphi$. A basis $\mathcal{B} = (u, z_2, \ldots, z_n)$ obtained by adjoining u to a basis of the kernel satisfies $Au = \alpha u$ and $Az_j = 0$. Its representing matrix is $\operatorname{diag}(\alpha, 0, \ldots, 0)$. If $\alpha = 0$, then $u \in \ker \varphi$. Choose w with $\varphi(w) = 1$ and extend u to a basis $(u, z_3, \ldots, z_n)$ of the kernel. The list $\mathcal{B} = (u, w, z_3, \ldots, z_n)$ is a basis: applying φ to a zero combination forces the coefficient of w to vanish; the remaining coefficients vanish by independence in the kernel. Now $Au = Az_j = 0$ and $Aw = u$, giving the matrix E_{12}. Two rank-one matrices with the same size and trace are therefore similar to the same displayed matrix, hence to one another.

13.6 Inner Product Spaces

RECALL AND TOOLS

Throughout this chapter, let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and let V be a vector space over $\mathbb{F}$. Write $\bar{\cdot} : \mathbb{F} \to \mathbb{F}$ for standard conjugation: that is, if $\mathbb{F} = \mathbb{C}$, then $\overline{a + ib} = a - ib$; if $\mathbb{F} = \mathbb{R}$, then $\overline{x} = x$ for every $x \in \mathbb{R}$.

RECALL AND TOOLS

Let V be a vector space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. An *inner product* on V is a map

$$\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$$

satisfying the following axioms for all $u, v, w \in V$ and all $\alpha, \beta \in \mathbb{F}$.

1. **Linearity in the first argument:**

$$\langle \alpha u + \beta v,\ w \rangle = \alpha \langle u, w \rangle + \beta \langle v, w \rangle.$$

2. **Conjugate symmetry (Hermitian symmetry):**

$$\langle u, v \rangle = \overline{\langle v, u \rangle}.$$

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3. **Positive definiteness:**

$$\langle v, v\rangle \in \mathbb{R}, \qquad \langle v, v\rangle \geq 0, \qquad \langle v, v\rangle = 0 \iff v = 0.$$

Formal observation: Linearity in the first argument and conjugate symmetry imply *conjugate linearity in the second argument*: for all $\alpha, \beta \in \mathbb{F}$ and all $u, v, w \in V$,

$$\langle u,\ \alpha v + \beta w\rangle = \overline{\alpha}\,\langle u, v\rangle + \overline{\beta}\,\langle u, w\rangle.$$

When $\mathbb{F} = \mathbb{R}$, this gives ordinary linearity in the second argument as well.

EXAMPLE 13.5

Standard examples of inner products:

- On $\mathbb{C}^n$ (column vectors $u, v \in \mathbb{C}^n$):

$$\langle u, v\rangle := u^T \overline{v}.$$

  On $\mathbb{R}^n$:

$$\langle u, v\rangle := u^T v.$$

- On $M_{m\times n}(\mathbb{C})$:

$$\langle A, B\rangle := \operatorname{tr}\big(A^T \overline{B}\big).$$

- On $\mathbb{R}_n[x]$ (or on $C([a, b], \mathbb{R})$), where $a < b$:

$$\langle p, q\rangle := \int_a^b p(x)\, q(x)\, dx.$$

DEFINITION 13.9

Let $(V, \langle\cdot,\cdot\rangle)$ be a vector space equipped with an inner product. The **norm** induced by the inner product is defined, for each $v \in V$, by

$$\|v\| := \sqrt{\langle v, v\rangle}.$$

Thus $\|v\| \in [0, \infty)$; in particular, $\|v\| = 0$ if and only if $v = 0$.

DEFINITION 13.10

Inner product space: If V is a vector space over $\mathbb{F}$ and $\langle\cdot,\cdot\rangle$ is an inner product on V, then the pair

$$\big(V, \langle\cdot,\cdot\rangle\big)$$

is called an **inner product space**.

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PROPOSITION 13.5

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. For all $n, m \in \mathbb{N}$, all $\alpha_1, \ldots, \alpha_n, \beta_1, \ldots, \beta_m \in \mathbb{F}$, and all $v_1, \ldots, v_n, u_1, \ldots, u_m \in V$,

$$\left\langle \sum_{i=1}^{n} \alpha_i v_i,\ \sum_{j=1}^{m} \beta_j u_j \right\rangle = \sum_{i=1}^{n} \sum_{j=1}^{m} \alpha_i\, \overline{\beta_j}\, \langle v_i, u_j \rangle.$$

When $\mathbb{F} = \mathbb{R}$, we have $\overline{\beta_j} = \beta_j$, so the formula reduces to

$$\left\langle \sum_{i=1}^{n} \alpha_i v_i,\ \sum_{j=1}^{m} \beta_j u_j \right\rangle = \sum_{i=1}^{n} \sum_{j=1}^{m} \alpha_i\, \beta_j\, \langle v_i, u_j \rangle.$$

13.7 Basic Exercises and Results

EXERCISE 13.26

Let $n \geq 1$. On $V = M_n(\mathbb{R})$, define

$$f(A, B) := \operatorname{tr}(AB).$$

Please determine for which values of n this is an inner product, and

WORKED SOLUTION

If $n = 1$, identify matrices with real numbers. Then $f(a, b) = ab$ is the standard inner product on $\mathbb{R}$. If $n \geq 2$, choose $A = E_{21}$: that is, $a_{21} = 1$ and all other entries are zero. Then $A \neq 0$, but $A^2 = 0$, so

$$f(A, A) = \operatorname{tr}(A^2) = 0.$$

Positive definiteness therefore fails, and the function is not an inner product when $n \geq 2$.

EXERCISE 13.27

Let $n \in \mathbb{N}$, and let $M \in M_n(\mathbb{R})$ be symmetric and positive definite; that is,

$$M^T = M, \qquad x^T M x > 0 \ \ \forall x \in \mathbb{R}^n \setminus \{0\}.$$

Define $\langle\cdot,\cdot\rangle_M : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ by

$$\langle x, y \rangle_M := x^T M y.$$

1. Please prove that $\langle\cdot,\cdot\rangle_M$ is an inner product on $\mathbb{R}^n$.
2. Please show that the induced norm is $\|x\|_M = \sqrt{x^T M x}$.
3. Physical example: suppose a linear mechanical system is described by a mass matrix M (in some system of generalized coordinates). Please show that its kinetic energy

$$T(v) = \frac{1}{2}\, v^T M v$$

equals $\frac{1}{2}\|v\|_M^2$.

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EXERCISE 13.27 CONTINUED

WORKED SOLUTION

1. We verify the inner product axioms.

   Linearity in the first argument: Let $x_1, x_2, y \in \mathbb{R}^n$ and $\alpha, \beta \in \mathbb{R}$. Then

   $$\langle \alpha x_1 + \beta x_2,\, y\rangle_M = (\alpha x_1 + \beta x_2)^T M y = \alpha x_1^T M y + \beta x_2^T M y = \alpha\langle x_1, y\rangle_M + \beta\langle x_2, y\rangle_M.$$

   Symmetry: For every $x, y \in \mathbb{R}^n$,

   $$\langle x, y\rangle_M = x^T M y.$$

   Since this is a real scalar, $(x^T M y)^T = x^T M y$, so

   $$\langle x, y\rangle_M = (x^T M y)^T = y^T M^T x = y^T M x = \langle y, x\rangle_M,$$

   where we used $M^T = M$.

   Positive definiteness: For every $x \in \mathbb{R}^n$,

   $$\langle x, x\rangle_M = x^T M x \geq 0,$$

   and $\langle x, x\rangle_M = 0$ if and only if $x = 0$, because M is positive definite.

   Hence $\langle \cdot, \cdot\rangle_M$ is an inner product on $\mathbb{R}^n$.

2. By the definition of the norm induced by an inner product, for every $x \in \mathbb{R}^n$,

   $$\|x\|_M := \sqrt{\langle x, x\rangle_M} = \sqrt{x^T M x}.$$

3. For every $v \in \mathbb{R}^n$,

   $$T(v) = \frac{1}{2} v^T M v = \frac{1}{2}\langle v, v\rangle_M = \frac{1}{2}\|v\|_M^2,$$

   so the kinetic energy is half the square of the norm induced by $\langle \cdot, \cdot\rangle_M$.

EXERCISE 13.28

Let $(V, \langle \cdot, \cdot\rangle)$ be an inner product space over $\mathbb{C}$, and let $u, v \in V$ satisfy $u \neq v$. Please show that there is $w \in V$ such that $\langle w, u\rangle \neq \langle w, v\rangle$.

WORKED SOLUTION

Choose $w = u - v$. Since $u \neq v$, the vector w is nonzero. Conjugate linearity in the second argument implies additivity, so

$$\langle w, u\rangle - \langle w, v\rangle = \langle w, u - v\rangle = \langle u - v, u - v\rangle = \|u - v\|^2 > 0.$$

Therefore $\langle w, u\rangle \neq \langle w, v\rangle$, as required.

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13.8 Norms and the Cauchy-Schwarz Inequality

DEFINITION 13.11

Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and let V be a vector space over $\mathbb{F}$. A **norm** on V is a function

$$\|\cdot\| : V \to [0, \infty)$$

satisfying the following axioms for all $u, v \in V$ and all $\alpha \in \mathbb{F}$:

1. **Positive definiteness:**
$$\|v\| \geq 0, \qquad \|v\| = 0 \iff v = 0.$$
2. **Homogeneity:**
$$\|\alpha v\| = |\alpha| \, \|v\|.$$
3. **Triangle inequality:**
$$\|u + v\| \leq \|u\| + \|v\|.$$

Here $|\alpha|$ denotes absolute value over $\mathbb{R}$ or modulus over $\mathbb{C}$. A vector space equipped with a norm is a **normed space**, denoted $(V, \|\cdot\|)$.

PROPOSITION 13.6

Cauchy-Schwarz inequality. For all $u, v \in V$,

$$|\langle u, v \rangle| \leq \|u\| \, \|v\|.$$

Equality holds if and only if u, v are linearly dependent: that is, there is $\alpha \in \mathbb{F}$ such that $u = \alpha v$ or $v = \alpha u$.

EXERCISE 13.29

Let $\omega, \zeta, \xi > 0$. Please prove the inequality

$$\frac{\omega}{\zeta + \xi} + \frac{\zeta}{\xi + \omega} + \frac{\xi}{\omega + \zeta} \geq \frac{(\omega + \zeta + \xi)^2}{2(\omega\zeta + \zeta\xi + \xi\omega)}.$$

WORKED SOLUTION

Use the standard inner product on $\mathbb{R}^3$, namely $\langle x, y \rangle = x^T y$.
Define the vectors

$$x = \left(\sqrt{\frac{\omega}{\zeta + \xi}}, \sqrt{\frac{\zeta}{\xi + \omega}}, \sqrt{\frac{\xi}{\omega + \zeta}} \right), \qquad y = \left(\sqrt{\omega(\zeta + \xi)}, \sqrt{\zeta(\xi + \omega)}, \sqrt{\xi(\omega + \zeta)} \right).$$

Then

$$\langle x, y \rangle = \sqrt{\frac{\omega}{\zeta + \xi}} \sqrt{\omega(\zeta + \xi)} + \sqrt{\frac{\zeta}{\xi + \omega}} \sqrt{\zeta(\xi + \omega)} + \sqrt{\frac{\xi}{\omega + \zeta}} \sqrt{\xi(\omega + \zeta)} = \omega + \zeta + \xi.$$

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EXERCISE 13.29 CONTINUED

Also,

$$\|x\|^2 = \langle x, x\rangle = \left(\sqrt{\frac{\omega}{\zeta+\xi}}\right)^2 + \left(\sqrt{\frac{\zeta}{\xi+\omega}}\right)^2 + \left(\sqrt{\frac{\xi}{\omega+\zeta}}\right)^2 = \frac{\omega}{\zeta+\xi} + \frac{\zeta}{\xi+\omega} + \frac{\xi}{\omega+\zeta},$$

and

$$\begin{aligned}\|y\|^2 &= \langle y, y\rangle \\ &= \left(\sqrt{\omega(\zeta+\xi)}\right)^2 + \left(\sqrt{\zeta(\xi+\omega)}\right)^2 + \left(\sqrt{\xi(\omega+\zeta)}\right)^2 \\ &= \omega(\zeta+\xi) + \zeta(\xi+\omega) + \xi(\omega+\zeta).\end{aligned}$$

Expand the last sum:

$$\omega(\zeta+\xi) + \zeta(\xi+\omega) + \xi(\omega+\zeta) = (\omega\zeta+\omega\xi) + (\zeta\xi+\zeta\omega) + (\xi\omega+\xi\zeta) = 2(\omega\zeta+\zeta\xi+\xi\omega).$$

The Cauchy-Schwarz inequality gives

$$\langle x, y\rangle^2 \le \|x\|^2 \|y\|^2,$$

and hence

$$(\omega+\zeta+\xi)^2 \le \left(\frac{\omega}{\zeta+\xi} + \frac{\zeta}{\xi+\omega} + \frac{\xi}{\omega+\zeta}\right) \cdot 2(\omega\zeta+\zeta\xi+\xi\omega).$$

Since $2(\omega\zeta+\zeta\xi+\xi\omega) > 0$, division yields

$$\frac{\omega}{\zeta+\xi} + \frac{\zeta}{\xi+\omega} + \frac{\xi}{\omega+\zeta} \ge \frac{(\omega+\zeta+\xi)^2}{2(\omega\zeta+\zeta\xi+\xi\omega)},$$

as required.

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EXERCISE 13.30 CHALLENGE

Let V be a complex inner product space, with the inner product linear in its first argument. Please prove that

$$4\langle u, v\rangle = \|u+v\|^2 - \|u-v\|^2 + i\big(\|u+iv\|^2 - \|u-iv\|^2\big).$$

Let $T : V \to V$ be complex-linear and satisfy $\|Tx\| = \|x\|$ for every x. Please deduce that $\langle Tu, Tv\rangle = \langle u, v\rangle$ for every u, v.

WORKED SOLUTION

Put $z = \langle u, v\rangle$. Expanding gives

$$\|u+v\|^2 = \|u\|^2 + \|v\|^2 + z + \overline{z}, \qquad \|u-v\|^2 = \|u\|^2 + \|v\|^2 - z - \overline{z}.$$

Their difference is $4\,\mathrm{Re}\, z$. Conjugate-linearity in the second argument gives $\langle u, iv\rangle = -iz$ and $\langle iv, u\rangle = i\overline{z}$. Consequently

$$\|u+iv\|^2 = \|u\|^2 + \|v\|^2 - iz + i\overline{z},$$
$$\|u-iv\|^2 = \|u\|^2 + \|v\|^2 + iz - i\overline{z}.$$

Their difference is $4\,\mathrm{Im}\, z$, proving the formula. Apply it to Tu, Tv. Complex linearity gives $Tu \pm Tv = T(u \pm v)$ and $Tu \pm iTv = T(u \pm iv)$. Norm preservation replaces all four norms by the original ones, so the inner product is preserved as well.

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TUTORIAL 14

Orthogonality, Best Approximation, and Gram-Schmidt

MOTIVATION AND GUIDING IDEA

An inner product adds geometric measurement to a vector space: we can ask whether two directions are perpendicular, measure size, and decompose a vector relative to a subspace. A different inner product changes the geometry even when the vectors and subspace remain fixed. Begin every computation by identifying the given inner product, and only then apply the formulas.

NOTATION AND CONVENTIONS

- $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ is the scalar field; $\overline{z}$ is complex conjugation, with $\overline{z} = z$ over the reals.
- A^T is the transpose and $A^* = \overline{A}^T$ is the conjugate transpose. For column vectors, the standard inner product is $\langle x, y \rangle = x^T \overline{y} = y^* x$.
- $\langle v, w \rangle$ denotes the given inner product, and $\|v\| = \sqrt{\langle v, v \rangle}$ is its induced norm. For a weighted inner product, we sometimes write $\langle v, w \rangle_G$ and $\|v\|_G$.
- $v \perp w$ means $\langle v, w \rangle = 0$; $S^\perp$ is the set of all vectors orthogonal to every element of S.
- δ_{ij} is the Kronecker delta: it is 1 when $i = j$ and 0 otherwise. Thus $\langle e_i, e_j \rangle = \delta_{ij}$ expresses orthonormality.
- P_W is the orthogonal projection map onto W; $P_W(v)$ is the projection of v onto W, and $v - P_W(v)$ is the orthogonal residual.
- $V = W \oplus W^\perp$ means that every $v \in V$ has a unique decomposition $v = w + z$ with $w \in W$ and $z \in W^\perp$.
- $[v]_B$ is the coordinate vector of v relative to the **ordered** basis B; the basis order determines the coordinate order.
- $G \succ 0$ and $G \succeq 0$ denote a positive definite Hermitian matrix and a positive semidefinite Hermitian matrix, respectively.

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DEFINITION 14.1

Let V be an inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. We use linearity in the first argument and conjugate linearity in the second. Thus, for $u \neq 0$, the projection coefficient onto $\operatorname{span}\{u\}$ is

$$\frac{\langle v, u\rangle}{\langle u, u\rangle},$$

with this order of the arguments in the numerator. The denominator is strictly positive. If V is finite dimensional and $W \leq V$, then $P_W(v)$ is uniquely characterized by the two conditions together:

$$P_W(v) \in W, \qquad v - P_W(v) \in W^{\perp}.$$

If $(e_1, \ldots, e_k)$ is an orthonormal basis of W, then $P_W(v) = \sum_{j=1}^{k} \langle v, e_j\rangle e_j$. For an orthogonal basis that is not normalized, divide each coefficient by $\|e_j\|^2$. Membership alone is insufficient, and orthogonality alone is also insufficient.

MOTIVATION AND GUIDING IDEA

We study how an inner product measures lengths and angles, how to choose an orthonormal basis, and how orthogonal projection decomposes a vector into a component in a subspace and a component perpendicular to it. We connect algebraic computation with geometry and compare different inner products.

PROPOSITION 14.1

Angle in a real space. If $\mathbb{F} = \mathbb{R}$ and $u, v \neq 0$, the number

$$\cos\zeta := \frac{\langle u, v\rangle}{\|u\|\|v\|}$$

satisfies $\cos\zeta \in [-1, 1]$. In particular, there is $\zeta \in [0, \pi]$ for which this equality holds.

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14.1 Orthogonality

DEFINITION 14.2

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space. Vectors $v, w \in V$ are **orthogonal** if $\langle v, w\rangle = 0$; we write $v \perp w$.

PROPOSITION 14.2

If $v \perp b$ for every $b \in B$, then $v \perp \operatorname{span}(B)$. Indeed, for any linear combination $s = \sum_{j=1}^{k} \alpha_j b_j$,

$$\langle v, s\rangle = \sum_{j=1}^{k} \overline{\alpha_j}\langle v, b_j\rangle = 0.$$

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DEFINITION 14.3

A set $B = \{v_1, \dots, v_n\} \subseteq V$ is **orthogonal** if

$$\langle v_i, v_j \rangle = 0 \qquad \text{for all } i \neq j.$$

If, in addition, $\|v_i\| = 1$ for every i, it is **orthonormal**. In this case,

$$\langle v_i, v_j \rangle = \delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

If B is also a basis of V, we call it an orthogonal basis or an orthonormal basis, respectively.

RECALL AND TOOLS

Normalization: an orthogonal set can be converted into an orthonormal set: if $\{v_1, \dots, v_n\}$ is orthogonal and contains no zero vector, then $\left\{ \frac{v_1}{\|v_1\|}, \dots, \frac{v_n}{\|v_n\|} \right\}$ is orthonormal. Indeed, for every $1 \leq i \leq n$, $\left\langle \frac{v_i}{\|v_i\|}, \frac{v_i}{\|v_i\|} \right\rangle = \frac{1}{\|v_i\|^2} \langle v_i, v_i \rangle = 1$.

PROPOSITION 14.3

- An **orthogonal** set containing no zero vector is linearly independent.
- An **orthonormal** set is linearly independent. (Since each of its elements has norm 1, it contains no zero vector.)
- The converse fails: for example, $\left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$ with the standard inner product is linearly independent but is neither orthogonal nor orthonormal.

EXERCISE 14.1

Suppose $u, v \in \mathbb{R}^n$ and $a = 4v - 3u$, $b = u - v$ are orthonormal. Please compute $\|u\|$, $\|v\|$, and $\langle u, v \rangle$, and please prove that u, v are linearly independent.

WORKED SOLUTION

From $u = v + b$ we obtain $a = v - 3b$, hence

$$v = a + 3b, \qquad u = a + 4b.$$

Using $\|a\| = \|b\| = 1$ and $\langle a, b \rangle = 0$ gives

$$\|v\|^2 = 1 + 9 = 10, \qquad \|u\|^2 = 1 + 16 = 17, \qquad \langle u, v \rangle = 1 + 12 = 13.$$

Thus $\|v\| = \sqrt{10}$ and $\|u\| = \sqrt{17}$. If $\alpha u + \beta v = 0$, then

$$(\alpha + \beta)a + (4\alpha + 3\beta)b = 0.$$

Independence of a, b gives $\alpha + \beta = 4\alpha + 3\beta = 0$. Subtracting three times the first equation from the second gives $\alpha = 0$, then $\beta = 0$.

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14.2 Coordinates in an Orthogonal or Orthonormal Basis

DEFINITION 14.4

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. Let $B = (b_1, \ldots, b_n)$ be an ordered basis of V.
Every $v \in V$ has a unique representation as a linear combination of the basis vectors: $v = \alpha_1 b_1 + \cdots + \alpha_n b_n$.
If B is an **orthogonal** basis, its coefficients are given by $\alpha_i = \frac{\langle v, b_i\rangle}{\|b_i\|^2}$.
If B is an **orthonormal** basis, then $\|b_i\|^2 = 1$, so $\alpha_i = \langle v, b_i\rangle$. Thus the coordinate vector relative to B is

$$[v]_B = \begin{pmatrix} \langle v, b_1\rangle \\ \vdots \\ \langle v, b_n\rangle \end{pmatrix}$$

EXERCISE 14.2

In $\mathbb{C}^3$ with $\langle x, y\rangle = x^T\overline{y}$, let

$$b_1 = \tfrac{1}{13}(5, 12i, 0)^T, \quad b_2 = \tfrac{1}{13}(12, -5i, 0)^T, \quad b_3 = (0, 0, 1)^T.$$

Please show that $B = (b_1, b_2, b_3)$ is an orthonormal basis and please find $[v]_B$ for $v = (-2, 3i, 4)^T$.

WORKED SOLUTION

The first two squared norms are $(25 + 144)/169 = 1$, and the third is 1. Also,

$$\langle b_1, b_2\rangle = \frac{5 \cdot 12 + (12i)(5i)}{169} = 0.$$

The products with b_3 vanish, so these three vectors form an orthonormal basis. With linearity in the first slot, the coordinates are

$$\langle v, b_1\rangle = \frac{-10 + (3i)(-12i)}{13} = 2, \qquad \langle v, b_2\rangle = \frac{-24 + (3i)(5i)}{13} = -3, \qquad \langle v, b_3\rangle = 4.$$

Thus $[v]_B = (2, -3, 4)^T$. Reconstruction gives

$$2b_1 - 3b_2 + 4b_3 = \left(\frac{10 - 36}{13}, \frac{24i + 15i}{13}, 4\right)^T = (-2, 3i, 4)^T.$$

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14.3 Orthogonal Complements and Orthogonal Projections

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DEFINITION 14.5

Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. For $S \subseteq V$, define the **orthogonal complement** of S by

$$S^{\perp} := \{\, v \in V : \ \forall s \in S,\ \langle v, s \rangle = 0 \,\}.$$

In particular, if $W \subseteq V$ is a subspace, then $W^{\perp}$ is the **orthogonal complement** of W.

DEFINITION 14.6

Let $U, W \subseteq V$ be subspaces. We say that U and W are **orthogonal** (written $U \perp W$) if

$$\forall u \in U \, \forall w \in W \qquad \langle u, w \rangle = 0.$$

PROPOSITION 14.4

Let $S \subseteq V$, and let $U, W \subseteq V$ be subspaces. The following statements hold:

1. $S^{\perp}$ is a subspace of V.
2. If $S \subseteq T \subseteq V$, then $T^{\perp} \subseteq S^{\perp}$.
3. $(\operatorname{span} S)^{\perp} = S^{\perp}$.
4. $W \cap W^{\perp} = \{0\}$.
5. $U \perp W \iff U \subseteq W^{\perp} \iff W \subseteq U^{\perp}$.
6. $(U + W)^{\perp} = U^{\perp} \cap W^{\perp}$.
7. If V is finite dimensional, then $(U \cap W)^{\perp} = U^{\perp} + W^{\perp}$.
8. If $U \perp W$, then $U + W = U \oplus W$.
9. $W \subseteq (W^{\perp})^{\perp}$. If V is finite dimensional, then $(W^{\perp})^{\perp} = W$.

RECALL AND TOOLS

For all projection results below, assume that V is finite dimensional. In infinite dimensions, closedness of W alone is insufficient in a general inner product space: the projection theorem holds when V is complete in the induced norm, that is, a Hilbert space, and W is closed. Alternatively, it suffices that W be finite dimensional, without assuming completeness of V.

DEFINITION 14.7

Suppose V is finite dimensional and $W \subseteq V$ is a subspace. The **orthogonal projection onto** W is the map $P_W : V \to W$ defined by requiring that, for every $v \in V$, $P_W(v) \in W$ be the unique vector satisfying

$$v - P_W(v) \in W^{\perp}.$$

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PROPOSITION 14.5

Suppose V is finite dimensional and $W \subseteq V$ is a subspace. The following hold:

1. For every $v \in V$, there are unique $w \in W$ and $z \in W^\perp$ such that
$$v = w + z, \qquad P_W(v) = w.$$
In particular, $V = W \oplus W^\perp$.
2. P_W is linear.
3. $$P_W^2 = P_W, \qquad \mathrm{Im}(P_W) = W, \qquad \ker(P_W) = W^\perp.$$
4. For every $v \in V$,
$$P_W(v) \perp \big(v - P_W(v)\big), \qquad \|v\|^2 = \|P_W(v)\|^2 + \|v - P_W(v)\|^2.$$
5. For all $u, v \in V$,
$$\langle P_W(u), v\rangle = \langle u, P_W(v)\rangle.$$
6. For every $v \in V$, $\|P_W(v)\| \le \|v\|$.

RECALL AND TOOLS

If $\dim V < \infty$ and $\{e_1, \ldots, e_k\}$ is an orthonormal basis of W, then for every $v \in V$,

$$P_W(v) = \sum_{i=1}^{k} \langle v, e_i\rangle\, e_i,$$

and, in particular,

$$W^\perp = \{\, v \in V : \ \langle v, e_i\rangle = 0 \ \forall i = 1, \ldots, k \,\}.$$

PROPOSITION 14.6

If $\dim V < \infty$ and $W \subseteq V$ is a subspace, then

$$\dim W + \dim W^\perp = \dim V.$$

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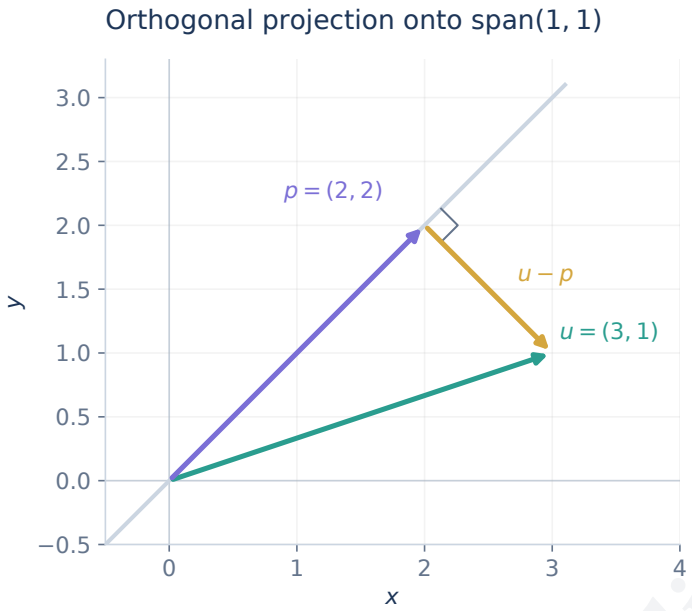


Figure 14.1. Orthogonal projection decomposes a vector into a component in the subspace and a perpendicular residual. The two components satisfy the Pythagorean identity.

EXERCISE 14.3

Let $y = (0, 10)^T$, $u = (3, 1)^T$, and $W = \text{span}\{u\}$. For $\langle x, z\rangle_M = x^T M z$ with $M = \text{diag}(1, 9)$, please compute $P_W^M(y)$ and please check the residual's orthogonality. Please compare with the standard inner product.

WORKED SOLUTION

The matrix M is positive definite because $x^T M x = x_1^2 + 9x_2^2 > 0$ for $x \neq 0$. The projection coefficient is

$$t_M = \frac{y^T M u}{u^T M u} = \frac{0 \cdot 3 + 9 \cdot 10}{3^2 + 9} = 5.$$

Thus $P_W^M(y) = (15, 5)^T$ and $r_M = (-15, 5)^T$. Indeed,

$$r_M^T M u = -15 \cdot 3 + 9 \cdot 5 = 0.$$

For the standard inner product,

$$t_I = \frac{y^T u}{u^T u} = \frac{10}{10} = 1, \qquad P_W^I(y) = (3, 1)^T, \qquad r_I = (-3, 9)^T.$$

Again $r_I^T u = -3 \cdot 3 + 9 = 0$. In each case the projection belongs to W and the residual is orthogonal to W for the chosen inner product.

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Additional practice

EXERCISE 14.4 FOUNDATION

On $\mathbb{R}^2$, let $\langle x, y\rangle_G = 3x_1y_1 + x_2y_2$. Let $u = (1,1)^T$, $v = (2,0)^T$, and $W = \operatorname{span}\{u\}$. Please compute $p = P_W(v)$ and $r = v - p$, and please check orthogonality and the Pythagorean identity.

WORKED SOLUTION

The matrix $G = \operatorname{diag}(3,1)$ is positive definite. We have

$$\langle v, u\rangle_G = 6, \qquad \langle u, u\rangle_G = 4,$$

so

$$p = \tfrac{3}{2}u = (3/2, 3/2)^T, \qquad r = (1/2, -3/2)^T.$$

We have $p \in W$ and $\langle r, u\rangle_G = 3/2 - 3/2 = 0$. Finally,

$$\|v\|_G^2 = 12, \quad \|p\|_G^2 = 3 \cdot \tfrac{9}{4} + \tfrac{9}{4} = 9, \quad \|r\|_G^2 = 3 \cdot \tfrac{1}{4} + \tfrac{9}{4} = 3,$$

and indeed $12 = 9 + 3$.

14.4 Best Approximation

NOTATION AND CONVENTIONS

Let V be an inner product space over $\mathbb{R}$ or $\mathbb{C}$, and let $W \leq V$. The distance from $v \in V$ to W is

$$d(v, W) = \inf\{\|v - w\| : w \in W\}.$$

The symbol inf means the greatest lower bound. Under the hypotheses below, an element of W attains this distance, so it is a minimum. Over $\mathbb{C}$, the inner product is linear in the first variable.

PROPOSITION 14.7

Suppose W is finite-dimensional. Every $v \in V$ has a unique closest element $p \in W$. It is the orthogonal projection $p = P_W v$, and for every $w \in W$,

$$\|v - w\|^2 = \|v - p\|^2 + \|p - w\|^2.$$

In particular, $d(v, W) = \|v - P_W v\|$.

Proof. Choose an orthonormal basis $e_1, \ldots, e_d$ of W. Such a basis can be obtained by the Gram-Schmidt process below. Define

$$p = \sum_{j=1}^{d} \langle v, e_j\rangle e_j.$$

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Then $p \in W$, and for every k,

$$\langle v - p, e_k \rangle = \langle v, e_k \rangle - \sum_{j=1}^{d} \langle v, e_j \rangle \langle e_j, e_k \rangle = 0.$$

Thus $v - p \perp W$. If $w \in W$, then $p - w \in W$, so

$$\begin{aligned} \|v - w\|^2 &= \|(v - p) + (p - w)\|^2 \\ &= \|v - p\|^2 + \|p - w\|^2 + 2\,\mathrm{Re}\langle v - p, p - w \rangle \\ &= \|v - p\|^2 + \|p - w\|^2. \end{aligned}$$

The second term is nonnegative and vanishes exactly when $w = p$. Hence p is the unique closest element. For $W = \{0\}$, the same calculation uses an empty sum and $p = 0$.

EXAMPLE 14.1

In $\mathbb{R}^3$ with its standard inner product, let

$$v = (3, 0, 2)^T, \qquad W = \{(a, b, a + b)^T : a, b \in \mathbb{R}\}.$$

The vector $n = (-1, -1, 1)^T$ is orthogonal to every element of W, since $-a - b + (a + b) = 0$. The component of v in the direction of n is

$$r = \frac{\langle v, n \rangle}{\langle n, n \rangle} n = \frac{-3 + 0 + 2}{3} n = \left(\frac{1}{3}, \frac{1}{3}, -\frac{1}{3} \right)^T.$$

Hence

$$p = v - r = \left(\frac{8}{3}, -\frac{1}{3}, \frac{7}{3} \right)^T.$$

Indeed $8/3 - 1/3 = 7/3$, so $p \in W$. Since $v - p = r \perp W$, this is the projection. We have

$$d(v, W)^2 = \|r\|^2 = \frac{1}{9} + \frac{1}{9} + \frac{1}{9} = \frac{1}{3}.$$

For every $w \in W$, the squared distance is $1/3 + \|p - w\|^2$, whose minimum is $1/3$.

EXERCISE 14.5

Let $v_1, \ldots, v_m \in V$, where $m \geq 1$, and let $W \leq V$ be finite-dimensional. Write $\bar{v} = \frac{1}{m} \sum_{j=1}^{m} v_j$. Please find the unique $w \in W$ minimizing

$$\sum_{j=1}^{m} \|v_j - w\|^2.$$

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EXERCISE 14.5 CONTINUED

WORKED SOLUTION

Write $v_j - w = (v_j - \bar{v}) + (\bar{v} - w)$. Expand each squared norm and sum:

$$\sum_{j=1}^{m} \|v_j - w\|^2 = \sum_{j=1}^{m} \|v_j - \bar{v}\|^2 + m\|\bar{v} - w\|^2 + 2\,\mathrm{Re}\left\langle \sum_{j=1}^{m} (v_j - \bar{v}), \bar{v} - w \right\rangle.$$

But $\sum_j (v_j - \bar{v}) = \sum_j v_j - m\bar{v} = 0$. Therefore

$$\sum_{j=1}^{m} \|v_j - w\|^2 = \sum_{j=1}^{m} \|v_j - \bar{v}\|^2 + m\|\bar{v} - w\|^2.$$

The first sum does not depend on w. By best approximation, the second term has the unique minimizer

$$w = P_W \bar{v}.$$

The minimum value is

$$\sum_{j=1}^{m} \|v_j - \bar{v}\|^2 + m\|\bar{v} - P_W \bar{v}\|^2.$$

EXERCISE 14.6

Let V be a finite-dimensional inner product space, and let $U, W \leq V$. Please prove that

$$d(x, U)^2 + d(x, W)^2 = \|x\|^2 \quad \text{for every } x \in V$$

if and only if $W = U^{\perp}$.

WORKED SOLUTION

First suppose $W = U^{\perp}$. Decompose $x = u + w$ with $u \in U$ and $w \in W$. The components are orthogonal, so

$$d(x, U)^2 + d(x, W)^2 = \|w\|^2 + \|u\|^2 = \|x\|^2.$$

Conversely, assume the identity for every x. For $u \in U$, we have $d(u, U) = 0$, hence $d(u, W)^2 = \|u\|^2$. Pythagoras for projection onto W gives

$$\|u\|^2 = \|P_W u\|^2 + d(u, W)^2.$$

Thus $\|P_W u\|^2 = 0$, so $P_W u = 0$ and $u \perp W$. We have proved $U \perp W$.
If $z \in (U + W)^{\perp}$, both projections of z vanish. The assumed identity gives

$$\|z\|^2 + \|z\|^2 = \|z\|^2,$$

so $z = 0$. Hence $(U + W)^{\perp} = \{0\}$. Finite dimension gives $U + W = V$. Since $U \perp W$, this is the orthogonal decomposition of V, and therefore $W = U^{\perp}$.

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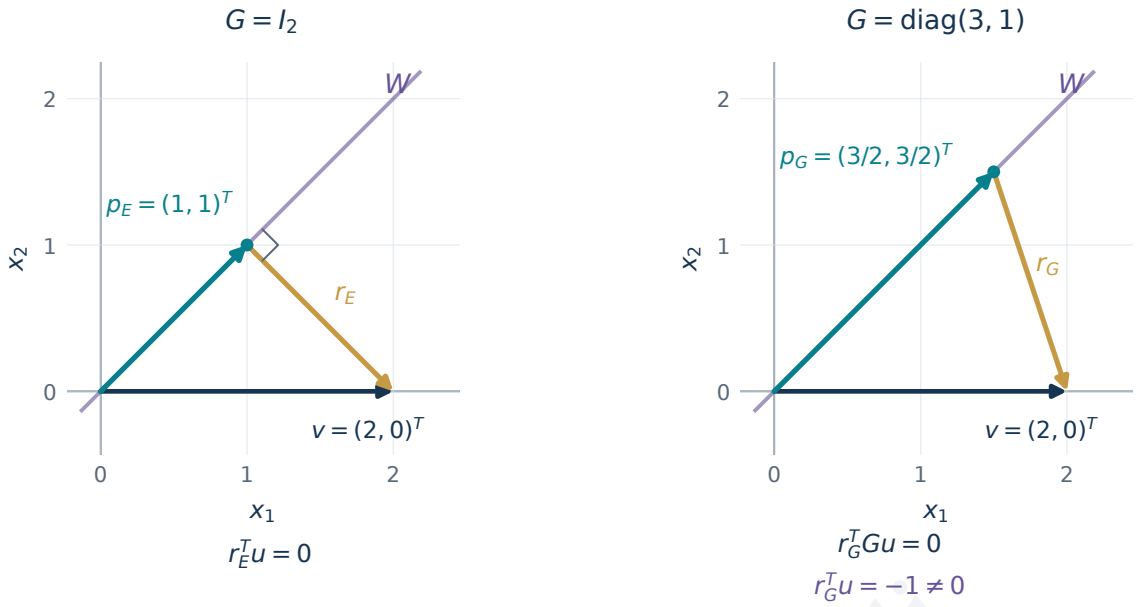


Figure 14.2. Both panels use $v = (2, 0)^T$ and $W = \mathrm{span}\{u\}$, where $u = (1, 1)^T$. On the left, the standard inner product gives $p_E = (1, 1)^T$; on the right, $\langle x, y \rangle_G = x^T G y$ with $G = \mathrm{diag}(3, 1)$ gives $p_G = (3/2, 3/2)^T$. Gold arrows are the residuals. On the right, $r_G^T G u = 0$, whereas $r_G^T u = -1 \neq 0$: orthogonality is with respect to the weighted inner product, so no Euclidean right-angle marker appears there.

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EXERCISE 14.7 CHALLENGE

Let V be a finite-dimensional inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and let $U, W \leq V$. Please write $P = P_U$ and $Q = P_W$, viewed as maps from V to itself.

1. Please prove that PQ is the orthogonal projection onto $U \cap W$ if and only if $PQ = QP$.
2. In $\mathbb{R}^2$ with its standard inner product, take $U = \text{span}\{(1,0)^T\}$ and $W = \text{span}\{(1,2)^T\}$. Please compute PQ and please check whether it is an orthogonal projection.

WORKED SOLUTION

1. Use $\langle Px, y\rangle = \langle x, Py\rangle$, valid for every orthogonal projection, and the corresponding identity for Q. First suppose $PQ = QP$, and set $R = PQ$. For each $x \in V$, the vector Rx belongs to U; since $Rx = QPx$, it also belongs to W. For every $z \in U \cap W$, we have $Pz = Qz = z$, so $Rz = z$. Furthermore,

$$\langle Rx, z\rangle = \langle PQx, z\rangle = \langle Qx, Pz\rangle = \langle x, QPz\rangle = \langle x, z\rangle.$$

Thus $x - Rx \perp U \cap W$. The two defining conditions for projection show that $R = P_{U\cap W}$. Conversely, suppose $R = PQ = P_{U\cap W}$. For all $x, y \in V$,

$$\langle x, QPy\rangle = \langle PQx, y\rangle = \langle Rx, y\rangle = \langle x, Ry\rangle = \langle x, PQy\rangle.$$

Hence $\langle x, (QP - PQ)y\rangle = 0$ for all x, y. For fixed y, choose $x = (QP - PQ)y$; positive definiteness gives $(QP - PQ)y = 0$. Therefore $QP = PQ$.

2. Here

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad Q = \frac{1}{5}\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}, \qquad PQ = \frac{1}{5}\begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}.$$

However,

$$(PQ)^2 = \frac{1}{25}\begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix} \neq PQ.$$

Thus PQ is not an orthogonal projection. Here $U \cap W = \{0\}$, whereas $PQ \neq 0$.

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EXERCISE 14.8 CHALLENGE

Let $a_1, \dots, a_m$ lie in an n-dimensional real inner product space, where $n \geq 1$, and assume $\|a_i - a_j\| = 1$ for all $i \neq j$. Please prove that $m \leq n+1$. Please construct an example attaining the bound in an n-dimensional subspace of $\mathbb{R}^{n+1}$.

WORKED SOLUTION

The case $m = 1$ is immediate. Otherwise put $v_i = a_i - a_m$ for $i < m$. Then $\|v_i\| = 1$, and for $i \neq j$,

$$1 = \|v_i - v_j\|^2 = 2 - 2\langle v_i, v_j \rangle.$$

Thus $\langle v_i, v_j \rangle = 1/2$. For real scalars c_i,

$$\left\| \sum_i c_i v_i \right\|^2 = \frac{1}{2} \sum_i c_i^2 + \frac{1}{2} \left(\sum_i c_i \right)^2 .$$

A zero combination therefore has every $c_i = 0$. The $m-1$ vectors are independent, giving $m - 1 \leq n$. For an example, let $\mathbf{1} = (1, \dots, 1)^T \in \mathbb{R}^{n+1}$ and take

$$H = \{x : x_1 + \cdots + x_{n+1} = 0\}, \qquad a_j = \frac{1}{\sqrt{2}} \left(e_j - \frac{1}{n+1} \mathbf{1} \right).$$

The space H has dimension n, and all $n+1$ points lie in it. For $i \neq j$, $\|a_i - a_j\|^2 = \|e_i - e_j\|^2/2 = 1$, so the bound is attained.

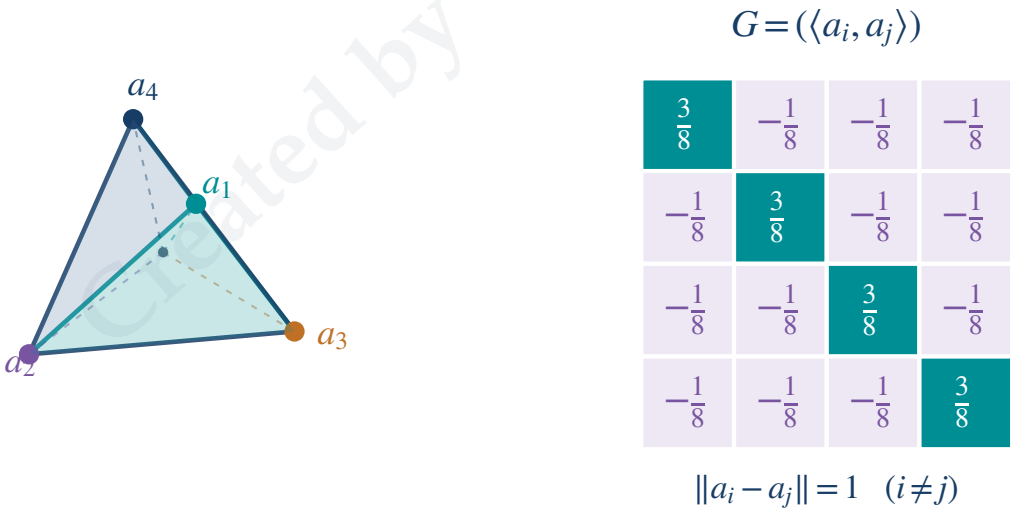


Figure 14.3. In dimension 3, four points can have all pairwise distances equal. The matrix records inner products of the vertices after their centre is placed at zero.

MOTIVATION AND GUIDING IDEA

We construct orthonormal bases by systematically subtracting projections and compute projections in polynomial spaces. We study the conditions for orthogonal diagonalization over the reals and unitary diagonalization over the complex numbers, and use an orthonormal eigenbasis to prove normality.

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14.5 Review and Notation

RECALL AND TOOLS

Throughout this tutorial, the inner product space is over $\mathbb{R}$ or $\mathbb{C}$; in the complex case, our convention is:

1. $\langle x, x\rangle \geq 0$, with $\langle x, x\rangle = 0 \iff x = 0$.
2. $\langle x, y\rangle = \overline{\langle y, x\rangle}$.
3. Linearity in the first argument: $\langle ax + by, z\rangle = a\langle x, z\rangle + b\langle y, z\rangle$.

It follows that the second argument is conjugate linear: $\langle x, ay + bz\rangle = \overline{a}\langle x, y\rangle + \overline{b}\langle x, z\rangle$. This is the same convention as in the preceding tutorial; over $\mathbb{R}$, the two linearity conventions coincide.

DEFINITION **14.8**

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space.

1. Vectors $u, v \in V$ are orthogonal if $\langle u, v\rangle = 0$.
2. A list $(e_1, \ldots, e_k)$ is orthonormal if $\langle e_i, e_j\rangle = \delta_{ij}$ for all i, j.
3. For each $x \in V$, define the norm $\|x\| := \sqrt{\langle x, x\rangle}$.

RECALL AND TOOLS

Orthogonal projection onto a line: if $u \neq 0$, the vector

$$\text{proj}_{\text{span}\{u\}}(v) := \frac{\langle v, u\rangle}{\langle u, u\rangle}\, u$$

is the unique vector of the form αu for which $v - \alpha u \perp u$.

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14.6 The Gram-Schmidt Algorithm

We present Gram-Schmidt with immediate normalization at each step, constructing an orthonormal list directly.

Let $v_1, \ldots, v_k \in V$ be linearly independent. Define recursively:

1. Set $w_1 := v_1$, and then $e_1 := \dfrac{w_1}{\|w_1\|}$.
2. For $j = 2, \ldots, k$, first define
$$w_j := v_j - \sum_{i=1}^{j-1} \langle v_j, e_i\rangle\, e_i,$$
and then normalize:
$$e_j := \frac{w_j}{\|w_j\|}.$$

The main conclusions are:

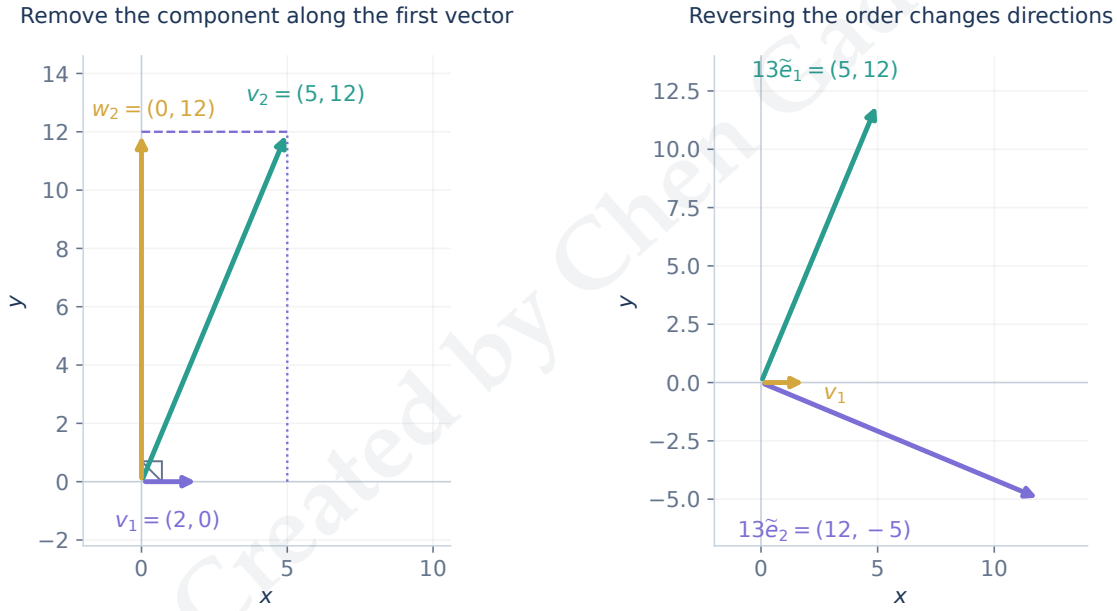
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1. $(e_1, \ldots, e_k)$ is orthonormal.
2. For every m, $\text{span}\{v_1, \ldots, v_m\} = \text{span}\{e_1, \ldots, e_m\}$.

Formal justification: assume inductively that $e_1, \ldots, e_{j-1}$ are orthonormal. For every $r < j$,

$$\begin{aligned}\langle w_j, e_r \rangle &= \left\langle v_j - \sum_{i=1}^{j-1} \langle v_j, e_i \rangle e_i,\ e_r \right\rangle \\ &= \langle v_j, e_r \rangle - \sum_{i=1}^{j-1} \langle v_j, e_i \rangle \langle e_i, e_r \rangle \\ &= \langle v_j, e_r \rangle - \langle v_j, e_r \rangle = 0.\end{aligned}$$

If $w_j = 0$, then v_j belongs to the span of $e_1, \ldots, e_{j-1}$, which by induction equals the span of $v_1, \ldots, v_{j-1}$. This contradicts the linear independence of the input. Thus $w_j \neq 0$, normalization is permitted, and $\|e_j\| = 1$. Also, $w_j \in \text{span}\{v_1, \ldots, v_j\}$, and the subtraction formula implies $v_j \in \text{span}\{e_1, \ldots, e_{j-1}, w_j\}$. Hence the corresponding prefix spans agree at every step.



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Figure 14.4. At each step, subtract from the next vector its components in the directions already constructed. The residual is orthogonal to the preceding directions; normalization produces the next unit vector.

EXERCISE 14.9

Please apply Gram-Schmidt with normalization to

$$v_1 = (5, 12, 0)^T, \qquad v_2 = (1, 5, 0)^T, \qquad v_3 = (2, -3, 4)^T.$$

Please check that the output is orthonormal.

WORKED SOLUTION

First $\|v_1\| = 13$, so $e_1 = (5, 12, 0)^T/13$. We have

$$\langle v_2, e_1 \rangle = \frac{5 + 60}{13} = 5, \qquad w_2 = v_2 - 5e_1 = \tfrac{1}{13}(-12, 5, 0)^T.$$

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EXERCISE 14.9 CONTINUED

The norm of w_2 is 1, so $e_2 = w_2$. At the third step,

$$\langle v_3, e_1 \rangle = \frac{10 - 36}{13} = -2, \qquad \langle v_3, e_2 \rangle = \frac{-24 - 15}{13} = -3.$$

Hence

$$w_3 = v_3 + 2e_1 + 3e_2 = (0, 0, 4)^T, \qquad e_3 = (0, 0, 1)^T.$$

The first two squared norms are $169/169 = 1$, and their inner product is $(-60 + 60)/169 = 0$. Both are orthogonal to e_3, whose norm is also 1.

EXERCISE 14.10

In $\mathbb{C}^2$ with $\langle x, y \rangle = x^T \overline{y}$, let $v_1 = (4, 3i)^T$ and $v_2 = (2, i)^T$. Please apply Gram-Schmidt with normalization.

WORKED SOLUTION

First $\|v_1\|^2 = 16 + 9 = 25$, so $e_1 = (4, 3i)^T/5$. With linearity in the first slot,

$$\langle v_2, e_1 \rangle = 2 \cdot \tfrac{4}{5} + i \cdot \tfrac{-3i}{5} = \tfrac{11}{5}.$$

Thus

$$w_2 = v_2 - \tfrac{11}{5} e_1 = \tfrac{1}{25}(6, -8i)^T, \qquad \|w_2\| = \tfrac{2}{5}, \qquad e_2 = \tfrac{1}{5}(3, -4i)^T.$$

Indeed, $\|e_2\|^2 = (9 + 16)/25 = 1$ and $\langle e_1, e_2 \rangle = (12 + (3i)(4i))/25 = 0$.

EXERCISE 14.11

On $\mathbb{R}[t]_{\leq 2}$, let $\langle p, q \rangle = \int_{-1}^{1} p(t) q(t)\, dt$. Please apply Gram-Schmidt to $(1, t, t^2)$, normalizing at every step.

WORKED SOLUTION

First $\|1\|^2 = 2$, so $e_1 = 1/\sqrt{2}$. Odd functions integrate to zero on a symmetric interval, hence

$$\langle t, e_1 \rangle = 0, \qquad \|t\|^2 = \int_{-1}^{1} t^2 dt = \tfrac{2}{3}, \qquad e_2 = \sqrt{\tfrac{3}{2}}\, t.$$

At the third step,

$$\langle t^2, e_1 \rangle = \tfrac{2}{3\sqrt{2}}, \qquad \langle t^2, e_2 \rangle = 0, \qquad w_3 = t^2 - \tfrac{1}{3}.$$

Its squared norm is

$$\|w_3\|^2 = \int_{-1}^{1} \left(t^4 - \tfrac{2}{3} t^2 + \tfrac{1}{9}\right) dt = \tfrac{2}{5} - \tfrac{4}{9} + \tfrac{2}{9} = \tfrac{8}{45}.$$

Therefore

$$e_3 = \sqrt{\tfrac{45}{8}} \left(t^2 - \tfrac{1}{3}\right) = \sqrt{\tfrac{5}{8}} (3t^2 - 1).$$

These are the first three Legendre polynomials, normalized for this inner product.

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EXERCISE 14.12

In $\mathbb{R}^2$, please compare Gram-Schmidt for the orders (v_1, v_2) and (v_2, v_1), where $v_1 = (2,0)^T$ and $v_2 = (5,12)^T$.

WORKED SOLUTION

The first order gives $e_1 = (1,0)^T$, followed by residual $w_2 = (0,12)^T$ and $e_2 = (0,1)^T$. For the reverse order,

$$\widetilde{e}_1 = \tfrac{1}{13}(5,12)^T, \qquad \langle v_1, \widetilde{e}_1 \rangle = \tfrac{10}{13},$$

so

$$\widetilde{w}_2 = (2,0)^T - \tfrac{10}{169}(5,12)^T = \tfrac{1}{169}(288,-120)^T.$$

Its norm is $24/13$, hence $\widetilde{e}_2 = (12,-5)^T/13$. Both outputs are orthonormal and span the same plane, but their directions differ.

EXERCISE 14.13

Let $V = \mathbb{C}[x]_{\leq 3}$ and define

$$\langle f, g \rangle = \sum_{t \in \{-3,-1,1,3\}} f(t)\overline{g(t)}.$$

Let $W = \mathbb{C}[x]_{\leq 2}$. Please prove that this is an inner product, please find an orthonormal basis of W, and please find $W^{\perp}$.

WORKED SOLUTION

Linearity in the first slot and conjugate symmetry follow from the sum. Also $\langle f, f \rangle = \sum_t |f(t)|^2 \geq 0$. Equality gives four distinct roots, so a polynomial of degree at most 3 must vanish identically. Apply Gram-Schmidt to $(1, x, x^2)$. Symmetry gives

$$\sum_t t = \sum_t t^3 = 0, \qquad \sum_t 1 = 4, \qquad \sum_t t^2 = 20, \qquad \sum_t t^4 = 164.$$

Hence

$$e_1 = \tfrac{1}{2}, \qquad e_2 = \tfrac{x}{\sqrt{20}}, \qquad w_3 = x^2 - \frac{20}{4} = x^2 - 5.$$

The values of w_3 are $4, -4, -4, 4$, so $\|w_3\|^2 = 64$ and $e_3 = (x^2 - 5)/8$. Thus (e_1, e_2, e_3) is an orthonormal basis of W. The polynomial

$$u(x) = 5x^3 - 41x$$

is odd, so it is orthogonal to $1, x^2$. Moreover,

$$\langle u, x \rangle = 5\sum_t t^4 - 41\sum_t t^2 = 5 \cdot 164 - 41 \cdot 20 = 0.$$

It is nonzero and orthogonal to W. Since $\dim W^{\perp} = 4 - 3 = 1$,

$$W^{\perp} = \operatorname{span}\{5x^3 - 41x\}.$$

Its values are $-12, 36, -36, 12$, so $\|u\|^2 = 2880$; division by $24\sqrt{5}$ normalizes it.

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EXERCISE 14.14

Continue with V, W and the preceding inner product. Let P_W be orthogonal projection onto W.

1. Please prove that P_W is linear and $P_W^2 = P_W$.
2. Please compute $P_W(x^3)$ using the orthonormal basis you found.
3. Define $(Tq)(x) = q(x+2) - q(x)$. Please find all $q \in W$ with $Tq \in W^\perp$, and please find the unique solution satisfying $\langle q, 1\rangle = 2$. Please determine whether $\|q\| = 1$ guarantees uniqueness.

WORKED SOLUTION

1. For the orthogonal decomposition $f = w + u$ with $w \in W$, $u \in W^\perp$, we have $P_W f = w$. Combining the decompositions of f, g gives $P_W(\alpha f + \beta g) = \alpha P_W f + \beta P_W g$. Since P_W is the identity on W, we have $P_W^2 = P_W$.
2. Symmetry gives $\langle x^3, e_1\rangle = \langle x^3, e_3\rangle = 0$, while
$$\langle x^3, e_2\rangle = \frac{164}{\sqrt{20}}.$$
Thus
$$P_W(x^3) = \frac{164}{\sqrt{20}}\frac{x}{\sqrt{20}} = \frac{41}{5}x.$$
The residual $x^3 - P_W(x^3) = (5x^3 - 41x)/5 \in W^\perp$ confirms the answer.
3. For $q(x) = ax^2 + bx + c$,
$$(Tq)(x) = a(x+2)^2 + b(x+2) + c - (ax^2 + bx + c) = 4ax + 4a + 2b.$$
Thus $Tq \in W$. Requiring $Tq \in W^\perp$ forces $Tq = 0$, hence $a = b = 0$. All solutions are constants $q = c$. The condition $\langle q, 1\rangle = 4c = 2$ gives $q = 1/2$. In contrast, $\|q\|^2 = 4|c|^2 = 1$ gives $|c| = 1/2$: there are two real solutions and infinitely many complex ones.

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Additional practice

EXERCISE 14.15 FOUNDATION

In $\mathbb{R}^3$ with the standard inner product, consider the list
$$v_1 = (1,1,0)^T, \qquad v_2 = (3,3,0)^T, \qquad v_3 = (1,0,1)^T.$$
Please apply Gram-Schmidt, explicitly handling any zero residual. Please find and please check an orthonormal basis of the input span.

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EXERCISE 14.15 FOUNDATION CONTINUED

WORKED SOLUTION

First, $\|v_1\| = \sqrt{2}$, so

$$e_1 = \frac{1}{\sqrt{2}}(1, 1, 0)^T.$$

For the second input,

$$\langle v_2, e_1 \rangle = 3\sqrt{2}, \qquad w_2 = v_2 - 3\sqrt{2}\, e_1 = 0.$$

We cannot normalize w_2; we skip it because $v_2 = 3v_1$. For the third input,

$$\langle v_3, e_1 \rangle = \frac{1}{\sqrt{2}}, \qquad w_3 = v_3 - \frac{1}{\sqrt{2}} e_1 = (1/2, -1/2, 1)^T.$$

We have $\|w_3\|^2 = 1/4 + 1/4 + 1 = 3/2$, so the second output vector is

$$e_2 = \frac{1}{\sqrt{6}}(1, -1, 2)^T.$$

A direct check gives

$$\|e_1\|^2 = \frac{1+1}{2} = 1, \qquad \|e_2\|^2 = \frac{1+1+4}{6} = 1, \qquad \langle e_1, e_2 \rangle = \frac{1-1}{\sqrt{12}} = 0.$$

The vectors v_1, v_3 are linearly independent: in $av_1 + bv_3 = 0$, the third coordinate gives $b = 0$, and then $a = 0$. Thus the input span has dimension 2, and (e_1, e_2) is its orthonormal basis. It cannot contain three orthonormal vectors, since those vectors would be linearly independent.

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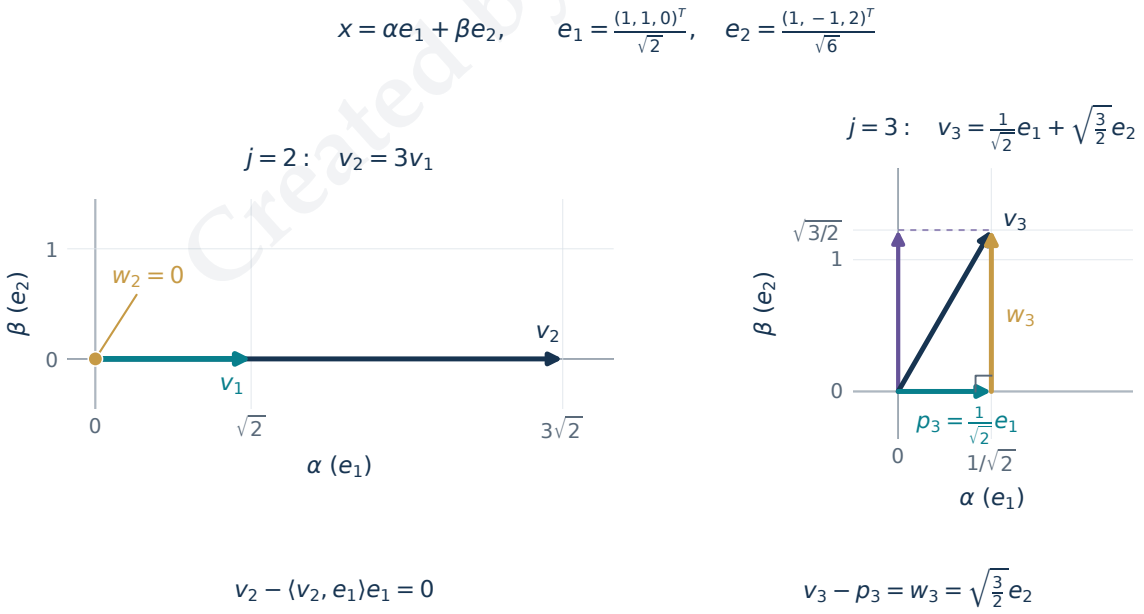


Figure 14.5. The axes α, β are coordinates in the orthonormal basis (e_1, e_2) of the plane $\text{span}\{v_1, v_3\}$: each point represents $x = \alpha e_1 + \beta e_2$. The input coordinates in this chart are $(\sqrt{2}, 0)^T$, $(3\sqrt{2}, 0)^T$, and $(1/\sqrt{2}, \sqrt{3/2})^T$, respectively. On the left, $v_2 = 3v_1$, so the residual $w_2 = 0$ must be skipped. On the right, subtract $p_3 = \langle v_3, e_1 \rangle e_1$ from v_3; the residual $w_3 = (1/2, -1/2, 1)^T = \sqrt{3/2}\, e_2$ is orthogonal to e_1.

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EXERCISE 14.16 EXAM QUESTION

1. Work in $M_{3\times4}(\mathbb{R})$. Define

$$C = \operatorname{diag}(2,0,1), \qquad D = \operatorname{diag}(2,0,0,3),$$

and

$$W = \{X \in M_{3\times4}(\mathbb{R}) : CXD = 0\}.$$

Please prove that W is a subspace of $M_{3\times4}(\mathbb{R})$.

2. On $M_{3\times4}(\mathbb{R})$, define

$$\langle X, Y\rangle = \operatorname{tr}(X^T QY), \qquad Q = \operatorname{diag}(2,3,5).$$

Please prove that this is an inner product. In particular, please verify linearity in the second argument.

3. Now work in $\mathbb{R}^4$ with the standard inner product. Let

$$M = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \qquad W = N(M), \qquad v = \begin{pmatrix} 2 \\ -1 \\ 4 \\ 3 \end{pmatrix}.$$

Please compute the orthogonal projection $P_W(v)$.

WORKED SOLUTION

1. Define $L : M_{3\times4}(\mathbb{R}) \longrightarrow M_{3\times4}(\mathbb{R})$ by $L(X) = CXD$. For all $\alpha, \beta \in \mathbb{R}$ and all X_1, X_2,

$$\begin{aligned} L(\alpha X_1 + \beta X_2) &= C(\alpha X_1 + \beta X_2)D \\ &= \alpha CX_1D + \beta CX_2D \\ &= \alpha L(X_1) + \beta L(X_2). \end{aligned}$$

Thus L is linear and $W = \ker L$, so W is a subspace. The condition can also be read directly: multiplication by C zeros the second row, and multiplication by D zeros the second and third columns. Hence

$$CXD = 0 \quad \Longleftrightarrow \quad x_{11} = x_{14} = x_{31} = x_{34} = 0.$$

These are homogeneous linear conditions on the entries of X.

2. Write $Q = \operatorname{diag}(q_1, q_2, q_3)$, where $(q_1, q_2, q_3) = (2, 3, 5)$. If $X = (x_{ij})$ and $Y = (y_{ij})$, computation of the diagonal entries gives

$$(X^T QY)_{jj} = \sum_{i=1}^{3} x_{ij} q_i y_{ij},$$

$$\langle X, Y\rangle = \sum_{j=1}^{4} \sum_{i=1}^{3} q_i x_{ij} y_{ij}.$$

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EXERCISE 14.16 EXAM QUESTION CONTINUED

For all $\alpha, \beta \in \mathbb{R}$,

$$\begin{aligned}\langle X, \alpha Y + \beta Z\rangle &= \sum_{j=1}^{4}\sum_{i=1}^{3} q_i x_{ij}(\alpha y_{ij} + \beta z_{ij}) \\ &= \alpha\langle X, Y\rangle + \beta\langle X, Z\rangle.\end{aligned}$$

This proves linearity in the second argument. Since all entries are real, the same calculation also proves linearity in the first argument. In addition,

$$\langle X, Y\rangle = \sum_{j=1}^{4}\sum_{i=1}^{3} q_i y_{ij} x_{ij} = \langle Y, X\rangle,$$

so symmetry holds. Finally,

$$\langle X, X\rangle = \sum_{j=1}^{4}(2x_{1j}^2 + 3x_{2j}^2 + 5x_{3j}^2) \geq 0.$$

Equality holds if and only if all entries x_{ij} vanish, or equivalently $X = 0$. We have proved linearity, symmetry, and positive definiteness; thus this is a real inner product.

3. Solve $Mx = 0$:

$$x_1 + x_3 = 0, \qquad x_2 + x_4 = 0.$$

Thus

$$W = \operatorname{span}\{w_1, w_2\}, \qquad w_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}, \quad w_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}.$$

The two vectors are linearly independent and orthogonal, and

$$\begin{aligned}\langle w_1, w_2\rangle &= 0, & \|w_1\|^2 &= \|w_2\|^2 = 2, \\ \langle v, w_1\rangle &= 2 - 4 = -2, & \langle v, w_2\rangle &= -1 - 3 = -4.\end{aligned}$$

The projection formula for an orthogonal basis therefore gives

$$\begin{aligned}P_W(v) &= \frac{\langle v, w_1\rangle}{\|w_1\|^2} w_1 + \frac{\langle v, w_2\rangle}{\|w_2\|^2} w_2 \\ &= -w_1 - 2w_2 = \begin{pmatrix} -1 \\ -2 \\ 1 \\ 2 \end{pmatrix}.\end{aligned}$$

As a check, $MP_W(v) = 0$, whereas

$$v - P_W(v) = \begin{pmatrix} 3 \\ 1 \\ 3 \\ 1 \end{pmatrix}$$

is orthogonal to both w_1 and w_2, since $3 - 3 = 0$ and $1 - 1 = 0$. The projection indeed belongs to W, and the residual is orthogonal to all of W.

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EXERCISE 14.17 EXAM QUESTION

Let $V = \mathbb{R}_4[x]$, the space of real polynomials of degree at most 4, including the zero polynomial. Define $L : V \to \mathbb{R}^3$ by

$$L(p) = (p(0), p(1), p'(1)).$$

Equip V with the inner product

$$\langle p, q \rangle = \int_0^1 p(t)q(t)\,dt + p(0)q(0) + p(1)q(1).$$

Please write $W = \ker L$ and $s(x) = 1 + x + x^2 + x^3 + 2x^4$.

1. Please prove that L is linear.
2. Please compute $\ker L$ and $\operatorname{Im} L$, and please find a basis of each.
3. Please compute the orthogonal projection $P_W(s)$ for the given inner product.

WORKED SOLUTION

1. Let $p, q \in V$ and $a, b \in \mathbb{R}$. By the linearity of evaluation and differentiation,

$$\begin{aligned} L(ap + bq) &= ((ap + bq)(0), (ap + bq)(1), (ap + bq)'(1)) \\ &= (ap(0) + bq(0), ap(1) + bq(1), ap'(1) + bq'(1)) \\ &= aL(p) + bL(q). \end{aligned}$$

   so L is linear.

2. The condition $p \in \ker L$ is equivalent to the three conditions

$$p(0) = 0, \qquad p(1) = 0, \qquad p'(1) = 0.$$

   The first says that x divides p. The last two say that 1 is a root of multiplicity at least 2: if $p(x) = (x-1)q(x)$, then $p'(x) = q(x) + (x-1)q'(x)$, so $p'(1) = q(1)$. Thus $(x-1)^2$ divides p. Since x and $(x-1)^2$ have no common root, and p has degree at most 4,

$$W = \{x(x-1)^2(a + bx) : a, b \in \mathbb{R}\}.$$

   Define

$$w_1 = x(x-1)^2 = x^3 - 2x^2 + x, \qquad w_2 = x^2(x-1)^2 = x^4 - 2x^3 + x^2.$$

   The polynomials w_1, w_2 span W and are linearly independent, for example because they have different degrees. Hence (w_1, w_2) is a basis of W, and $\dim W = 2$.

   To determine the image, exhibit polynomials mapped to the three standard basis vectors:

$$h_1 = (x-1)^2, \qquad h_2 = 2x - x^2, \qquad h_3 = x^2 - x.$$

   A direct calculation gives

$$L(h_1) = (1, 0, 0), \qquad L(h_2) = (0, 1, 0), \qquad L(h_3) = (0, 0, 1).$$

   Thus $\operatorname{Im} L = \mathbb{R}^3$, with the standard basis as a basis of the image. As a dimension check,

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EXERCISE 14.17 EXAM QUESTION CONTINUED

$\dim\ker L + \dim \operatorname{Im} L = 2 + 3 = 5 = \dim V$.

3. Seek $p = P_W(s)$ in the form $p = aw_1 + bw_2$. The definition of orthogonal projection requires $p \in W$ and $s - p \in W^\perp$. Since w_1, w_2 form a basis of W, it suffices to require

$$\langle s - p, w_1\rangle = 0, \qquad \langle s - p, w_2\rangle = 0.$$

Every $w \in W$ satisfies $w(0) = w(1) = 0$, so the endpoint terms in the inner product vanish whenever one factor is w_1 or w_2. First compute the inner products between the basis vectors:

$$\begin{aligned}
\langle w_1, w_1\rangle &= \int_0^1 (t^2 - 4t^3 + 6t^4 - 4t^5 + t^6)\,dt \\
&= \frac{1}{3} - 1 + \frac{6}{5} - \frac{2}{3} + \frac{1}{7} = \frac{1}{105}, \\
\langle w_1, w_2\rangle &= \int_0^1 (t^3 - 4t^4 + 6t^5 - 4t^6 + t^7)\,dt \\
&= \frac{1}{4} - \frac{4}{5} + 1 - \frac{4}{7} + \frac{1}{8} = \frac{1}{280}, \\
\langle w_2, w_2\rangle &= \int_0^1 (t^4 - 4t^5 + 6t^6 - 4t^7 + t^8)\,dt \\
&= \frac{1}{5} - \frac{2}{3} + \frac{6}{7} - \frac{1}{2} + \frac{1}{9} = \frac{1}{630}.
\end{aligned}$$

Multiplying out the polynomials also gives

$$\begin{aligned}
\langle s, w_1\rangle &= \int_0^1 (2t^7 - 3t^6 + t^5 - t^2 + t)\,dt = \frac{2}{8} - \frac{3}{7} + \frac{1}{6} - \frac{1}{3} + \frac{1}{2} = \frac{13}{84}, \\
\langle s, w_2\rangle &= \int_0^1 (2t^8 - 3t^7 + t^6 - t^3 + t^2)\,dt = \frac{2}{9} - \frac{3}{8} + \frac{1}{7} - \frac{1}{4} + \frac{1}{3} = \frac{37}{504}.
\end{aligned}$$

The orthogonality conditions are therefore

$$\begin{pmatrix} 1/105 & 1/280 \\ 1/280 & 1/630 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 13/84 \\ 37/504 \end{pmatrix}.$$

Multiplication by 2520 gives the system

$$24a + 9b = 390, \qquad 9a + 4b = 185.$$

Multiply the first equation by 4 and the second by 9, and subtract:

$$(96a + 36b) - (81a + 36b) = 1560 - 1665 \quad\Longrightarrow\quad 15a = -105.$$

Hence $a = -7$; the second equation gives $-63 + 4b = 185$, so $b = 62$. We conclude that

$$\begin{aligned}
P_W(s) &= -7x(x-1)^2 + 62x^2(x-1)^2 \\
&= 62x^4 - 131x^3 + 76x^2 - 7x.
\end{aligned}$$

The polynomial belongs to W by the factorization in the first line, and the two inner product equations ensure that its difference from s is orthogonal to all of W. These are exactly the two conditions that uniquely characterize the orthogonal projection.

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EXERCISE 14.18 CHALLENGE

Let $v_1, \ldots, v_k$ be independent in a real inner product space, and define

$$u_i = \sum_{j=1}^{i} a_{ji} v_j, \qquad a_{ii} > 0.$$

Please prove that normalized Gram-Schmidt produces the same orthonormal list from $(u_1, \ldots, u_k)$ and $(v_1, \ldots, v_k)$. Please then allow any $a_{ii} \neq 0$ and please describe the relation between the two outputs.

WORKED SOLUTION

For each i, the first i vectors of the two lists span the same subspace. One inclusion follows from the definition; the reverse follows by successively solving for v_i using $a_{ii} \neq 0$. In particular the u_i are independent. Let e_i be the output for the v_i, with residual

$$r_i = v_i - \sum_{j<i} \langle v_i, e_j \rangle e_j, \qquad e_i = r_i / \|r_i\|.$$

Both Gram-Schmidt runs subtract the orthogonal projection onto the same previous span. The terms $a_{ji} v_j$ for $j < i$ lie in that span and disappear from the residual of u_i. Its residual is therefore $a_{ii} r_i$. If $a_{ii} > 0$, normalization gives exactly e_i. For arbitrary nonzero diagonal coefficients, normalization instead gives

$$f_i = \frac{a_{ii} r_i}{|a_{ii}| \|r_i\|} = \frac{a_{ii}}{|a_{ii}|} e_i.$$

Earlier sign changes do not affect orthogonal projections or previous spans, so this relation holds at every step.

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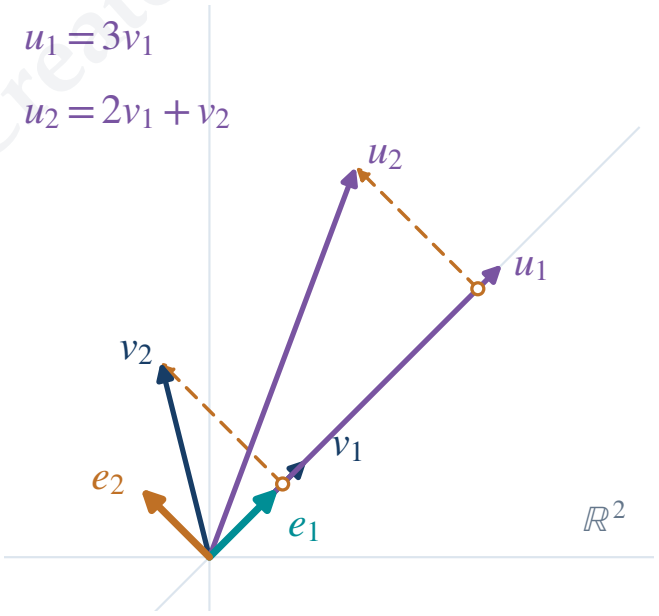


Figure 14.6. Here $u_1 = 3v_1$ and $u_2 = 2v_1 + v_2$. Both lists give the same unit vectors e_1, e_2 under Gram--Schmidt. Dashed segments show the second-step residuals.

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TUTORIAL 15

Completion: Orthogonal Matrices, the Spectral Theorem, and SVD

15.1 Orthogonal Matrices and the Spectral Theorem

DEFINITION 15.1

A real square matrix Q is orthogonal if $Q^TQ = I$. Equivalently, its columns form an orthonormal basis, or $Q^{-1} = Q^T$. In particular, $\|Qx\| = \|x\|$ for the standard inner product. For a complex matrix, $A^* = \overline{A}^T$ denotes the conjugate transpose. A matrix U is unitary if $U^*U = I$. The adjoint map satisfies $\langle Tv, w\rangle = \langle v, T^*w\rangle$ and is represented by A^* in an orthonormal basis.

PROPOSITION 15.1

On a finite-dimensional complex inner product space, every self-adjoint map has an orthonormal eigenbasis, and all its eigenvalues are real. This is the spectral theorem for self-adjoint maps. Thus for every Hermitian matrix $A = A^*$, there are a unitary matrix U and a real diagonal matrix D such that

$$U^*AU = D, \qquad A = UDU^*.$$

Column j of U is an eigenvector of A belonging to d_{jj}. Over $\mathbb{R}$, every real symmetric matrix $A = A^T$ has a real orthonormal eigenbasis; hence there is an orthogonal matrix Q such that

$$Q^TAQ = D, \qquad A = QDQ^T.$$

Diagonalization by an orthogonal matrix is called **orthogonal diagonalization**; over the complex numbers, we use the term **unitary diagonalization**.

RECALL AND TOOLS

To find an orthogonal diagonalization of a real symmetric matrix, find a basis of each eigenspace, apply Gram-Schmidt within each eigenspace as needed, and normalize. Eigenvectors of a symmetric matrix belonging to distinct eigenvalues are already orthogonal. Place the unit vectors in the columns of Q, and place the corresponding eigenvalues in the same order on the diagonal of D. Over the complex numbers, the same construction applies to a Hermitian matrix, using U^* in place of Q^T.

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15.2 Linear Maps on Inner Product Spaces

EXERCISE 15.1

Let $(V, \langle\cdot,\cdot\rangle)$ be a real inner product space of dimension $n \geq 3$. Let $v, w \in V$ be orthonormal: that is, $\langle v, v\rangle = \langle w, w\rangle = 1$ and $\langle v, w\rangle = 0$. Define the linear map $T : V \to V$ by

$$T(u) = \langle u, v\rangle\, w + \langle u, w\rangle\, v.$$

1. Please find all eigenvalues of T, all eigenvectors, and the characteristic polynomial of T.
2. Please prove that $\|T(u)\| \leq \|u\|$ for every $u \in V$, and please find all u for which equality holds.

WORKED SOLUTION

1. Start with the eigenvalue equation and compare components using an orthogonal decomposition:

   $$T(u) = \lambda u.$$

   Decompose each $u \in V$ using the orthogonal direct sum

   $$V = \text{span}\{v, w\} \oplus (\text{span}\{v, w\})^{\perp}.$$

   Here $\{v, w\}^{\perp}$ denotes the orthogonal complement of $\text{span}\{v, w\}$. Thus there are unique $\alpha, \beta \in \mathbb{R}$ and $y \in \{v, w\}^{\perp}$ such that

   $$u = \alpha v + \beta w + y.$$

   Now compute $T(u)$: since $y \perp v, w$, we have $\langle y, v\rangle = \langle y, w\rangle = 0$, so

   $$\langle u, v\rangle = \langle \alpha v + \beta w + y,\ v\rangle = \alpha, \qquad \langle u, w\rangle = \langle \alpha v + \beta w + y,\ w\rangle = \beta.$$

   Hence

   $$T(u) = \alpha w + \beta v.$$

   Substitute into the eigenvalue equation $T(u) = \lambda u$ to obtain

   $$\alpha w + \beta v = \lambda(\alpha v + \beta w + y).$$

   Compare components in the direct sum $\text{span}\{v, w\} \oplus \{v, w\}^{\perp}$. The component in $\{v, w\}^{\perp}$ on the left is zero, so necessarily

   $$\lambda y = 0.$$

   Consequently:

   - If $\lambda \neq 0$, then $y = 0$, so $u \in \text{span}\{v, w\}$.
   - If $\lambda = 0$, then $T(u) = 0$, and $\alpha w + \beta v = 0$ gives $\alpha = \beta = 0$, so $u = y \in \{v, w\}^{\perp}$.

   Now consider $\lambda \neq 0$. Then $u = \alpha v + \beta w$ and $y = 0$. The eigenvalue equation becomes

   $$\alpha w + \beta v = \lambda(\alpha v + \beta w).$$

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EXERCISE 15.1 CONTINUED

Comparing coefficients of v and w gives the system

$$\beta = \lambda\alpha, \qquad \alpha = \lambda\beta.$$

Substitution gives $\alpha = \lambda(\lambda\alpha) = \lambda^2\alpha$, so

$$(\lambda^2 - 1)\alpha = 0.$$

If $\alpha = 0$, then $\beta = \lambda\alpha = 0$ as well, giving $u = 0$, contrary to the definition of an eigenvector. Thus $\alpha \neq 0$, and

$$\lambda^2 - 1 = 0 \quad \Longleftrightarrow \quad \lambda \in \{1, -1\}.$$

Now determine the eigenvectors:

- For $\lambda = 1$, $\beta = \alpha$, so $u = \alpha(v + w)$; hence

$$V_1 = \text{span}\{v + w\}.$$

- For $\lambda = -1$, $\beta = -\alpha$, so $u = \alpha(v - w)$; hence

$$V_{-1} = \text{span}\{v - w\}.$$

- For $\lambda = 0$, we obtained $u \in \{v, w\}^\perp$, so

$$V_0 = \{v, w\}^\perp.$$

In particular, there are no other eigenvalues, because the eigenvalue equation forced $\lambda \in \{0, 1, -1\}$.

To compute the characteristic polynomial, choose an orthonormal basis of V of the form

$$B = (v, w, e_3, \ldots, e_n),$$

where $e_3, \ldots, e_n$ is an orthonormal basis of $\{v, w\}^\perp$. Then

$$T(v) = w, \qquad T(w) = v, \qquad T(e_i) = 0 \ \ (i = 3, \ldots, n).$$

The matrix representing T relative to B is therefore

$$[T]_B = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix}.$$

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EXERCISE 15.1 CONTINUED

Hence its characteristic polynomial is

$$p_T(\lambda) = \det(\lambda I - [T]_B) = \det \begin{pmatrix} \lambda & -1 & 0 & \cdots & 0 \\ -1 & \lambda & 0 & \cdots & 0 \\ 0 & 0 & \lambda & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda \end{pmatrix} = (\lambda^2 - 1)\lambda^{n-2}.$$

2. Set $\alpha := \langle u, v \rangle$ and $\beta := \langle u, w \rangle$, and define

$$s := \alpha v + \beta w \in \text{span}\{v, w\}.$$

Since v, w are orthonormal,

$$\|s\|^2 = \langle s, s \rangle = \alpha^2 + \beta^2 = \langle u, v \rangle^2 + \langle u, w \rangle^2 = \|T(u)\|^2.$$

Also,

$$\langle u, s \rangle = \alpha \langle u, v \rangle + \beta \langle u, w \rangle = \alpha^2 + \beta^2 = \|s\|^2.$$

By the Cauchy-Schwarz inequality,

$$|\langle u, s \rangle| \leq \|u\| \, \|s\|.$$

Since $|\langle u, s \rangle| = \|s\|^2$, we obtain

$$\|s\|^2 \leq \|u\| \, \|s\|.$$

If $s = 0$, then $\|T(u)\| = \|s\| = 0 \leq \|u\|$. If $s \neq 0$, divide by $\|s\|$ to obtain

$$\|T(u)\| = \|s\| \leq \|u\|.$$

To check equality, write $u = s + y$ with $y \in \{v, w\}^\perp$. By the Pythagorean identity,

$$\|u\|^2 = \|s\|^2 + \|y\|^2 = \|T(u)\|^2 + \|y\|^2.$$

Thus $\|T(u)\| = \|u\|$ if and only if $y = 0$, or equivalently $u \in \text{span}\{v, w\}$. This argument also includes $s = 0$, in which case equality requires $u = 0$.

EXERCISE 15.2

Let $n \geq 3$ be a natural number and let $a \in \mathbb{R}$. Define $A_n(a) \in M_n(\mathbb{R})$ by

$$(A_n(a))_{ij} = \begin{cases} 1, & i = j, \\ a, & i \neq j. \end{cases}$$

Recall the definition: a Hermitian matrix $H \in M_n(\mathbb{C})$ is positive semidefinite if $x^* H x \geq 0$ for every $x \in \mathbb{C}^n$, and positive definite if $x^* H x > 0$ for every $x \neq 0$.

1. For every a, please find all eigenvalues of $A_n(a)$ and their multiplicities.
2. Please determine for which values of a the matrix $A_n(a)$ is positive definite, and for which values it is positive semidefinite.

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EXERCISE 15.2 CONTINUED

3. Please find $\operatorname{rank}(A_n(a))$ and $\ker(A_n(a))$ in the two boundary cases where $A_n(a)$ is positive semidefinite but not positive definite.

In addition, let $B \in M_n(\mathbb{C})$ and please write $B^* = \overline{B}^T$. Define

$$H := B^*B + I.$$

(4) Please prove that H is positive definite and hence invertible.

(5) Please prove that, for every $x \in \mathbb{C}^n$,

$$x^*H^{-1}x \leq \|x\|^2,$$

and also prove that

$$\det(H) \geq 1,$$

with equality if and only if $B = 0$.

WORKED SOLUTION

Write $e = (1, 1, \ldots, 1)^T \in \mathbb{R}^n$. We have the decomposition

$$A_n(a) = (1-a)I + a\,ee^T,$$

because every entry of ee^T equals 1.

1. Note that $ee^Te = (e^Te)\,e = n\,e$, so

$$A_n(a)e = (1-a)e + a\,ee^Te = (1-a)e + an\,e = \big(1 + (n-1)a\big)e,$$

Thus e is an eigenvector with eigenvalue $\lambda_1 = 1 + (n-1)a$.
Now let $x \in \mathbb{R}^n$ satisfy $e^Tx = 0$. Then $ee^Tx = e(e^Tx) = 0$, so

$$A_n(a)x = (1-a)x + a \cdot 0 = (1-a)x.$$

Consequently, every $x \in e^\perp \setminus \{0\}$ is an eigenvector with eigenvalue $\lambda_2 = 1 - a$. Since $\mathbb{R}^n = \operatorname{span}\{e\} \oplus e^\perp$, these are all eigenvalues. If $a \neq 0$, then $\lambda_1 \neq \lambda_2$, so

$$\begin{aligned} &\lambda_1 = 1 + (n-1)a && \text{with algebraic and geometric multiplicity } 1, \\ &\lambda_2 = 1 - a && \text{with algebraic and geometric multiplicity } n-1. \end{aligned}$$

If $a = 0$, then $A_n(0) = I$, so its only eigenvalue is 1, with algebraic and geometric multiplicity n.

2. The matrix $A_n(a)$ is real symmetric and hence Hermitian. It is therefore positive definite exactly when all its eigenvalues are strictly positive, and positive semidefinite exactly when all its eigenvalues are nonnegative. Thus:

$$A_n(a) \succ 0 \iff \begin{cases} 1-a > 0, \\ 1 + (n-1)a > 0, \end{cases} \iff -\frac{1}{n-1} < a < 1,$$

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EXERCISE 15.2 CONTINUED

and

$$A_n(a) \succeq 0 \iff \begin{cases} 1-a \geq 0, \\ 1+(n-1)a \geq 0, \end{cases} \iff -\frac{1}{n-1} \leq a \leq 1.$$

3. If $a = 1$, then $\lambda_2 = 0$ has multiplicity $n-1$, and $\lambda_1 = n$. Hence

$$\operatorname{rank}(A_n(1)) = 1, \qquad \ker(A_n(1)) = e^{\perp} = \{x \in \mathbb{R}^n : \sum_{i=1}^{n} x_i = 0\}.$$

If $a = -\frac{1}{n-1}$, then $\lambda_1 = 0$ has multiplicity 1, and $\lambda_2 = 1 - a = \frac{n}{n-1} > 0$ has multiplicity $n-1$. Hence

$$\operatorname{rank}\Big(A_n\big(-\tfrac{1}{n-1}\big)\Big) = n-1, \qquad \ker\Big(A_n\big(-\tfrac{1}{n-1}\big)\Big) = \operatorname{span}\{e\}.$$

Now turn to $H = B^*B + I$.

(4) Let $x \in \mathbb{C}^n$. Then

$$x^*Hx = x^*(B^*B)x + x^*Ix = \|Bx\|^2 + \|x\|^2.$$

Hence $x^*Hx \geq 0$ for every x, and $x^*Hx > 0$ for every $x \neq 0$, since $\|x\|^2 > 0$. Thus $H \succ 0$, so H is invertible.

(5) From $H = I + B^*B$, we obtain $H - I = B^*B \succeq 0$; thus $x^*Hx \geq x^*Ix = \|x\|^2$ for every x. Let $\lambda_{\min}$ be the smallest eigenvalue of H. The preceding inequality gives $\lambda_{\min} \geq 1$. Every eigenvalue of H^{-1} is therefore at most 1, so for every x,

$$x^*H^{-1}x \leq \|x\|^2.$$

For the determinant inequality, write the eigenvalues of H as $\lambda_1, \ldots, \lambda_n$. Since $H \succ 0$, we have $\lambda_i > 0$; since $\lambda_{\min} \geq 1$, in fact $\lambda_i \geq 1$ for every i. Hence

$$\det(H) = \prod_{i=1}^{n} \lambda_i \geq 1.$$

Equality holds if and only if $\lambda_i = 1$ for every i: equivalently, $H = I$, or $B^*B = 0$. Finally, $B^*B = 0$ implies $\|Bx\|^2 = x^*B^*Bx = 0$ for every x. Thus $Bx = 0$ for all x, so $B = 0$.

EXERCISE 15.3 CHALLENGE

Let $A, B \in M_n(\mathbb{R})$ be symmetric matrices.

1. Please prove that there is an orthogonal matrix Q for which both Q^TAQ and Q^TBQ are diagonal if and only if $AB = BA$.

2. Please find such a matrix for

$$A = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 5 & 2 & 0 \\ 2 & 5 & 0 \\ 0 & 0 & 6 \end{pmatrix},$$

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EXERCISE 15.3 CHALLENGE CONTINUED

and also give the diagonal matrix representing AB in the same basis.

WORKED SOLUTION

1. If $Q^TAQ = D_A$ and $Q^TBQ = D_B$ are diagonal, then $A = QD_AQ^T$ and $B = QD_BQ^T$. Since diagonal matrices commute,

$$AB = QD_AD_BQ^T = QD_BD_AQ^T = BA.$$

Conversely, suppose $AB = BA$. By the spectral theorem, $\mathbb{R}^n$ is the orthogonal direct sum of the eigenspaces of A. If $x \in E_\lambda(A)$, then

$$A(Bx) = B(Ax) = B(\lambda x) = \lambda Bx,$$

so $Bx \in E_\lambda(A)$. Thus B preserves every eigenspace of A. On each such space, the restriction of B is self-adjoint for the induced inner product: for all x, y in that space,

$$\langle Bx, y\rangle = \langle x, By\rangle,$$

because $B^T = B$. By the spectral theorem on that space, choose an orthonormal basis of eigenvectors of B. Each vector is also an eigenvector of A belonging to λ. Combining these bases over all eigenvalues of A gives an orthonormal basis of $\mathbb{R}^n$ common to both matrices. Placing it in the columns of Q gives the required diagonalizations, including the case of repeated eigenvalues.

2. Choose

$$q_1 = \frac{1}{\sqrt{2}}(1,1,0)^T, \qquad q_2 = \frac{1}{\sqrt{2}}(1,-1,0)^T, \qquad q_3 = (0,0,1)^T.$$

Their norms are 1 and their pairwise inner products are 0. Furthermore,

$$Aq_1 = -q_1, \quad Aq_2 = -q_2, \quad Aq_3 = 4q_3, \qquad Bq_1 = 7q_1, \quad Bq_2 = 3q_2, \quad Bq_3 = 6q_3.$$

Thus, for

$$Q = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

we obtain

$$Q^TAQ = \operatorname{diag}(-1,-1,4), \qquad Q^TBQ = \operatorname{diag}(7,3,6),$$
$$Q^TABQ = \operatorname{diag}(-7,-3,24).$$

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EXERCISE 15.4 CHALLENGE

Let $A \in M_n(\mathbb{R})$ be symmetric and positive definite, meaning $x^T Ax > 0$ for every $x \neq 0$. Please prove that there is a unique symmetric positive definite matrix B satisfying $B^2 = A$.

WORKED SOLUTION

The spectral theorem gives an orthogonal matrix Q with

$$A = QDQ^T, \qquad D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

Every λ_j is positive, because a corresponding unit eigenvector q_j satisfies $\lambda_j = q_j^T Aq_j > 0$. Define

$$B = Q \operatorname{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n})Q^T.$$

It is symmetric and satisfies $B^2 = A$. For $x \neq 0$, put $y = Q^T x \neq 0$. Then $x^T Bx = \sum_j \sqrt{\lambda_j} y_j^2 > 0$. For uniqueness, let C be another symmetric positive definite square root. Since $CA = C^3 = AC$, it preserves every eigenspace $E_\lambda(A)$. Its restriction to that eigenspace is symmetric and positive definite, so it has an orthonormal eigenbasis. For a vector with $Cv = \mu v$, positivity gives $\mu > 0$, and

$$\mu^2 v = C^2 v = Av = \lambda v.$$

Thus $\mu = \sqrt{\lambda}$. On each eigenspace of A, both B and C act as multiplication by $\sqrt{\lambda}$. These eigenspaces span $\mathbb{R}^n$, proving $C = B$.

15.3 Singular Value Decomposition

NOTATION AND CONVENTIONS

In this part, $A \in M_{m\times n}(\mathbb{R})$ with $m, n \geq 1$, and all inner products are standard. A square Q is orthogonal if $Q^T Q = QQ^T = I$. Its columns form an orthonormal basis.

DEFINITION 15.2

A **singular value decomposition**, abbreviated SVD, is a factorization

$$A = U\Sigma V^T,$$

where $U \in M_m(\mathbb{R})$ and $V \in M_n(\mathbb{R})$ are orthogonal, and $\Sigma \in M_{m\times n}(\mathbb{R})$ has numbers

$$\sigma_1 \geq \cdots \geq \sigma_r > 0$$

in positions $(1,1), \ldots, (r,r)$ and zero elsewhere. Here $r = \operatorname{rank} A$; if $A = 0$, the positive list is empty and $\Sigma = 0$. The numbers σ_i are the **positive singular values**. When including zeros, we list $\min\{m, n\}$ singular values, with their multiplicities.

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RECALL AND TOOLS

Write the columns of U as $u_1, \dots, u_m$, and those of V as $v_1, \dots, v_n$. The factorization gives

$$Av_i = \sigma_i u_i \quad (1 \le i \le r), \qquad Av_i = 0 \quad (r < i \le n),$$

so

$$A = \sum_{i=1}^{r} \sigma_i u_i v_i^T.$$

For example, $x = \sum_{i=1}^{n} c_i v_i$ is mapped to

$$Ax = \sum_{i=1}^{r} \sigma_i c_i u_i.$$

The basis V describes directions in the domain, and U describes directions in the codomain. Thus $U^T AV = \Sigma$. These are two separate basis choices, even when A is square; similarity uses the same basis change on both sides.

PROPOSITION 15.2

Every real matrix admits an SVD. Indeed, $A^T A$ is symmetric, and for every $x \in \mathbb{R}^n$,

$$x^T A^T Ax = (Ax)^T (Ax) = \|Ax\|^2 \ge 0.$$

The spectral theorem gives an orthonormal eigenbasis v_i with nonnegative eigenvalues λ_i. Order the positive eigenvalues first. For each $\lambda_i > 0$, set

$$\sigma_i = \sqrt{\lambda_i}, \qquad u_i = \frac{Av_i}{\sigma_i}.$$

If $\lambda_i, \lambda_j > 0$, then

$$u_i^T u_j = \frac{v_i^T A^T Av_j}{\sigma_i \sigma_j} = \frac{\lambda_j v_i^T v_j}{\sigma_i \sigma_j} = \begin{cases} 1 & i = j, \\ 0 & i \ne j. \end{cases}$$

If $\lambda_i = 0$, then $\|Av_i\|^2 = v_i^T A^T Av_i = 0$, so $Av_i = 0$. Thus $Av_1, \dots, Av_r$ are linearly independent and span $\operatorname{Im} A$, so there are exactly $r = \operatorname{rank} A \le m$ positive eigenvalues. Extend $u_1, \dots, u_r$ to an orthonormal basis of $\mathbb{R}^m$. The corresponding matrices U, V satisfy $AV = U\Sigma$, hence $A = U\Sigma V^T$.

The positive singular values are therefore the positive square roots of the positive eigenvalues of $A^T A$. The matrix $A^T A$ also has $n - r$ zero eigenvalues; if $n > m$, some of these are absent from the shorter list of $\min\{m, n\}$ singular values.

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EXAMPLE 15.1

Consider the rectangular matrix

$$A = \begin{pmatrix} 2 & 4 \\ -1 & -2 \\ 2 & 4 \end{pmatrix}.$$

The second column is twice the first, so its rank is 1. Also,

$$A^T A = \begin{pmatrix} 4+1+4 & 8+2+8 \\ 8+2+8 & 16+4+16 \end{pmatrix} = \begin{pmatrix} 9 & 18 \\ 18 & 36 \end{pmatrix}.$$

Hence

$$\det(tI - A^T A) = (t-9)(t-36) - 324 = t(t-45).$$

Its singular values are $3\sqrt{5}, 0$. Choose

$$v_1 = \frac{1}{\sqrt{5}}\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \qquad v_2 = \frac{1}{\sqrt{5}}\begin{pmatrix} -2 \\ 1 \end{pmatrix}.$$

Direct multiplication gives

$$Av_1 = \frac{1}{\sqrt{5}}(10, -5, 10)^T = 3\sqrt{5}\,\frac{1}{3}(2, -1, 2)^T, \qquad Av_2 = 0.$$

An orthonormal basis of the codomain is

$$u_1 = \frac{1}{3}(2, -1, 2)^T, \quad u_2 = \frac{1}{\sqrt{5}}(1, 2, 0)^T, \quad u_3 = \frac{1}{3\sqrt{5}}(-4, 2, 5)^T.$$

The numerator dot products are $2 - 2 = 0$, $-8 - 2 + 10 = 0$, and $-4 + 4 = 0$, and their norms are $3, \sqrt{5}, 3\sqrt{5}$. For $U = (u_1\ u_2\ u_3)$ and $V = (v_1\ v_2)$,

$$\Sigma = \begin{pmatrix} 3\sqrt{5} & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \qquad U\Sigma V^T = 3\sqrt{5}\,u_1 v_1^T = \begin{pmatrix} 2 & 4 \\ -1 & -2 \\ 2 & 4 \end{pmatrix}.$$

EXERCISE 15.5

Let $A = U\Sigma V^T \in M_{m\times n}(\mathbb{R})$ be an SVD of rank r. Define

$$B = \sum_{i=1}^{r} \frac{1}{\sigma_i} v_i u_i^T \in M_{n\times m}(\mathbb{R}).$$

For every $b \in \operatorname{Im} A$, please prove that Bb is the unique solution of $Ax = b$ with minimum norm. Please prove that B does not depend on the chosen decomposition.

WORKED SOLUTION

The vectors $u_1, \ldots, u_r$ form an orthonormal basis of the image. Thus, for $b \in \operatorname{Im} A$,

$$b = \sum_{i=1}^{r} \beta_i u_i, \qquad \beta_i = u_i^T b.$$

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EXERCISE 15.5 CONTINUED

Writing $x = \sum_{i=1}^{n} c_i v_i$ gives $Ax = \sum_{i=1}^{r} \sigma_i c_i u_i$. The equation $Ax = b$ is equivalent to $c_i = \beta_i/\sigma_i$ for $i \leq r$, with the other coefficients free. Every solution therefore has the form

$$x = \sum_{i=1}^{r} \frac{\beta_i}{\sigma_i} v_i + \sum_{i=r+1}^{n} c_i v_i = Bb + z, \qquad z \in \ker A.$$

The two sums are orthogonal, so

$$\|x\|^2 = \sum_{i=1}^{r} \frac{\beta_i^2}{\sigma_i^2} + \sum_{i=r+1}^{n} c_i^2 = \|Bb\|^2 + \|z\|^2.$$

The minimum occurs exactly when every $c_i = 0$ for $i > r$, or $z = 0$. Hence Bb is the unique minimum-norm solution. Now take an arbitrary $b \in \mathbb{R}^m$. There is a unique orthogonal decomposition $b = b_{\parallel} + b_{\perp}$, with $b_{\parallel} \in \operatorname{Im} A$ and $b_{\perp} \in (\operatorname{Im} A)^{\perp}$. The formula for B gives $Bb_{\perp} = 0$. Thus $Bb = Bb_{\parallel}$ is the unique minimum-norm solution of $Ax = b_{\parallel}$. Both the projection $b_{\parallel}$ and the unique solution are determined by A, b, so every SVD gives the same Bb. This holds for every b, proving that B itself is independent of the decomposition. The matrix B is called the Moore-Penrose pseudoinverse and is denoted by A^{+}. When $A = 0$, the sum is empty and $A^{+} = 0$, as required.

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EXERCISE 15.6

Let $A \in M_n(\mathbb{R})$ satisfy $A^T = -A$. Please prove that each positive singular value of A has even multiplicity. Please deduce that rank A is even.

WORKED SOLUTION

We have $A^T A = -A^2$. Let $\sigma > 0$ be a singular value, and set

$$E = \ker(A^T A - \sigma^2 I), \qquad d = \dim E.$$

The dimension d is the multiplicity of σ: the symmetric matrix $A^T A$ is diagonalizable, so the multiplicity of σ^2 equals its eigenspace dimension. For $x \in E$, we have $-A^2 x = \sigma^2 x$. Moreover,

$$-A^2(Ax) = A(-A^2 x) = \sigma^2 Ax,$$

so $Ax \in E$. Define $Jx = Ax/\sigma$ on E. Then

$$J^2 x = \frac{A^2 x}{\sigma^2} = -x.$$

In any basis of E, the real matrix $M = [J]$ satisfies $M^2 = -I_d$. Taking determinants yields

$$(\det M)^2 = \det(-I_d) = (-1)^d.$$

The left side is nonnegative, so d must be even. Every positive singular value therefore occurs an even number of times. The rank is the number of positive singular values counted with multiplicities, so it is a sum of even integers and is even.

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APPENDIX A

Further Topics: Polynomials, Adjoints, and Unitary Diagonalization

MOTIVATION AND GUIDING IDEA

Gram-Schmidt replaces a list of directions with an orthonormal list spanning the same space, simplifying coordinate and projection computations. Orthogonal diagonalization imposes an additional requirement: the basis vectors must also be eigenvectors. For diagonalization, therefore, begin with the eigenspaces and apply the algorithm within each one; applying it to an arbitrary basis does not generally produce eigenvectors.

NOTATION AND CONVENTIONS

- v_j is an input vector, w_j is the residual after subtracting projections, and e_i is a unit vector already obtained. Never divide by $\|w_j\|$ when $w_j = 0$.
- $\text{span}\{v_1, \ldots, v_j\}$ is the span of the input prefix; it equals the span of the output vectors obtained up to that step.
- $\mathbb{R}_n[x]$ and $\mathbb{C}_n[x]$ denote polynomials of degree at most n, including the zero polynomial. Evaluation values and integrals are computed according to the inner product specified in the question.
- $E_\lambda(T) = \ker(T - \lambda I)$ is the eigenspace belonging to λ; an eigenvector is a **nonzero** element of this space.
- T^* is the adjoint map; $A^* = \overline{A}^T$ is the conjugate transpose. The identity $[T^*]_B = [T]_B^*$ requires an orthonormal basis B.
- $T = T^*$ means self-adjointness; $TT^* = T^*T$ means normality.
- $Q^TQ = QQ^T = I$ describes a real orthogonal matrix; $U^*U = UU^* = I$ describes a complex unitary matrix. In particular, $Q^{-1} = Q^T$ and $U^{-1} = U^*$.
- In $Q^TAQ = D$ or $U^*AU = D$, column j of the change-of-basis matrix is an eigenvector belonging to d_{jj}. The column order must match the order of the eigenvalues on the diagonal.

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DEFINITION A.1

Conditions for the tools. In an inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, Gram-Schmidt applies to any finite list, including a dependent list: after obtaining $e_1, \ldots, e_r$, compute

$$w_j = v_j - \sum_{i=1}^{r} \langle v_j, e_i \rangle e_i.$$

If $w_j = 0$, skip that input vector; otherwise append $e_{r+1} = w_j / \|w_j\|$. The resulting list is an orthonormal basis of the input span. For a linearly independent input, no vector is skipped. All projection coefficients use the convention of linearity in the first argument.
If V is finite dimensional and $T : V \to V$ is linear, there is a unique linear map $T^* : V \to V$ satisfying $\langle Tv, w \rangle = \langle v, T^* w \rangle$ for all v, w. Over $\mathbb{C}$, a map is normal if and only if it has an orthonormal eigenbasis. If it is self-adjoint, all its eigenvalues are real. Over $\mathbb{R}$, every symmetric matrix admits an orthogonal diagonalization; normality alone does not suffice for a real orthogonal diagonalization. For example,

$$R = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}, \qquad RR^T = R^T R = 4I,$$

but $R^2 = -4I$, so R has no real eigenvector. When constructing an orthonormal eigenbasis, apply Gram-Schmidt **within each eigenspace** so that the linear combinations remain in that eigenspace.

A.1 Adjoints, Normality, and Diagonalization in an Orthonormal Basis

DEFINITION A.2

Let V be a finite-dimensional inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and let $T : V \to V$ be linear. The **adjoint** T^* is the unique linear map satisfying

$$\langle Tv, w \rangle = \langle v, T^* w \rangle \qquad \text{for all } v, w \in V.$$

In an orthonormal basis, if A represents T, then $A^* = \overline{A}^T$ represents T^*. We call T **self-adjoint** if $T^* = T$, and **normal** if $TT^* = T^*T$. A matrix $U \in M_n(\mathbb{C})$ is **unitary** if $U^*U = UU^* = I$; its columns form an orthonormal basis of $\mathbb{C}^n$. Over the reals, a square matrix Q is **orthogonal** if $Q^TQ = QQ^T = I$.

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A.2 Polynomials over the Complex Numbers

DEFINITION A.3

The notation $\mathbb{C}[X]$ denotes the polynomials in the variable X with complex coefficients:

$$p(X) = a_0 + a_1 X + \cdots + a_n X^n, \qquad a_j \in \mathbb{C}.$$

Two polynomials are equal when the coefficients of each power agree. Addition adds the coefficients of each power; multiplication by a complex number multiplies every coefficient. Polynomial multiplication expands brackets using $X^j X^k = X^{j+k}$ and collects equal powers. If $a_n \neq 0$, the degree is $\deg p = n$, and a_n is its leading coefficient. A monic polynomial has leading coefficient 1. The zero polynomial has every coefficient zero; we will not assign it a numerical degree here. The phrase *degree at most* n will also allow the zero polynomial.

NOTE

For nonzero polynomials p, q, we have $\deg(pq) = \deg p + \deg q$: the leading coefficient of the product is the product of the two nonzero leading coefficients. In particular, a product of nonzero polynomials is not the zero polynomial.

PROPOSITION A.1

For $p, d \in \mathbb{C}[X]$ with $d \neq 0$, there are unique polynomials q, r such that

$$p = dq + r, \qquad r = 0 \text{ or } \deg r < \deg d.$$

These are the quotient and remainder when dividing p by d. Long division repeatedly subtracts a multiple of d that cancels the leading term, so the degree decreases. For uniqueness, if $p = dq + r = dq' + r'$, then $d(q - q') = r' - r$. If $q - q' \neq 0$, the left side has degree at least $\deg d$, whereas the right side is zero or has smaller degree. Thus $q = q'$ and then $r = r'$.

PROPOSITION A.2

A number $a \in \mathbb{C}$ is a root of p when $p(a) = 0$. We have

$$p(a) = 0 \quad \Longleftrightarrow \quad p(X) = (X - a)q(X)$$

for some polynomial q.

Proof.
On division by $X - a$, the remainder is a constant r, so $p(X) = (X - a)q(X) + r$. Substitution at $X = a$ gives $r = p(a)$, proving both directions.

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PROPOSITION A.3

A nonzero polynomial of degree n has at most n distinct roots.

Proof.
For $n = 0$, the polynomial is a nonzero constant and has no roots. If there is a root a, write $p = (X - a)q$ with $\deg q = n - 1$. Every other root $b \neq a$ satisfies $0 = (b - a)q(b)$, so $q(b) = 0$. Induction gives at most $n - 1$ such roots, or at most n including a.

DEFINITION A.4

For $p \neq 0$ and an integer $m \geq 1$, a root a has multiplicity m if

$$p(X) = (X - a)^m q(X), \qquad q(a) \neq 0.$$

Thus exactly m factors $X - a$ can be removed. We do not assign a finite root multiplicity to the zero polynomial.

PROPOSITION A.4

The fundamental theorem of algebra. Every nonconstant complex polynomial has a complex root. Consequently, if $\deg p = n \geq 1$ and its leading coefficient is c, then

$$p(X) = c(X - \lambda_1) \cdots (X - \lambda_n).$$

In this list, each root occurs according to its multiplicity. We will use this theorem without proof.

NOTE

For $p(X) = \sum_{k=0}^{n} a_k X^k$, define $\overline{p}(X) = \sum_{k=0}^{n} \overline{a_k} X^k$ by conjugating the coefficients. Then $\overline{p}(\overline{z}) = \overline{p(z)}$. If p has real coefficients, then $\overline{p} = p$, so the conjugate $\overline{a}$ of every root a is also a root. Multiplicity is preserved as well: from $p = (X - a)^m q$ with $q(a) \neq 0$, coefficient conjugation gives

$$p(X) = (X - \overline{a})^m \overline{q}(X), \qquad \overline{q}(\overline{a}) = \overline{q(a)} \neq 0.$$

DEFINITION A.5

The formal derivative of $p(X) = \sum_{k=0}^{n} a_k X^k$ is

$$p'(X) = \sum_{k=1}^{n} k a_k X^{k-1}.$$

This is an operation on coefficients. The product rule $(fg)' = f'g + fg'$ holds: for $j, k \geq 1$, it is the identity $(X^j X^k)' = (j+k)X^{j+k-1} = jX^{j-1}X^k + X^j kX^{k-1}$; if one exponent is zero, the rule follows because a constant has derivative zero. Expansion and addition give the rule for all polynomials.

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PROPOSITION A.5

For $p \neq 0$, the root a has multiplicity at least 2 if and only if $p(a) = p'(a) = 0$.

Proof.
If $p(a) = 0$, write $p = (X - a)q$. The product rule gives

$$p' = q + (X - a)q', \qquad p'(a) = q(a).$$

Thus $p'(a) = 0$ is equivalent to q also being divisible by $X - a$, which means that p is divisible by $(X - a)^2$.

EXERCISE A.1

Please factor $P(X) = (X - 1)^4 + 4$ into linear factors over $\mathbb{C}$, and please find all its roots and their multiplicities.

WORKED SOLUTION

Set $Y = X - 1$. A difference of squares gives

$$Y^4 + 4 = (Y^2 + 2)^2 - (2Y)^2 = (Y^2 - 2Y + 2)(Y^2 + 2Y + 2).$$

Returning to X,

$$P(X) = (X^2 - 4X + 5)(X^2 + 1) = \big((X - 2)^2 + 1\big)(X^2 + 1).$$

Therefore

$$P(X) = (X - i)(X + i)(X - 2 - i)(X - 2 + i).$$

The roots are $i, -i, 2 + i, 2 - i$. They are distinct, so each has multiplicity 1.

EXERCISE A.2

Please find a real-coefficient polynomial of the smallest possible degree for which $1 + 2i$ is a root of multiplicity 2 and $p(0) = 50$. Please determine whether such a polynomial of smallest degree is unique.

WORKED SOLUTION

The conjugate $1 - 2i$ must also be a root of multiplicity 2. The two roots are distinct, so the degree is at least 4. In degree 4, the polynomial must be

$$p(X) = c(X - 1 - 2i)^2(X - 1 + 2i)^2 = c(X^2 - 2X + 5)^2.$$

The condition $p(0) = 50$ gives $25c = 50$, hence $c = 2$. Thus

$$p(X) = 2(X^2 - 2X + 5)^2 = 2X^4 - 8X^3 + 28X^2 - 40X + 50.$$

This polynomial satisfies all conditions. The root $1 + 2i$ has multiplicity exactly 2 because $1 + 2i \neq 1 - 2i$. The value at 0 determines c, so the polynomial of smallest degree is unique.

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EXERCISE A.3

Let $n \geq 0$ be an integer. Please find the remainder when X^{4n+3} is divided by $X^2 - 2X + 2$.

WORKED SOLUTION

The divisor has roots $\alpha = 1 + i$ and $\beta = 1 - i$. The remainder has the form $r(X) = aX + b$. Substituting the roots gives

$$a\alpha + b = \alpha^{4n+3}, \qquad a\beta + b = \beta^{4n+3}.$$

We compute

$$(1 \pm i)^2 = \pm 2i, \qquad (1 \pm i)^4 = -4, \qquad (1 \pm i)^3 = -2 \pm 2i.$$

Hence

$$a(1 + i) + b = (-4)^n(-2 + 2i), \qquad a(1 - i) + b = (-4)^n(-2 - 2i).$$

Subtracting gives $2ia = 4i(-4)^n$, so $a = 2(-4)^n$. Substitution in either equation gives $a + b = -2(-4)^n$, hence $b = -4(-4)^n$. The remainder is

$$r(X) = 2(-4)^n(X - 2).$$

It has degree less than 2 and the required values at both roots. Uniqueness of quotient and remainder gives the result.

EXERCISE A.4

Please find all $p \in \mathbb{C}[X]$ of degree at most 3 satisfying

$$p(i) = p'(i) = 0, \qquad p(-i) = 2, \qquad p'(-i) = 0.$$

WORKED SOLUTION

The conditions at i imply $p = (X - i)^2(aX + b)$ for some $a, b \in \mathbb{C}$. The value $p(-i) = 2$ gives

$$(-2i)^2(-ia + b) = 2 \quad \Longrightarrow \quad b - ia = -\frac{1}{2}.$$

Differentiation gives

$$p'(X) = 2(X - i)(aX + b) + a(X - i)^2.$$

Thus $p'(-i) = 0$ becomes

$$-4i(b - ia) - 4a = 0.$$

Substituting $b - ia = -1/2$ gives $2i - 4a = 0$, so $a = i/2$. Now $b - i(i/2) = -1/2$, hence $b = -1$. The unique polynomial is

$$p(X) = (X - i)^2 \left(\frac{i}{2}X - 1\right) = 1 + \frac{i}{2}(X^3 + 3X).$$

Indeed, $p(i) = 0$, $p(-i) = 2$, and $p'(X) = \frac{3i}{2}(X^2 + 1)$ vanishes at both points. Uniqueness also follows directly: the difference of two solutions would be divisible by $(X - i)^2(X + i)^2$, a degree-4 polynomial, although its degree is at most 3. The difference must therefore be zero.

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EXERCISE A.5

Let $p \in \mathbb{C}[X]$ have degree $n \geq 1$. Please prove that, except for at most $n-1$ values of $w \in \mathbb{C}$, the equation $p(z) = w$ has exactly n distinct solutions. Please find the exceptional values for $p(X) = X^3 - 3X$.

WORKED SOLUTION

The derivative p' has degree $n-1$, so it has at most $n-1$ distinct roots. Let

$$S = \{p(c) : c \in \mathbb{C},\ p'(c) = 0\}$$

be the set of possible exceptional values. Its number of elements is at most the number of roots of p', so $|S| \leq n-1$. If $w \notin S$, the polynomial $p - w$ has no repeated root: such a root c would satisfy both $p(c) = w$ and $p'(c) = 0$, contradicting the choice of w. The fundamental theorem of algebra gives n roots counted with multiplicity. They are all simple, hence n distinct solutions. Conversely, if $w \in S$, the polynomial $p-w$ has a repeated root and therefore fewer than n distinct roots. For the given example,

$$p'(X) = 3(X^2 - 1), \qquad p(1) = -2, \qquad p(-1) = 2.$$

The exceptional values are exactly $-2, 2$. Indeed,

$$p(X) - 2 = (X-2)(X+1)^2, \qquad p(X) + 2 = (X+2)(X-1)^2.$$

Each exceptional value has two distinct solutions; every other value has three.

EXERCISE A.6

Let $p(X) = X^n + a_{n-1}X^{n-1} + \cdots + a_0$ be monic, with $n \geq 1$, and set $a_n = 1$. Suppose all its roots lie on the unit circle. Please prove that

$$|a_0| = 1, \qquad a_k = a_0\overline{a_{n-k}} \quad (0 \leq k \leq n).$$

Please determine whether these coefficient conditions guarantee that all roots lie on the unit circle.

WORKED SOLUTION

Write $p(X) = (X-\lambda_1)\cdots(X-\lambda_n)$, listing each root according to its multiplicity, with $|\lambda_j| = 1$. The constant coefficient is

$$a_0 = (-1)^n \lambda_1 \cdots \lambda_n,$$

so $|a_0| = 1$. Define a new polynomial by reversing and conjugating the coefficients:

$$p^{\#}(X) = \sum_{k=0}^{n} \overline{a_{n-k}} X^k = (1 - \overline{\lambda_1}X)\cdots(1 - \overline{\lambda_n}X).$$

Since $\overline{\lambda_j}\lambda_j = 1$, each factor satisfies

$$1 - \overline{\lambda_j}X = -\overline{\lambda_j}(X - \lambda_j).$$

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EXERCISE A.6 CONTINUED

Consequently,

$$p^{\#}(X) = (-1)^n \overline{\lambda_1 \cdots \lambda_n}\, p(X) = \overline{a_0}\, p(X).$$

Comparison of coefficients of X^k gives $\overline{a_{n-k}} = \overline{a_0} a_k$. Multiplying by a_0 and using $a_0\overline{a_0} = 1$ gives the required identity. The conditions are not sufficient. For example,

$$p(X) = X^2 - \frac{5}{2}X + 1 = (X-2)\left(X - \frac{1}{2}\right)$$

satisfies them, but neither root lies on the unit circle.

EXERCISE A.7

Let V be a finite-dimensional complex inner product space, and let $T : V \to V$ be linear. Suppose every eigenvector of $T + T^*$ is also an eigenvector of $T - T^*$. Please prove that T is normal.

WORKED SOLUTION

Write

$$H := T + T^*, \qquad K := T - T^*.$$

We aim to find an orthonormal basis in which both maps act by scalar multiplication on every basis vector. First compute the adjoints:

$$H^* = (T + T^*)^* = T^* + (T^*)^* = T^* + T = H,$$

and

$$K^* = (T - T^*)^* = T^* - (T^*)^* = T^* - T = -K.$$

Thus H is self-adjoint. If $V = \{0\}$, the result is immediate. Otherwise, the spectral theorem provides an orthonormal basis $B = (e_1, \ldots, e_n)$ and real numbers $h_1, \ldots, h_n$ such that

$$He_j = h_j e_j \qquad (1 \le j \le n).$$

Every e_j is an eigenvector of H, so the assumption provides $k_j \in \mathbb{C}$ such that

$$Ke_j = k_j e_j.$$

The number k_j may depend on j; we need not assume that the eigenvalues of H are distinct. In the basis B, both matrices are diagonal:

$$[H]_B = D_H := \operatorname{diag}(h_1, \ldots, h_n), \qquad [K]_B = D_K := \operatorname{diag}(k_1, \ldots, k_n).$$

For every j,

$$HKe_j = H(k_j e_j) = k_j h_j e_j, \qquad KHe_j = K(h_j e_j) = h_j k_j e_j.$$

Scalars commute, so $HKe_j = KHe_j$ for every basis vector. Hence $HK = KH$ on all of V. Now recover T and T^*:

$$H + K = 2T, \qquad H - K = 2T^*, \qquad T = \frac{H+K}{2}, \quad T^* = \frac{H-K}{2}.$$

Compute the two products separately, without assuming in advance that the factors com-

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EXERCISE A.7 CONTINUED

mute:

$$\begin{aligned} TT^* &= \frac{1}{4}(H+K)(H-K) \\ &= \frac{1}{4}\big(HH - HK + KH - KK\big) \\ &= \frac{1}{4}\big(H^2 - HK + KH - K^2\big) \\ &= \frac{1}{4}(H^2 - K^2), \end{aligned}$$

where the last line uses $HK = KH$. Similarly,

$$\begin{aligned} T^*T &= \frac{1}{4}(H-K)(H+K) \\ &= \frac{1}{4}\big(HH + HK - KH - KK\big) \\ &= \frac{1}{4}\big(H^2 + HK - KH - K^2\big) \\ &= \frac{1}{4}(H^2 - K^2). \end{aligned}$$

Thus $TT^* = T^*T$, so T is normal.

We also make explicit the connection with unitary diagonalization and the order of the matrices. First choose a fixed orthonormal basis of V, write $A = [T]$ in that basis, and form U from the coordinate columns of $e_1, \dots, e_n$ in the same basis. Then U is unitary, and

$$U^*(A + A^*)U = D_H, \qquad U^*(A - A^*)U = D_K.$$

Adding the identities and dividing by 2 gives

$$U^*AU = D := \frac{1}{2}(D_H + D_K), \qquad A = UDU^*.$$

Thus $[T]_B = D$ and $[T^*]_B = D^*$. For diagonal matrices,

$$(DD^*)_{ij} = \sum_{\ell=1}^{n} d_{i\ell}\overline{d_{j\ell}} = \begin{cases} |d_{ii}|^2, & i = j, \\ 0, & i \neq j, \end{cases}$$

and

$$(D^*D)_{ij} = \sum_{\ell=1}^{n} \overline{d_{\ell i}}\, d_{\ell j} = \begin{cases} |d_{ii}|^2, & i = j, \\ 0, & i \neq j. \end{cases}$$

We can therefore check the equality again in matrix form:

$$\begin{aligned} AA^* &= (UDU^*)(UD^*U^*) = UD(U^*U)D^*U^* = U(DD^*)U^*, \\ A^*A &= (UD^*U^*)(UDU^*) = UD^*(U^*U)DU^* = U(D^*D)U^*. \end{aligned}$$

The two results agree. The basis that diagonalizes the self-adjoint part also diagonalizes the other part by the hypothesis, and consequently diagonalizes T itself.

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EXERCISE A.8

Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and $A \in M_n(\mathbb{F})$. For any matrix C, please write $C^* := \overline{C}^T$; over $\mathbb{R}$, $C^* = C^T$. Following the convention that the inner product is linear in its first argument, define

$$\langle x, y \rangle := y^* x = x^T \overline{y}.$$

1. Please prove that $\langle Av, w \rangle = \langle v, A^* w \rangle$ for all $v, w \in \mathbb{F}^n$.
2. Let $G \in M_n(\mathbb{F})$ be Hermitian and positive definite. Define

   $$\langle x, y \rangle_G := y^* G x.$$

   Please prove that this is an inner product, and please show that there is a unique matrix A^{*_G} satisfying

   $$\langle Av, w \rangle_G = \langle v, A^{*_G} w \rangle_G \quad \text{for all } v, w \in \mathbb{F}^n,$$

   and that $A^{*_G} = G^{-1} A^* G$.

WORKED SOLUTION

1. The identity $(A^* w)^* = w^* A$ gives

   $$\langle Av, w \rangle = w^* A v = (A^* w)^* v = \langle v, A^* w \rangle.$$

2. Linearity in the first argument follows from the linearity of matrix multiplication. Since $G^* = G$,

   $$\overline{\langle y, x \rangle_G} = \overline{x^* G y} = (x^* G y)^* = y^* G^* x = y^* G x = \langle x, y \rangle_G.$$

   Furthermore, $\langle x, x \rangle_G = x^* G x > 0$ for every $x \neq 0$, and $\langle 0, 0 \rangle_G = 0$. The matrix G is invertible: otherwise $Gx = 0$ for some $x \neq 0$ would imply $x^* G x = 0$. Set $B := G^{-1} A^* G$. Then

   $$B^* = (G^{-1} A^* G)^* = G A G^{-1},$$

   and hence

   $$\langle v, Bw \rangle_G = (Bw)^* G v = w^* B^* G v = w^* G A v = \langle Av, w \rangle_G.$$

   For uniqueness, if C also satisfies the identity, then $w^*(C^* G - GA)v = 0$ for all v, w. Choosing $v = e_j$ and $w = e_i$ shows that every entry of $C^* G - GA$ is zero. Thus

   $$C^* G = GA \implies C^* = GAG^{-1} \implies C = G^{-1} A^* G = B.$$

A.3 Unitary Diagonalization

NOTATION AND CONVENTIONS

The following exercises are over $\mathbb{C}$, and $A^* = \overline{A}^T$ is the conjugate transpose. A matrix U is unitary when $U^* U = U U^* = I$. A unitary diagonalization has the form $U^* A U = D$ with D diagonal. The inner product is linear in its first variable.

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PROPOSITION A.6

A matrix $A \in M_n(\mathbb{C})$ is unitarily diagonalizable if and only if it is normal, meaning $AA^* = A^*A$.

Proof.
If $A = UDU^*$, then $A^* = UD^*U^*$, so

$$AA^* = UDD^*U^*, \qquad A^*A = UD^*DU^*.$$

Diagonal matrices commute, giving $AA^* = A^*A$. Conversely, suppose A is normal, and set

$$H = \frac{A + A^*}{2}, \qquad K = \frac{A - A^*}{2i}.$$

We have $H^* = H$ and

$$K^* = \frac{-1}{2i}(A^* - A) = K,$$

so both are Hermitian. Compute both products:

$$HK = \frac{1}{4i}\big(A^2 - AA^* + A^*A - (A^*)^2\big),$$
$$KH = \frac{1}{4i}\big(A^2 + AA^* - A^*A - (A^*)^2\big).$$

Normality gives $HK = KH$. By the Hermitian spectral theorem, the space is the orthogonal direct sum of the eigenspaces of H. If $Hx = \lambda x$, then

$$H(Kx) = K(Hx) = \lambda Kx,$$

so K preserves each eigenspace of H. Its restriction to each such space is Hermitian, since $\langle Kx, y\rangle = \langle x, Ky\rangle$ there. Choose an orthonormal eigenbasis of K in each space. Their union is a common orthonormal eigenbasis of H, K, and hence of $A = H + iK$. Its basis matrix U is unitary and satisfies $U^*AU = D$ diagonal. For $n = 0$, the empty basis proves the statement immediately.

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EXERCISE A.9 CHALLENGE

Consider

$$A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}.$$

Please compute A^2. Please find a diagonalization $P^{-1}AP = D$, and please determine whether a unitary diagonalization also exists. Please determine whether $A^2 = I$ together with all eigenvalues on the unit circle forces A to be unitary.

WORKED SOLUTION

Direct multiplication gives

$$A^2 = \begin{pmatrix} 1 & 2-2 \\ 0 & 1 \end{pmatrix} = I_2.$$

Eigenvectors for $1, -1$ are $(1,0)^T$ and $(-1,1)^T$, respectively. Thus take

$$P = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \quad P^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Then

$$AP = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \qquad P^{-1}AP = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

However,

$$A^*A = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}, \qquad AA^* = \begin{pmatrix} 5 & -2 \\ -2 & 1 \end{pmatrix}.$$

They differ, so A is not normal and has no unitary diagonalization. In particular, $A^*A \neq I_2$, so A is not unitary despite $A^2 = I_2$ and its eigenvalues $1, -1$ lying on the unit circle. Its eigendirections are not orthogonal: $\langle (1,0), (-1,1) \rangle = -1$.

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EXERCISE A.10 CHALLENGE

Let $U \in M_n(\mathbb{C})$ be unitary, with $n \geq 1$.

1. Please prove that there is a unitary matrix S satisfying $S^2 = U$.
2. If U has n distinct eigenvalues, please find the number of unitary matrices S satisfying $S^2 = U$.
3. Please construct infinitely many distinct unitary matrices satisfying $S^2 = -I_2$.

WORKED SOLUTION

1. A unitary matrix is normal, so $U = V \operatorname{diag}(\lambda_1, \ldots, \lambda_n)V^*$ with V unitary. If $Uq_j = \lambda_j q_j$ and $q_j \neq 0$, then
$$\|q_j\| = \|Uq_j\| = |\lambda_j|\|q_j\|,$$
so $|\lambda_j| = 1$. Write $\lambda_j = e^{i\theta_j}$, and set
$$S = V \operatorname{diag}(e^{i\theta_1/2}, \ldots, e^{i\theta_n/2})V^*.$$
Every diagonal entry has modulus 1, giving $S^*S = SS^* = I$ and $S^2 = U$.

2. Every square root commutes with U, since $SU = S^3 = US$. Thus S preserves every eigenspace of U. Distinct eigenvalues make these spaces one-dimensional, so in the same eigenbasis
$$S = V \operatorname{diag}(\mu_1, \ldots, \mu_n)V^*, \qquad \mu_j^2 = \lambda_j.$$
Each nonzero λ_j has exactly two distinct complex square roots, both of modulus 1. The choices are independent, giving exactly 2^n such matrices.

3. For $0 \leq t < \pi$, let $v_t = (\cos t, \sin t)^T$ and let $P_t = v_t v_t^T$ be the orthogonal projection onto its span. We have $P_t^2 = P_t = P_t^*$. Set
$$S_t = i(2P_t - I_2) = i\begin{pmatrix} \cos(2t) & \sin(2t) \\ \sin(2t) & -\cos(2t) \end{pmatrix}.$$
Since $(2P_t - I_2)^2 = 4P_t^2 - 4P_t + I_2 = I_2$,
$$S_t^2 = -I_2, \qquad S_t^* S_t = (-i)(i)(2P_t - I_2)^2 = I_2.$$
The lines $\operatorname{span}\{v_t\}$ are distinct for $t \in [0, \pi)$, so their projections P_t and the matrices S_t are distinct. This gives an infinite family.

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EXERCISE A.11 CHALLENGE

Let $A, B \in M_n(\mathbb{C})$ be normal.

1. Please prove that there is a unitary Q for which both Q^*AQ and Q^*BQ are diagonal if and only if $AB = BA$.

2. Please find such a simultaneous diagonalization for

$$A = \operatorname{diag}(1+i, 1+i, -2i), \qquad B = \begin{pmatrix} 2 & i & 0 \\ -i & 2 & 0 \\ 0 & 0 & 1+i \end{pmatrix}.$$

Please also compute Q^*ABQ.

WORKED SOLUTION

1. If $A = QD_AQ^*$ and $B = QD_BQ^*$, then

$$AB = QD_AD_BQ^* = QD_BD_AQ^* = BA.$$

Conversely, suppose $AB = BA$. Unitarily diagonalize $A = VDV^*$, and set $C = V^*BV$. Entrywise, $DC = CD$ gives

$$(\lambda_i - \lambda_j)c_{ij} = 0.$$

Thus C is block diagonal, with one block per distinct eigenvalue of A. Normality of B gives $CC^* = C^*C$, so every block is normal. Unitarily diagonalize each block, and combine the changes into a block-diagonal unitary matrix W. Then W^*CW is diagonal, while $W^*DW = D$, since each block of D is scalar. The matrix $Q = VW$ gives both diagonalizations.

2. The diagonal matrix A is normal. Both blocks of B are normal: the first is Hermitian and the second is scalar. The matrices commute because the first block of A is $(1+i)I_2$. Choose

$$q_1 = \frac{1}{\sqrt{2}}(i, 1, 0)^T, \quad q_2 = \frac{1}{\sqrt{2}}(-i, 1, 0)^T, \quad q_3 = (0, 0, 1)^T.$$

Their norms are 1, and $\langle q_1, q_2 \rangle = (i \cdot i + 1)/2 = 0$; also q_3 is orthogonal to both. Direct multiplication gives

$$Bq_1 = 3q_1, \qquad Bq_2 = q_2, \qquad Bq_3 = (1+i)q_3.$$

For $Q = (q_1\ q_2\ q_3)$, we obtain

$$\begin{aligned} Q^*AQ &= \operatorname{diag}(1+i, 1+i, -2i), \\ Q^*BQ &= \operatorname{diag}(3, 1, 1+i), \\ Q^*ABQ &= \operatorname{diag}(3+3i, 1+i, 2-2i). \end{aligned}$$

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EXERCISE A.12 CHALLENGE

Let $A \in M_n(\mathbb{C})$ be normal, and suppose $\|A^2x\| = \|Ax\|$ for every $x \in \mathbb{C}^n$. Please prove that A^*A is an orthogonal projection matrix and that $\operatorname{Im} A = (\ker A)^\perp$. Please prove that the restriction of A to $\operatorname{Im} A$ is unitary. Please check whether normality can be dropped from the conclusion about A^*A.

WORKED SOLUTION

Write $A = UDU^*$ in an orthonormal eigenbasis. For a unit eigenvector q_j with eigenvalue λ_j, the assumption gives

$$|\lambda_j|^2 = \|A^2 q_j\| = \|Aq_j\| = |\lambda_j|.$$

Thus $|\lambda_j|$ is 0 or 1. With $\varepsilon_j = |\lambda_j|^2$,

$$A^*A = U \operatorname{diag}(\varepsilon_1, \ldots, \varepsilon_n) U^*, \qquad \varepsilon_j \in \{0, 1\}.$$

This matrix is Hermitian and satisfies $(A^*A)^2 = A^*A$, so it is an orthogonal projection. The kernel of A is spanned by the eigenvectors with $\lambda_j = 0$, and its image by the remaining eigenvectors: for $\lambda_j \neq 0$,

$$q_j = A(\lambda_j^{-1} q_j).$$

Orthonormality gives $\operatorname{Im} A = (\ker A)^\perp$. On the image, the restriction is diagonal with all entries of modulus 1, hence unitary. If the image is the zero space, the empty basis gives the assertion. Without normality, take

$$C = \begin{pmatrix} 1 & 3 \\ 0 & 0 \end{pmatrix}.$$

Then $C^2 = C$, so $\|C^2x\| = \|Cx\|$ for every x. However,

$$C^*C = \begin{pmatrix} 1 & 3 \\ 3 & 9 \end{pmatrix}, \qquad (C^*C)^2 = 10C^*C \neq C^*C.$$

Thus C^*C is not a projection.

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EXERCISE A.13 CHALLENGE

For a matrix $M = (m_{jk}) \in M_n(\mathbb{C})$, define its Frobenius norm by

$$\|M\|_F^2 = \sum_{j,k=1}^{n} |m_{jk}|^2 = \mathrm{tr}(M^*M).$$

Let $A \in M_n(\mathbb{C})$ be normal and invertible.

1. Please find the unitary matrix U minimizing $\|A - U\|_F$, and please prove that it is unique.
2. Please compute it and the minimum distance for

$$A = \frac{1}{2}\begin{pmatrix} -3+2i & 3+2i \\ 3+2i & -3+2i \end{pmatrix}.$$

Hint: please use an orthonormal eigenbasis of A, and please use $|w_{jj}| \le 1$ for every unitary matrix W.

WORKED SOLUTION

The Frobenius norm is unchanged by multiplication by unitary matrices. For example, if V is unitary,

$$\|V^*MV\|_F^2 = \mathrm{tr}(V^*M^*VV^*MV) = \mathrm{tr}(V^*M^*MV) = \mathrm{tr}(M^*M).$$

The last step uses $\mathrm{tr}(XY) = \mathrm{tr}(YX)$, obtained by summing the entries $\sum_{j,k} x_{jk}y_{kj}$ in either order. Write $A = VDV^*$ with $D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ and every $\lambda_j \neq 0$. For each unitary U, set $W = V^*UV$, which is also unitary. Then

$$\begin{aligned} \|A - U\|_F^2 &= \|D - W\|_F^2 \\ &= \sum_j |\lambda_j|^2 + n - 2\,\mathrm{Re}\sum_j \lambda_j \overline{w_{jj}} \\ &\ge \sum_j |\lambda_j|^2 + n - 2\sum_j |\lambda_j| \\ &= \sum_j (|\lambda_j| - 1)^2. \end{aligned}$$

Here $\sum_{j,k} |w_{jk}|^2 = n$ because each column of W is a unit vector; in particular $|w_{jj}| \le 1$. Equality is attained by

$$W_0 = \mathrm{diag}\left(\frac{\lambda_1}{|\lambda_1|}, \ldots, \frac{\lambda_n}{|\lambda_n|}\right), \qquad U_0 = VW_0V^*.$$

For uniqueness, every bound $\mathrm{Re}(\lambda_j\overline{w_{jj}}) \le |\lambda_j|$ must be an equality. As $\lambda_j \neq 0$, this forces $w_{jj} = \lambda_j/|\lambda_j|$, hence $|w_{jj}| = 1$. The squared moduli in that row sum to 1, so every other entry in the row is zero. Thus $W = W_0$ and $U = U_0$. In the numerical case, take

$$V = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad V^*AV = \mathrm{diag}(2i, -3).$$

Therefore

$$U_0 = V\,\mathrm{diag}(i, -1)V^* = \frac{1}{2}\begin{pmatrix} -1+i & 1+i \\ 1+i & -1+i \end{pmatrix},$$

and the distance is $\sqrt{(2-1)^2 + (3-1)^2} = \sqrt{5}$.

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A.4 Matrices and Linear Maps: Reference Table

NOTATION AND CONVENTIONS

In this table, V, W are finite-dimensional spaces over $\mathbb{F}$, with dimensions n, m and ordered bases B, C. For $T : V \to W$, write $A = [T]_C^B \in M_{m\times n}(\mathbb{F})$. Rows involving a subspace U assume $U \leq V$, and $[U]_B = \{[u]_B : u \in U\}$ denotes its coordinate set. Square-matrix properties refer to $T : V \to V$, using the same basis in domain and codomain. Inner-product rows are over $\mathbb{R}$ or $\mathbb{C}$; identities involving A^* use orthonormal bases, except for the row explicitly introducing the matrix G. The symbol I denotes an identity, and 0 denotes a zero vector, map or matrix as appropriate. The symbols ker, Im, Row and Col denote kernel, image, row space and column space. For the zero space the basis is empty, rank and nullity are 0, and the empty matrix has determinant 1 and trace 0.

Concept	Maps and spaces	Matrix form
Vector and coordinates	$v = \sum_j x_j b_j$; the coordinates are its coefficients in the basis B.	$[v]_B = (x_1, \ldots, x_n)^T$.
Linear combination	A vector $\sum_j \alpha_j v_j$ with scalars $\alpha_j \in \mathbb{F}$.	If the columns of M are $[v_j]_B$, then $[\sum_j \alpha_j v_j]_B = M\alpha$.
Span	$\operatorname{span}\{v_j\}$ is the set of all linear combinations of the list.	The column space of its coordinate matrix.
Linear independence	$\sum_j \alpha_j v_j = 0$ forces every $\alpha_j = 0$.	The columns of M are independent exactly when $\ker M = \{0\}$.
Basis	An independent list spanning the space.	The coordinate matrix of a basis of $\mathbb{F}^n$ is square and invertible.
Dimension	$\dim V$ is the number of vectors in any basis of V.	$\dim \mathbb{F}^n = n$; $\dim M_{m\times n}(\mathbb{F}) = mn$.
Linear map	$T(\alpha v + \beta w) = \alpha T(v) + \beta T(w)$.	$[T(v)]_C = A[v]_B$.
Representing matrix	Column j gives the coordinates of $T(b_j)$ in C.	$A = [T]_C^B$. $A_{\cdot j} = [T(b_j)]_C$.
Composition	$S \circ T$ applies T first and then S.	$[S \circ T]_D^B = [S]_D^C [T]_C^B$; the intermediate bases must agree.
Kernel	$\ker T = \{v \in V : T(v) = 0\}$.	$[\ker T]_B = \ker A$; the solutions of $Ax = 0$.
Image	$\operatorname{Im} T = \{T(v) : v \in V\}$.	$[\operatorname{Im} T]_C = \operatorname{Col} A$.
Rank	$\operatorname{rank} T = \dim \operatorname{Im} T$.	$\operatorname{rank} A = \dim \operatorname{Col} A = \dim \operatorname{Row} A$.
Nullity	$\operatorname{nullity} T = \dim \ker T$.	$\operatorname{nullity} A = n - \operatorname{rank} A$.
Rank-nullity	$\dim \ker T + \dim \operatorname{Im} T = n$.	The number of free variables plus the rank equals the number of columns.
Injectivity	$T(v) = T(w)$ implies $v = w$; equivalently $\ker T = \{0\}$.	The columns of A are independent; $\operatorname{rank} A = n$.
Surjectivity	$\operatorname{Im} T = W$.	The columns of A span $\mathbb{F}^m$; $\operatorname{rank} A = m$.
Isomorphism	A linear map that is both injective and surjective.	$m = n$ and A is invertible.
Inverse	$T^{-1}T = I_V$ and $TT^{-1} = I_W$.	$[T^{-1}]_B^C = A^{-1}$; $AA^{-1} = A^{-1}A = I$. A square matrix is singular when it is not invertible, equivalently $\det A = 0$.

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Concept	Maps and spaces	Matrix form
Row space	The subspace spanned by the rows, viewed as vectors in $\mathbb{F}^n$.	Row $A \leq \mathbb{F}^n$; row operations preserve it.
Column space	The subspace spanned by the columns, in $\mathbb{F}^m$.	Col $A = \mathrm{Im}(x \mapsto Ax)$.
Row equivalence	A sequence of invertible row operations; applied to an augmented system, they preserve its solutions.	$B = EA$ with E invertible, a product of elementary matrices.
Change of basis	P converts B' coordinates into B coordinates.	$[v]_B = P[v]_{B'}$; column j is $[b'_j]_B$.
Similarity	Two matrices of the same endomorphism $T : V \to V$ in different bases.	$D = P^{-1}AP$ with P invertible.
Trace	A change of basis preserves the sum of diagonal entries.	$\mathrm{tr}\, A = \sum_j a_{jj}$; $\mathrm{tr}(XY) = \mathrm{tr}(YX)$ when both products are defined and square.
Determinant	A scalar attached to an endomorphism, independent of the chosen basis.	$\det A$; $\det(AB) = \det A \det B$. Over $\mathbb{R}$, its absolute value is the volume scale factor.
Minor and cofactor	For $n \geq 1$, $A_{\widehat{i},\widehat{j}}$ is obtained by deleting row i and column j.	$C_{ij} = (-1)^{i+j} \det A_{\widehat{i},\widehat{j}}$.
Classical adjugate	The transpose of the cofactor matrix; here $n \geq 1$.	$\mathrm{adj}(A) = (C_{ij})^T$; $A\,\mathrm{adj}(A) = \det(A)I$.
Eigenvalue and eigenvector	$T(v) = \lambda v$ with $v \neq 0$.	$Ax = \lambda x$ with $x \neq 0$; $\det(A - \lambda I) = 0$.
Eigenspace	$E_\lambda(T) = \ker(T - \lambda I_V)$.	$E_\lambda(A) = \ker(A - \lambda I_n)$.
Characteristic polynomial	Independent of basis; its roots in the field are the eigenvalues.	$p_A(t) = \det(tI_n - A)$; a monic polynomial of degree n.
Algebraic multiplicity	The multiplicity of λ as a root of the characteristic polynomial.	If $p_A(t) = (t - \lambda)^r q(t)$ and $q(\lambda) \neq 0$, the multiplicity is r.
Geometric multiplicity	The dimension of $E_\lambda(T)$.	$\dim \ker(A - \lambda I)$; between 1 and the algebraic multiplicity for an eigenvalue λ.
Diagonalization	Existence of a basis of eigenvectors of T.	$P^{-1}AP = D$ diagonal. Equivalent to splitting of the characteristic polynomial over the field and equality of both multiplicities at every eigenvalue.
Cayley-Hamilton	Substituting the endomorphism into its characteristic polynomial gives the zero map.	$p_A(A) = 0$; a constant term c becomes cI_n.
Minimal polynomial	The monic polynomial of least degree annihilating the endomorphism.	$m_A(A) = 0$ and $m_A \mid p_A$; for $n \geq 1$.
Nilpotence	$T^k = 0$ for some $k \geq 1$.	$A^k = 0$ for some $k \geq 1$; every eigenvalue is zero.

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Concept	Maps and spaces	Matrix form
Idempotence	$T^2 = T$; the map acts as the identity on its image.	$A^2 = A$; projection onto the image along the kernel.
Involution	$T^2 = I_V$; applying the map twice recovers the input.	$A^2 = I_n$; $A^{-1} = A$.
Zero and identity	The zero map sends every vector to zero; I_V fixes every vector.	$0_{m\times n}$ has only zero entries. The diagonal of I_n consists of ones, and all other entries are zero.
Diagonal matrix	In the basis B, $T(b_j) = d_j b_j$ for every j.	$D = \operatorname{diag}(d_1, \ldots, d_n)$; every off-diagonal entry is zero.
Triangular matrix	For $1 \le k \le n$, upper triangular preserves every $\operatorname{span}\{b_1, \ldots, b_k\}$; lower triangular preserves every $\operatorname{span}\{b_k, \ldots, b_n\}$.	Upper: $a_{ij} = 0$ for $i > j$. Lower: $a_{ij} = 0$ for $i < j$.
Scalar matrix	$T(v) = \lambda v$ for every $v \in V$, for a fixed $\lambda \in \mathbb{F}$.	$A = \lambda I_n$. Its matrix is the same in every basis of V.
Invariant subspace	$U \le V$ satisfies $T(U) \subseteq U$. A basis of U can be extended to a basis of V.	In such a basis, partitioned as $k + (n-k)$ with $k = \dim U$: $\begin{pmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{pmatrix}$.
Elementary matrix	Represents one elementary row operation. A row-scaling factor must be nonzero; every such operation is invertible.	E is obtained from I_m by one operation. The product EA applies that operation to the rows of A.
Permutation matrix	Permutes the basis vectors: $T(e_j) = e_{\sigma(j)}$, where σ permutes $\{1, \ldots, n\}$.	Every row and column has one entry equal to 1 and all others zero. $P^{-1} = P^T$.
Transpose	Interchange the row and column indices.	$(A^T)_{ij} = a_{ji}$; $(AB)^T = B^T A^T$.
Conjugate transpose	Over $\mathbb{R}$ or $\mathbb{C}$, transpose and conjugate every entry.	$(A^*)_{ij} = \overline{a_{ji}}$; $(AB)^* = B^* A^*$.
Adjoint map	$\langle Tv, w\rangle_W = \langle v, T^* w\rangle_V$ for all v, w.	In orthonormal bases: $[T^*]_B^C = A^*$.
Inner product	Linear in its first variable, conjugate symmetric and positive definite.	In an orthonormal basis: $\langle v, w\rangle = [w]^*[v]$.
Inner-product matrix	In a basis $B = (b_j)$, set $G_{ij} = \langle b_j, b_i\rangle$. Then $\langle v, w\rangle = [w]_B^* G [v]_B$.	In the same basis, for $T : V \to V$: $[T^*]_B = G^{-1} A^* G$.
Induced norm	$\|v\| = \sqrt{\langle v, v\rangle}$.	In an orthonormal basis: $\|v\|^2 = \sum_j |x_j|^2$.
Orthogonality	$v \perp w$ means $\langle v, w\rangle = 0$.	In an orthonormal basis: $[w]^*[v] = 0$.
Orthonormal basis	$\langle e_i, e_j\rangle = \delta_{ij}$; $\delta_{ij} = 1$ if $i = j$, and zero otherwise.	The basis matrix Q satisfies $Q^* Q = I$.
Orthogonal complement	$U^\perp = \{v : \langle v, u\rangle = 0 \text{ for all } u \in U\}$.	If the columns of M span U in orthonormal coordinates, then $[U^\perp]_B = \ker M^*$.

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Concept	Maps and spaces	Matrix form
Orthogonal projection	$P_U v \in U$ and $v - P_U v \in U^\perp$.	$P^2 = P = P^*$; if the columns of Q are an orthonormal basis of U, then $P = QQ^*$.
Gram-Schmidt	Subtract projections onto preceding directions and normalize each nonzero residual.	For independent input: $w_k = v_k - \sum_{j<k} \langle v_k, e_j \rangle e_j$; $e_k = w_k / \|w_k\|$.
Linear isometry	$\langle Tv, Tw \rangle = \langle v, w \rangle$ for all v, w.	In orthonormal bases: $A^*A = I_n$, even when A is rectangular.
Unitarity	An invertible inner-product-preserving endomorphism over $\mathbb{C}$.	$A^*A = AA^* = I_n$; $A^{-1} = A^*$.
Orthogonal matrix	An invertible isometry of a real inner product space.	$A^TA = AA^T = I_n$; $A^{-1} = A^T$.
Self-adjoint and Hermitian	$T = T^*$; every eigenvalue is real.	$A = A^*$ in an orthonormal basis. Over $\mathbb{R}$: $A = A^T$.
Symmetry	The algebraic matrix condition $A^T = A$. Over $\mathbb{R}$, it represents self-adjointness in an orthonormal basis.	Over $\mathbb{C}$, $A^T = A$ differs from $A^* = A$.
Skew-adjointness	$T^* = -T$; over $\mathbb{C}$ every eigenvalue is purely imaginary.	$A^* = -A$ in an orthonormal basis.
Skew-symmetry	The matrix condition $A^T = -A$; in characteristic different from 2, the diagonal is zero.	Over $\mathbb{R}$, in an orthonormal basis, it represents skew-adjointness.
Normality	$TT^* = T^*T$.	$AA^* = A^*A$ in an orthonormal basis; over $\mathbb{C}$, equivalent to unitary diagonalizability.
Positive definiteness	$T = T^*$ and $\langle Tv, v \rangle > 0$ for all $v \neq 0$.	$A = A^*$ and $x^*Ax > 0$ for all $x \neq 0$; every eigenvalue is positive.
Positive semidefiniteness	$T = T^*$ and $\langle Tv, v \rangle \geq 0$ for all v.	$A = A^*$ and $x^*Ax \geq 0$ for all x; every eigenvalue is nonnegative.
Unitary diagonalization	Existence of an orthonormal eigenbasis in a complex inner product space.	$U^*AU = D$ diagonal with U unitary; equivalent to normality.
Orthogonal diagonalization	Existence of a real orthonormal eigenbasis.	$Q^TAQ = D$ real diagonal with Q orthogonal; equivalent to symmetry of $A \in M_n(\mathbb{R})$.
Frobenius norm	A norm on matrix space: the square root of the sum of squared moduli of all entries.	$\|A\|_F^2 = \sum_{i,j} \|a_{ij}\|^2 = \operatorname{tr}(A^*A)$.

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APPENDIX B

Logic and Proof Methods

B.1 Logic

This appendix introduces some foundations of mathematical logic. We will study propositions, quantifiers, and logical connectives, and see how they are used to construct proofs and assess mathematical claims.

B.1.1 Propositions

A proposition is a statement to which exactly one of the truth values ``true'' and ``false'' can be assigned unambiguously. Examples of propositions include:

- Rome is the capital of Italy. (A true proposition.)
- $3^2 = 8$. (A false proposition.)

Consider the statement that the number of grains of sand on Earth exceeds 10^{24}. We may not know its truth value, and checking it would be difficult. Nevertheless, under an agreed meaning of the statement, it is either true or false; our lack of knowledge does not itself prevent it from being a proposition.

Examples of expressions that are not propositions include:

- How are you? (A question is not a proposition.)
- Please close the door! (An instruction is not a proposition.)
- $x^3 > x$. (No truth value is determined until the meaning or quantification of x has been specified.)

The sentence ``The Mona Lisa is a beautiful painting'' expresses a subjective judgement. To treat it as a formal proposition, one would first need to specify a precise meaning of ``beautiful.'' In mathematical logic, we work with statements whose meanings and domains of discourse have been specified.

B.1.2 Quantifiers

- Consider ``He is the prime minister.'' Without specifying the referent, this does not determine a proposition. Similarly, $n > 4$ does not determine a truth value when n is unspecified. To bind variables in statements, we use two logical symbols called **quantifiers**.

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RECALL AND TOOLS

- **For every**, denoted by $\forall$. For example, ``Every natural number is greater than 4'' is written as
$$\forall n \in \mathbb{N}\ (n > 4).$$
- **There exists**, denoted by $\exists$. For example, ``There exists a natural number greater than 4'' is written as
$$\exists n \in \mathbb{N}\ (n > 4).$$

Specify the domain of each quantifier, either explicitly or through a clearly established context. The following example explains why this matters.

- The expression $\forall x\ x^3 \geq x$ is incomplete without an agreed domain. The operations and comparison may not even be defined for the objects under consideration. In contrast, the following are propositions because the domain is stated:
  - $\forall x \in \mathbb{N}\ x^3 \geq x$ is true, since $x^3 \geq x$ for every natural number x.
  - $\forall x \in \mathbb{Q}\ x^3 \geq x$ is false. Choosing $x = \frac{1}{3}$ gives $x^3 = \frac{1}{27} < \frac{1}{3} = x$, a counterexample.

EXERCISE **B.1**

Please write the following statement in mathematical notation: a set is contained in another set if and only if every element of the first set also belongs to the second set.

WORKED SOLUTION

$$A \subseteq B \iff \forall a \in A,\ a \in B.$$

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Logical Connectives: Conjunction, Disjunction, and Negation

Conjunction ($\wedge$): both conditions hold.

- Saying that ``our room is large and well lit'' means that ``our room is large'' *and* ``our room is well lit.'' The conjunction is true exactly when both statements are true. If either statement is false, the conjunction is false.

Truth table: The table gives the truth value of $A \wedge B$ for every possible pair of truth values of A, B.

A	B	$A \wedge B$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction ($\vee$): logical ``or.''

- $A \vee B$ is true when at least one of A, B is true, and false when both are false. This is **inclusive** ``or'': it permits both statements to be true.

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Truth table:

A	B	$A \vee B$
T	T	T
T	F	T
F	T	T
F	F	F

Negation ($\neg$): reversing a truth value.

- For example, the negation of ``Every multiple of 6 is a multiple of 3'' is ``Not every multiple of 6 is a multiple of 3,'' or equivalently, ``There exists a multiple of 6 that is not a multiple of 3.'' A proposition and its negation have opposite truth values: one is true exactly when the other is false.

Truth table:

A	$\neg A$
T	F
F	T

EXERCISE B.2

Please use logical quantifiers to write the following proposition and its negation: there exists a natural number M such that $n \leq M$ for every natural number n.

WORKED SOLUTION

The proposition is

$$\exists M \in \mathbb{N}\, \forall n \in \mathbb{N},\ n \leq M.$$

Its negation is

$$\forall M \in \mathbb{N}\, \exists n \in \mathbb{N},\ n > M.$$

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Implication ($\Rightarrow$) and Equivalence ($\Leftrightarrow$)

**Implication ($\Rightarrow$): ``if … then …."** Consider the statement ``If Tamar received a final grade of at least 75, then Tamar passed the course.'' Let A denote the assumption and B the conclusion.

- If A is true and B is true, the implication is true.
- If A is true and B is false, the implication is false.
- If A is false and B is true, the implication is true.
- If A is false and B is false, the implication is true.

This example illustrates the meaning of implication.

The implication is **false** only when A, the assumption, is true and B, the conclusion, is false. In every other case it is true.

Truth table:

A	B	$A \Rightarrow B$
T	T	T
T	F	F
F	T	T
F	F	T

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Example: The proposition

$$4 = 7 \Rightarrow 6 = 9$$

is **true**: its assumption is false, even though its conclusion is also false.

Important remarks. For a statement of the form ``If $P$, then $Q$,'' we ask whether $Q$ must hold whenever $P$ holds. Everyday language often uses ``if … then …'' causally (for example, ``If you eat, you will feel full''). Formal implication has the truth-table meaning given above; it need not express a causal connection.

Negating an implication. The implication fails exactly when P is true and Q is false:

$$\neg(P \Rightarrow Q) \equiv P \wedge \neg Q.$$

In words: The statement $P \Rightarrow Q$ says that whenever P is true, Q is true as well. To negate it, identify the case in which it fails: P is true but Q is false.

Equivalence ($\Leftrightarrow$): ``if and only if.''

- $P \Leftrightarrow Q$ is true exactly when P, Q have the same truth value: either both are true or both are false.

Truth table:

A	B	$A \Leftrightarrow B$
T	T	T
T	F	F
F	T	F
F	F	T

Proof Techniques

B.1.3 Proof Templates

NOTE

The following templates summarize proof methods that have already appeared and will be used throughout the course.

Universal Statements

RECALL AND TOOLS

A universal statement asserts that every object satisfying a property p also satisfies a property q. It can be written as an implication: if an object satisfies p, then it satisfies q.
To prove the statement, take an arbitrary object and assume only that it satisfies p. Show that q follows. Be careful not to impose additional assumptions on the object.
To disprove the statement, find an object satisfying p but not q. One such counterexample is enough to disprove a universal statement.

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EXERCISE B.3

Please prove that

$$\forall x \in \mathbb{R},\ (x > 3) \Rightarrow (x^2 > 9).$$

WORKED SOLUTION

Let $x \in \mathbb{R}$ satisfy $x > 3$. Since x is positive, multiply both sides of the inequality by x without reversing its direction:

$$x \cdot x > 3 \cdot x.$$

Hence

$$x^2 > 3x.$$

Also, $x > 3$ implies $3x > 9$. By transitivity,

$$x^2 > 9.$$

We have therefore shown that

$$(x > 3) \Rightarrow (x^2 > 9).$$

Because x was arbitrary, the conclusion holds universally:

$$\forall x \in \mathbb{R},\ (x > 3) \Rightarrow (x^2 > 9).$$

EXERCISE B.4

Please prove by contradiction: if $A \setminus B = \varnothing$, then $A \subseteq B$.

WORKED SOLUTION

Assume that $A \setminus B = \varnothing$ and, for a contradiction, that $A \not\subseteq B$. Then some $x \in A$ satisfies $x \notin B$, so $x \in A \setminus B$. This contradicts the assumption that the difference is empty. Therefore $A \subseteq B$.

EXERCISE B.5

Please disprove the following statement by a counterexample: $x^3 > x$ for every $x \in \mathbb{R}$.

WORKED SOLUTION

The statement is false. Take $x = \frac{2}{3}$. Then

$$\left(\frac{2}{3}\right)^3 = \frac{8}{27} < \frac{2}{3},$$

so the proposed strict inequality fails.

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